Computer Architecture

Lecture 24: On-Chip Networks & Router Design

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ETH Zürich

Fall 2025

12 December 2025

1st SAFARI Conference, 2025



Rooms: HG D16.2 & ETZ E7

https://youtube.com/live/QtS7v8aVGLQ

1st SAFARI Conference, 2025

Program at a Glance

Time	Speaker	Affiliation	Title	Location	Livestream
08:30	Minesh Patel	Rutgers University	Split Write DRAM: Reducing DRAM Access Latency by Mitigating Read-Write Interference	HG D16.2	YouTube
09.05	Abdullah Giray Yaglikci	CISPA	Making DRAM Available Again: Safely Mitigating RowHammer at Low Performance and Energy Overheads in Modern DRAM-based Systems	HG D16.2	YouTube
09:40	Yu (Lenny) Liang	Inria-Paris	Co-Designing Operating Systems and Processing-in-Memory for Next-Generation Mobile Devices	HG D16.2	YouTube
10:15		Coffee Break			
10:45	Can Firtina	University of Maryland, College Park	Enabling Real-Time Analysis of Human Genomes via New Algorithms and Architectures	HG D16.2	YouTube
11:20	Jawad Haj-Yahya	Rivos Inc.	Feeding the Beast: The Critical Challenges of Powering Next-Gen AI accelerators	HG D16.2	YouTube
11:55	Haiyu Mao	King's College London	From Microarchitectures to Systems: Designing Holistic Data-Centric Architectures for AI Applications	HG D16.2	YouTube
12:30		Lunch Break			
14:00	Juan Gómez Luna	NVIDIA	Accelerating Access to Data Storage and Services in the Age of AI	ETZ E7	YouTube
14:35	Lois Orosa	Galicia Supercomputing Center (CESGA)	Computer Architecture Research in the Galicia Supercomputing Center: Challenges and Opportunities	ETZ E7	YouTube
15:00	Christina Giannoula	Max Planck Institute for Software Systems (MPI-SWS)	Building the Machine Learning Software Stack for Processing-In-Memory Architectures	ETZ E7	YouTube
15:45		Coffee Break			
16:00	Geraldo F. Oliveira	SAFARI Research Group, ETH Zurich	New Tools, Programming Models, and System Support for Processing-in-Memory Architectures	ETZ E7	YouTube
16:35	Saugata Ghose	University of Illinois Urbana-Champaign	Multi-Domain Architectures for Processing-Using-Memory	ETZ E7	YouTube
17:10	Nastaran Hajinazar	Intel Labs	Virtual Memory Management: The XPU perspective	ETZ E7	YouTube
17:35	Gagandeep Singh	AMD	The Convergence of AI, Supercomputing, and Biology: A View from AMD	ETZ E7	YouTube

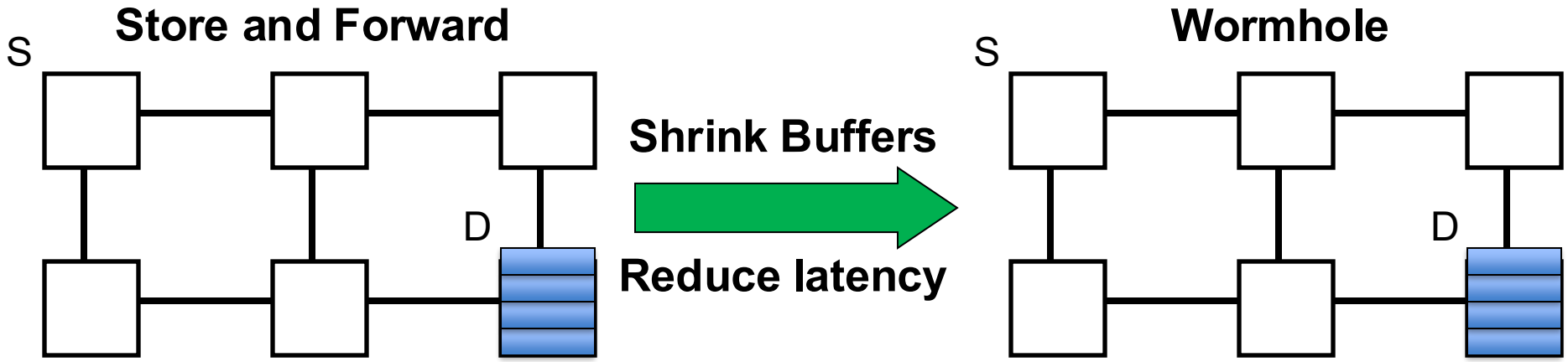
https://events.safari.ethz.ch/SafariConference25/doku.php?id=start

Recall: Interconnection Network Basics

- Topology
  - Specifies the way switches are wired
  - Affects routing, reliability, throughput, latency, building ease
- Routing (algorithm)
  - How does a message get from source to destination
  - Static or adaptive
- Buffering and Flow Control
  - What do we store within the routers & links?
    - Entire packets, parts of packets, etc?
  - How do we throttle during oversubscription?
  - Tightly coupled with routing strategy

Buffered Flow Control

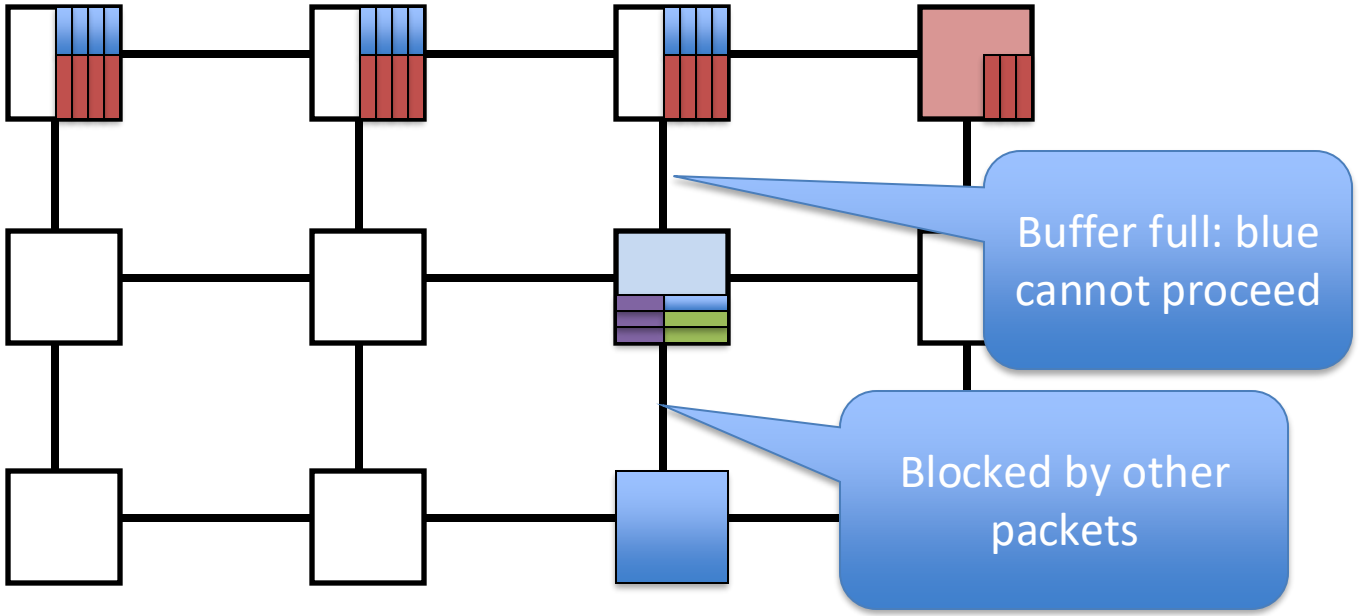
Review: Buffered Flow Control



Any other issues?

Head-of-Line Blocking

Use Virtual Channels



Recall: Communicating Buffer Availability

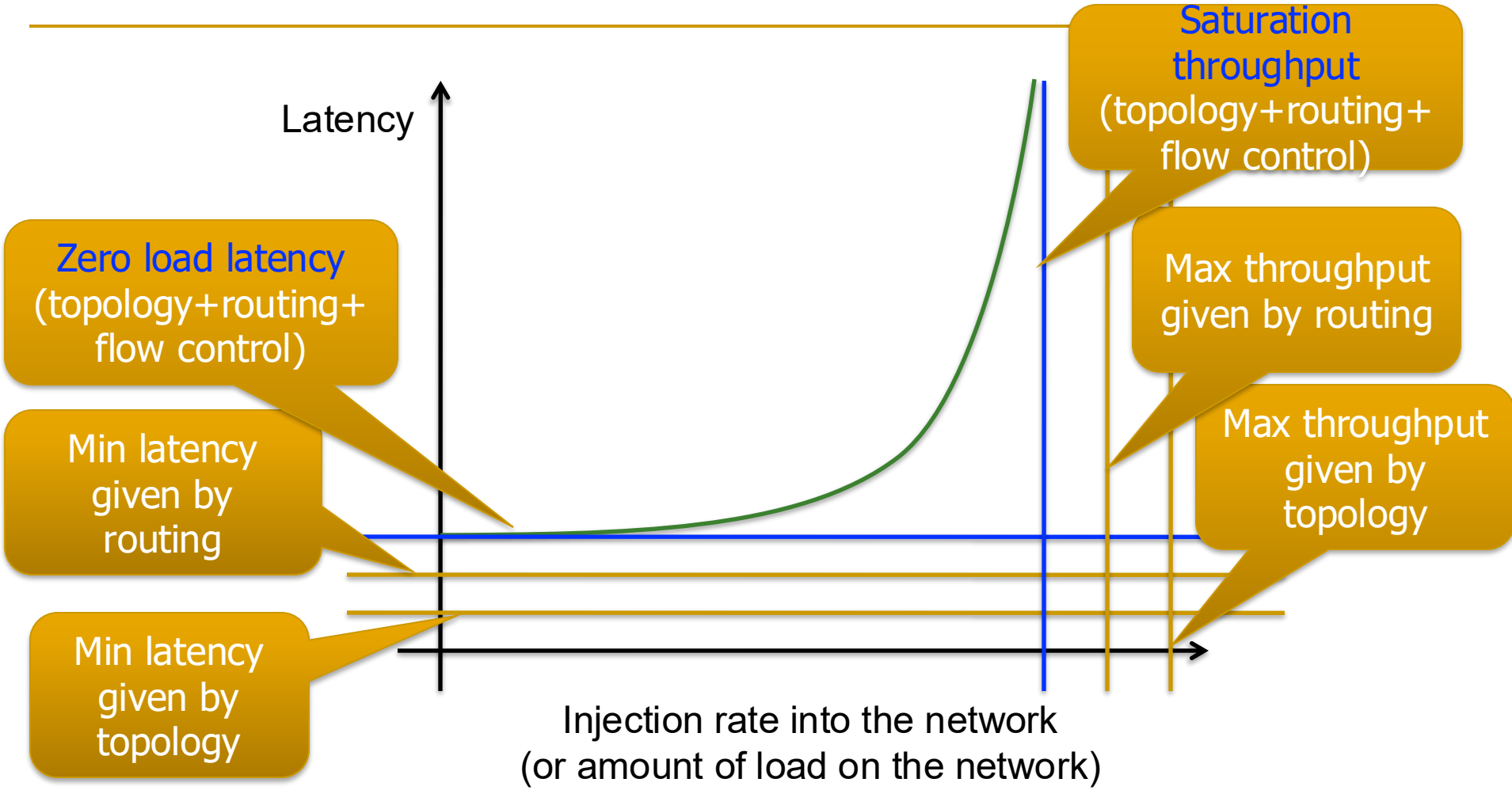
- Credit-based flow control
  - Upstream knows how many buffers are downstream
  - Downstream passes back credits to upstream
  - Significant upstream signaling (esp. for small flits)

- On/Off (XON/XOFF) flow control
  - Downstream has on/off signal to upstream

- ACK/NACK flow control
  - Upstream optimistically sends downstream
  - Buffer cannot be deallocated until ACK/NACK received
  - Inefficiently utilizes buffer space

Interconnection Network Performance

Interconnection Network Performance



Saturation throughput: Injection rate at which latency asymptotes
"Zero load" latency: Latency with no contention

Ideal Latency

- Ideal latency
  - Solely due to wire delay between source and destination

$$T_{ideal} = \frac{D}{v} + \frac{L}{b}$$

  - D = Manhattan distance
    - The distance between two points measured along axes at right angles.
  - v = propagation velocity
  - L = packet size
  - b = channel bandwidth

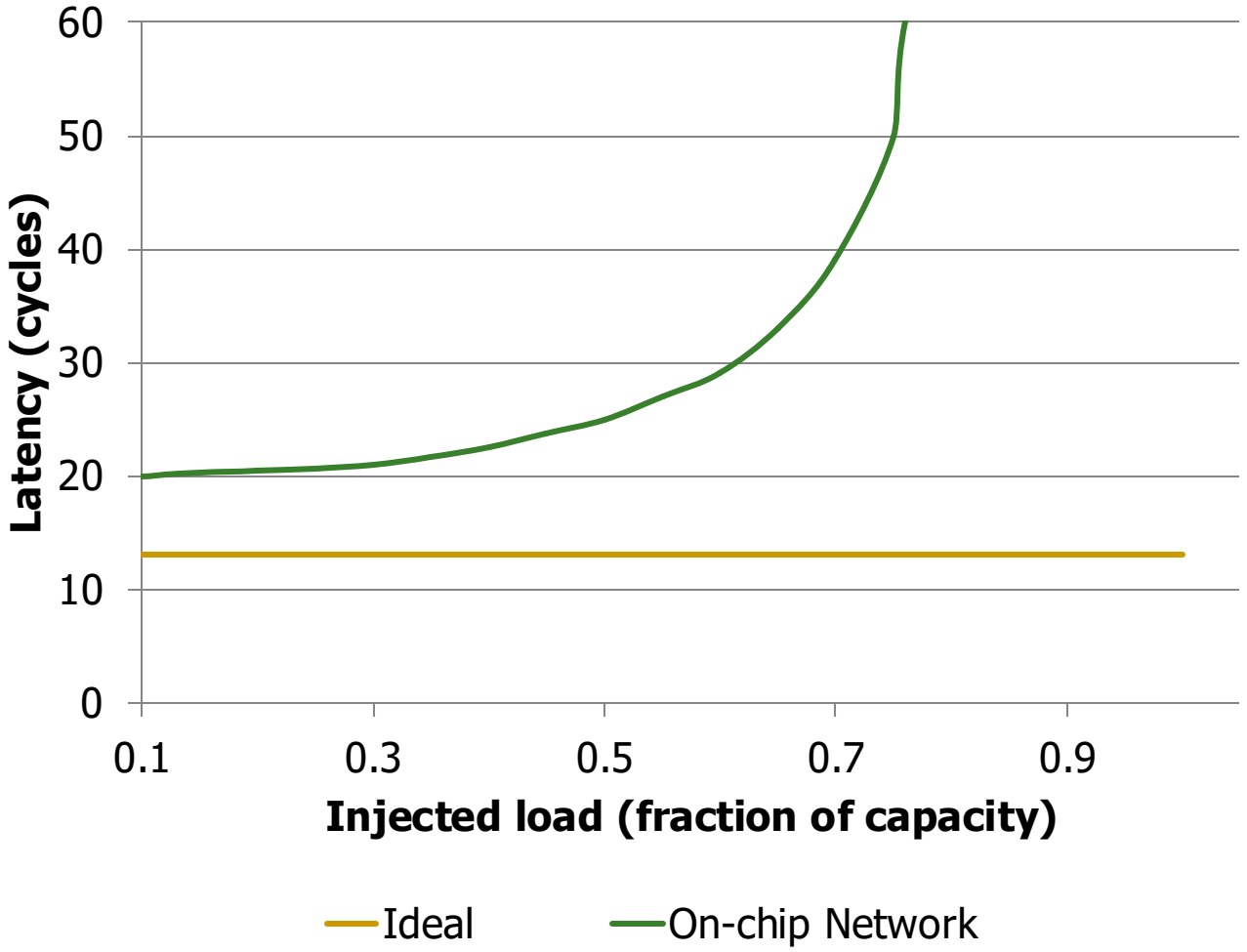
Actual Latency

- Dedicated wiring impractical
  - Long wires segmented with insertion of routers → More delay

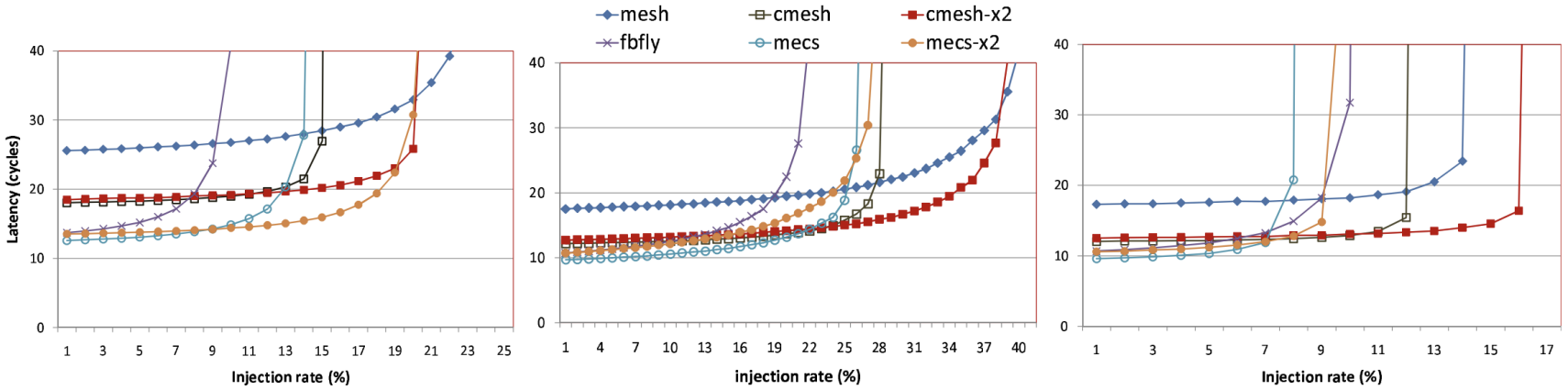
$$T_{actual} = \frac{D}{v} + \frac{L}{b} + H \cdot T_{router} + T_c$$

  - D = Manhattan distance
  - v = propagation velocity
  - L = packet size
  - b = channel bandwidth
  - H = hops
  - T_{router} = router latency
  - T_c = latency due to contention

Load-Latency Curve

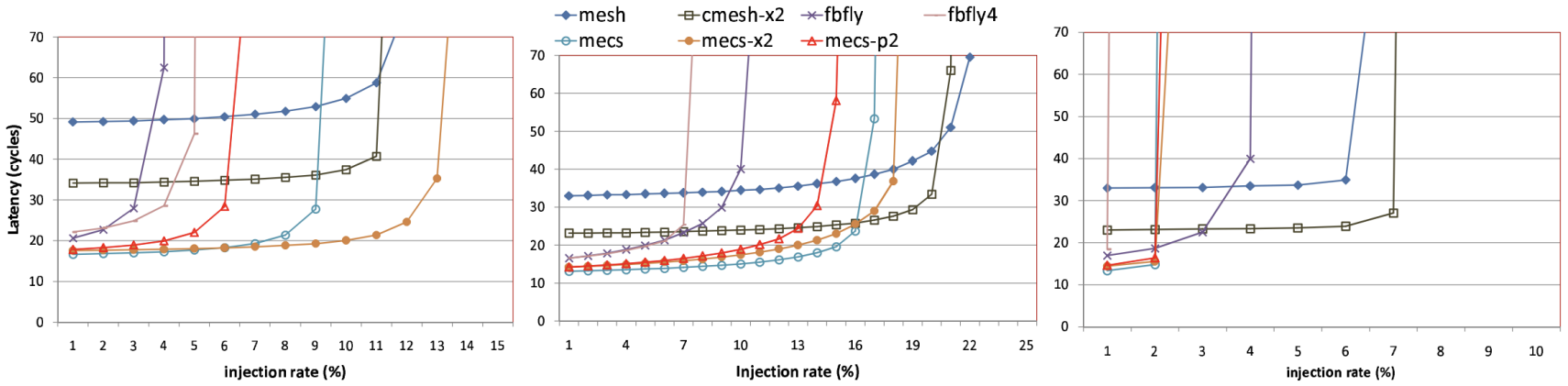


Load-Latency Curve Examples



(a) Bit Complement Traffic (b) Uniform Random Traffic (c) Transpose Traffic

Figure 4. Load-latency graphs for 64-node mesh, CMesh, flattened butterfly and MECS topologies.

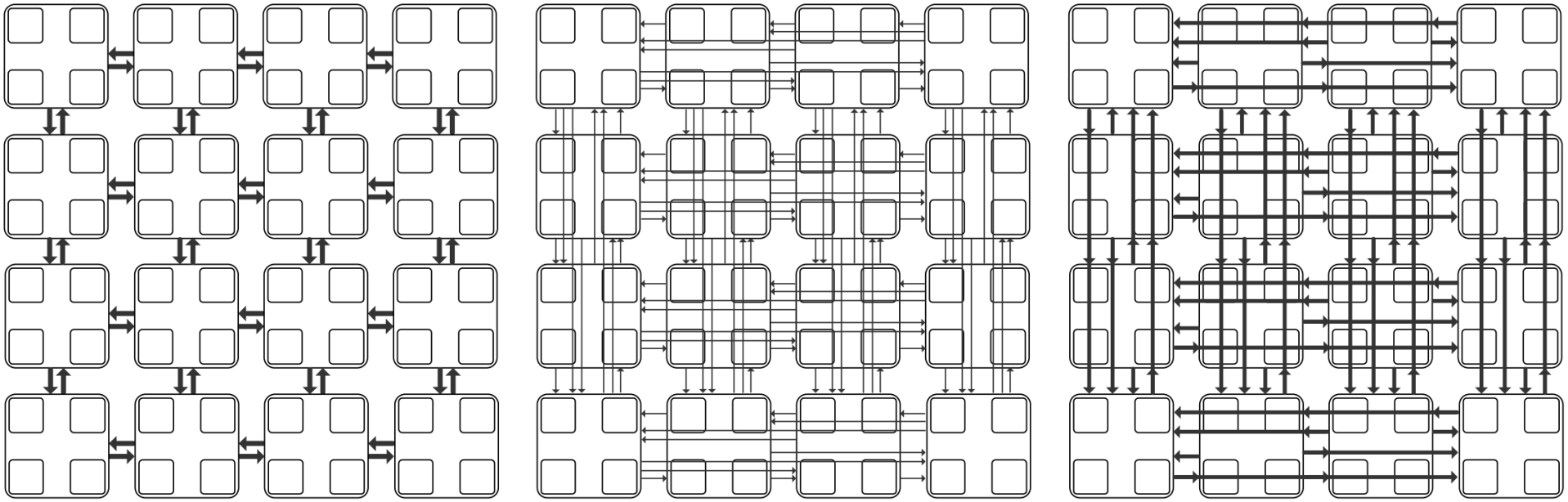


(a) Bit Complement Traffic (b) Uniform Random Traffic (c) Transpose Traffic

Figure 5. Load-latency graphs for 256-node mesh, CMesh, flattened butterfly and MECS topologies.

Grot+, "Express Cube Topologies for On-Chip Interconnects," HPCA 2009.

Examined Topologies in Prior Slide



(a) Concentrated mesh (b) Flattened butterfly (c) MECS

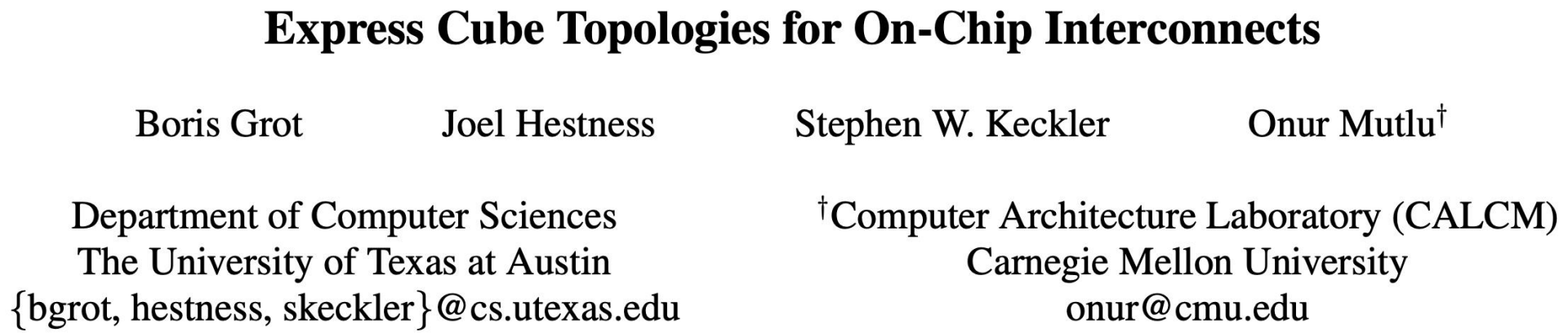
Figure 1. Concentrated Mesh, Flattened Butterfly and MECS topologies for a 64-terminal network.

Different topologies work differently for different communication patterns

Grot+, "Express Cube Topologies for On-Chip Interconnects," HPCA 2009.

Multi-Drop Express Channels (MECS)

- Boris Grot, Joel Hestness, Stephen W. Keckler, and Onur Mutlu, **"Express Cube Topologies for On-Chip Interconnects"** *Proceedings of the 15th International Symposium on High-Performance Computer Architecture (**HPCA**)*, pages 163-174, Raleigh, NC, February 2009. Slides (ppt)



Express Cube Topologies for On-Chip Interconnects

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Kilo-NoC Building on MECS

- Boris Grot, Joel Hestness, Stephen W. Keckler, and Onur Mutlu, **"Kilo-NOC: A Heterogeneous Network-on-Chip Architecture for Scalability and Service Guarantees"**
  *Proceedings of the 38th International Symposium on Computer Architecture (**ISCA**)*, San Jose, CA, June 2011. Slides (pptx)
  One of the 12 computer architecture papers of 2011 selected as Top Picks by IEEE Micro.

Kilo-NOC: A Heterogeneous Network-on-Chip Architecture for Scalability and Service Guarantees

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Kilo-NoC Building on MECS

- Boris Grot, Joel Hestness, Stephen W. Keckler, and Onur Mutlu, **"A QoS-Enabled On-Die Interconnect Fabric for Kilo-Node Chips"** *IEEE Micro*, *Special Issue: Micro's Top Picks from 2011 Computer Architecture Conferences (**MICRO TOP PICKS**)*, Vol. 32, No. 3, May/June 2012.



A QoS-Enabled On-Die Interconnect Fabric for Kilo-Node Chips

To meet rapidly growing performance demands and energy constraints, future chips will likely feature thousands of on-die resources. Existing network-on-chip solutions weren't designed for scalability and will be unable to meet future interconnect demands. A hybrid network-on-chip architecture called Kilo-NoC co-optimizes topology, flow control, and quality of service to achieve significant gains in efficiency.

Network Performance Metrics

- Packet latency (avg/max)
- Round trip latency (avg/max)
- Saturation throughput
- Application-level performance: execution time
- System performance: job throughput
  - Affected by interference among threads/applications

Rethinking Communication Networks

Rethinking Communication Networks

- New architectures, applications, technologies demand revisiting/rethinking how we handle communication

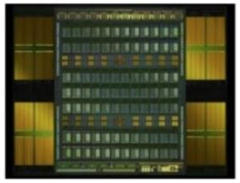
- On chip (versus off chip)
  - Communication between on-chip units

- Processing in memory/storage systems
  - Communication between in-memory processors
  - Communication between memory-centric and compute-centric units

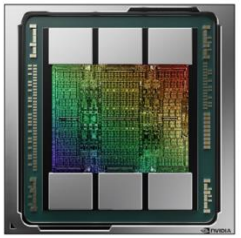
- New & evolving workloads
  - Adapting to new demands: AI/ML, Genomics, disaggregated systems

- New technologies
  - 3D stacking, monolithic 3D, optical, wireless?, chiplets?

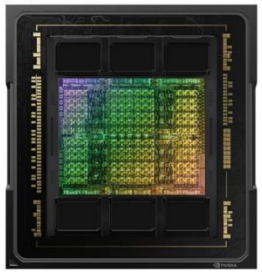
Evolution of GPU Off-Chip Networks (I)



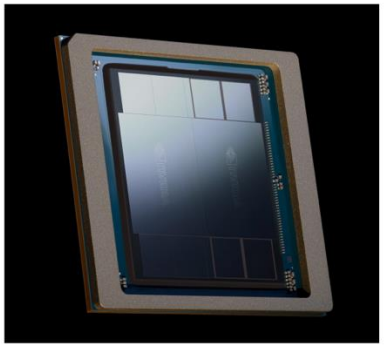
Volta
>21 billion transistors
815mm^2
TSMC 12nm FFN



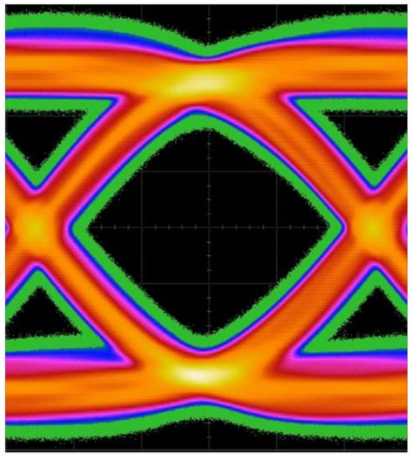
Ampere
>54 billion transistors
826 mm^2
TSMC N7



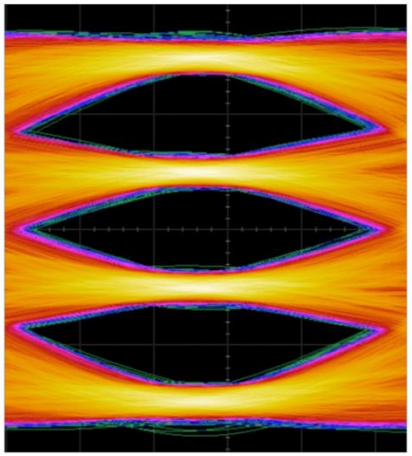
Hopper
>80 billion transistors
814 mm^2
TSMC 4N



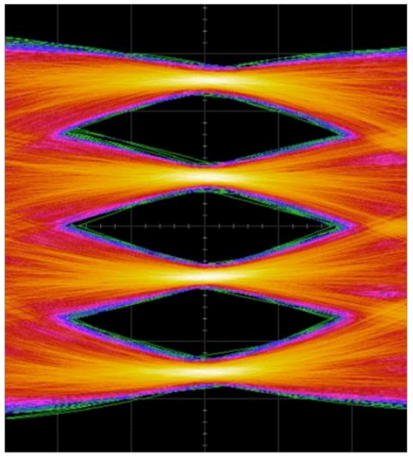
Blackwell
>208 billion transistors
>1600 mm^2
TSMC 4NP



Ampere | NVLink3
12 NVLinks | 50GB/s each
x4@50Gbps-NRZ
600GB/s total

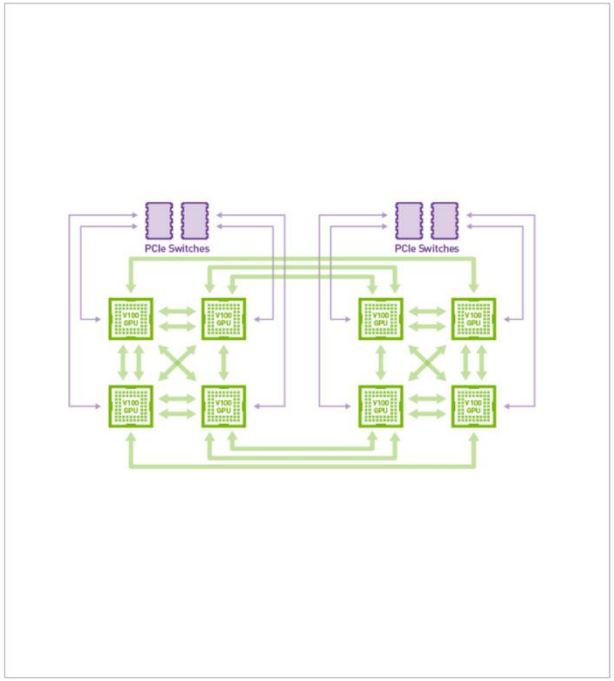


Hopper | NVLink4
18 NVLinks | 50GB/s each
x2@100Gbps-PAM4
900GB/s total

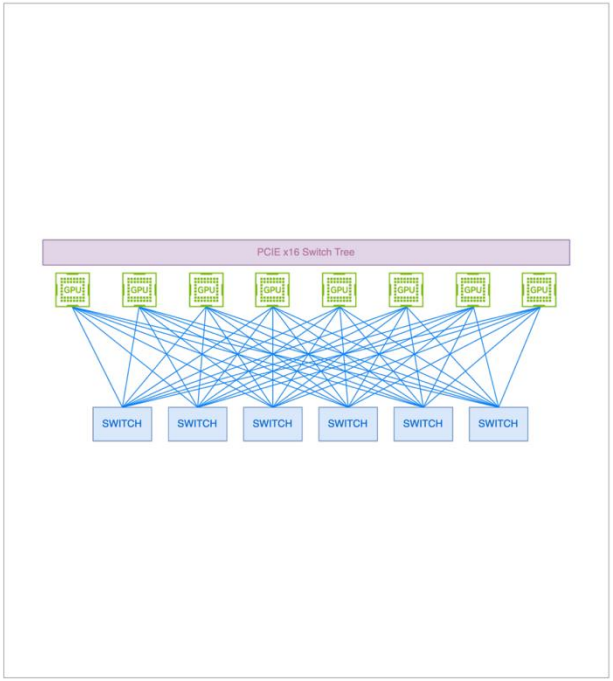


Blackwell | NVLink5
18 NVLinks | 100GB/s each
x2@200Gbps-PAM4
1800GB/s total

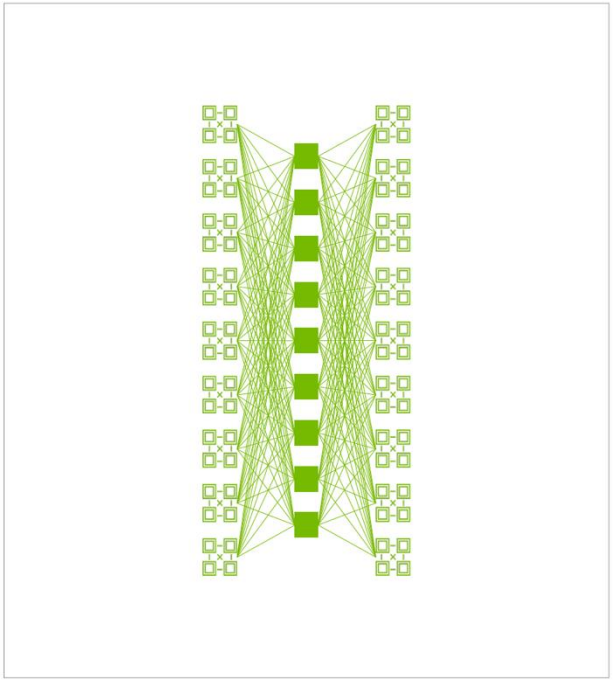
Evolution of GPU Off-Chip Networks (II)



2016
Hybrid Cube Mesh NVLink technology



2022
3rd Gen NVLink Switch
All-to-all connection among NVLink domain of 8 GPU

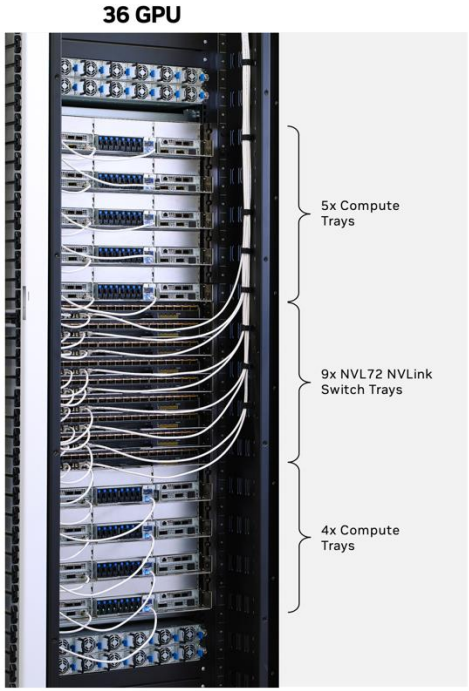


2024
4th Gen NVLink Switch Chip
All-to-all connection among NVLink domain of 72 GPU

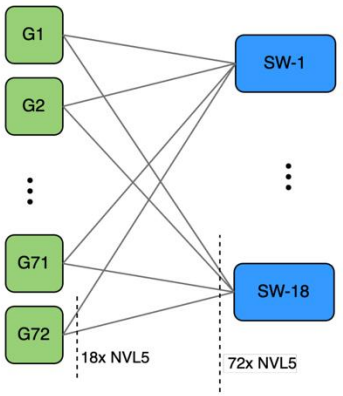
Evolution of GPU Off-Chip Networks (III)

GB200 NVL72 & NVL36

Delivering new unit of compute



GB200 NVL72	36 Grace CPUs 72 Blackwell GPUs Fully connected NVLink Switch rack
Training	720 PFLOPs
Inference	1,440 PFLOPs
NVL Model Size	27 Trillion params
Multi-Node Bandwidth	130 TB/s
Multi-Node All-Reduce	260 TB/s
GB200 NVL36	18 Grace CPUs 36 Blackwell GPUs Fully connected NVLink Switch rack



- 72 GPUs fully connected to 18 NVLink Switch
- 18 NVL5 ports per GPU
- 72 NVL5 ports per Switch

Rethinking Communication Networks

- New architectures, applications, technologies demand revisiting/rethinking how we handle communication

- On chip (versus off chip)
  - Communication between on-chip units

- Processing in memory/storage systems
  - Communication between in-memory processors
  - Communication between memory-centric and compute-centric units

- New & evolving workloads
  - Adapting to new demands: AI/ML, Genomics, disaggregated systems

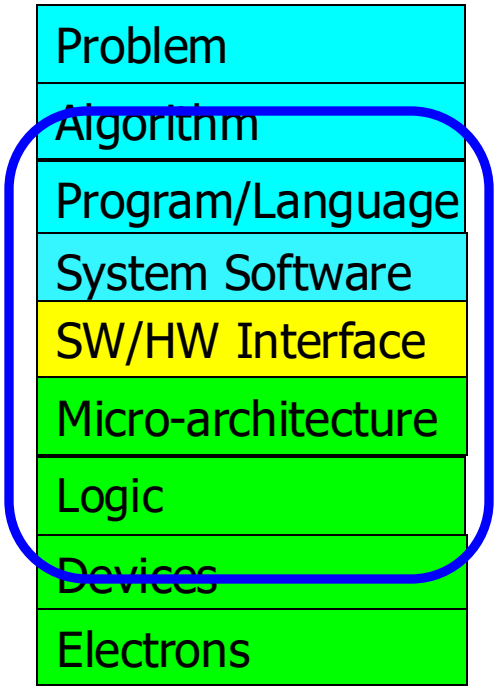
- New technologies
  - 3D stacking, monolithic 3D, optical, wireless?, chiplets?

Axiom

To achieve the highest efficiency, performance, robustness:

we must take the expanded view

of computer architecture

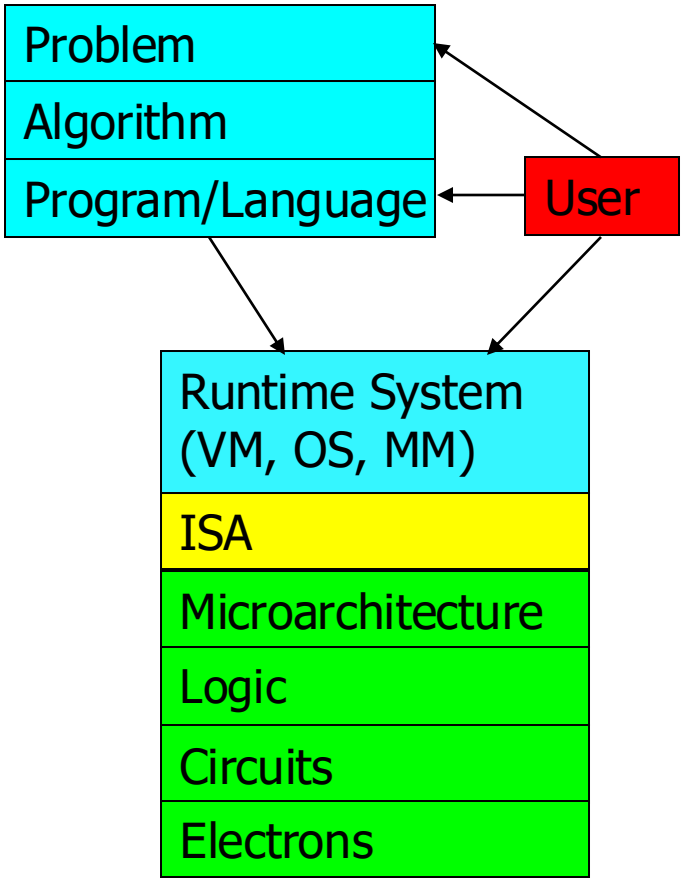


Co-design across the hierarchy: Algorithms to devices

Specialize as much as possible within the design goals

Recall: Computer Architecture Today (II)

Many new demands from the top (Look Up)

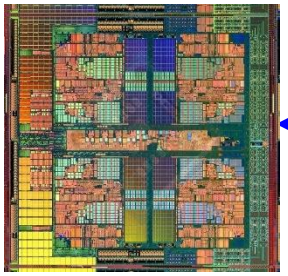


Fast changing demands and personalities of users (Look Up)

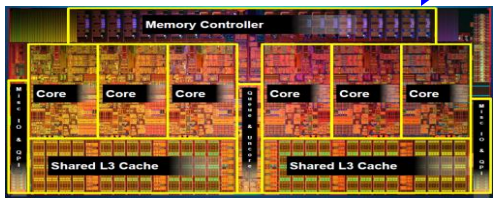
Many new issues at the bottom (Look Down)

Recall: Computer Architecture Today (III)

- Computing landscape is very different from 10-20 years ago
- Both UP (software and humanity trends) and DOWN (technologies and their issues), FORWARD and BACKWARD, and the resulting requirements and constraints



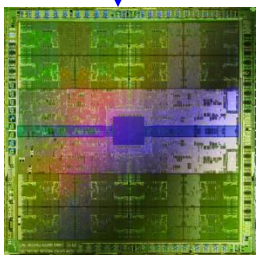
Heterogeneous Processors and Accelerators



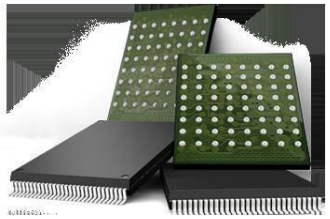


Hybrid Main Memory





General Purpose GPUs

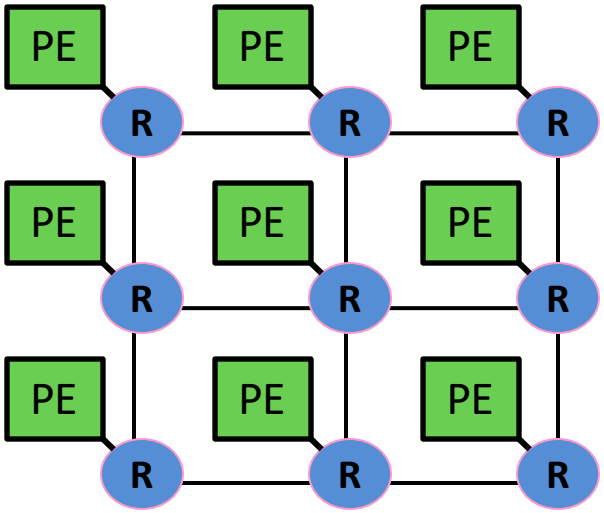


Persistent Memory/Storage

Every component and its interfaces, as well as entire system designs are being re-examined

An Exploration into On-Chip Networks

On-Chip Networks



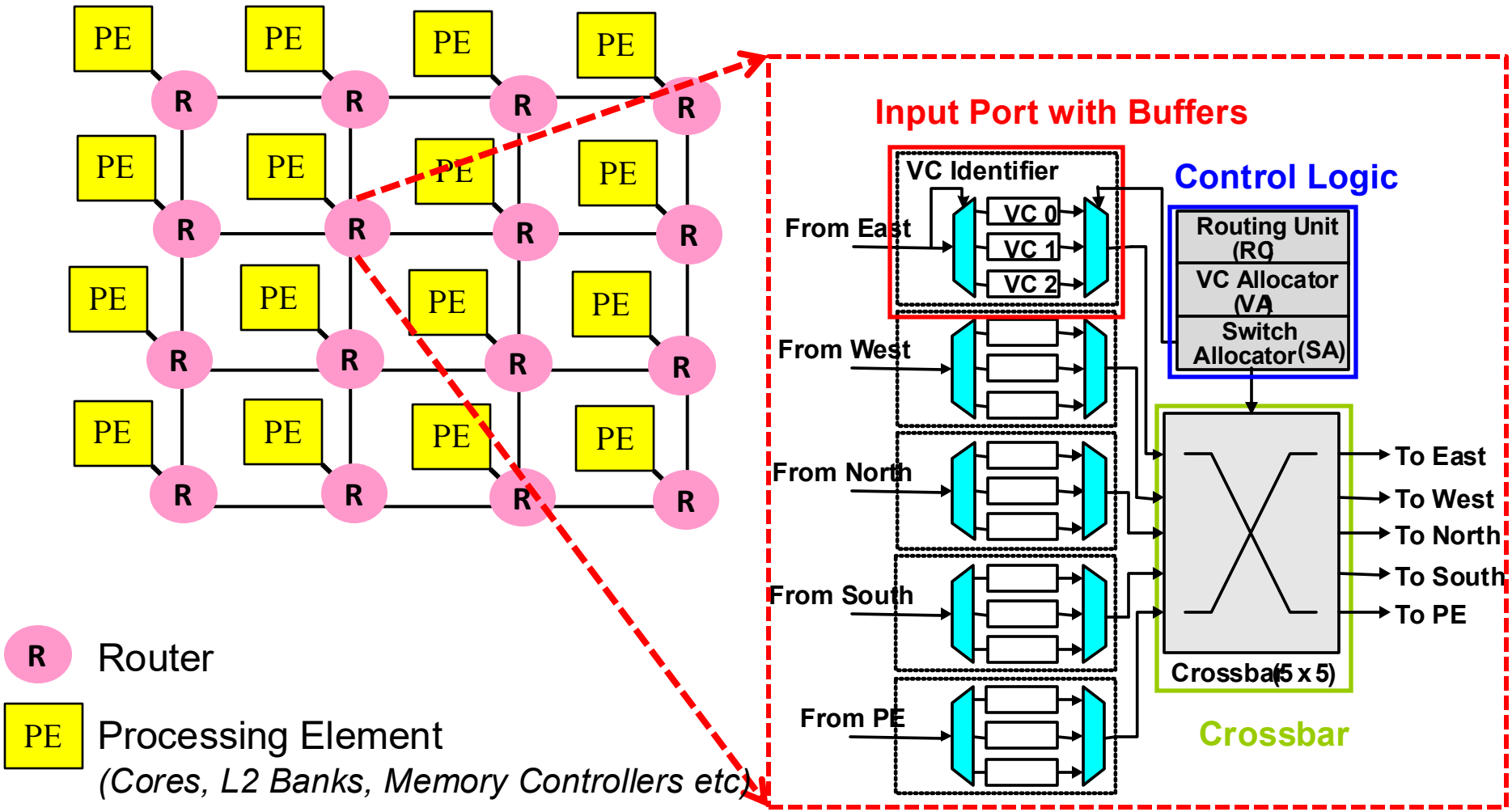
R Router

PE Processing Element (Cores, L2 Banks, Memory Controllers, etc)

- Connect **cores, caches, memory controllers, etc**
  - Buses and crossbars are not scalable
- **Usually packet switched**
- **2D mesh:** Commonly used topology
- **XY Routing with FIFO or Round robin port arbitration** common
- **Virtual channel buffering** common

- Primarily serve **cache misses** and **memory requests**

On-Chip Networks



Multiple On-Chip Networks (in Tilera)

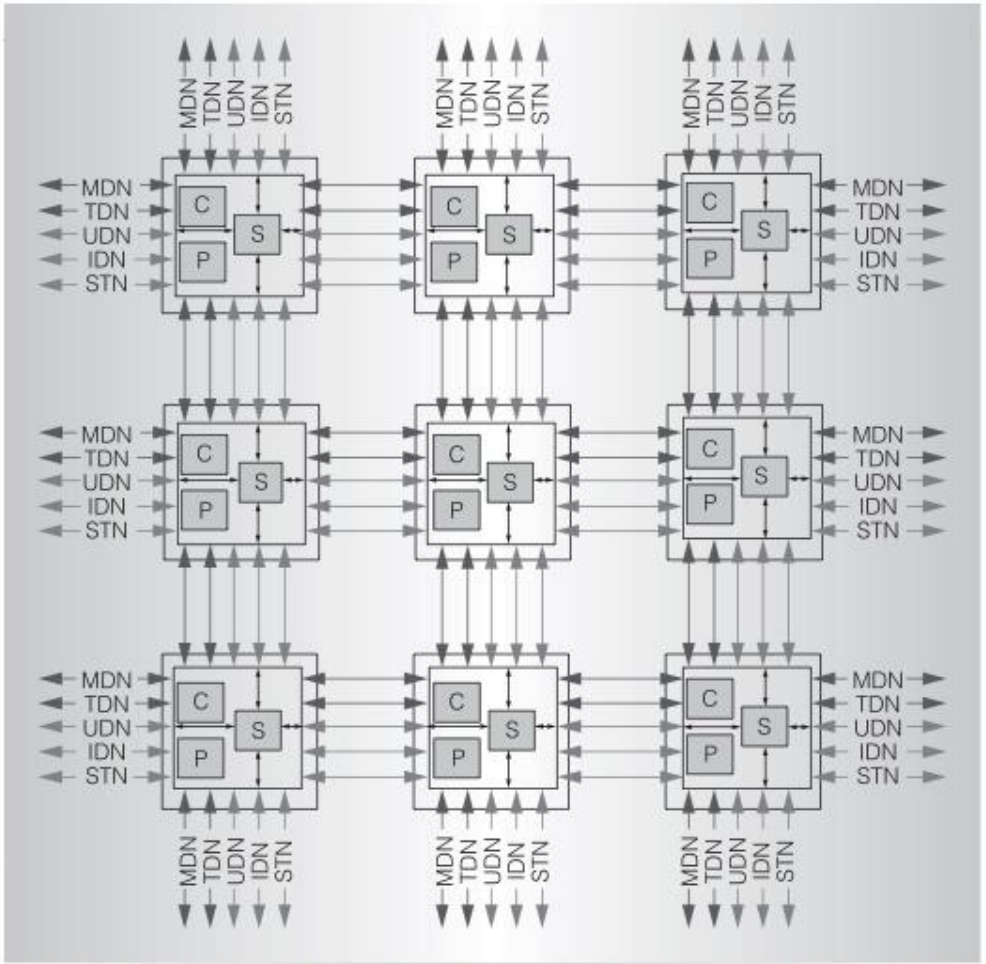


Figure 3. A 3 × 3 array of tiles connected by networks. (MDN: memory dynamic network; TDN: tile dynamic network; UDN: user dynamic network; IDN: I/O dynamic network; STN: static network.)

- Topology: 2D Mesh
- Five networks
- Four packet switched
  - Dimension order routing, wormhole flow control
  - TDN: Cache request packets
  - MDN: Response packets
  - IDN: I/O packets
  - UDN: Core to core messaging
- One circuit switched
  - STN: Low-latency, high-bandwidth static network
  - Streaming data

Wentzlaff et al., "On-Chip Interconnection Architecture of the Tile Processor," IEEE Micro 2007.

On-Chip vs. Off-Chip Interconnects

- On-chip advantages
  - Low latency between cores
  - No pin constraints
  - Rich & low-power wiring resources
  - → Very high bandwidth
  - → Simpler (global) coordination

- On-chip constraints/disadvantages
  - 2D substrate limits easy-to-implement topologies
  - Energy/power consumption a key concern
    - Complex algorithms undesirable
    - Large buffers undesirable
  - Logic area & metal layers constrain use of wiring resources

On-Chip vs. Off-Chip Interconnects (II)

- Cost
  - Off-chip: Channels, pins, connectors, cables
  - On-chip: Cost is storage and switches (wires are plentiful)
    - Leads to networks with many wide channels, less buffering
- Channel characteristics
  - On chip short distance → low latency
  - On chip RC lines → need repeaters every 1-2mm
    - Can put logic in repeaters
- Workloads
  - Off-chip: Large-scale parallel application multi-chip traffic
  - On-chip: Multi-core & accelerator cache/memory traffic

On-Chip vs. Off-Chip Tradeoffs

- Dally & Towles, "Route Packets, Not Wires: On-Chip Interconnection Networks," DAC 2001.

3 Challenges in architecture and design

While the same principles apply to interconnection networks at all scales, on-chip networks have a number of characteristics that make their design quite different than the inter-chip (and inter-board) networks that have been designed for years. In particular, wires and pins are more abundant than in inter-chip networks and buffers space is less abundant. These differences enable a number of new network topologies, flow control methods, and other techniques. In particular, we identify three areas that are ripe for future research:

3.1 What topologies are best matched to the abundant wiring resources available on chip?

On chip networks have enormous wiring resources at their disposal. In the example network described above, there can be up to 6,000 wires on each metal layer crossing each edge of a tile. It is quite easy to achieve over 24,000 'pins' crossing the four edges of a tile. In contrast, inter-chip networks have historically been pin limited, required to limit the connections of one router chip to far less than 1,000 total pins. This large, 24:1, difference between router pin limitations allows the designer to trade wiring resources for network performance, making a qualitative difference in network architecture.

3.2 What flow control methods reduce buffer count and hence router overhead?

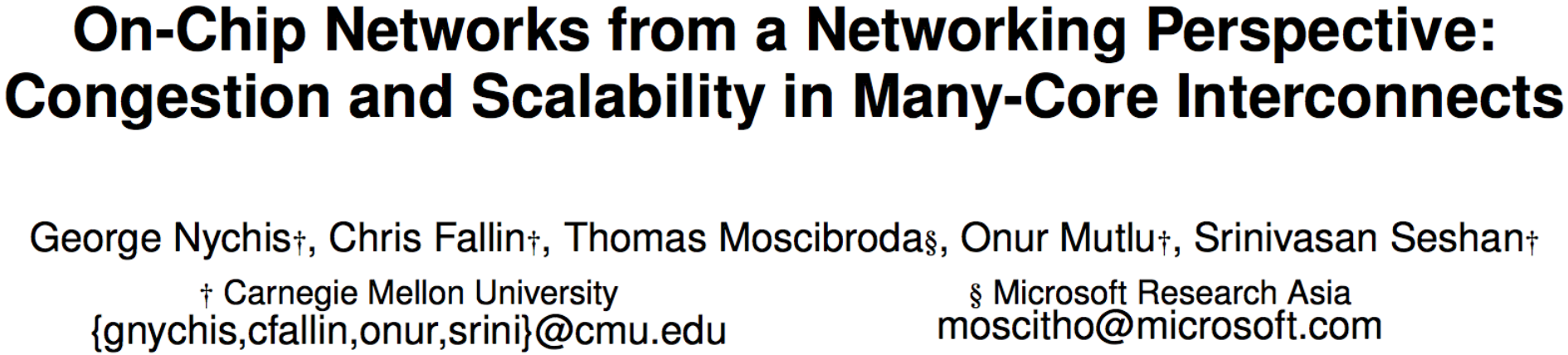
Buffer space in an on-chip router directly impacts the area overhead of the network and thus must be kept to a minimum. In contrast, most inter-chip network routers are pin limited and thus have ample room for very large buffers.

3.3 What circuits best exploit the structured wiring of on-chip networks?

Much of the advantage of on-chip networks derives from the regular, structured nature of their wiring. As described below, the well controlled electrical parameters of this wiring enable the use of high-performance circuits such as pulsed low-swing drivers and receivers to reduce power dissipation, reduce latency, and increase repeater spacing. While these transceivers yield big performance

On-Chip vs. Off-Chip Tradeoffs

- George Nychis, Chris Fallin, Thomas Moscibroda, Onur Mutlu, and Srinivasan Seshan,
  "On-Chip Networks from a Networking Perspective: Congestion and Scalability in Many-core Interconnects"
  *Proceedings of the 2012 ACM SIGCOMM Conference (**SIGCOMM**)*, Helsinki, Finland, August 2012. Slides (pptx)



On-Chip Networks from a Networking Perspective: Congestion and Scalability in Many-Core Interconnects

George Nychis†, Chris Fallin†, Thomas Moscibroda§, Onur Mutlu†, Srinivasan Seshan†

† Carnegie Mellon University
{gnychis,cfallin,onur,srini}@cmu.edu

§ Microsoft Research Asia
moscitho@microsoft.com

On-Chip vs. Off-Chip Tradeoffs (II)

- George Nychis, Chris Fallin, Thomas Moscibroda, and Onur Mutlu, **"Next Generation On-Chip Networks: What Kind of Congestion Control Do We Need?"** *Proceedings of the 9th ACM Workshop on Hot Topics in Networks (**HOTNETS**)*, Monterey, CA, October 2010. Slides (ppt) (key)

Next Generation On-Chip Networks: What Kind of Congestion Control Do We Need?

George Nychis†, Chris Fallin†, Thomas Moscibroda§, Onur Mutlu†

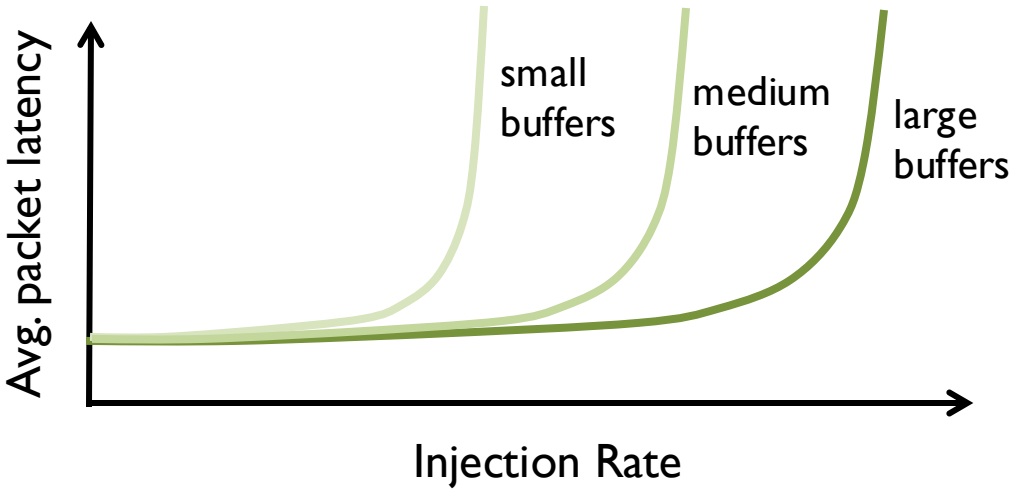
†Carnegie Mellon University
{gnychis,cfallin,onur}@cmu.edu

§Microsoft Research
moscitho@microsoft.com

Buffering and Flow Control in On-Chip Networks

Buffers in NoC Routers

- Buffers are necessary for high network throughput
  → buffers increase total available bandwidth in network



Buffers in NoC Routers

- Buffers are necessary for high networ
  → buffers increase total availab



- Buffers consume sig
  - Dynamic e
  - Static
- Buff icy
  - ment
  - cation
  - flow control





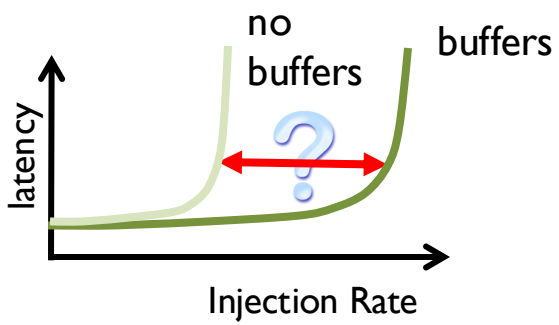
- quire significant chip area
  - E.g., in TRIPS prototype chip, input buffers occupy 75% of total on-chip network area [Gratz et al, ICCD' 06]



Can we get rid of buffers...?

Going Bufferless…?

- How much throughput do we lose?
  → How is latency affected?



- Up to what injection rates can we use bufferless routing?
  → Are there realistic scenarios in which an NoC operates at injection rates below the threshold?



- Can we achieve energy reduction?
  → If so, how much…?

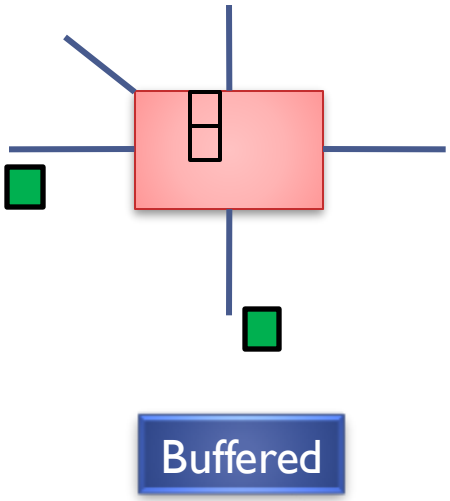
- Can we reduce area, complexity, etc…?

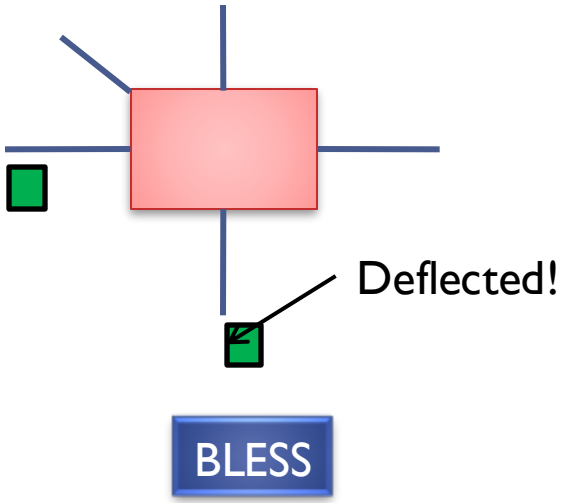
Answers in our paper (ISCA'09)!

BLESS: Bufferless Routing

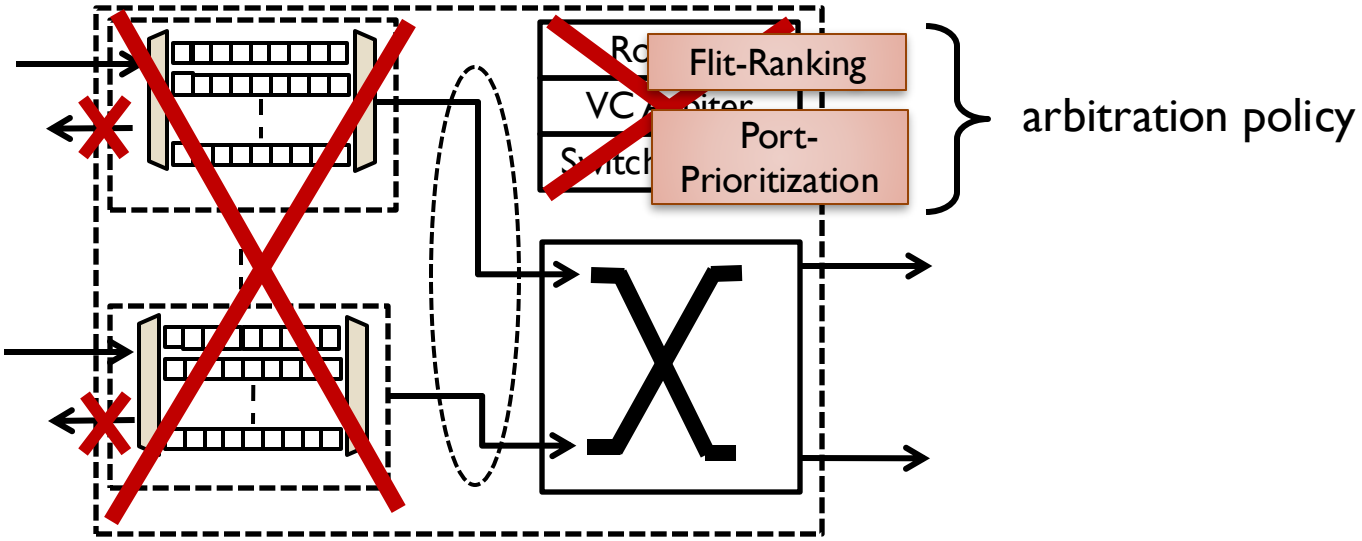
- Always forward *all* incoming flits to some output port
- If no productive direction is available, send to another direction
- → packet is deflected
- → Hot-potato routing [Baran' 62]







BLESS: Bufferless Routing



Flit-Ranking

1. Create a ranking over all incoming flits

Port-Prioritization

2. For a given flit in this ranking, find the best free output-port
   Apply to each flit in order of ranking

FLIT-BLESS: Flit-Level Routing

- Each flit is routed independently.
- Oldest-first arbitration (other policies evaluated in paper)

Flit-Ranking	1. Oldest-first ranking
Port-Prioritization	2. Assign flit to productive port, if possible. Otherwise, assign to non-productive port.

- Network Topology:

→ Can be applied to most topologies (Mesh, Torus, Hypercube, Trees, …)

1) #output ports ¸ #input ports at every router

2) every router is reachable from every other router

- Flow Control & Injection Policy:

→ Completely local, inject whenever input port is free

- Absence of Deadlocks: every flit is always moving
- Absence of Livelocks: with oldest-first ranking

BLESS: Advantages & Disadvantages

Advantages

- No buffers
- Purely local flow control
- Simplicity
  - no credit-flows
  - no virtual channels
  - simplified router design
- No deadlocks, livelocks
- Adaptivity
  - packets are deflected around congested areas!
- Router latency reduction
- Area savings

Disadvantages

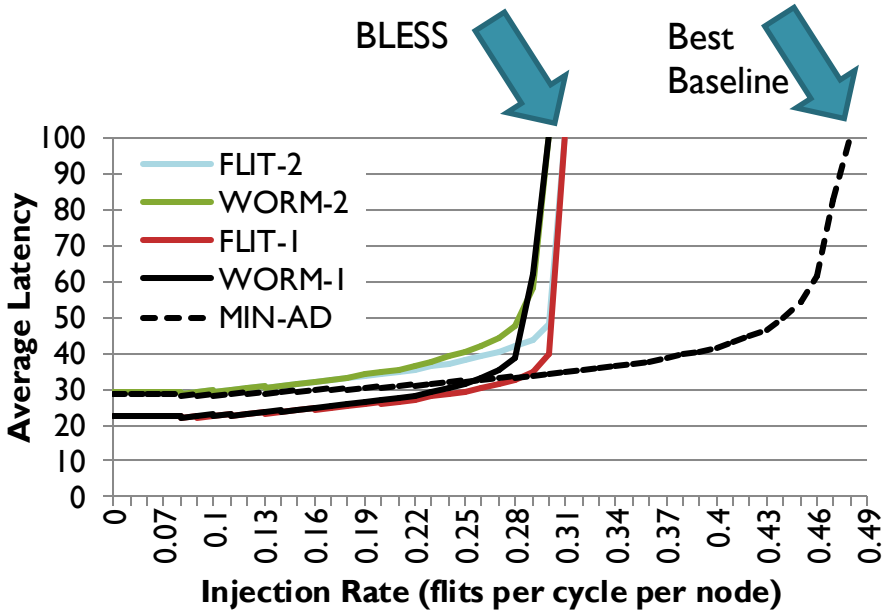
- Increased latency
- Reduced bandwidth
- Increased buffering at receiver
- Header information at each flit
- Oldest-first arbitration complex
- QoS becomes difficult

Impact on energy…?



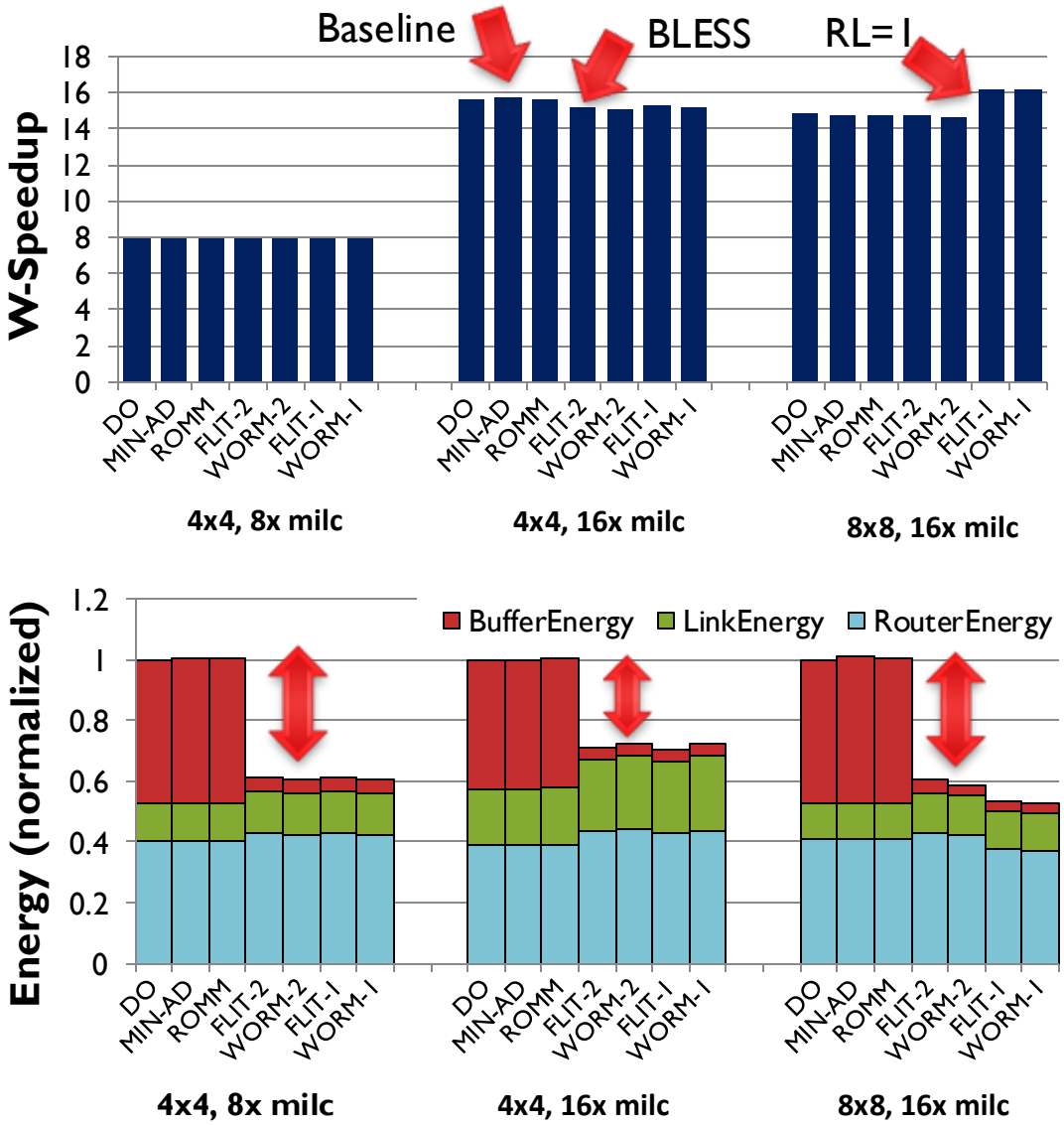
Evaluation – Synthetic Traces

- First, the bad news ☺
- Uniform random injection
- BLESS has significantly lower saturation throughput compared to buffered baseline.

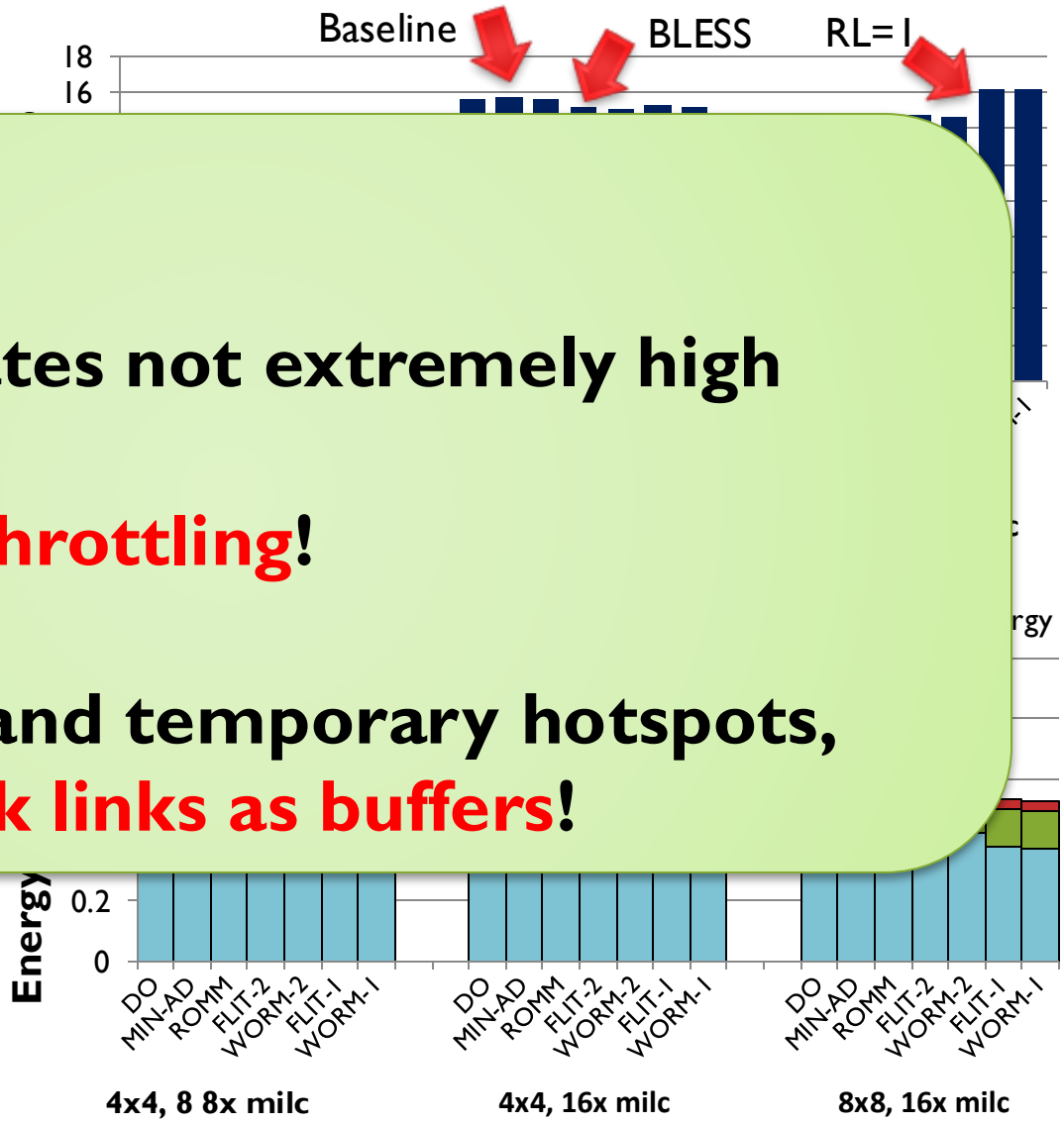


Evaluation – Homogenous Case Study

- milc benchmarks (moderately intensive)
- Perfect caches!
- Very little performance degradation with BLESS (less than 4% in dense network)
- With router latency 1, BLESS can even outperform baseline (by ~10%)
- Significant energy improvements (almost 40%)



Evaluation – Homogenous Case Study



Observations:

1) **Injection rates not extremely high on average**
   → **self-throttling!**

2) **For bursts and temporary hotspots, use network links as buffers!**

- Significant energy improvements (almost 40%)

BLESS Conclusions

- For a very wide range of applications and network settings, buffers are not needed in NoC
  - Significant energy savings
    (32% even in dense networks and perfect caches)
  - Area-savings of 60%
  - Simplified router and network design (flow control, etc…)
  - Performance slowdown is minimal (can even increase!)

➢ A strong case for a rethinking of NoC design!

- Future research:
  - Support for quality of service, different traffic classes, energy-management, etc…

Bufferless Deflection Routing in NoCs

- Thomas Moscibroda and Onur Mutlu,
  "A Case for Bufferless Routing in On-Chip Networks"
  *Proceedings of the 36th International Symposium on Computer Architecture (**ISCA**)*, pages 196-207, Austin, TX, June 2009. Slides (pptx)

A Case for Bufferless Routing in On-Chip Networks

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Better Bufferless Networks

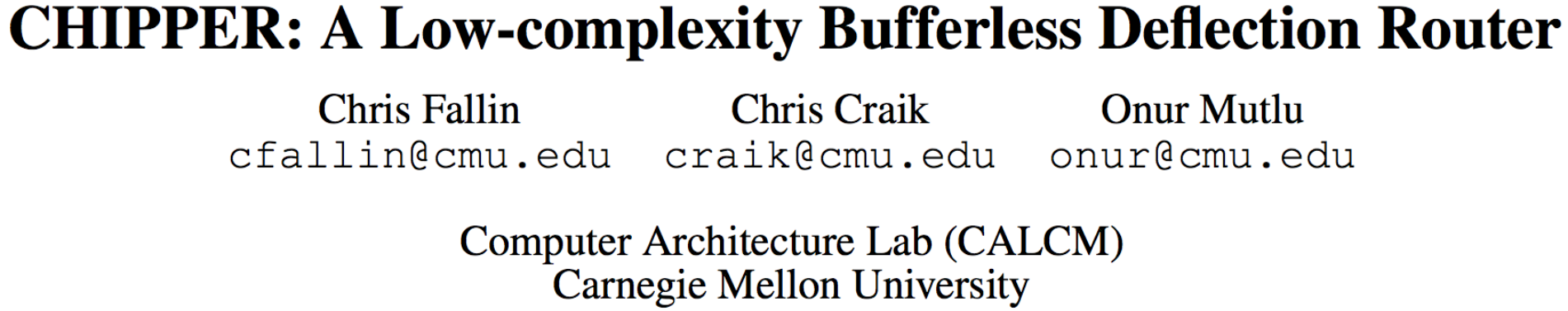
Can We Do Better?

Issues In Bufferless Deflection Routing

- Livelock
- Resulting Router Complexity
- Performance & Congestion at High Loads
- Quality of Service and Fairness
- Chris Fallin, Greg Nazario, Xiangyao Yu, Kevin Chang, Rachata Ausavarungnirun, and Onur Mutlu,
  "Bufferless and Minimally-Buffered Deflection Routing"
  Invited Book Chapter in Routing Algorithms in Networks-on-Chip, pp. 241-275, Springer, 2014.

Low-Complexity Bufferless Routing

- Chris Fallin, Chris Craik, and Onur Mutlu,
  "CHIPPER: A Low-Complexity Bufferless Deflection Router"
  *Proceedings of the 17th International Symposium on High-Performance Computer Architecture (**HPCA**)*, pages 144-155, San Antonio, TX, February 2011. Slides (pptx)



CHIPPER: A Low-complexity Bufferless Deflection Router

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Carnegie Mellon University

CHIPPER: A Low-Complexity Bufferless Deflection Router

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*Proceedings of the 17th International Symposium on High-Performance Computer Architecture (**HPCA**)*, pages 144-155, San Antonio, TX, February 2011. Slides (pptx)

SAFARI Carnegie Mellon

Motivation

- Recent work has proposed bufferless deflection routing (BLESS [Moscibroda, ISCA 2009])
  - Energy savings: ~40% in total NoC energy
  - Area reduction: ~40% in total NoC area
  - Minimal performance loss: ~4% on average
  - Unfortunately: unaddressed complexities in router
    - ➔ long critical path, large reassembly buffers
- **Goal**: obtain these benefits while simplifying the router in order to **make bufferless NoCs practical.**

Problems that Bufferless Routers Must Solve

1. Must provide livelock freedom

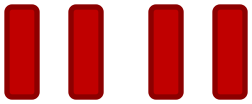
➔ A packet should not be deflected forever

2. Must reassemble packets upon arrival

Flit: atomic routing unit

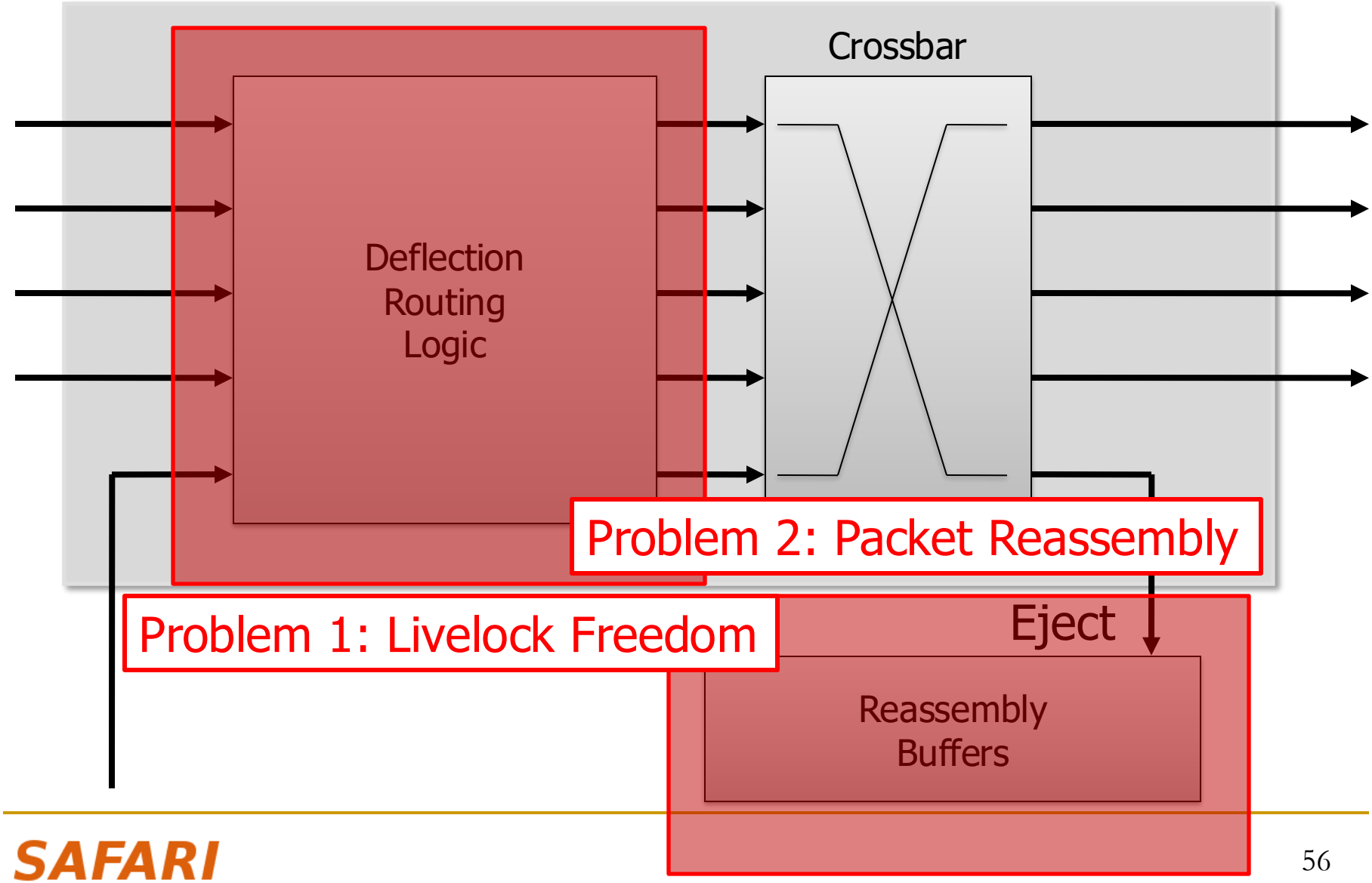


Packet: one or multiple flits



0 1 2 3

A Bufferless Router: A High-Level View



Complexity in Bufferless Deflection Routers

1. Must provide livelock freedom

Flits are sorted by age, then assigned in age order to output ports

➔ **43% longer critical path than buffered router**

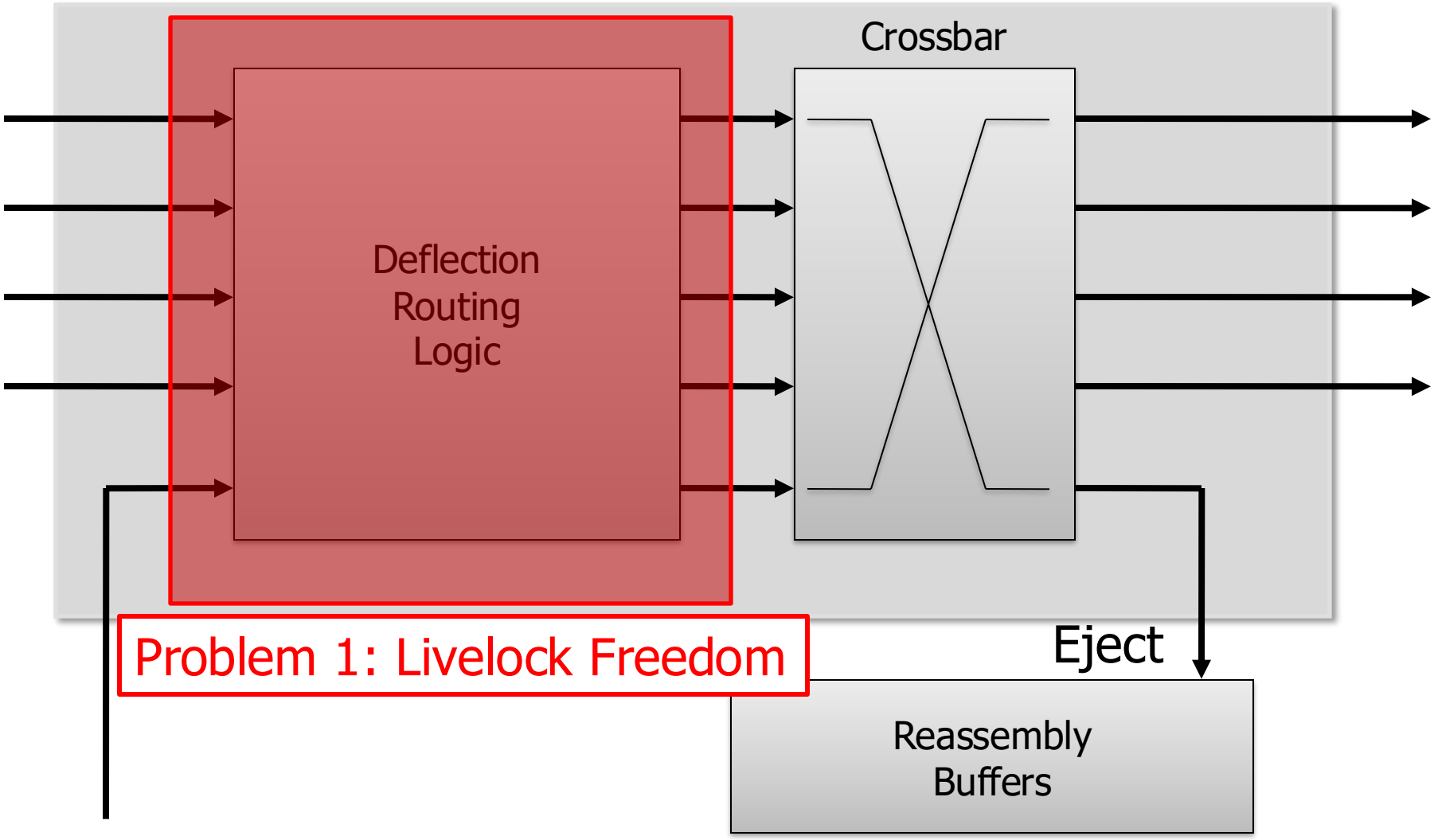
2. Must reassemble packets upon arrival

Reassembly buffers must be sized for worst case

➔ **4KB per node**

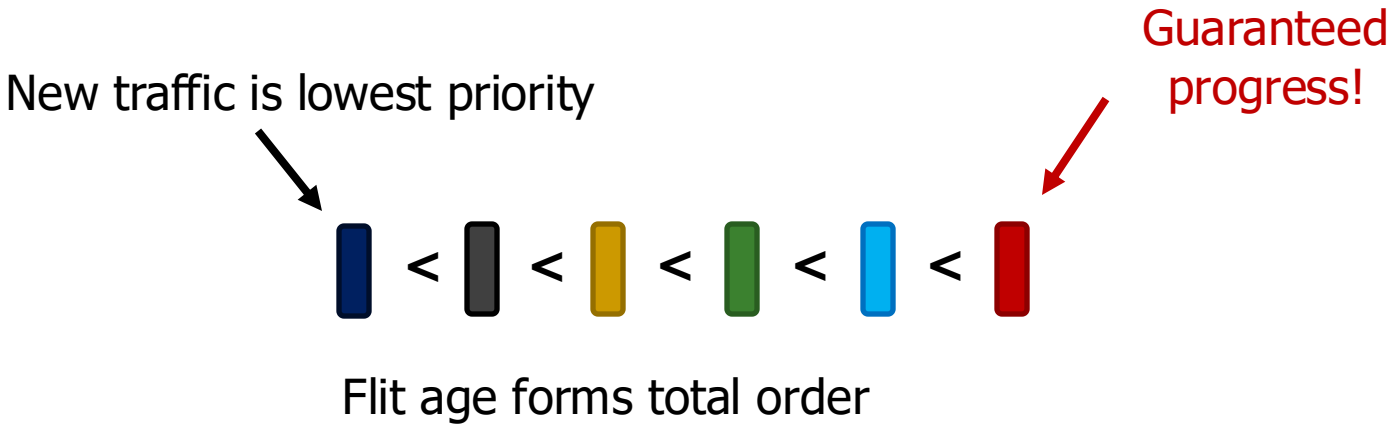
(8x8, 64-byte cache block)

Problem 1: Livelock Freedom



Livelock Freedom in Previous Work

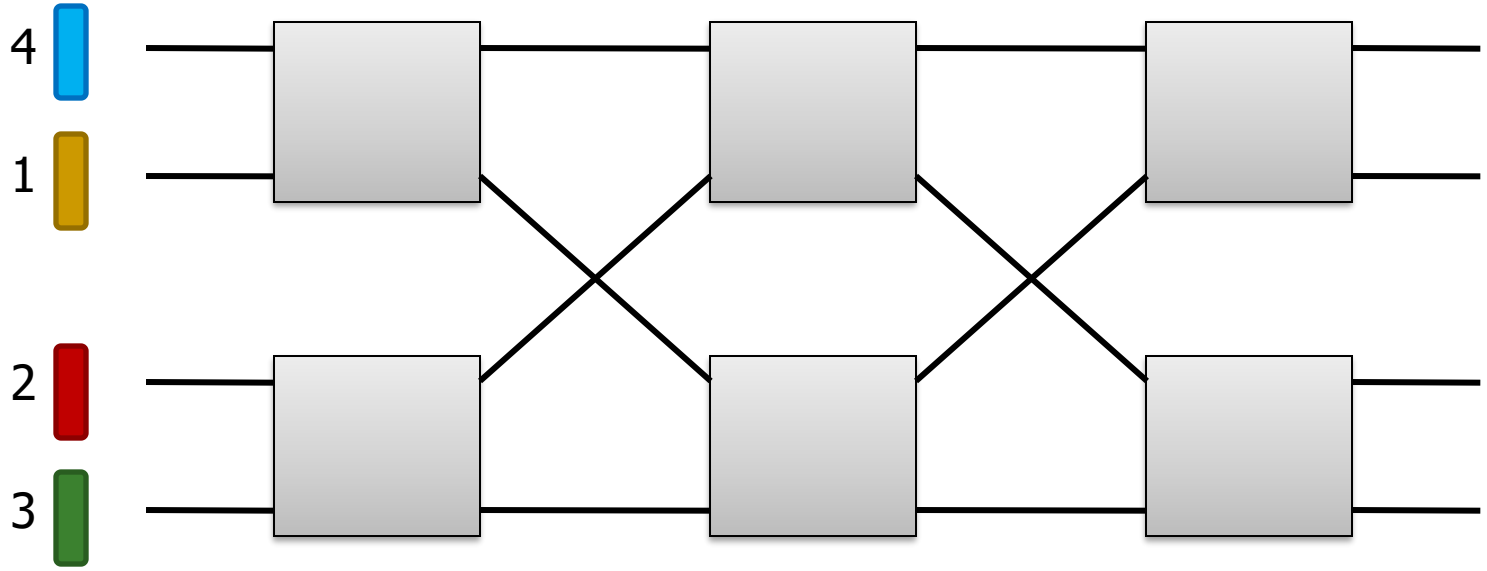
- What stops a flit from deflecting forever?
- All flits are timestamped
- Oldest flits are assigned their desired ports
- Total order among flits



- But what is the cost of this?

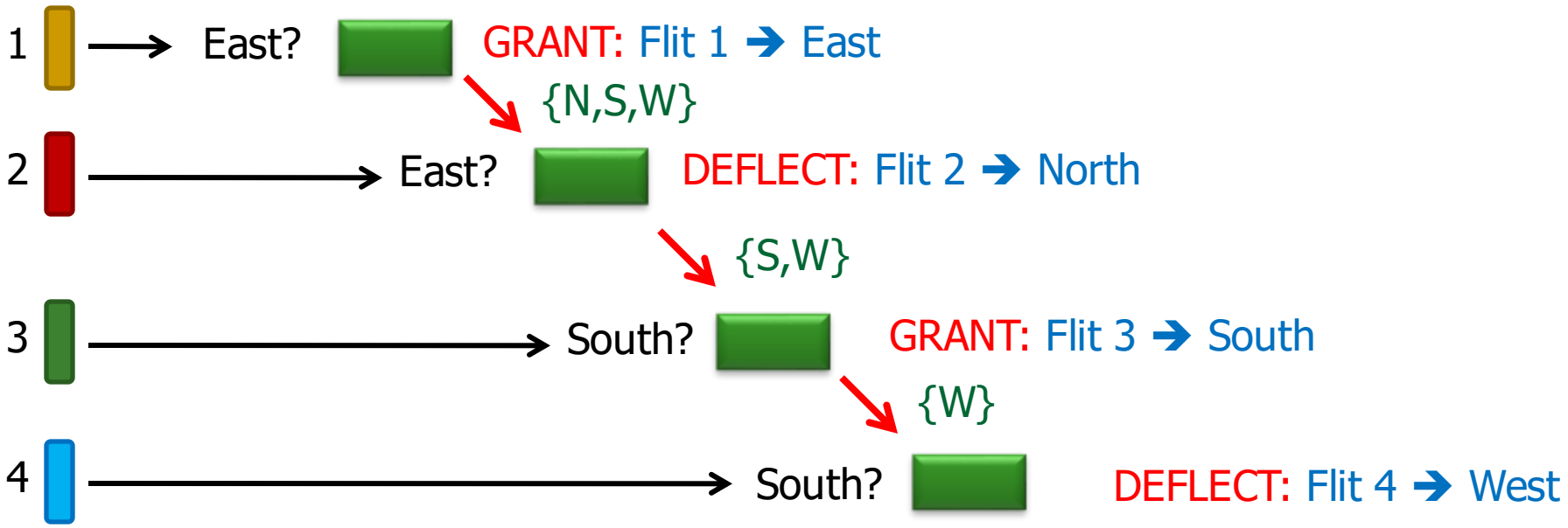
Age-Based Priorities are Expensive: Sorting

- Router must sort flits by age: long-latency sort network
  - **Three comparator stages** for 4 flits



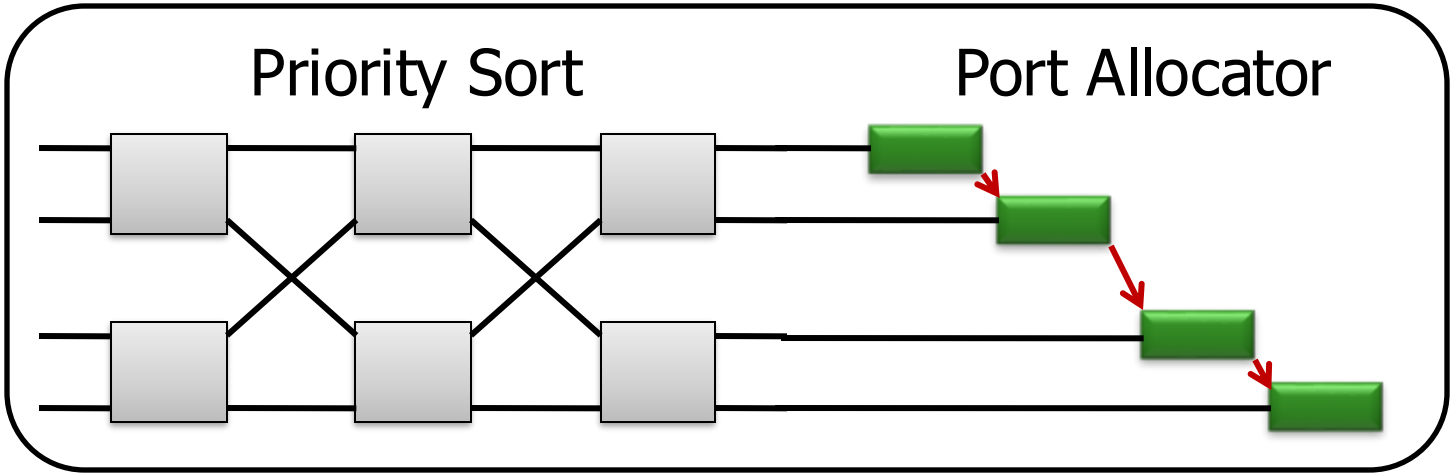
Age-Based Priorities Are Expensive: Allocation

- After sorting, flits assigned to output ports in priority order
- Port assignment of younger flits depends on that of older flits
  - **sequential dependence** in the port allocator



Age-Based Priorities Are Expensive

- Overall, **deflection routing logic** based on **Oldest-First** has a **43% longer critical path** than a buffered router



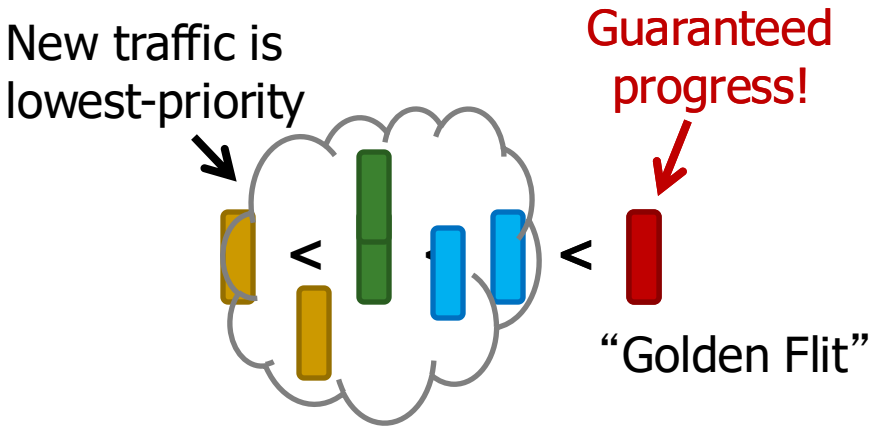
- Question: is there a cheaper way to route while guaranteeing livelock-freedom?

Solution: Golden Packet for Livelock Freedom

- What is *really necessary* for livelock freedom?

  Key Insight: No total order. It is enough to:

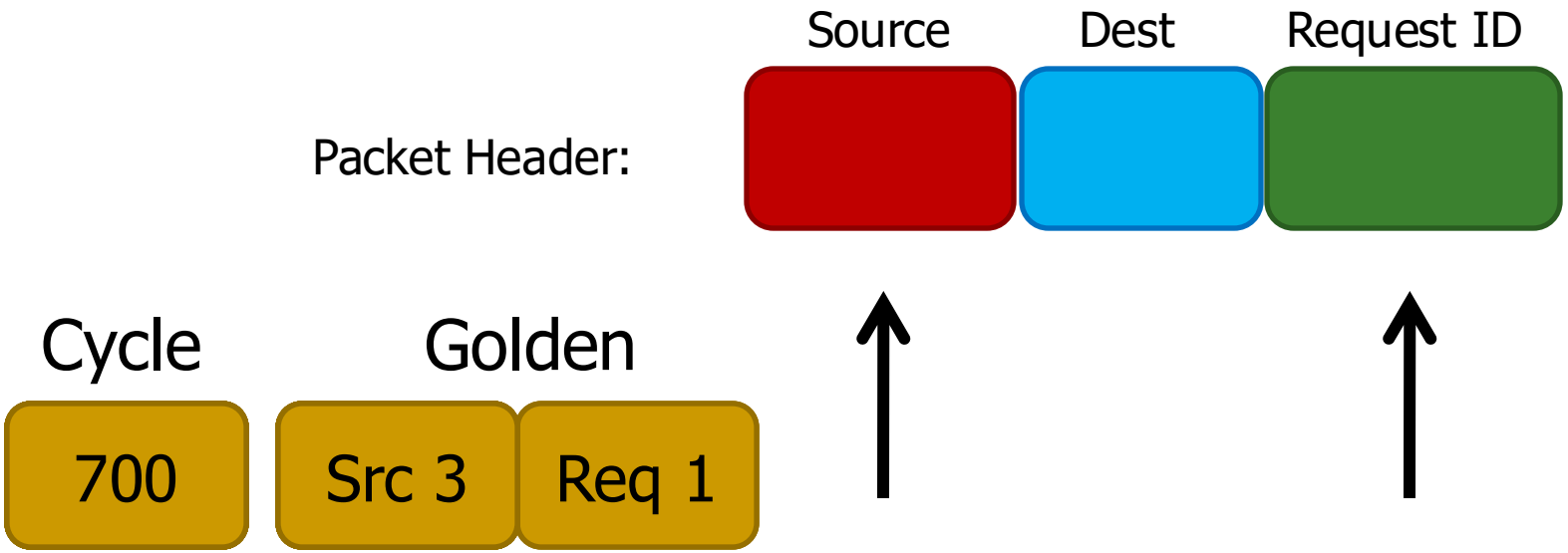
  1. Pick one flit to prioritize until arrival
  2. Ensure any flit is eventually picked



Flit age forms total order
partial ordering is sufficient!

Which Packet is Golden?

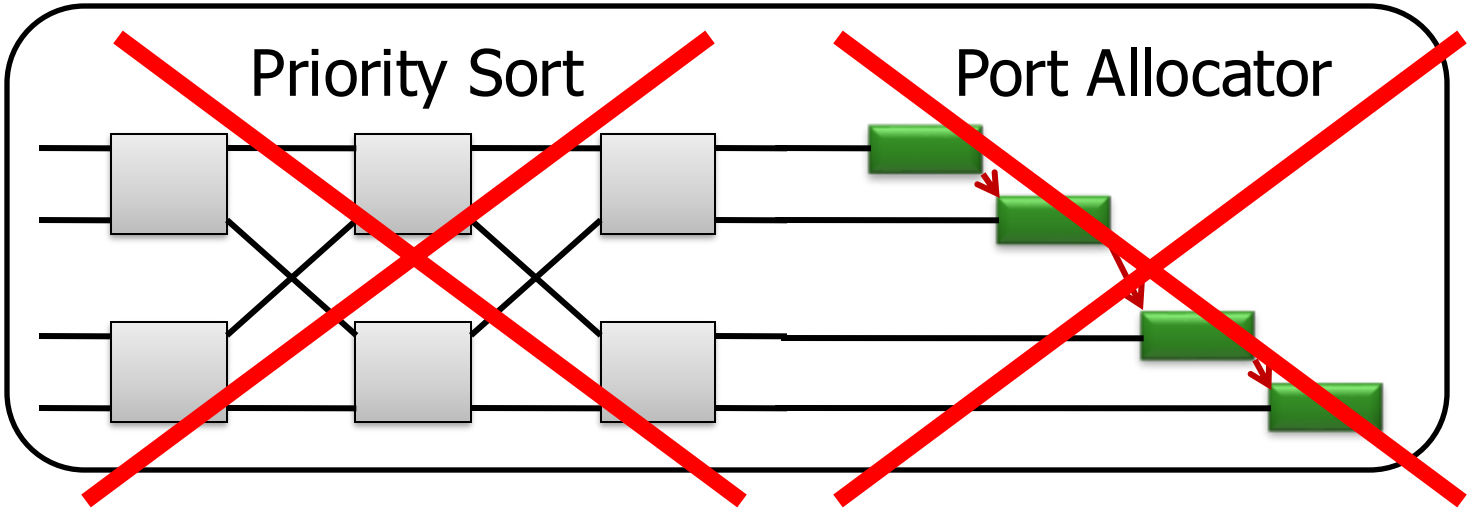
- We select the **Golden Packet** so that:
  1. a given packet stays golden long enough to ensure arrival
     → maximum no-contention latency
  2. the selection rotates through all possible packet IDs
     → static rotation schedule for simplicity



What Does Golden Flit Routing Require?

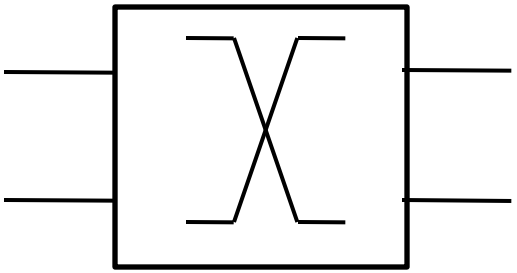
- Only **need** to properly route the Golden Flit

- **First Insight:** no need for full sort
- **Second Insight:** no need for sequential allocation



Golden Flit Routing With Two Inputs

- Let's route the Golden Flit in a two-input router first

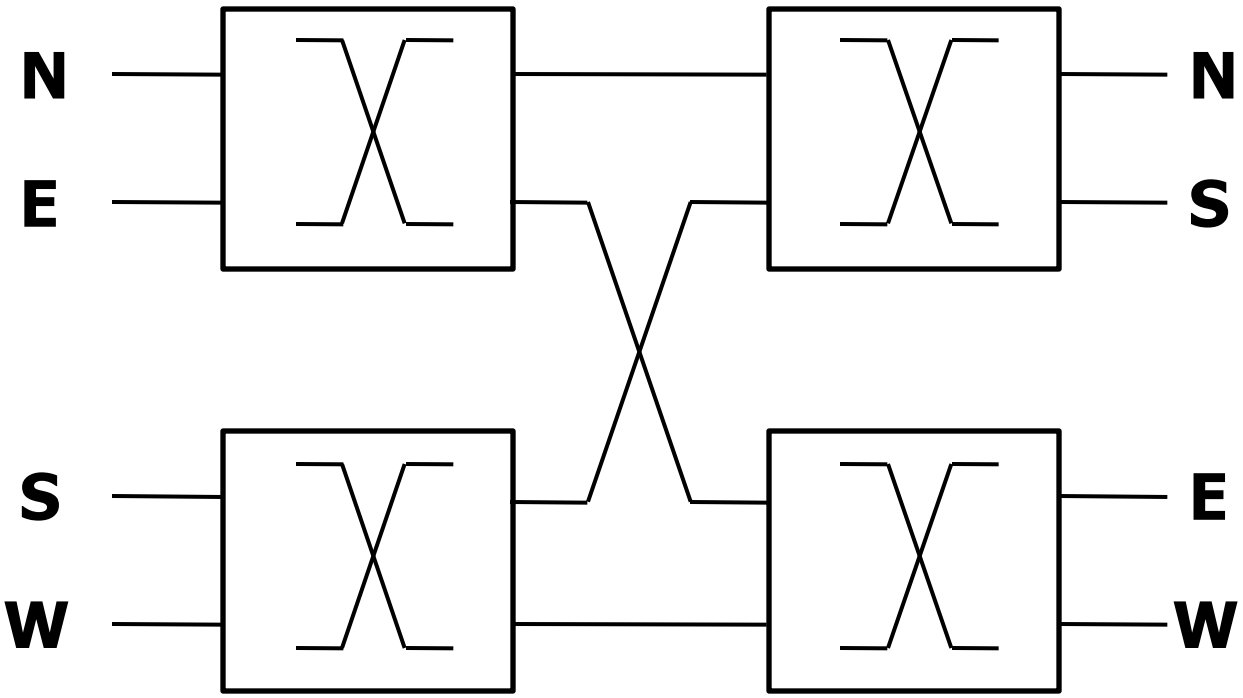


- **Step 1**: pick a "winning" flit: Golden Flit, else random
- **Step 2**: steer the winning flit to its desired output and deflect other flit

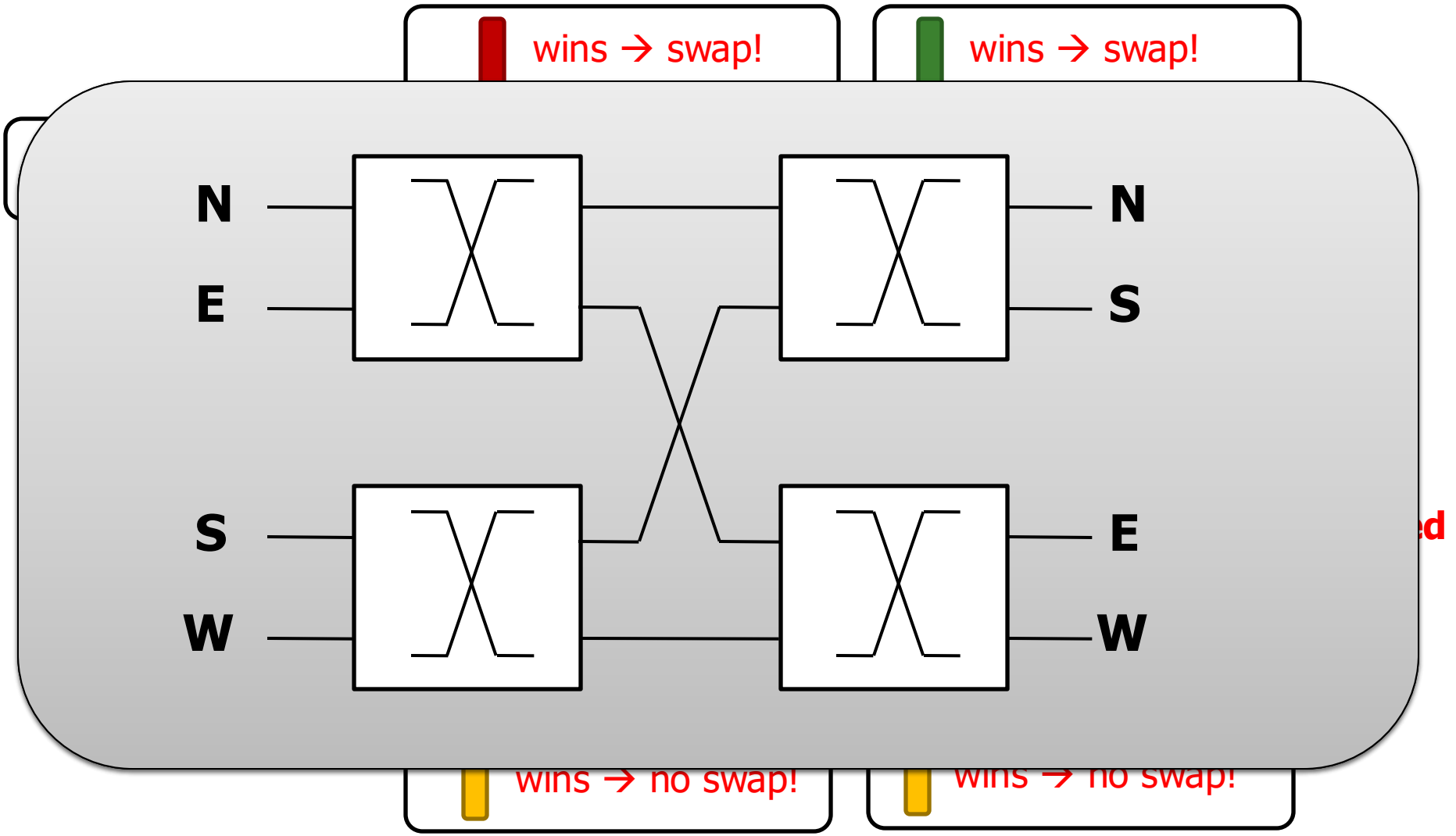
➔ **Golden Flit is always routed toward its destination**

Golden Flit Routing with Four Inputs

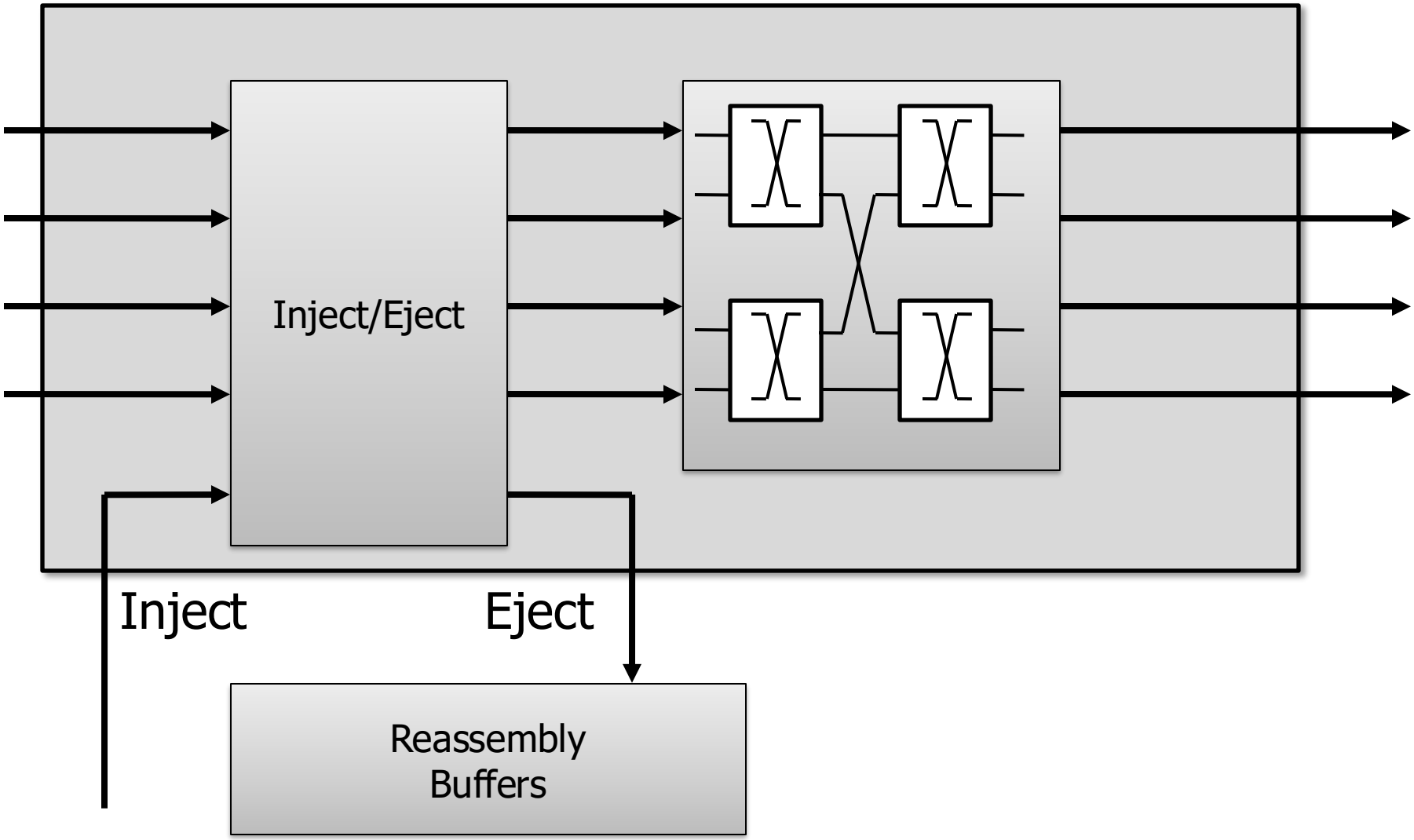
- Each block makes decisions independently!
  - **Deflection is a distributed decision**



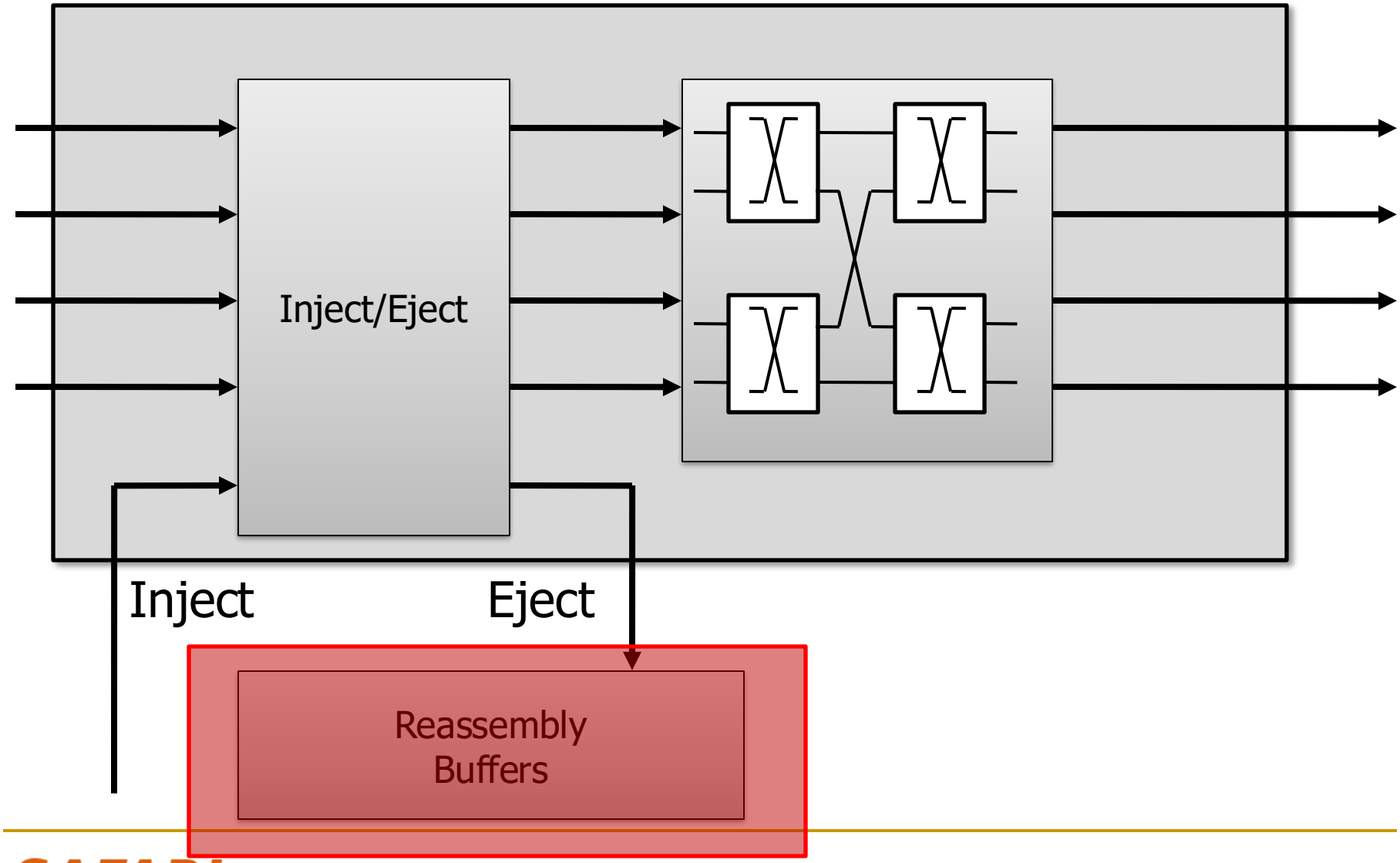
Permutation Network Operation



Permutation Network-based Pipeline

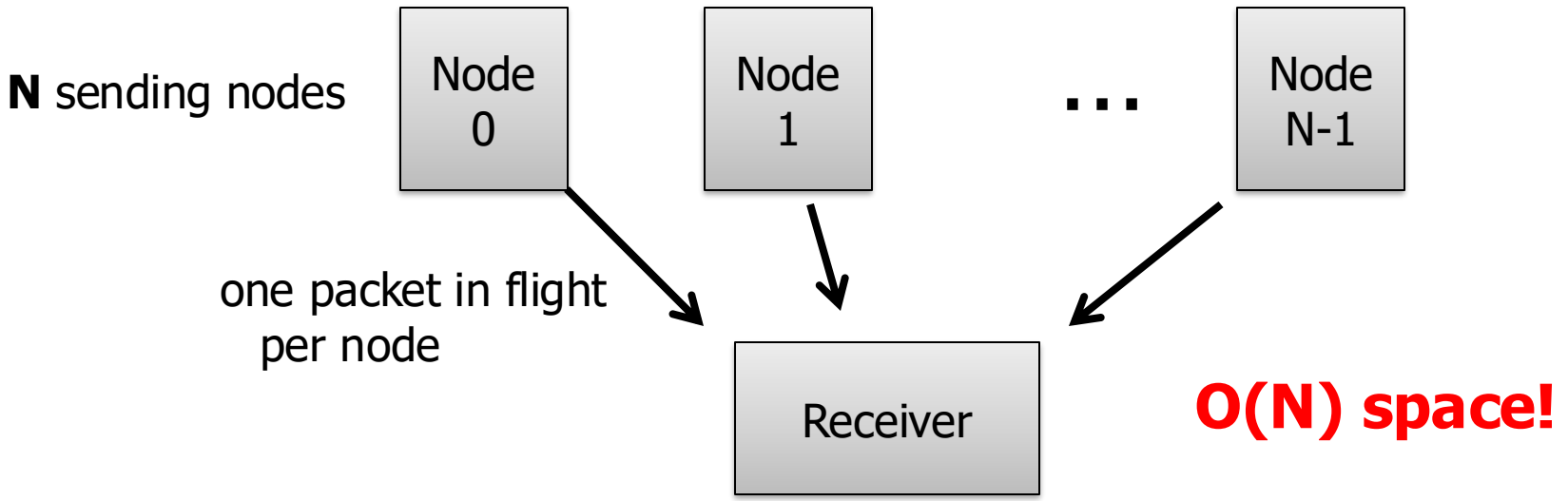


Problem 2: Packet Reassembly



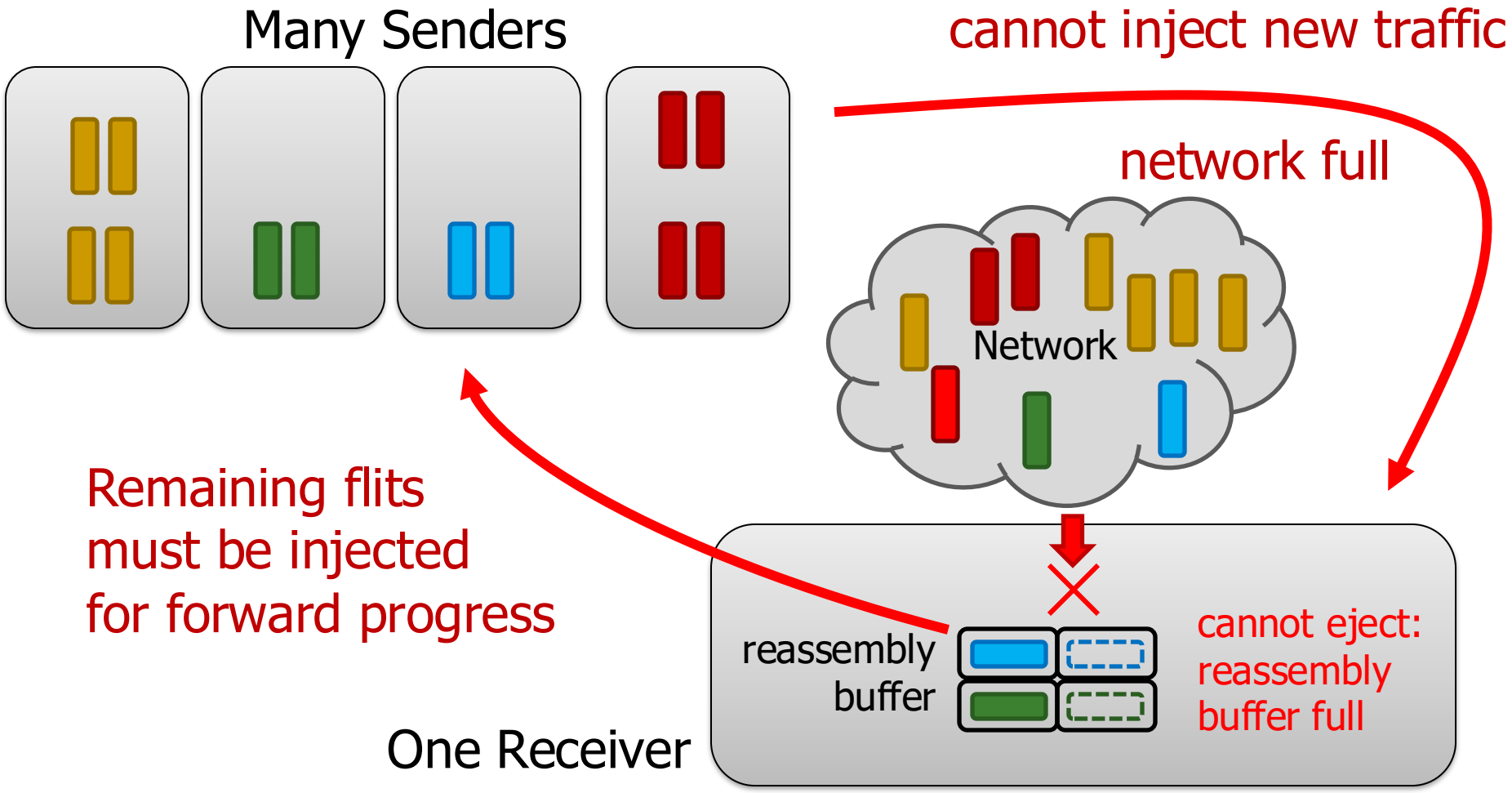
Reassembly Buffers are Large

- **Worst case**: every node sends a packet to one receiver
- Why can't we make reassembly buffers smaller?



Small Reassembly Buffers Cause Deadlock

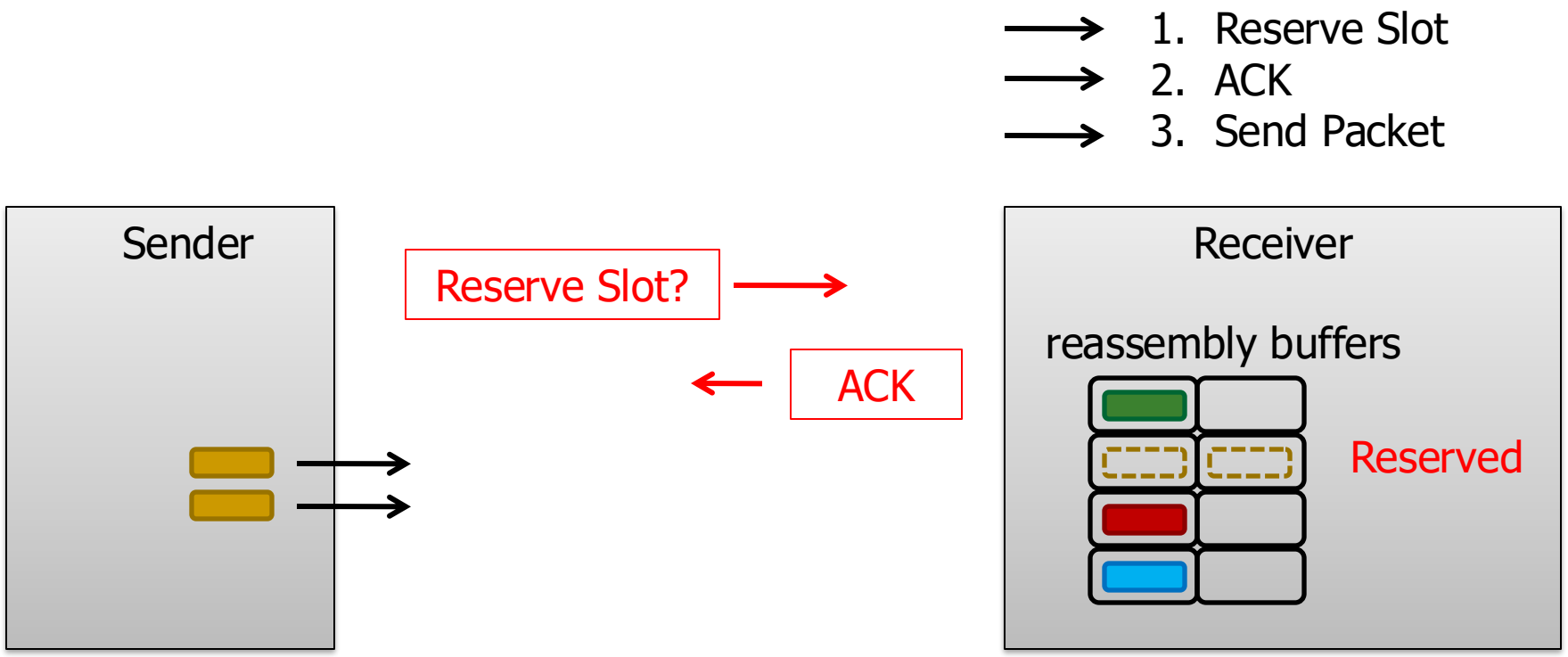
- What happens when reassembly buffer is too small?



Reserve Space to Avoid Deadlock?

- What if every sender asks permission from the receiver before it sends?

➔ **adds additional delay to every request**



Escaping Deadlock with Retransmissions

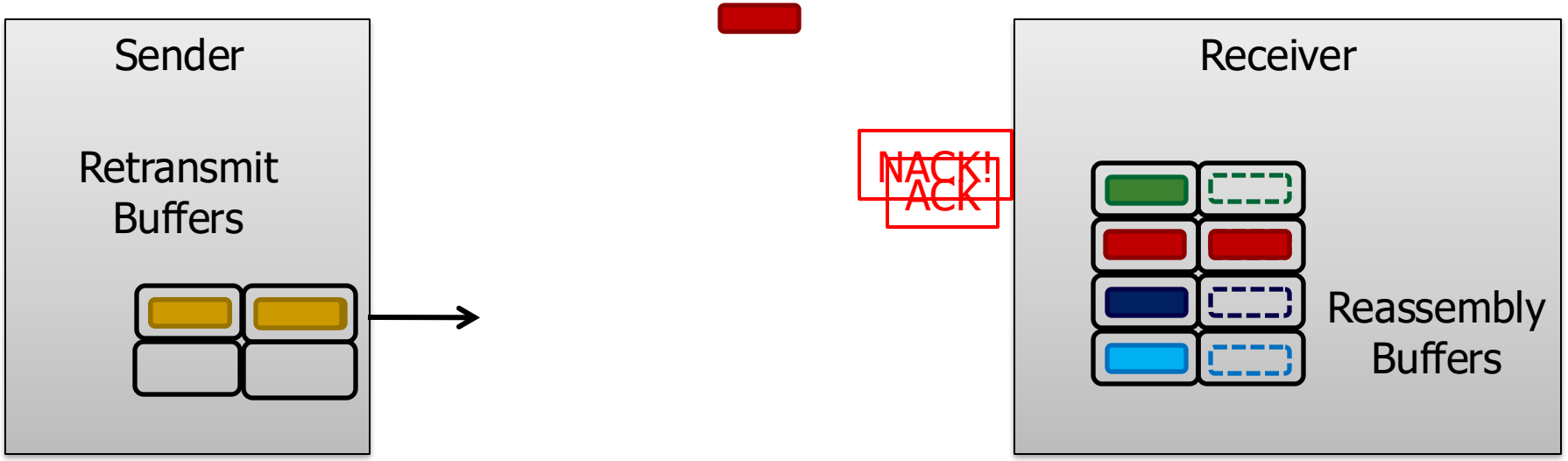
- Sender is optimistic instead: assume buffer is free
  - If not, receiver **drops** and NACKs; sender **retransmits**

→ **no additional delay in best case**

→ **transmit buffering overhead for all packets**

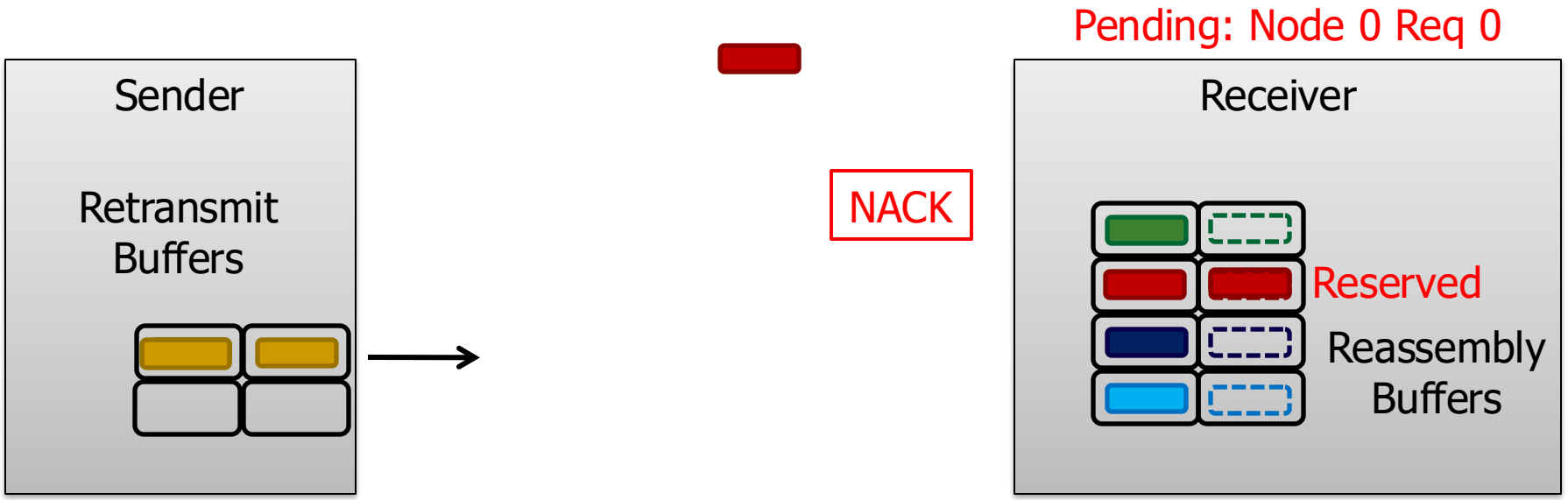
→ **potentially many retransmits**

1. Send (2 flits)
2. Drop, NACK
3. Other packet completes
4. Retransmit packet
5. ACK
6. Sender frees data

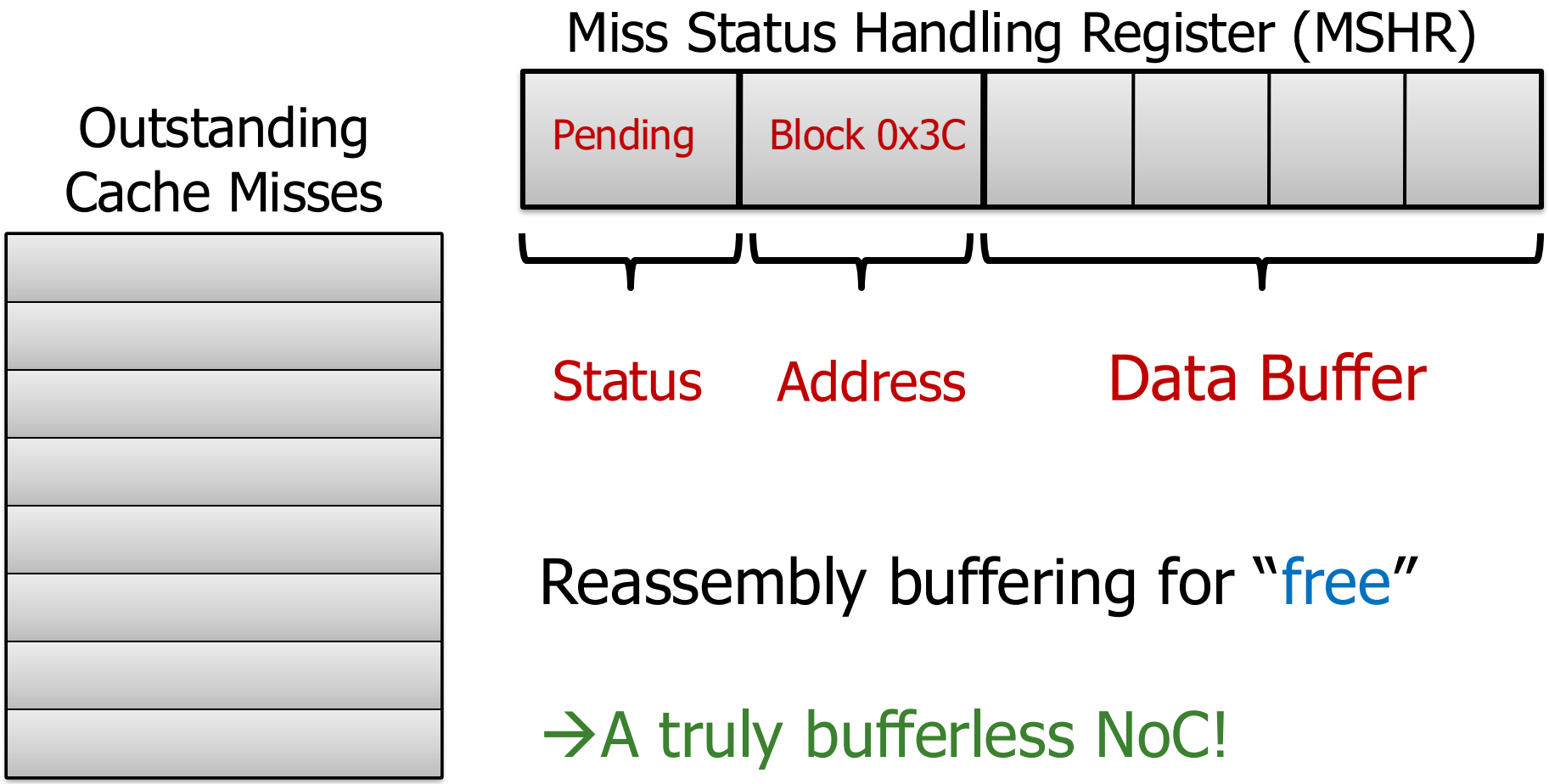


Solution: Retransmitting Only Once

- **Key Idea:** Retransmit only when space becomes available.
  - → Receiver drops packet if full; notes which packet it drops
  - → When space frees up, receiver reserves space so retransmit is successful
  - → Receiver notifies sender to retransmit



Use MSHRs as Reassembly Buffers

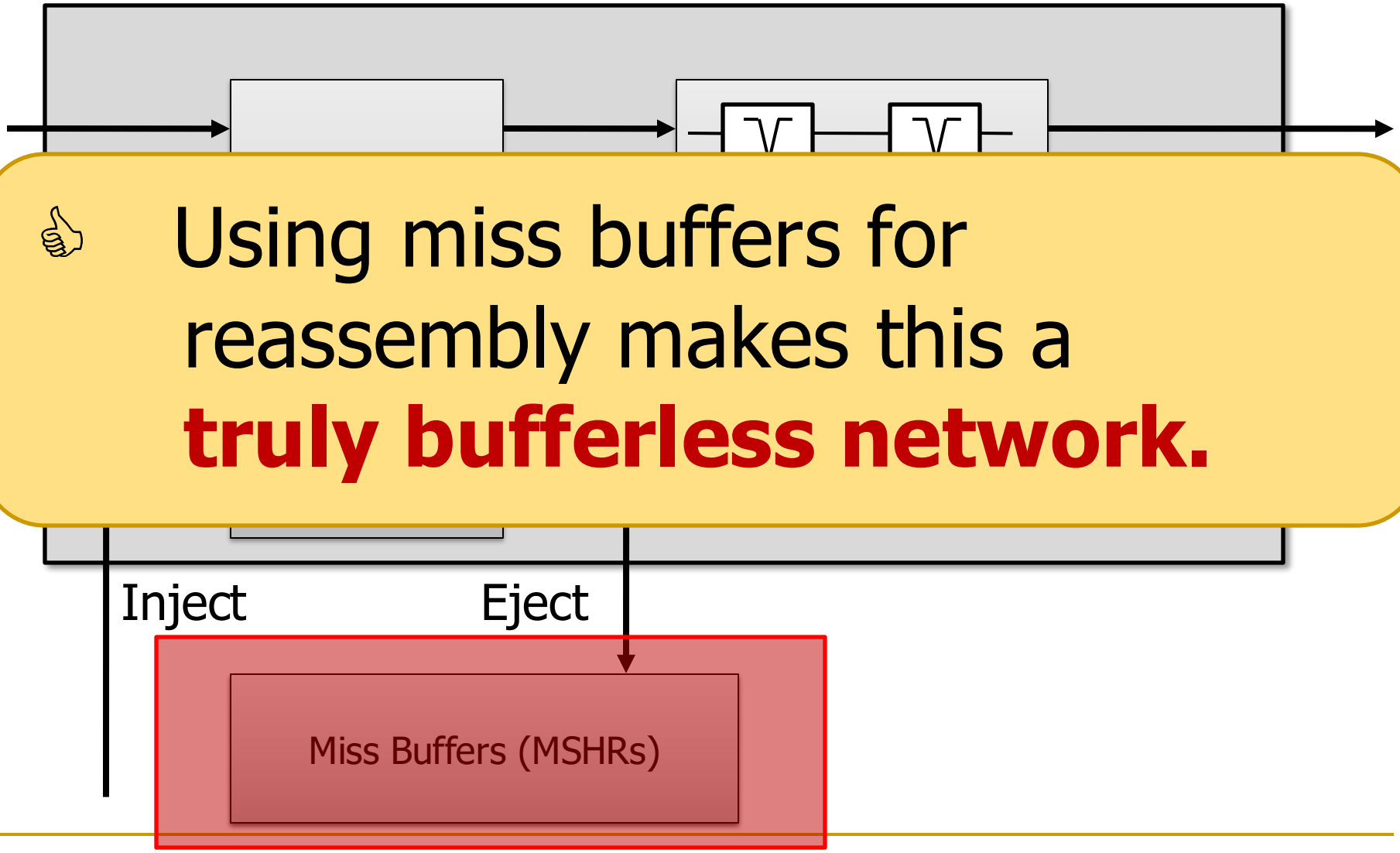


Reassembly buffering for "free"

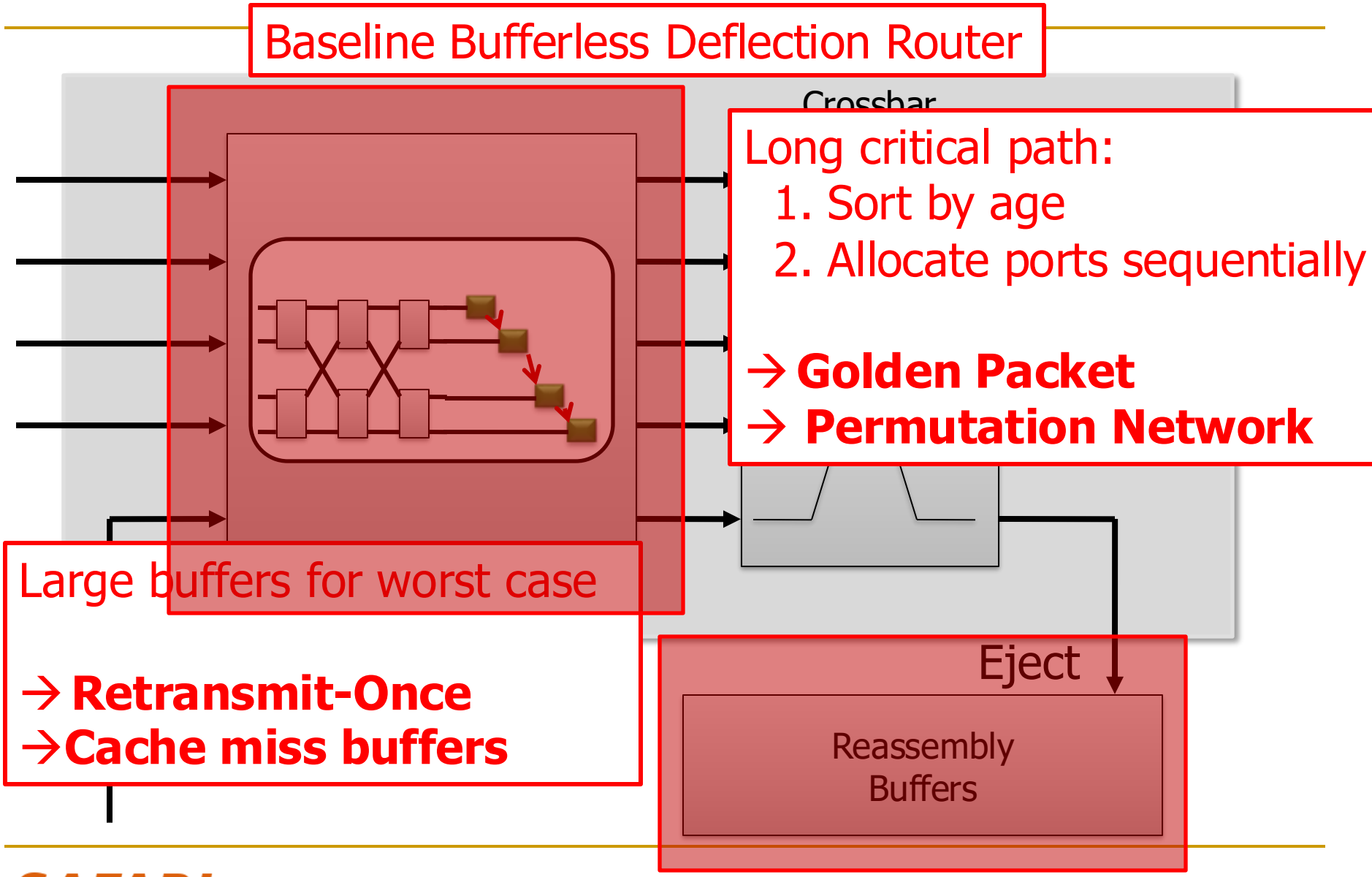
→A truly bufferless NoC!

Kroft, "Lockup-Free Instruction Fetch/Prefetch Cache Organization," ISCA 1981.

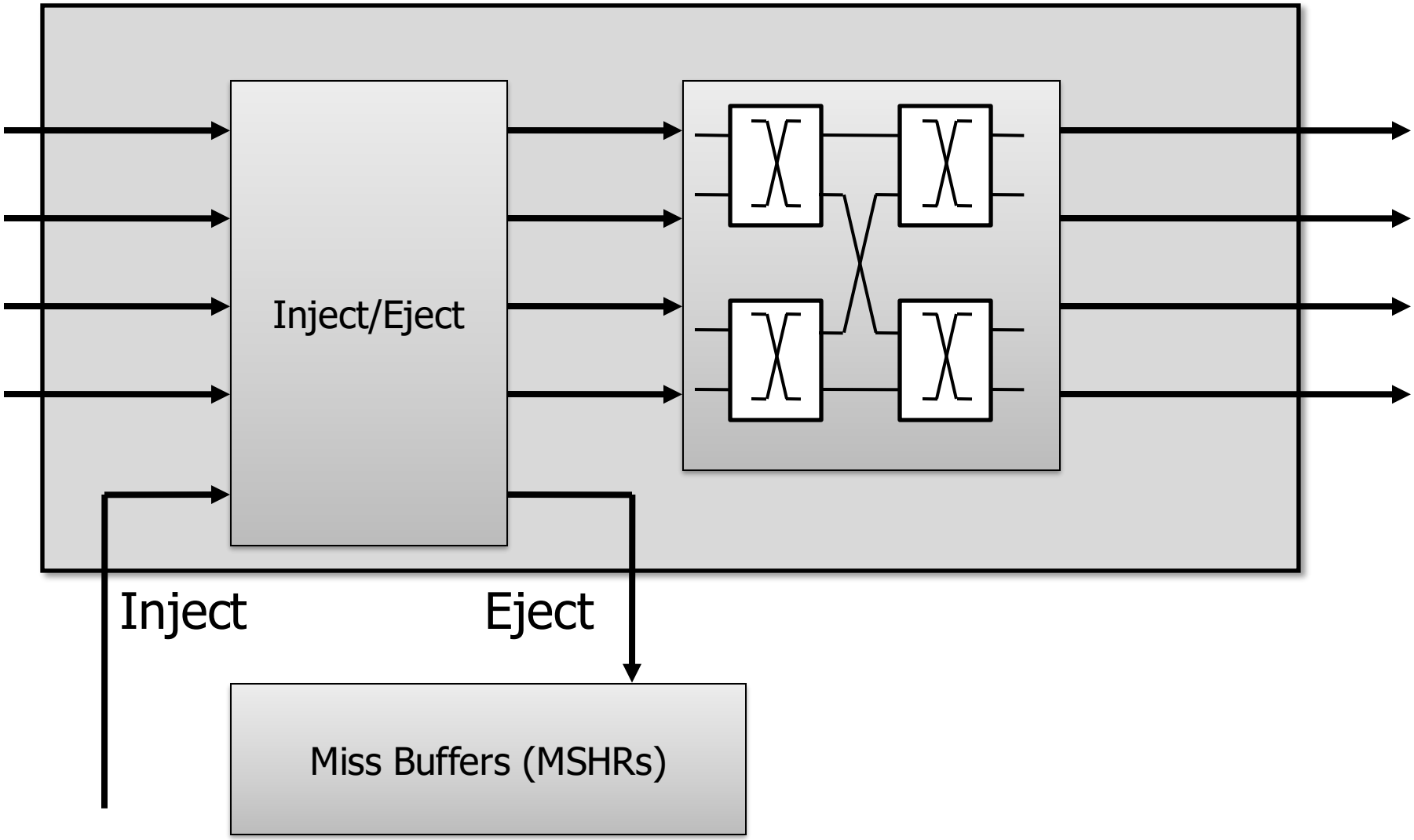
Using MSHRs as Reassembly Buffers



CHIPPER: Cheap Interconnect Partially-Permuting Router



CHIPPER: Cheap Interconnect Partially-Permuting Router



EVALUATION

Methodology

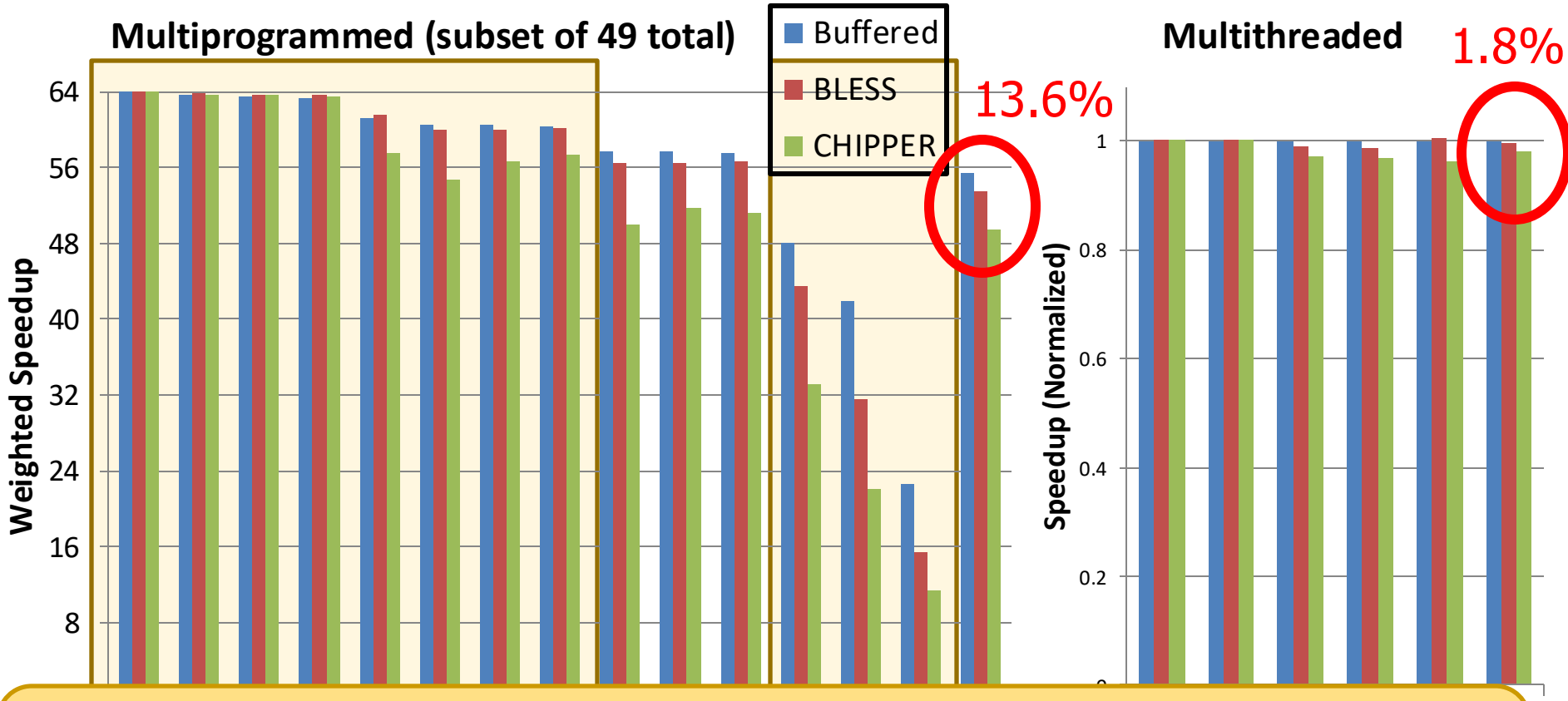
- **Multiprogrammed** workloads: CPU2006, server, desktop
  - 8x8 (64 cores), 39 homogeneous and 10 mixed sets
- **Multithreaded** workloads: SPLASH-2, 16 threads
  - 4x4 (16 cores), 5 applications
- **System configuration**
  - Buffered baseline: 2-cycle router, 4 VCs/channel, 8 flits/VC
  - Bufferless baseline: 2-cycle latency, FLIT-BLESS
  - Instruction-trace driven, closed-loop, 128-entry OoO window
  - 64KB L1, perfect L2 (stresses interconnect), XOR mapping

Methodology

- **Hardware modeling**
  - Verilog models for CHIPPER, BLESS, buffered logic
    - Synthesized with commercial 65nm library
  - ORION for crossbar, buffers and links

- **Power**
  - Static and dynamic power from hardware models
  - Based on event counts in cycle-accurate simulations

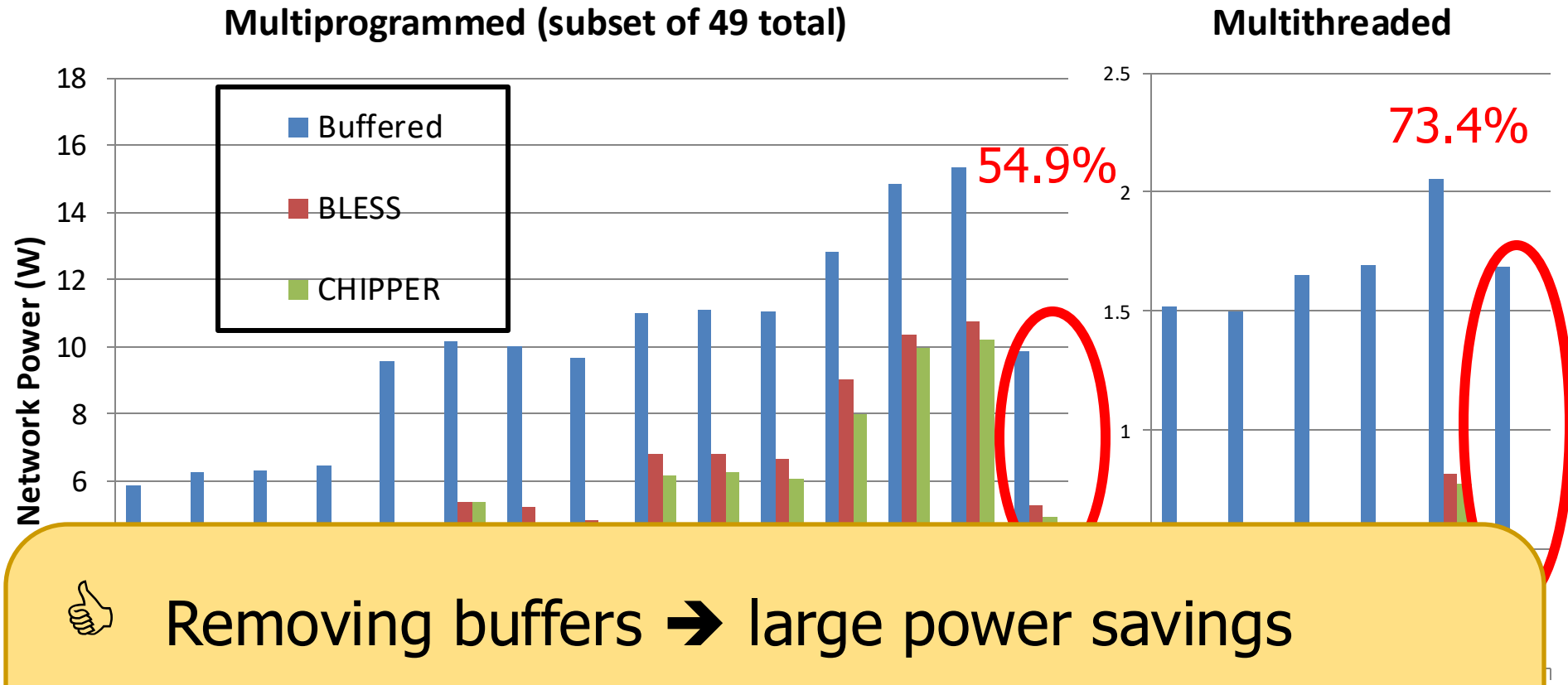
Results: Performance Degradation



Small loss for low-to-medium-intensity workloads

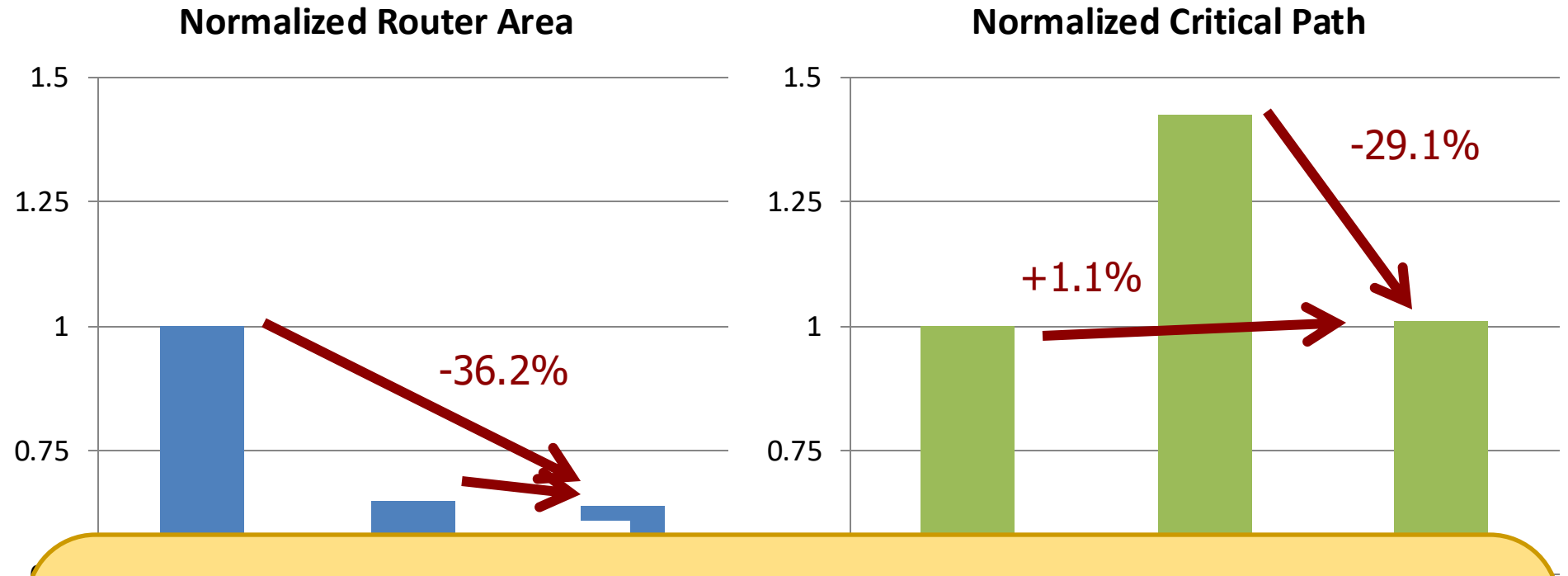
49.8%

Results: Power Reduction



- Removing buffers ➔ large power savings
- Slight savings from BLESS to CHIPPER

Results: Area and Critical Path Reduction



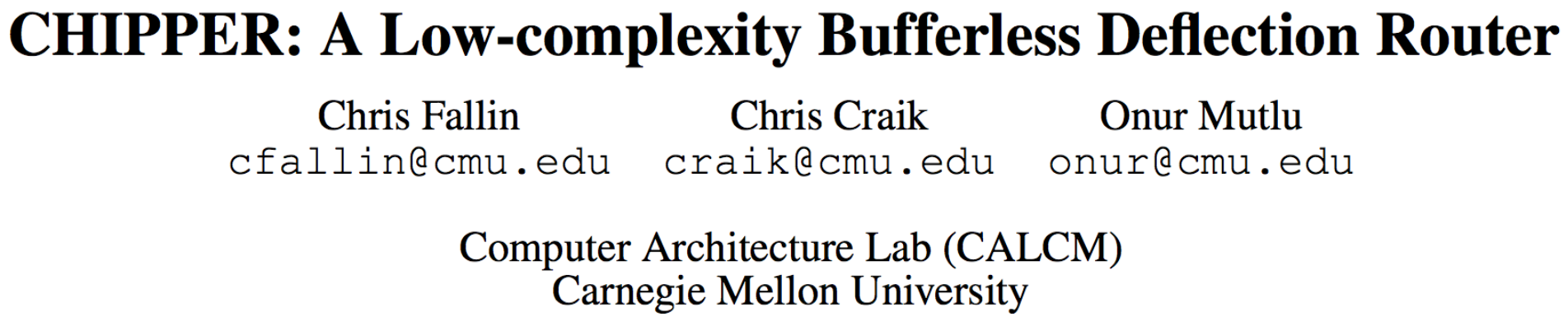
- **CHIPPER maintains area savings** of BLESS
- Critical path **becomes competitive** to buffered

CHIPPER Router: Conclusions

- Two key issues in bufferless deflection routing
  - livelock freedom and packet reassembly
- Bufferless deflection routers were high-complexity and impractical
  - Oldest-first prioritization → long critical path in router
  - No end-to-end flow control for reassembly → prone to deadlock with reasonably-sized reassembly buffers
- CHIPPER is a new, practical bufferless deflection router
  - Golden packet prioritization → short critical path in router
  - Retransmit-once protocol → deadlock-free packet reassembly
  - Cache miss buffers as reassembly buffers → truly bufferless network
- CHIPPER frequency comparable to buffered routers at much lower area and power cost, and minimal performance loss

More on CHIPPER

- Chris Fallin, Chris Craik, and Onur Mutlu, **"CHIPPER: A Low-Complexity Bufferless Deflection Router"** *Proceedings of the 17th International Symposium on High-Performance Computer Architecture (**HPCA**)*, pages 144-155, San Antonio, TX, February 2011. Slides (pptx)
  - An extended version as *SAFARI Technical Report*, TR-SAFARI-2010-001, Carnegie Mellon University, December 2010.



CHIPPER: A Low-complexity Bufferless Deflection Router

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Chris Craik craik@cmu.edu
Onur Mutlu onur@cmu.edu

Computer Architecture Lab (CALCM)
Carnegie Mellon University

Better Bufferless Networks

Can We Do Better?

Minimally-Buffered Deflection Routing

- Bufferless deflection routing offers reduced power & area
- But, high deflection rate hurts performance at high load

- **MinBD** (Minimally-Buffered Deflection Router) introduces:
  - Side buffer to hold **only** flits that would have been deflected
  - Dual-width ejection to address ejection bottleneck
  - Two-level prioritization to avoid unnecessary deflections

- MinBD yields reduced power (31%) & reduced area (36%) relative to **buffered** routers
- MinBD yields improved performance (8.1% at high load) relative to **bufferless** routers → closes half of perf. gap

- MinBD has the **best energy efficiency** of all evaluated designs with **competitive performance**

Minimally-Buffered Deflection Routing

- Chris Fallin, Greg Nazario, Xiangyao Yu, Kevin Chang, Rachata Ausavarungnirun, and Onur Mutlu,
  "MinBD: Minimally-Buffered Deflection Routing for Energy-Efficient Interconnect"
  *Proceedings of the 6th ACM/IEEE International Symposium on Networks on Chip (**NOCS**)*, Lyngby, Denmark, May 2012. Slides (pptx) (pdf)
  One of the five papers nominated for the Best Paper Award by the Program Committee.

MinBD: Minimally-Buffered Deflection Routing for Energy-Efficient Interconnect

Chris Fallin, Greg Nazario, Xiangyao Yu†, Kevin Chang, Rachata Ausavarungnirun, Onur Mutlu

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†Tsinghua University & Carnegie Mellon University
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SAFARI Carnegie Mellon University

Bufferless Deflection Routing

- **Key idea**: Packets are never buffered in the network. When two packets contend for the same link, one is **deflected.**

- Removing **buffers** yields significant benefits
  - Reduces **power** (CHIPPER: reduces NoC power by 55%)
  - Reduces **die area** (CHIPPER: reduces NoC area by 36%)

- But, at **high network utilization** (load), bufferless deflection routing causes **unnecessary link & router traversals**
  - Reduces network throughput and application performance
  - Increases dynamic power

- **Goal**: Improve high-load performance of low-cost deflection networks by reducing the deflection rate.

Outline: This Talk

- **Motivation**
- **Background**: Bufferless Deflection Routing
- **MinBD**: Reducing Deflections
  - Addressing Link Contention
  - Addressing the Ejection Bottleneck
  - Improving Deflection Arbitration
- **Results**
- **Conclusions**

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Issues in Bufferless Deflection Routing

- **Correctness**: Deliver all packets without livelock
  - **CHIPPER**[1]: **Golden Packet**
  - Globally prioritize one packet until delivered
- **Correctness**: Reassemble packets without deadlock
  - **CHIPPER**[1]: **Retransmit-Once**
- **Performance**: Avoid performance degradation at high load
  - **MinBD**

[1] Fallin et al., "CHIPPER: A Low-complexity Bufferless Deflection Router", HPCA 2011.

Key Performance Issues

1. **Link contention**: no buffers to hold traffic → any link contention causes a deflection

   → use side buffers

2. **Ejection bottleneck**: only one flit can eject per router per cycle → simultaneous arrival causes deflection

   → eject up to 2 flits/cycle

3. **Deflection arbitration**: practical (fast) deflection arbiters deflect unnecessarily

   → new priority scheme (silver flit)

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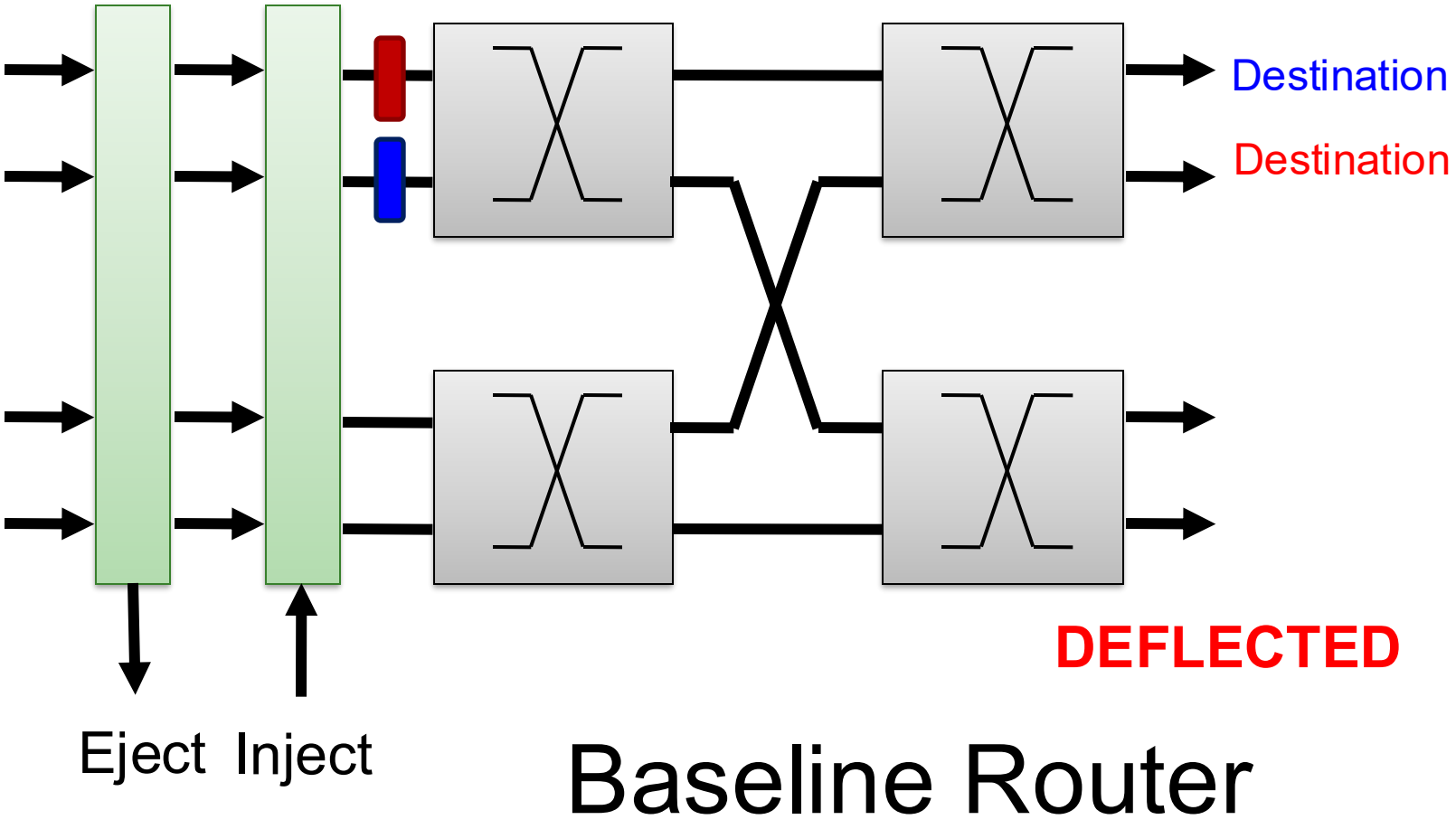
Addressing Link Contention

- **Problem 1**: Any link contention causes a deflection

- Buffering a flit can avoid deflection on contention
- But, **input buffers** are expensive:
  - All flits are buffered on every hop → high dynamic energy
  - Large buffers necessary → high static energy and large area

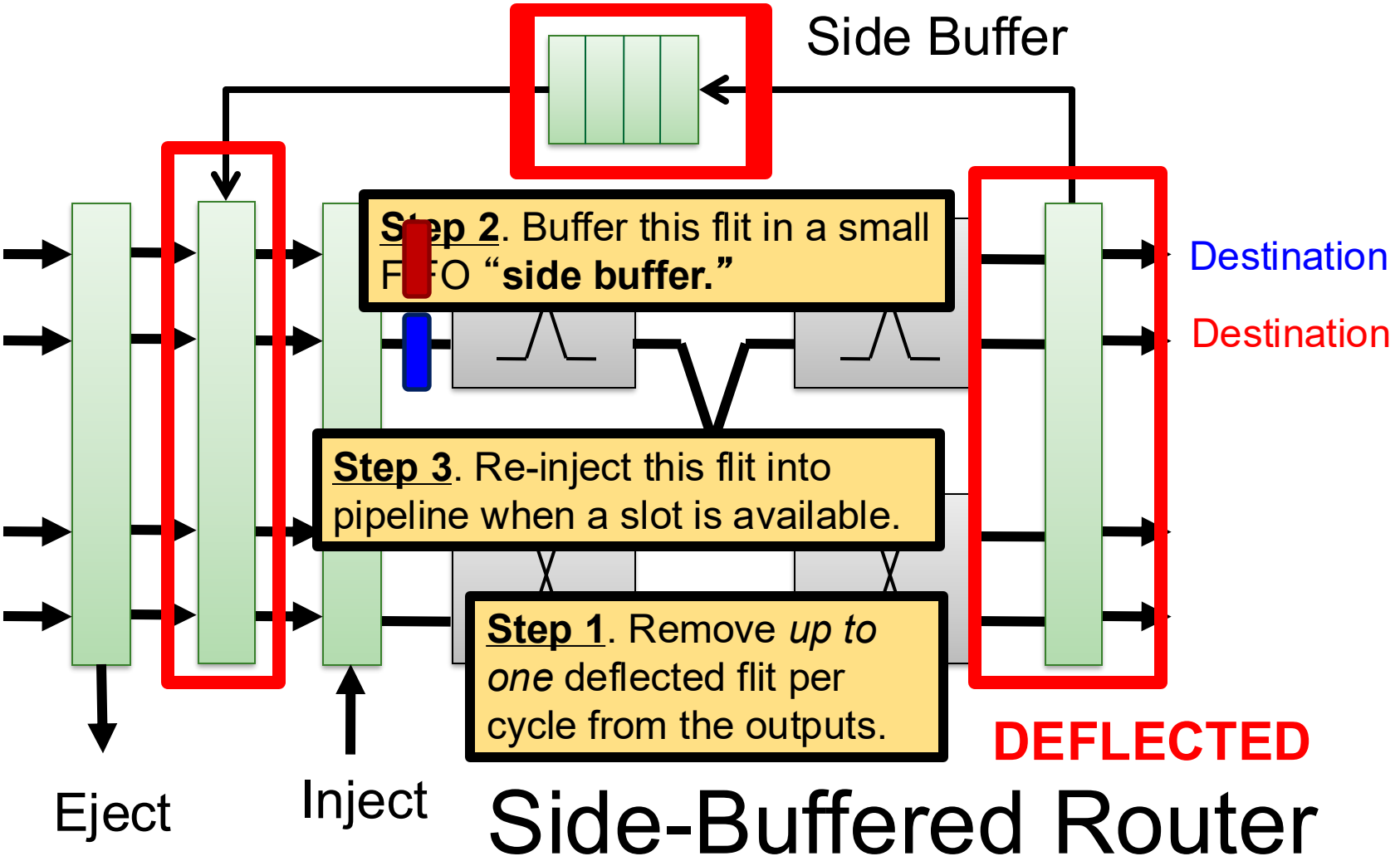
- **Key Idea 1**: add a small buffer to a bufferless deflection router to buffer **only** flits that would have been deflected

How to Buffer Deflected Flits



[1] Fallin et al., "CHIPPER: A Low-complexity Bufferless Deflection Router", HPCA 2011.

How to Buffer Deflected Flits



Why Could A Side Buffer Work Well?

- Buffer some flits and deflect other flits at per-flit level
  - Relative to **bufferless routers**, deflection rate reduces (need not deflect all contending flits)
    → 4-flit buffer reduces deflection rate by 39%
  - Relative to **buffered routers**, buffer is more efficiently used (need not buffer all flits)
    → similar performance with 25% of buffer space

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Addressing the Ejection Bottleneck

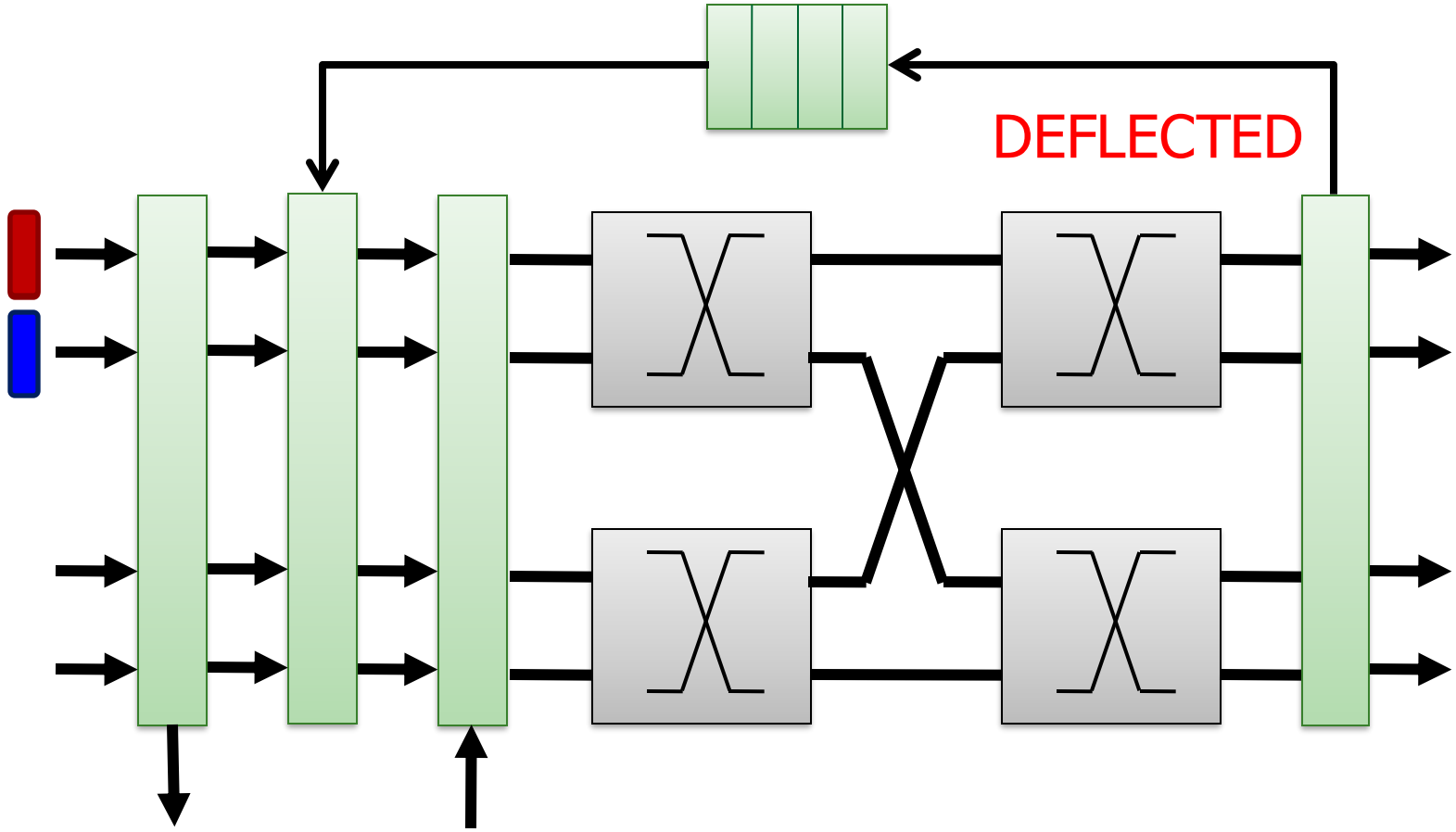
- **Problem 2**: Flits deflect unnecessarily because only one flit can **eject** per router per cycle

- In 20% of all ejections, ≥ 2 flits could have ejected
  → all but one flit must **deflect** and try again
  → these deflected flits cause additional contention

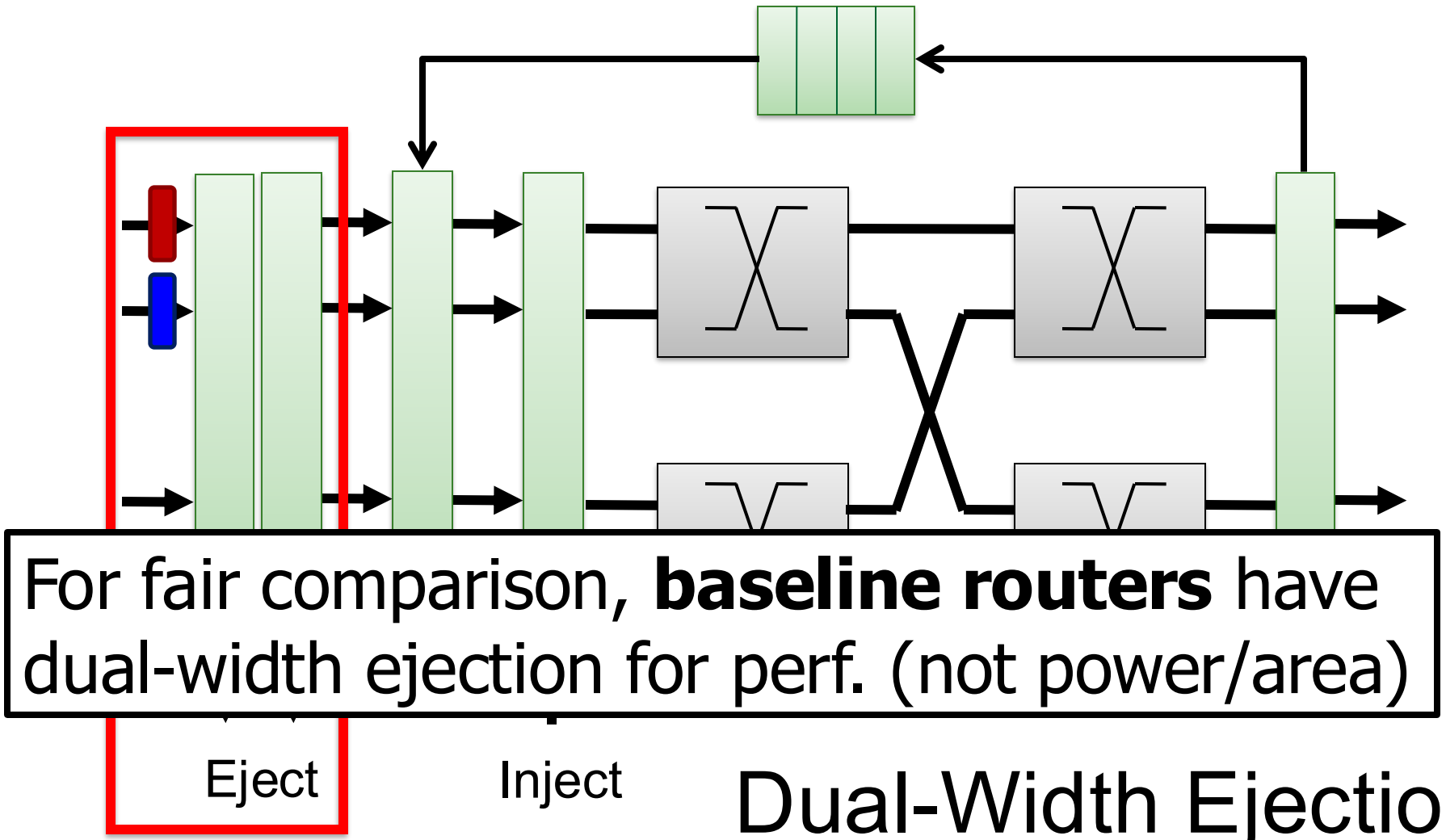
- Ejection width of 2 flits/cycle reduces deflection rate 21%

- **Key idea 2**: Reduce deflections due to a single-flit ejection port by allowing **two flits** to eject per cycle

Addressing the Ejection Bottleneck



Addressing the Ejection Bottleneck



Outline: This Talk

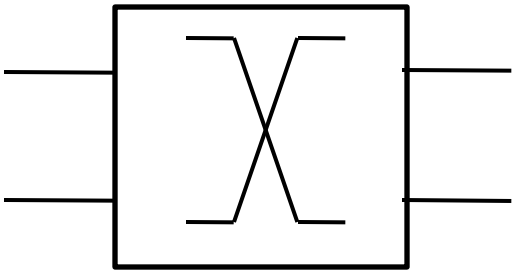
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Improving Deflection Arbitration

- **Problem 3**: Deflections occur unnecessarily because fast arbiters must use simple priority schemes
- Age-based priorities (several past works): full priority order gives fewer deflections, but requires slow arbiters
- State-of-the-art deflection arbitration (Golden Packet & two-stage permutation network)
  - Prioritize one packet globally (**ensure forward progress**)
  - Arbitrate other flits randomly (**fast critical path**)
- Random common case leads to uncoordinated arbitration

Fast Deflection Routing Implementation

- Let's route in a two-input router first:

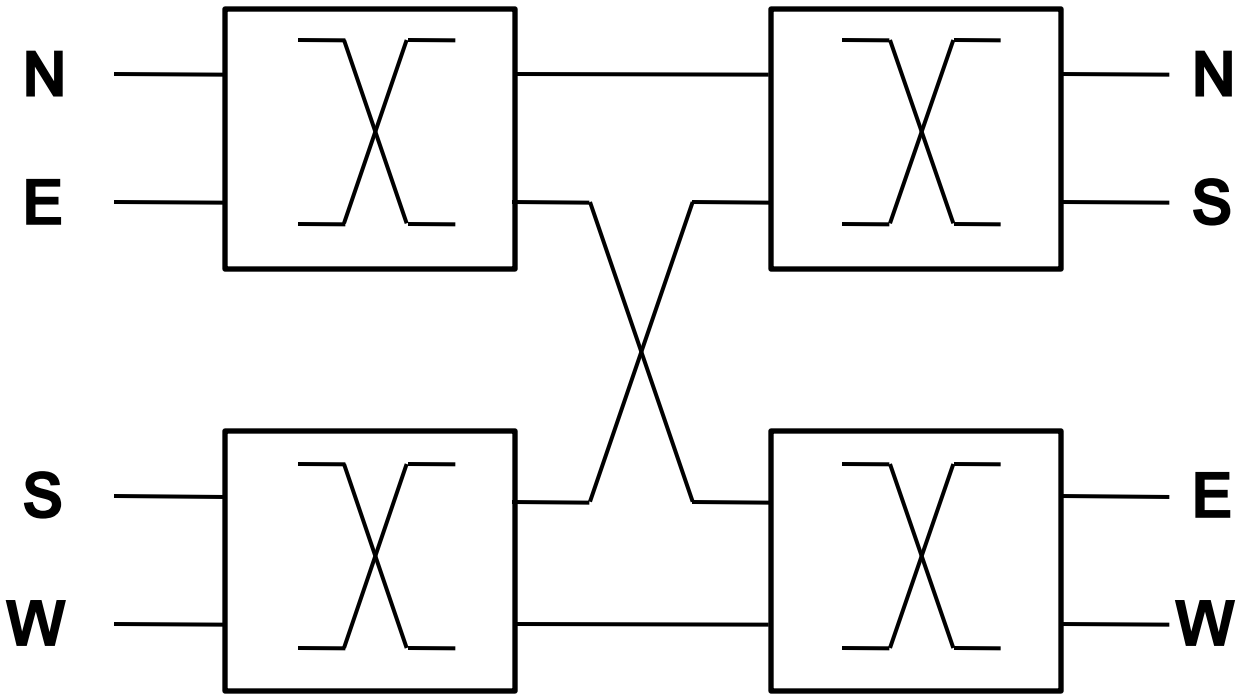


- **Step 1**: pick a "winning" flit (Golden Packet, else random)
- **Step 2**: steer the winning flit to its desired output and deflect other flit

➔ **Highest-priority flit always routes to destination**

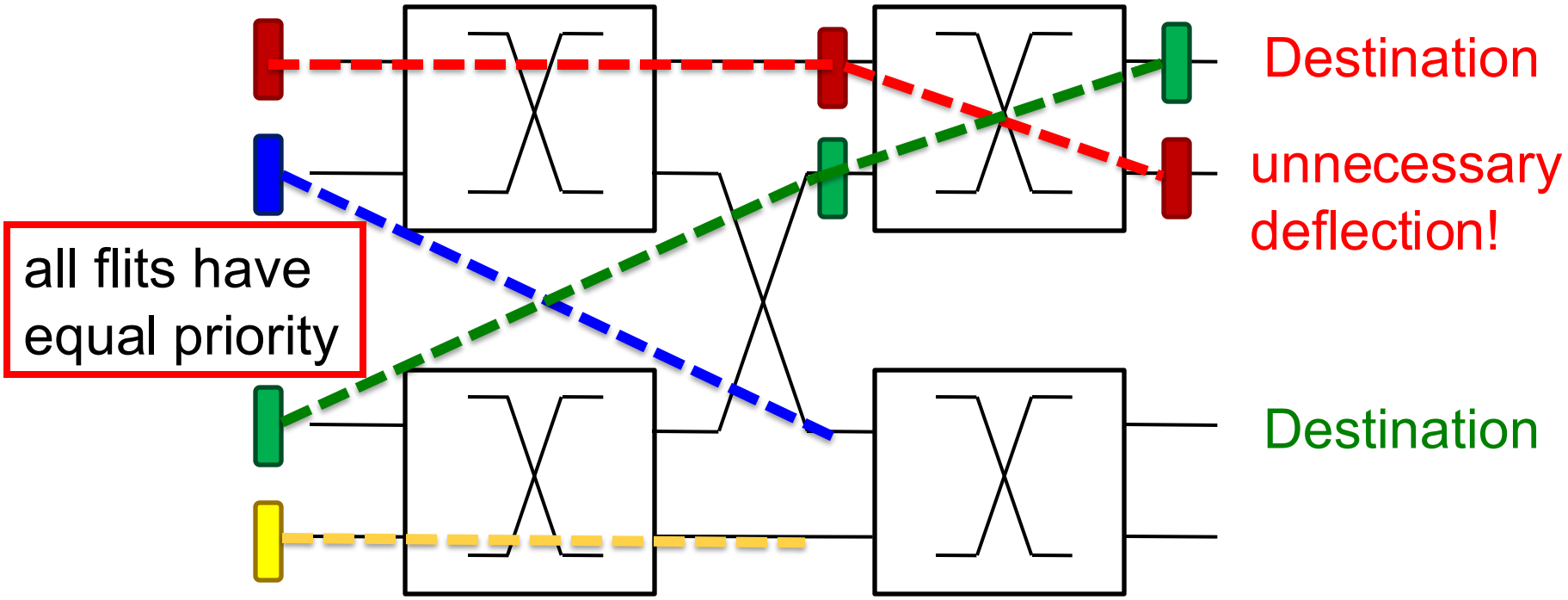
Fast Deflection Routing with Four Inputs

- Each block makes decisions independently
  - **Deflection is a distributed decision**



Unnecessary Deflections in Fast Arbiters

- How does lack of coordination cause unnecessary deflections?
  1. No flit is golden (pseudorandom arbitration)
  2. Red flit wins at first stage
  3. Green flit loses at first stage (must be deflected now)
  4. Red flit loses at second stage; Red and Green are deflected



Improving Deflection Arbitration

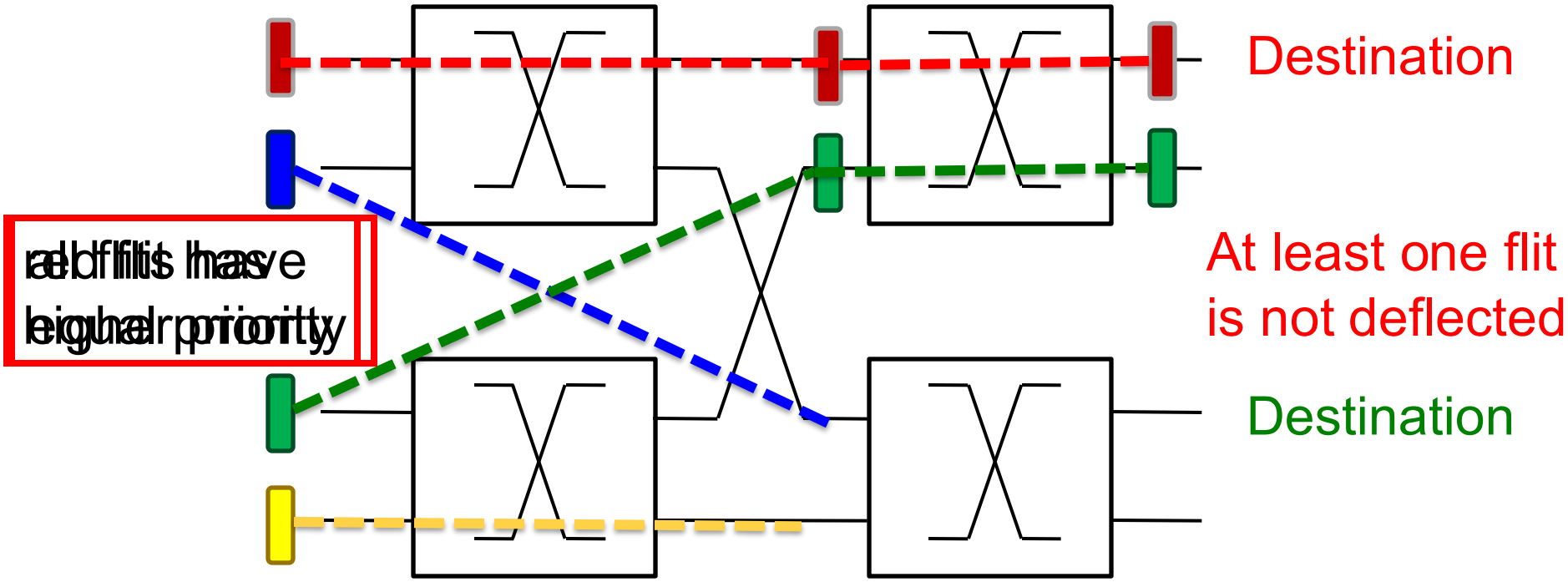
- **Key idea 3: Add a priority level** and prioritize one flit to ensure at least one flit is not deflected in each cycle

- **Highest priority:** one Golden Packet in network
  - Chosen in static round-robin schedule
  - Ensures correctness

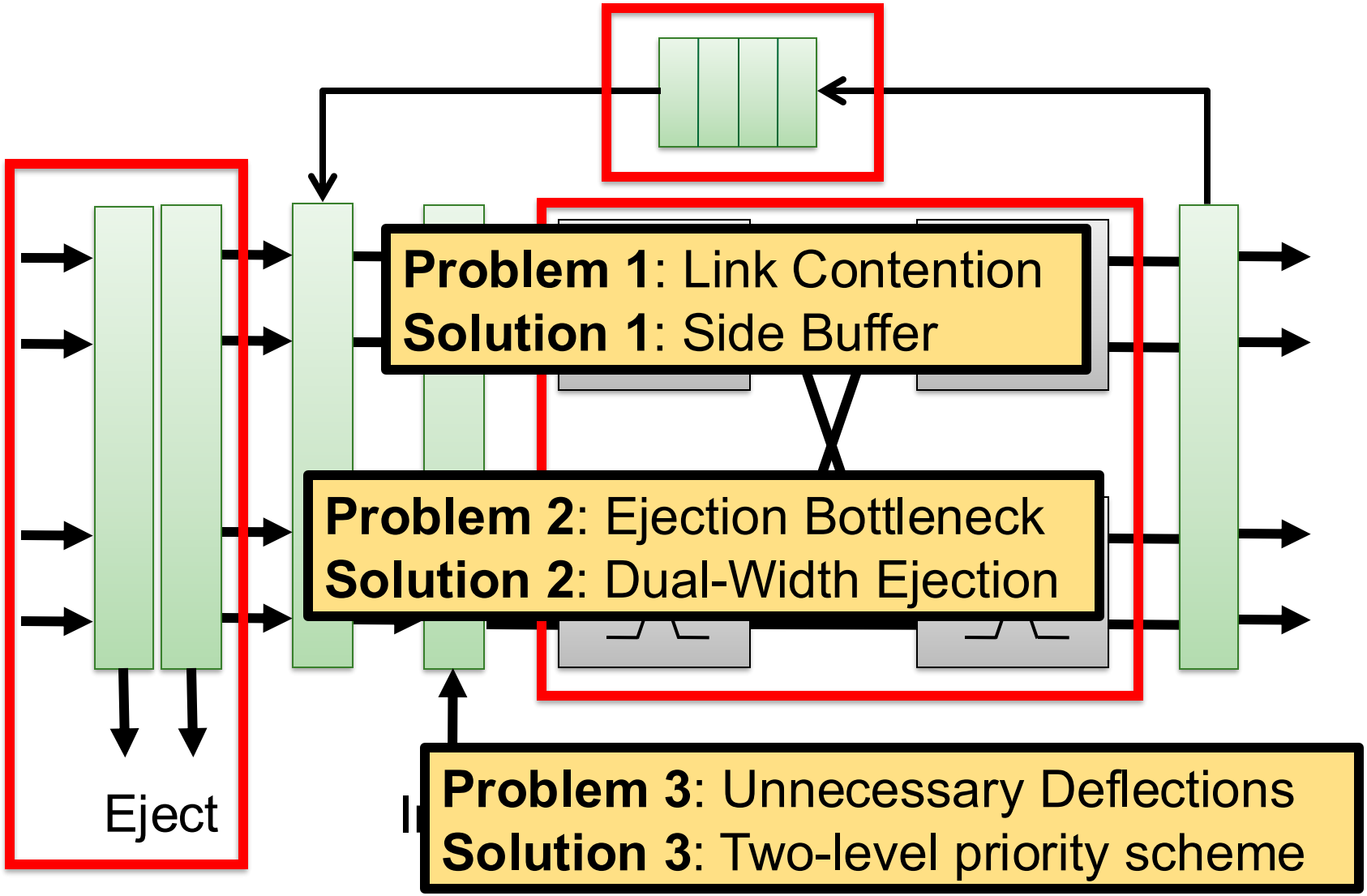
- **Next-highest priority**: one silver flit per router per cycle
  - Chosen pseudo-randomly & local to one router
  - Enhances performance

Adding A Silver Flit

- Randomly picking a silver flit ensures **one flit is not deflected**
  1. No flit is golden but Red flit is silver
  2. Red flit wins at first stage (silver)
  3. Green flit is deflected at first stage
  4. Red flit wins at second stage (silver); not deflected



Minimally-Buffered Deflection Router



Outline: This Talk

- **Motivation**
- **Background**: Bufferless Deflection Routing
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Methodology: Simulated System

- **Chip Multiprocessor Simulation**
  - **64-core** and **16-core** models
  - **Closed-loop** core/cache/NoC cycle-level model
  - Directory cache coherence protocol (SGI Origin-based)
  - 64KB L1, perfect L2 (stresses interconnect), XOR-mapping
  - Performance metric: **Weighted Speedup**
    (similar conclusions from network-level latency)
  - Workloads: multiprogrammed SPEC CPU2006
    - 75 randomly-chosen workloads
    - Binned into network-load categories by average injection rate

Methodology: Routers and Network

- **Input-buffered** virtual-channel router
  - 8 VCs, 8 flits/VC [Buffered(8,8)]: large buffered router
  - 4 VCs, 4 flits/VC [Buffered(4,4)]: typical buffered router
  - 4 VCs, 1 flit/VC [Buffered(4,1)]: smallest deadlock-free router
  - All power-of-2 buffer sizes up to (8, 8) for perf/power sweep
- **Bufferless deflection** router: **CHIPPER**[1]
- **Bufferless-buffered hybrid** router: **AFC**[2]
  - Has input buffers and deflection routing logic
  - Performs coarse-grained (multi-cycle) mode switching
- **Common parameters**
  - 2-cycle router latency, 1-cycle link latency
  - 2D-mesh topology (16-node: 4x4; 64-node: 8x8)
  - Dual ejection assumed for baseline routers (for perf. only)

[1]Fallin et al., "CHIPPER: A Low-complexity Bufferless Deflection Router", HPCA 2011.
[2]Jafri et al., "Adaptive Flow Control for Robust Performance and Energy", MICRO 2010.

Methodology: Power, Die Area, Crit. Path

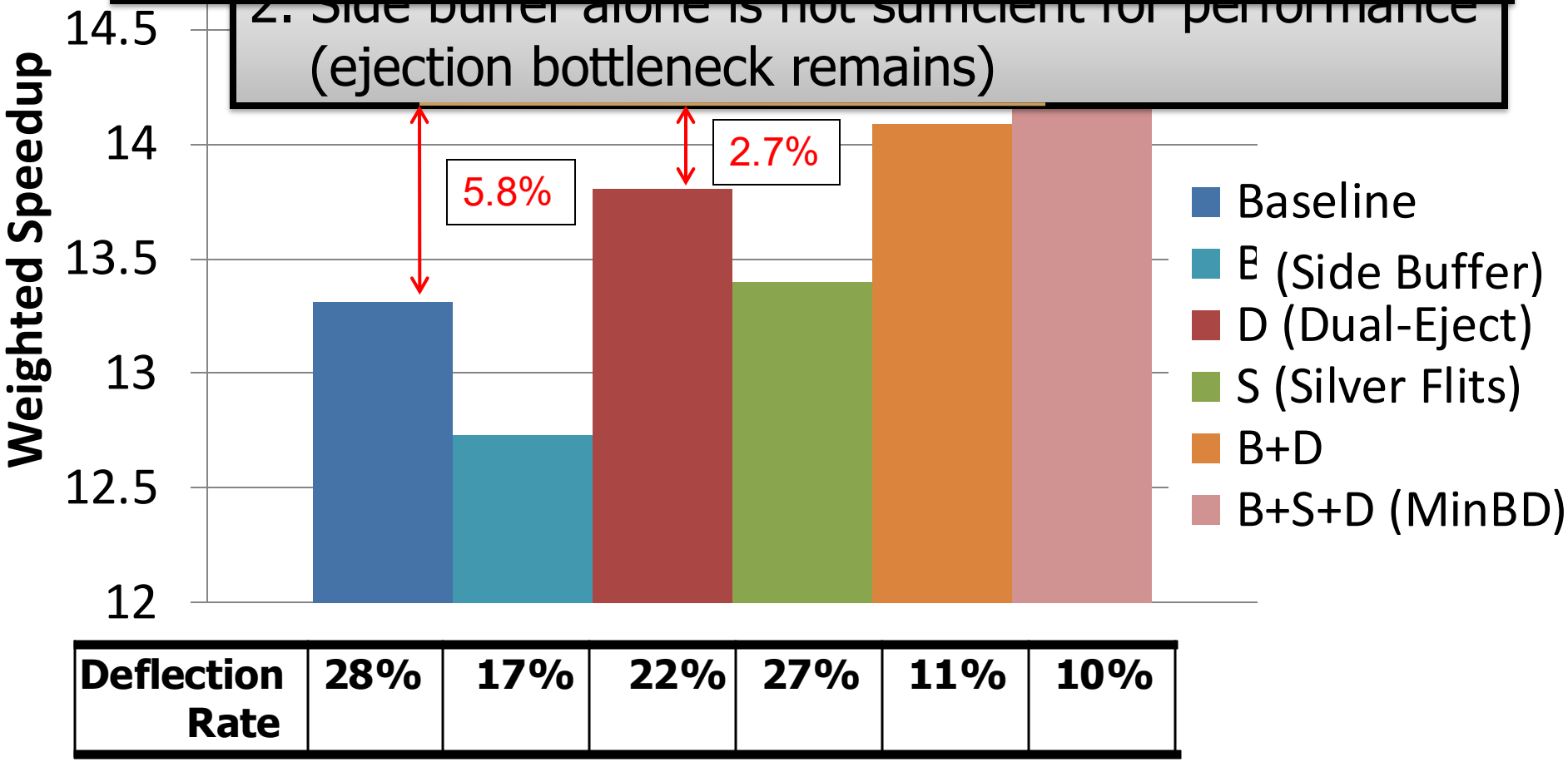
- **Hardware modeling**
  - Verilog models for CHIPPER, MinBD, buffered control logic
    - Synthesized with commercial 65nm library
  - ORION 2.0 for datapath: crossbar, muxes, buffers and links

- **Power**
  - Static and dynamic power from hardware models
  - Based on event counts in cycle-accurate simulations
  - Broken down into buffer, link, other

Reduced Deflections & Improved Perf.

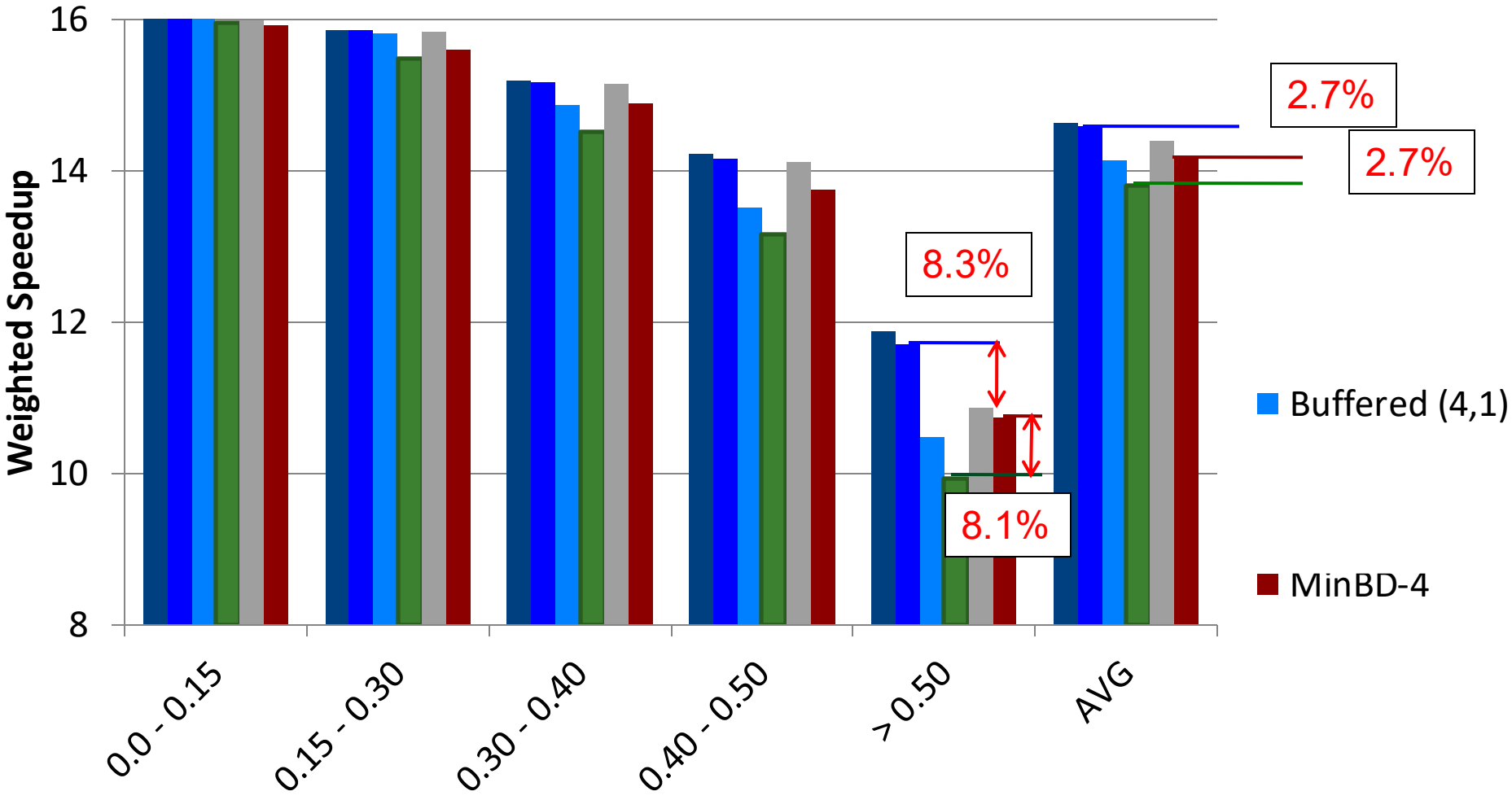
3. Overall, **5.8%** over baseline, **2.7%** over dual-eject by reducing deflections **64%** / **54%**

2. Side buffer alone is not sufficient for performance (ejection bottleneck remains)



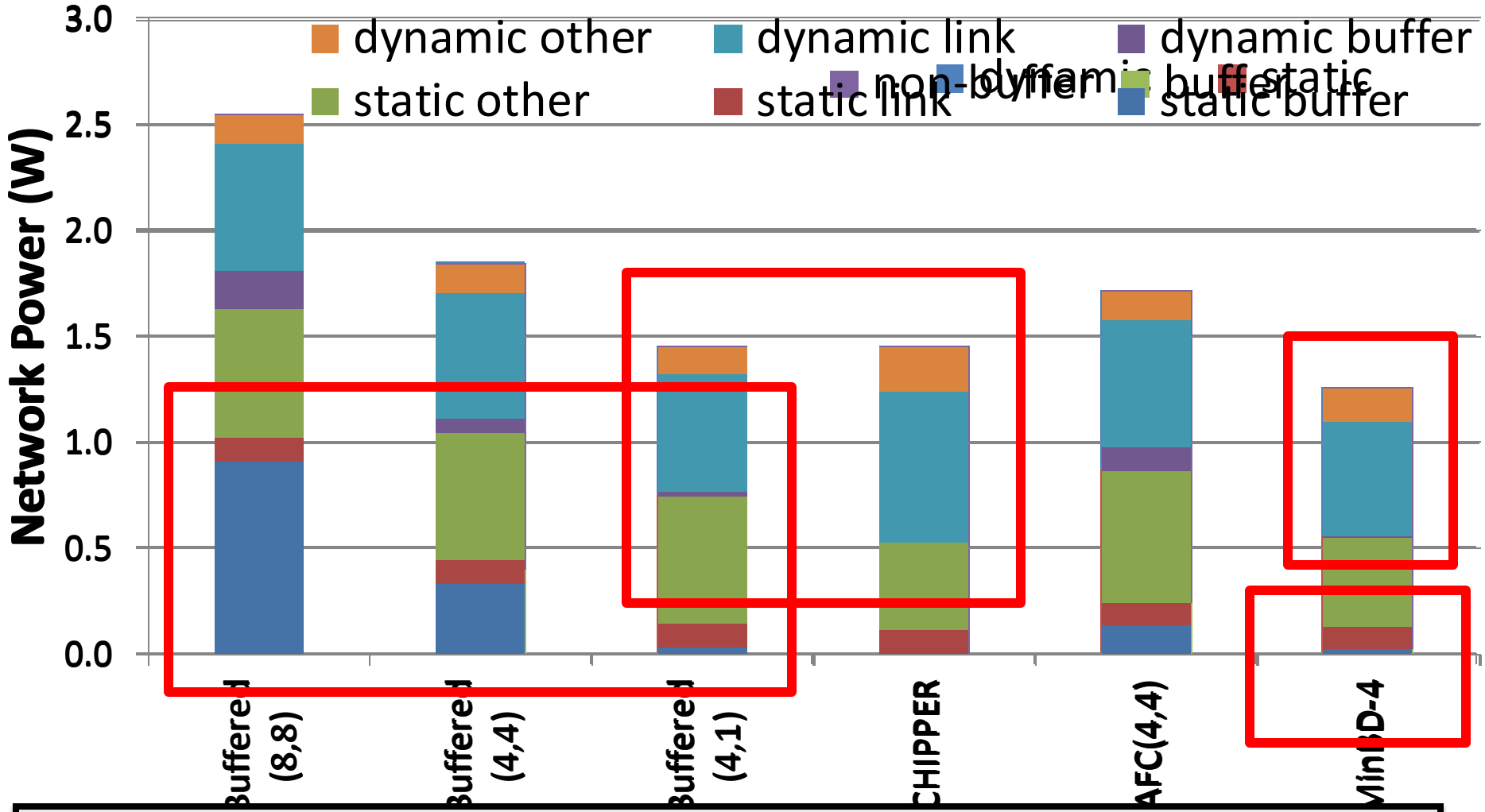
Deflection Rate	28%	17%	22%	27%	11%	10%

Overall Performance Results



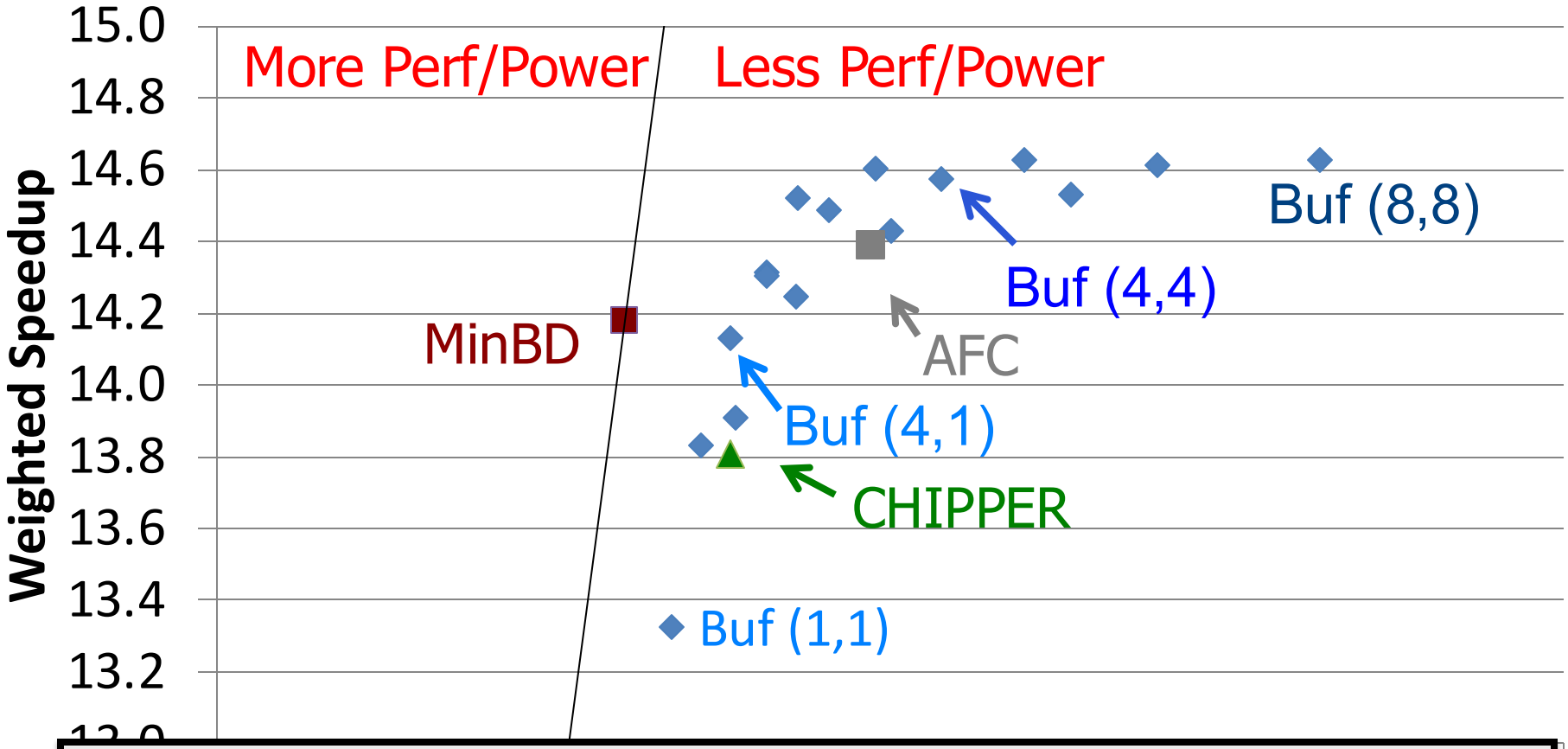
- Similar perf. to Buffered (4,1) @ 25% of buffering space
- Within **2.7%** of Buffered (4,4) (**8.3%** at high load)

Overall Power Results



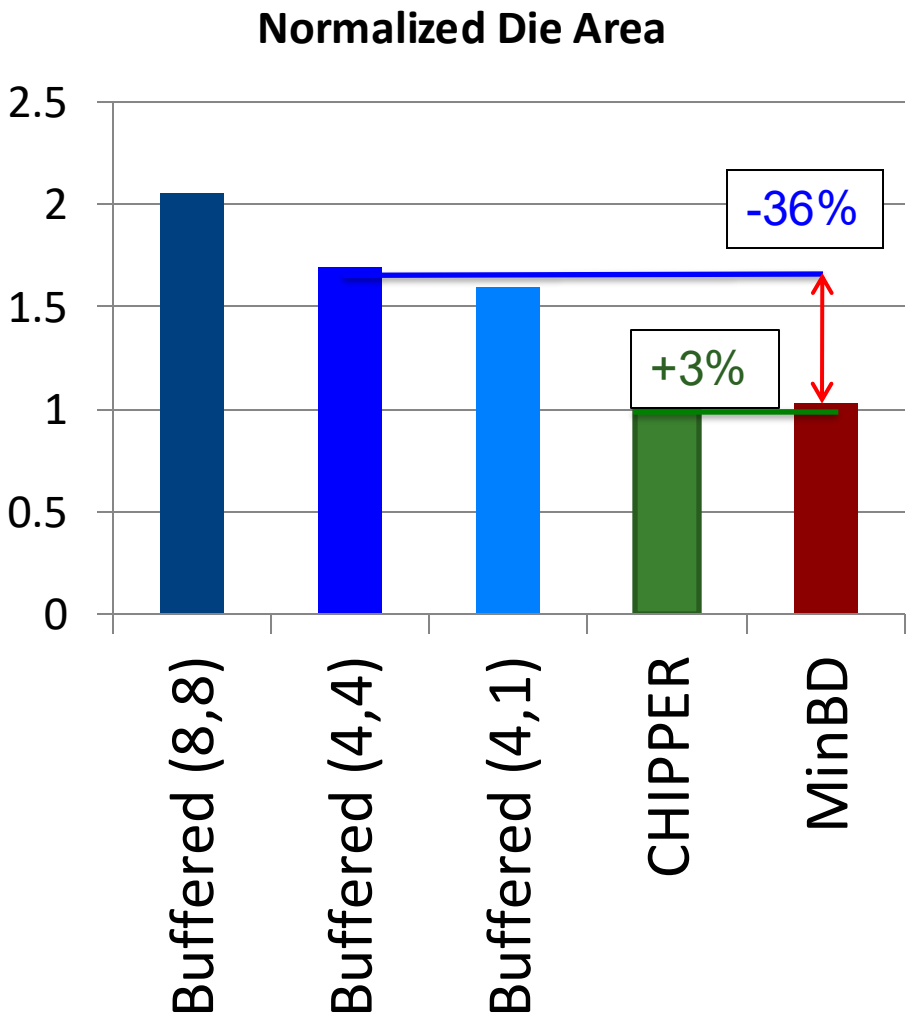
- Dynamic power increases with deflection routing
- Buffers are significant fraction of power in baseline routers
- Dynamic power reduces in MinBD relative to CHIPPER

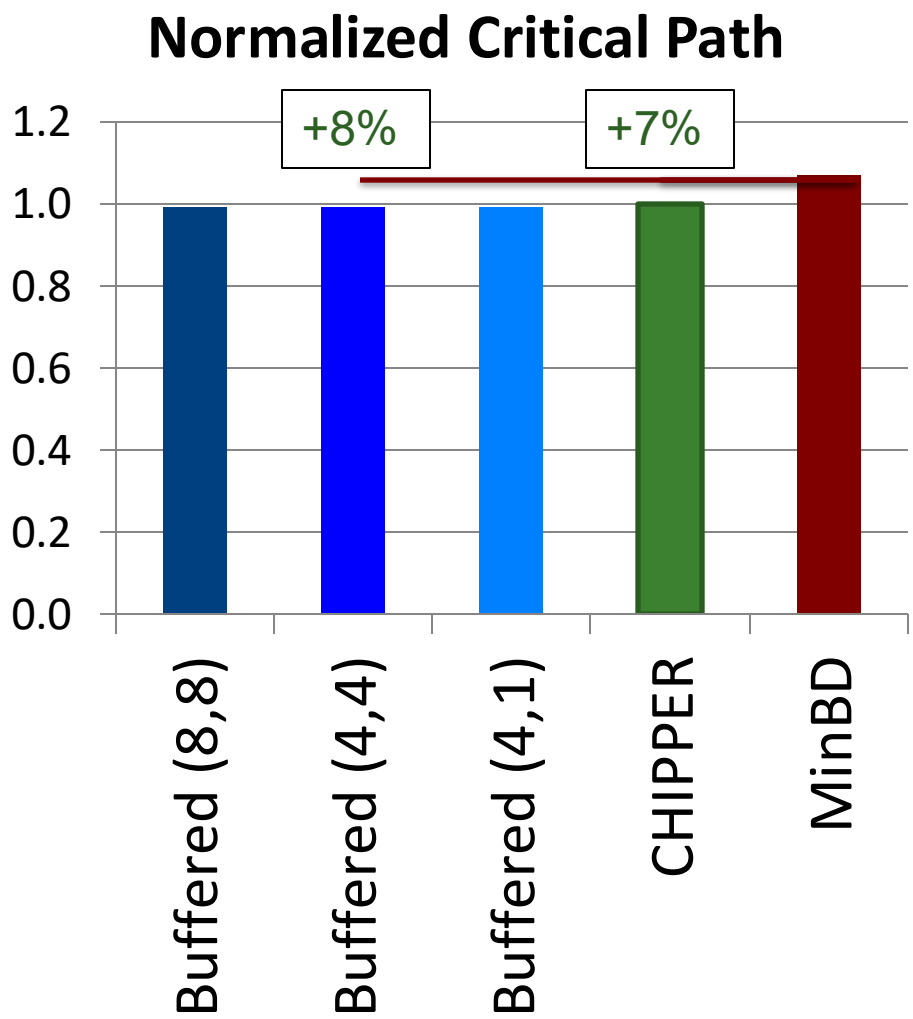
Performance-Power Spectrum



- Most **energy-efficient** (perf/watt) of any evaluated network router design

Die Area and Critical Path





- Only **3%** area increase over CHIPPER (4-flit buffer)
- Reduces area by **36%** from Buffered (4,4)
- Increases by **7%** over CHIPPER, **8%** over Buffered (4,4)

MinBD Router: Conclusions

- Bufferless deflection routing offers reduced power & area
- But, high deflection rate hurts performance at high load

- **MinBD** (Minimally-Buffered Deflection Router) introduces:
  - Side buffer to hold **only** flits that would have been deflected
  - Dual-width ejection to address ejection bottleneck
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- MinBD yields reduced power (31%) & reduced area (36%) relative to **buffered** routers
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  "MinBD: Minimally-Buffered Deflection Routing for Energy-Efficient Interconnect"
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†Tsinghua University & Carnegie Mellon University
yxythu@gmail.com

Better Bufferless Networks

Can We Do Better?

HAT: Heterogeneous Adaptive Throttling for On-Chip Networks

Kevin Chang, Rachata Ausavarungnirun, Chris Fallin, and Onur Mutlu,

"HAT: Heterogeneous Adaptive Throttling for On-Chip Networks"

*Proceedings of the 24th International Symposium on Computer Architecture and High Performance Computing (**SBAC-PAD**)*, New York, NY, October 2012. Slides (pptx) (pdf)

Carnegie Mellon University SAFARI

Executive Summary

- **Problem:** Packets contend in on-chip networks (NoCs), causing congestion, thus reducing performance
- **Observations:**

  1) Some applications are more sensitive to network latency than others
  2) Applications must be throttled differently to achieve peak performance
- **Key Idea: Heterogeneous Adaptive Throttling (HAT)**
  1) Application-aware source throttling
  2) Network-load-aware throttling rate adjustment
- **Result:** Improves performance and energy efficiency over state-of-the-art source throttling policies

Source Throttling in Bufferless NoCs

- Kevin Chang, Rachata Ausavarungnirun, Chris Fallin, and Onur Mutlu, **"HAT: Heterogeneous Adaptive Throttling for On-Chip Networks"** *Proceedings of the 24th International Symposium on Computer Architecture and High Performance Computing (**SBAC-PAD**)*, New York, NY, October 2012. Slides (pptx) (pdf)

HAT: Heterogeneous Adaptive Throttling for On-Chip Networks

Kevin Kai-Wei Chang, Rachata Ausavarungnirun, Chris Fallin, Onur Mutlu
Carnegie Mellon University
{kevincha,rachata,cfallin,onur}@cmu.edu

"Bufferless" Hierarchical Rings

- Rachata Ausavarungnirun, Chris Fallin, Xiangyao Yu, Kevin Chang, Greg Nazario, Reetuparna Das, Gabriel Loh, and Onur Mutlu,
  "Design and Evaluation of Hierarchical Rings with Deflection Routing"
  *Proceedings of the 26th International Symposium on Computer Architecture and High Performance Computing (**SBAC-PAD**)*, Paris, France, October 2014. [Slides (pptx) (pdf)] [Source Code]

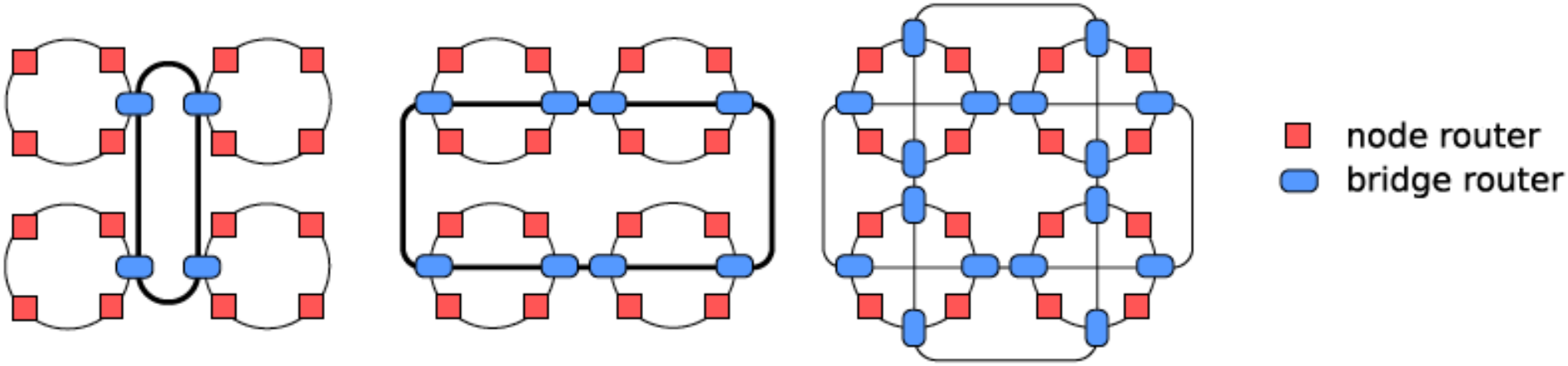
Describes the design and implementation of a **mostly-bufferless hierarchical ring**

Design and Evaluation of Hierarchical Rings with Deflection Routing

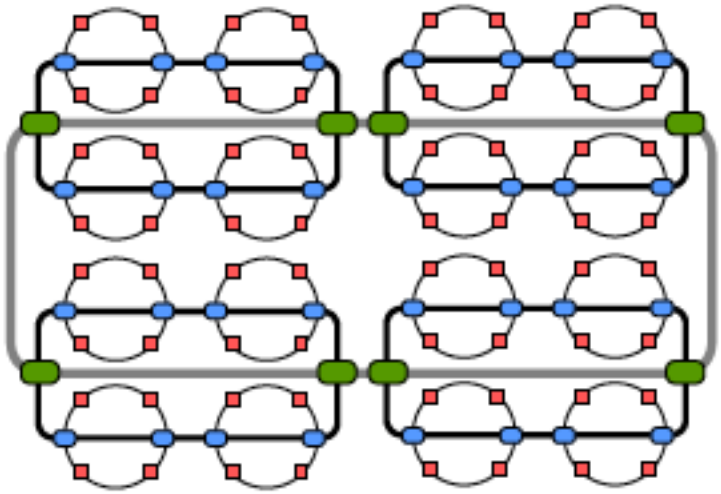
Rachata Ausavarungnirun Chris Fallin Xiangyao Yu† Kevin Kai-Wei Chang
Greg Nazario Reetuparna Das§ Gabriel H. Loh‡ Onur Mutlu

Carnegie Mellon University §University of Michigan †MIT ‡Advanced Micro Devices, Inc.

Hierarchical Rings



(a) 4-, 8-, and 16-bridge hierarchical ring topologies.



(b) Three-level hierarchy (8x8).

Hierarchical Rings: State of the Art

- Rachata Ausavarungnirun, Chris Fallin, Xiangyao Yu, Kevin Chang, Greg Nazario, Reetuparna Das, Gabriel Loh, and Onur Mutlu, **"A Case for Hierarchical Rings with Deflection Routing: An Energy-Efficient On-Chip Communication Substrate"** ***Parallel Computing*** ***(PARCO)***, 2016. arXiv.org version, February 2016.

A case for hierarchical rings with deflection routing: An energy-efficient on-chip communication substrate

Rachata Ausavarungnirun[a,*], Chris Fallin[a], Xiangyao Yu[b], Kevin Kai-Wei Chang[a], Greg Nazario[a], Reetuparna Das[c], Gabriel H. Loh[d], Onur Mutlu[a]

[a] *Carnegie Mellon University, United States*
[b] *University of Michigan, United States*
[c] *Massachusetts Institute of Technology, United States*
[d] *Advanced Micro Devices, United States*

Idea Implemented in Real Systems (I)

Application Defined On-chip Networks for Heterogeneous Chiplets: An Implementation Perspective

Tianqi Wang[1,*], Fan Feng[1,*], Shaolin Xiang[1,*], Qi Li[1], and Jing Xia[1,**]

[1]*Huawei*

https://ieeexplore.ieee.org/document/9773184/



https://ieeexplore.ieee.org/abstract/document/9444893

Idea Implemented in Real Systems (II)

Application Defined On-chip Networks for Heterogeneous Chiplets: An Implementation Perspective

Tianqi Wang[1,*], Fan Feng[1,*], Shaolin Xiang[1,*], Qi Li[1], and Jing Xia[1,**]

[1]*Huawei*

https://ieeexplore.ieee.org/document/9773184/

3.4 Bufferless Multi-Ring NoC

As mentioned above, several development teams provide highlights from their viewpoint. The requirements for the NoC are listed:

- Application: The NoC must provide high bandwidth (larger than 15TB/s) and maintain low latency for cross-chiplets requests at the same time.
- Architecture: The architecture of NoC should be flexible. Each two NoC transactions are independent and stateless.
- Physical Implementation: NoC design prefers to a simplified circuit structure and tries to improve distance per cycle.

3.4.1 Heterogeneous-Chiplet Consideration

As mentioned in Section 2.1, our products cover several various application domains based on the combination of heterogeneous chiplets.

A chiplet is a relatively tightly coupled system. However, transceivers, ethernet, and domain-specific accelerators that

A Review of Bufferless Interconnects

- Chris Fallin, Greg Nazario, Xiangyao Yu, Kevin Chang, Rachata Ausavarungnirun, and Onur Mutlu,
  "Bufferless and Minimally-Buffered Deflection Routing"
  Invited Book Chapter in Routing Algorithms in Networks-on-Chip, pp. 241-275, Springer, 2014.

Chapter 1

Bufferless and Minimally-Buffered Deflection Routing

Chris Fallin, Greg Nazario, Xiangyao Yu, Kevin Chang, Rachata Ausavarungnirun, Onur Mutlu

Summary of Eight Years of Research

Energy-Efficient Deflection-based On-chip Networks: Topology, Routing, Flow Control

Rachata Ausavarungnirun[b], Onur Mutlu[a]

SAFARI Research Group

[a]*ETH Zürich*
[b]*King Mongkut's University of Technology North Bangkok*

Abstract

As the number of cores scales to tens and hundreds, the energy consumption of routers across various types of on-chip networks in chip muiltiprocessors (CMPs) increases significantly. A major source of this energy consumption comes from the input buffers inside Network-on-Chip (NoC) routers, which are traditionally designed to maximize performance. To mitigate this high energy cost, many works propose bufferless router designs that utilize deflection routing to resolve port contention. While this approach is able to maintain high performance relative to its buffered counterparts at low network traffic, the bufferless router design suffers performance degradation under high network load.

In order to maintain high performance and energy efficiency under *both* low and high network loads, this chapter discusses critical drawbacks of traditional bufferless designs and describes recent research works focusing on two major modifications to improve the overall performance of the traditional bufferless network-on-chip design. The first modification is a minimally-buffered design that introduces limited buffering inside critical parts of the on-chip network in order to reduce the number of deflections. The second modification is a hierarchical bufferless interconnect design that aims to further improve performance by limiting the number of hops each packet needs to travel while in the network. In both approaches, we discuss design tradeoffs and provide evaluation results based on common CMP configurations with various network topologies to show the effectiveness of each proposal.

Keywords: network-on-chip, deflection routing, topology, bufferless router, energy efficiency, high-performance computing, computer architecture, emerging technologies, latency, low-latency computing

https://arxiv.org/pdf/2112.02516.pdf

Bufferless Interconnects in Real Systems

Application Defined On-chip Networks for Heterogeneous Chiplets: An Implementation Perspective

Tianqi Wang[1,*], Fan Feng[1,*], Shaolin Xiang[1,*], Qi Li[1], and Jing Xia[1,**]

[1]*Huawei*

THEME ARTICLE: COMMERCIAL PRODUCTS 2021

Kunpeng 920: The First 7-nm Chiplet-Based 64-Core ARM SoC for Cloud Services

Jing Xia, Chuanning Cheng, Xiping Zhou, Yuxing Hu, and Peter Chun, *HiSilicon Technologies Company, Ltd., Shenzhen, 518129, China*

More Readings

- Studies of congestion and congestion control in on-chip vs. internet-like networks

- George Nychis, Chris Fallin, Thomas Moscibroda, Onur Mutlu, and Srinivasan Seshan,
"On-Chip Networks from a Networking Perspective: Congestion and Scalability in Many-core Interconnects"
*Proceedings of the 2012 ACM SIGCOMM Conference (**SIGCOMM**)*, Helsinki, Finland, August 2012. Slides (pptx)

- George Nychis, Chris Fallin, Thomas Moscibroda, and Onur Mutlu,
"Next Generation On-Chip Networks: What Kind of Congestion Control Do We Need?"
*Proceedings of the 9th ACM Workshop on Hot Topics in Networks (**HOTNETS**)*, Monterey, CA, October 2010. Slides (ppt) (key)

On-Chip vs. Off-Chip Congestion Control

- George Nychis, Chris Fallin, Thomas Moscibroda, Onur Mutlu, and Srinivasan Seshan,
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On-Chip Networks from a Networking Perspective: Congestion and Scalability in Many-Core Interconnects

George Nychis†, Chris Fallin†, Thomas Moscibroda§, Onur Mutlu†, Srinivasan Seshan†

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On-Chip vs. Off-Chip Congestion Control

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Next Generation On-Chip Networks: What Kind of Congestion Control Do We Need?

George Nychis†, Chris Fallin†, Thomas Moscibroda§, Onur Mutlu†

†Carnegie Mellon University
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moscitho@microsoft.com

Summary of Study [SIGCOMM 2012]

- Highlighted a traditional networking problem in a new context
  - Unique design requires novel solution
- Showed congestion limits efficiency and scalability, and that self-throttling nature of cores prevents congestion collapse
- Showed on-chip congestion control requires application-awareness
- Our application-aware congestion controller provided:
  - A more efficient network-layer (reduced latency)
  - Improvements in system throughput (by 27%)
  - Effectively scale the CMP (shown for up to 4096 cores)

Heterogeneous Networks

- Asit K. Mishra, Onur Mutlu, and Chita R. Das,
  "A Heterogeneous Multiple Network-on-Chip Design: An Application-Aware Approach"
  *Proceedings of the 50th Design Automation Conference (**DAC**)*,
  Austin, TX, June 2013. Slides (pptx) Slides (pdf)

A Heterogeneous Multiple Network-On-Chip Design: An Application-Aware Approach

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Packet Scheduling

Packet Scheduling

- Which packet to choose for a given output port?
  - Router needs to prioritize between competing flits
  - Which input port?
  - Which virtual channel?
  - Which application's packet?

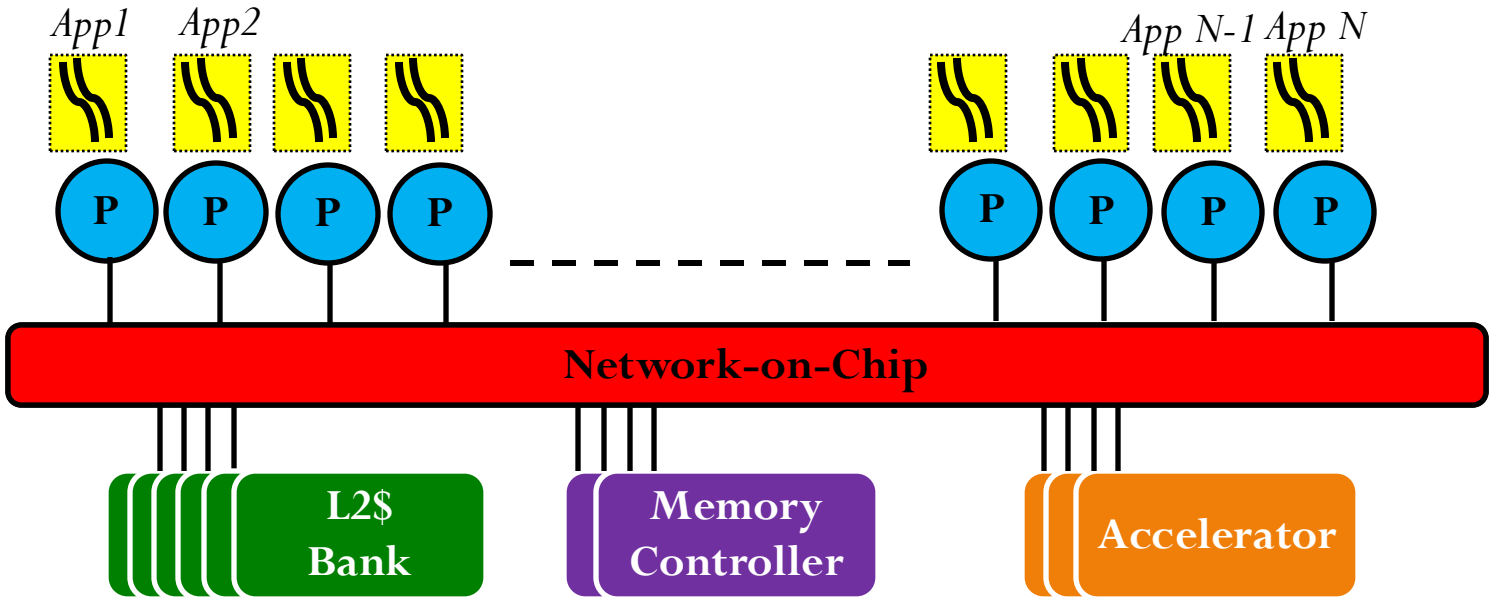
- Common strategies
  - Round robin across virtual channels
  - Oldest packet first (or an approximation)
  - Prioritize some virtual channels over others

- Better policies in a multi-core environment
  - Use application characteristics

Application-Aware Packet Scheduling

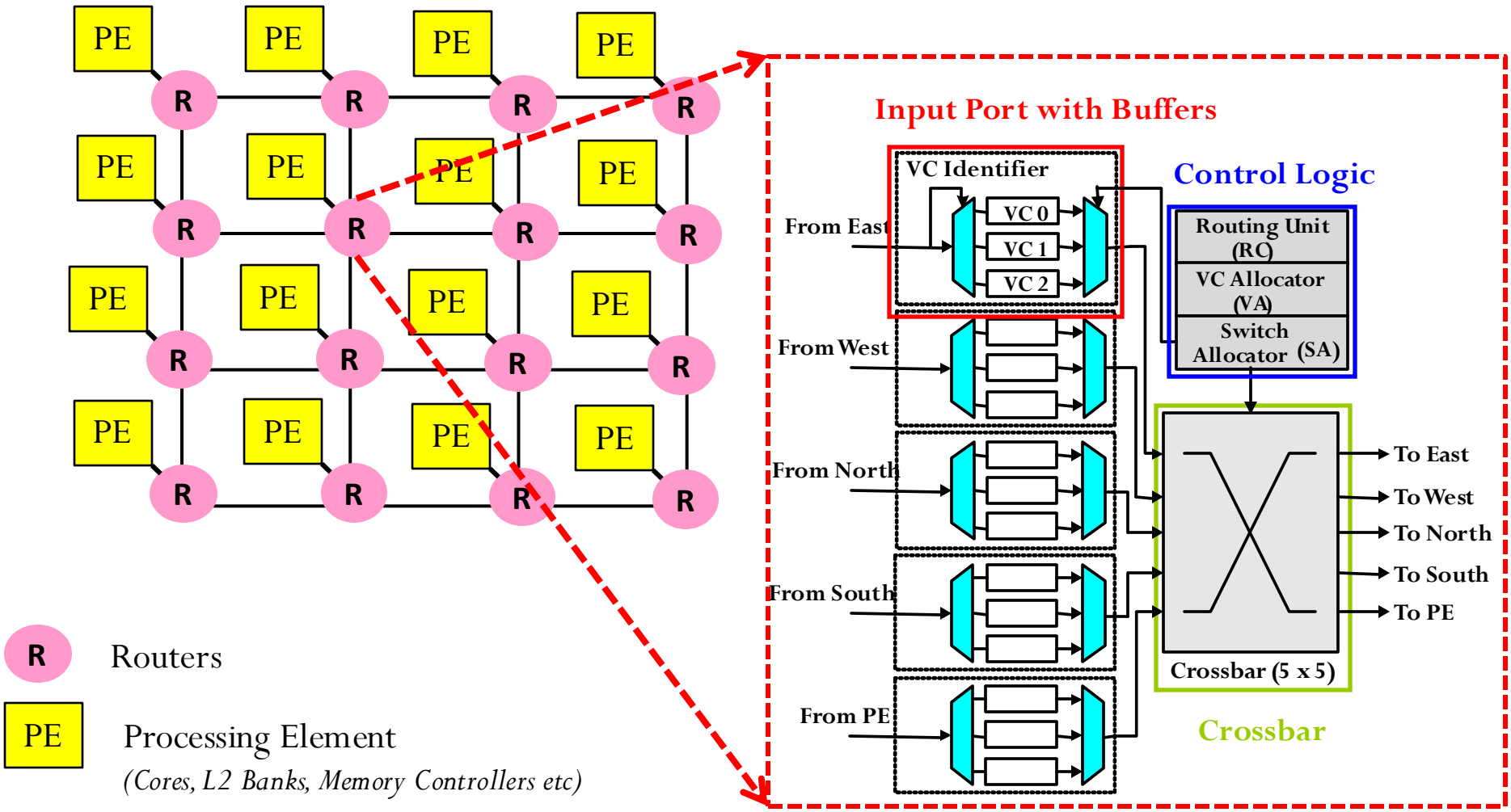
Das et al., "Application-Aware Prioritization Mechanisms for On-Chip Networks," MICRO 2009.

The Problem: Packet Scheduling

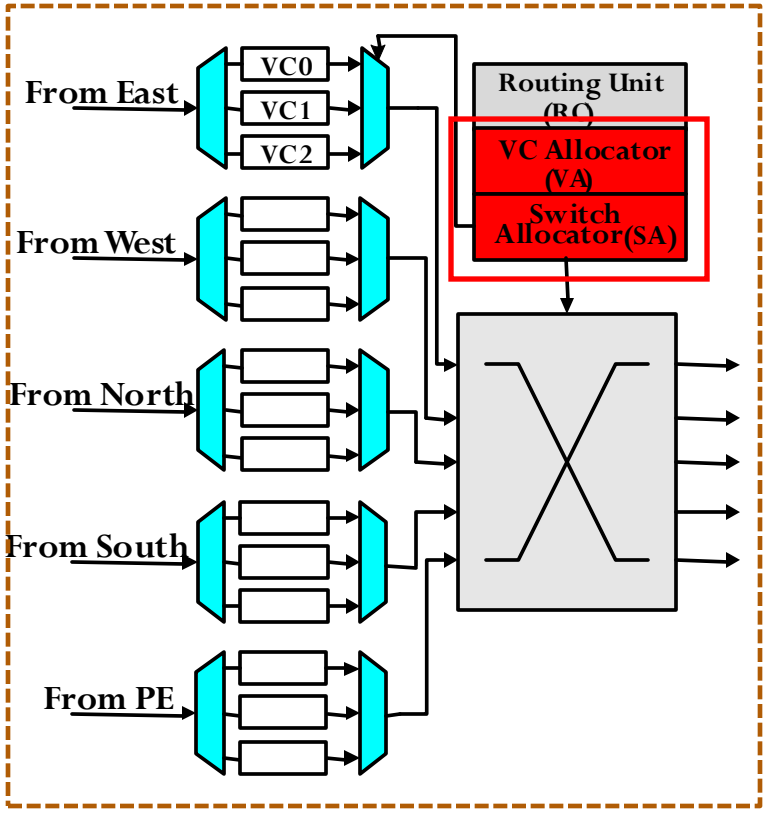


Network-on-Chip is a critical resource shared by multiple applications

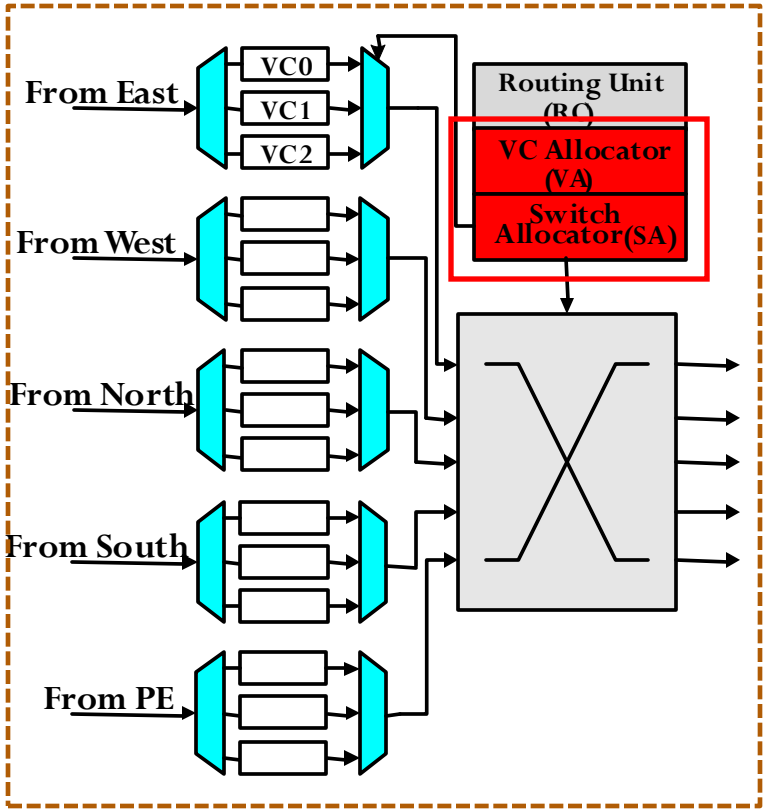
The Problem: Packet Scheduling



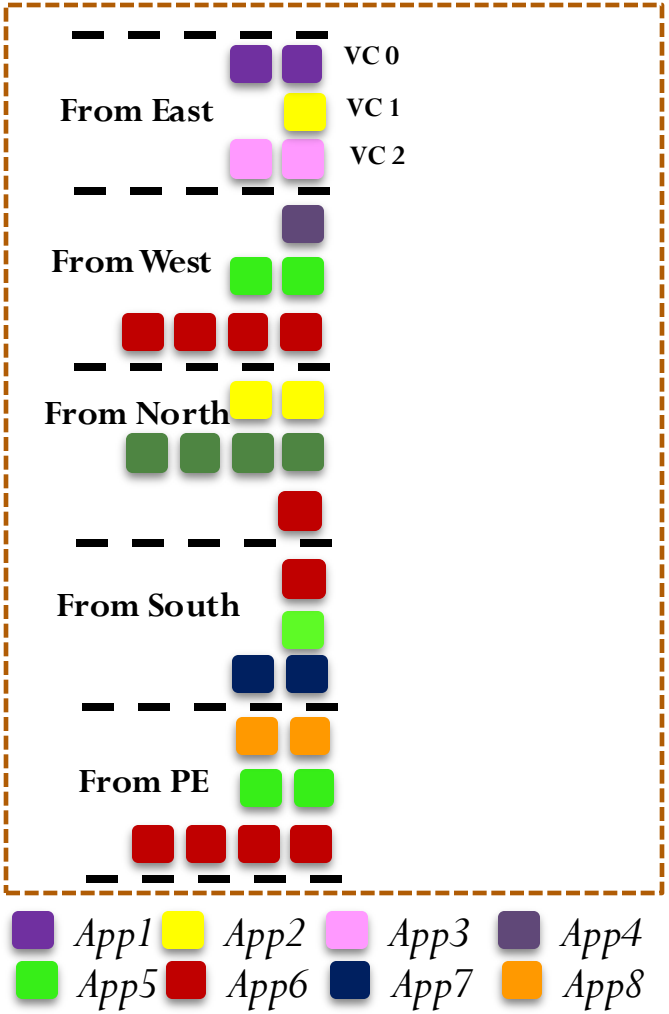
The Problem: Packet Scheduling



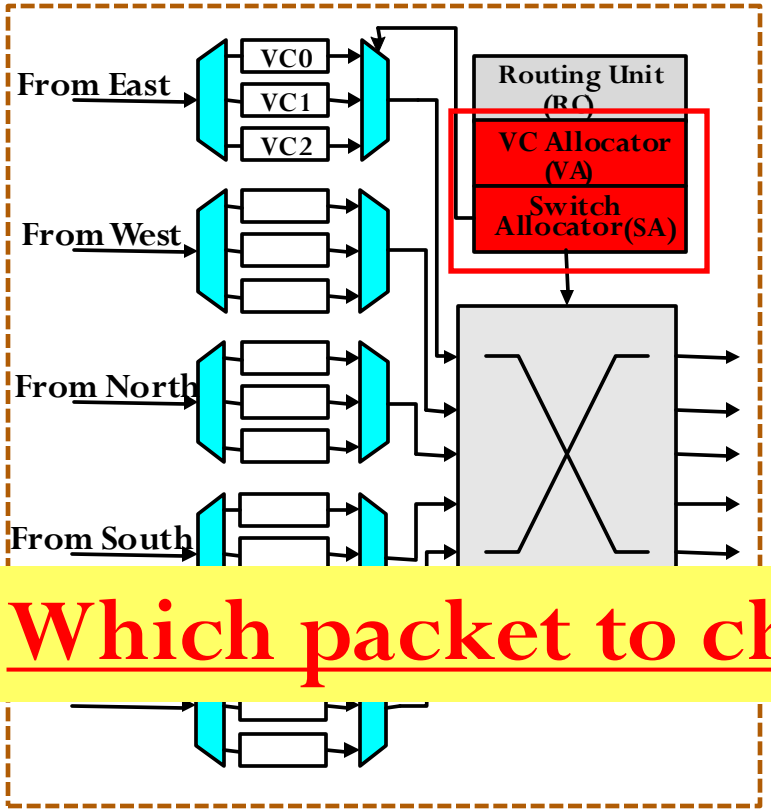
The Problem: Packet Scheduling



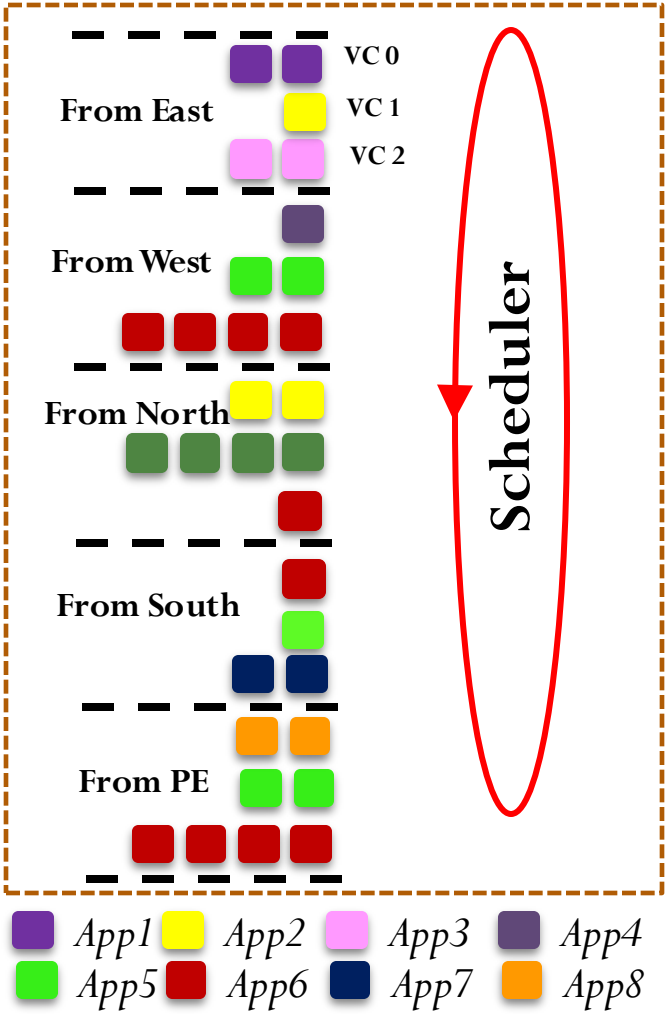
Conceptual View



The Problem: Packet Scheduling



Conceptual View

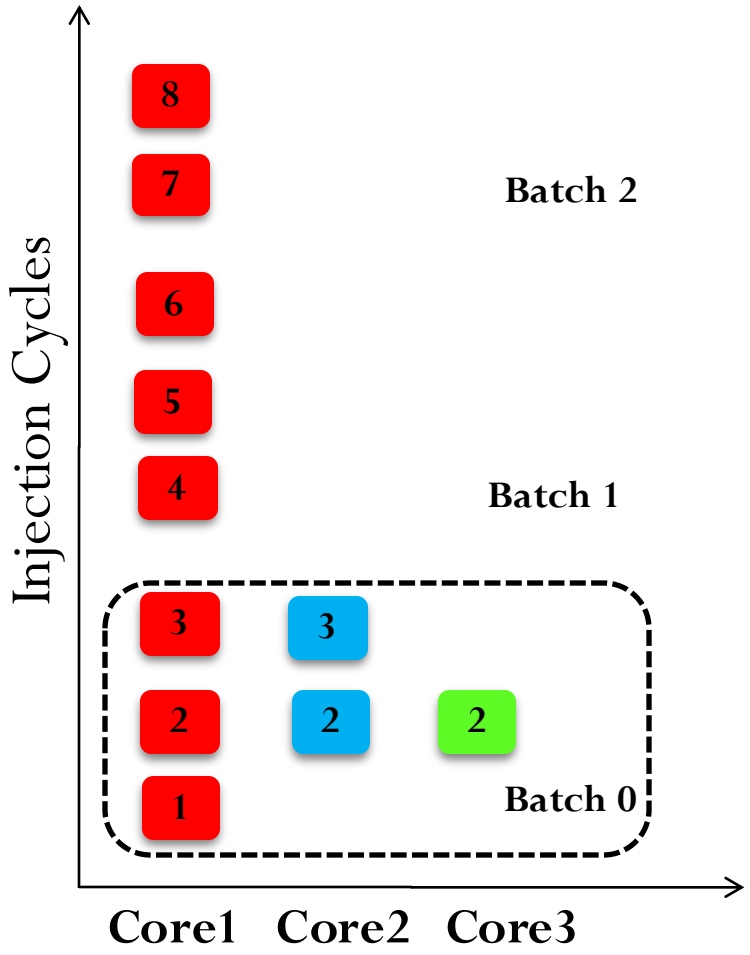


Which packet to choose?

The Problem: Packet Scheduling

- Existing scheduling policies
  - Round Robin
  - Age
- Problem 1: Local to a router
  - Lead to contradictory decision making between routers: packets from one application may be prioritized at one router, to be delayed at next.
- Problem 2: Application oblivious
  - Treat all applications packets equally
  - But applications are heterogeneous
- Solution : Application-aware global scheduling policies.

STC Scheduling Example



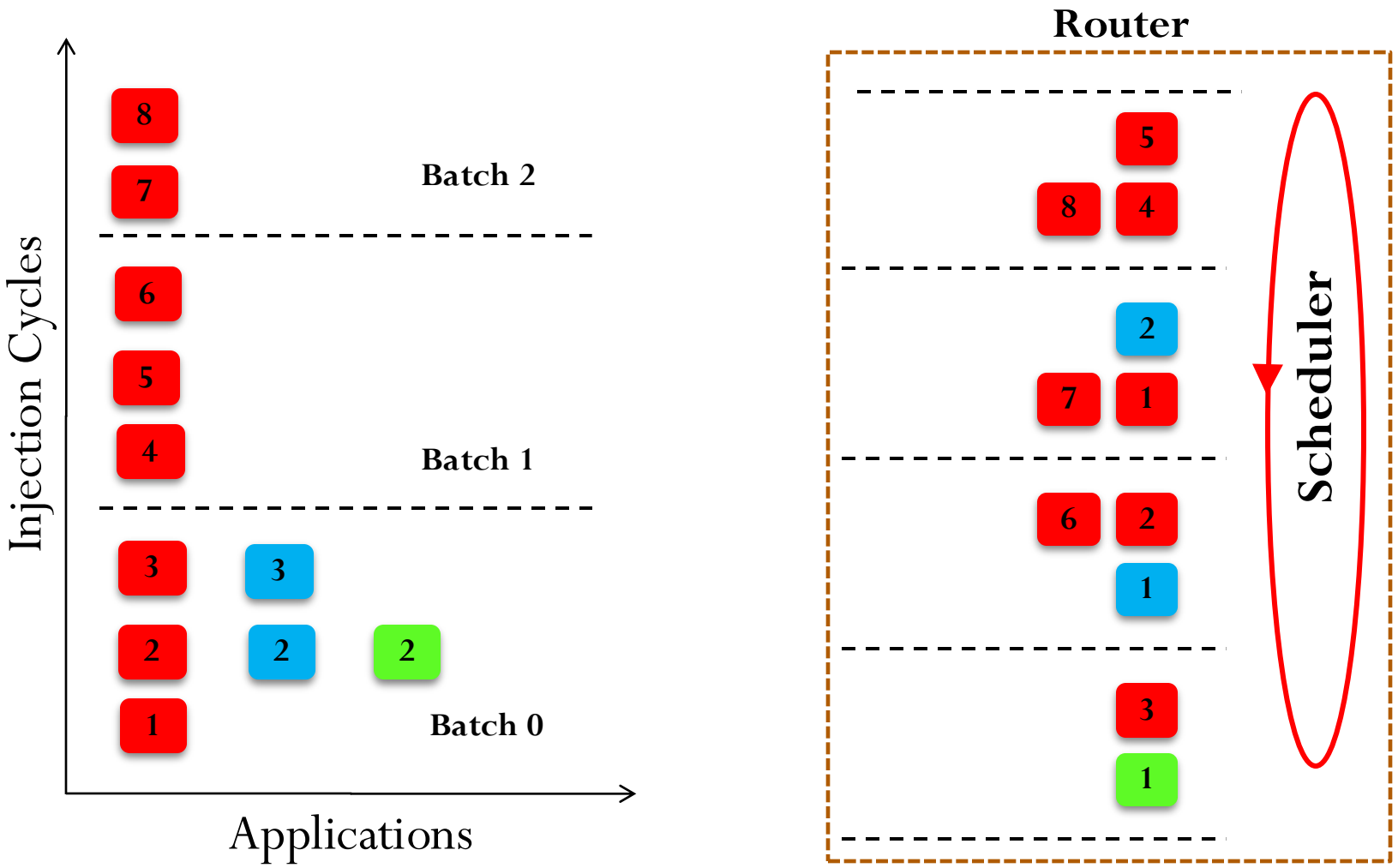
Batching interval length = 3 cycles

Ranking order =

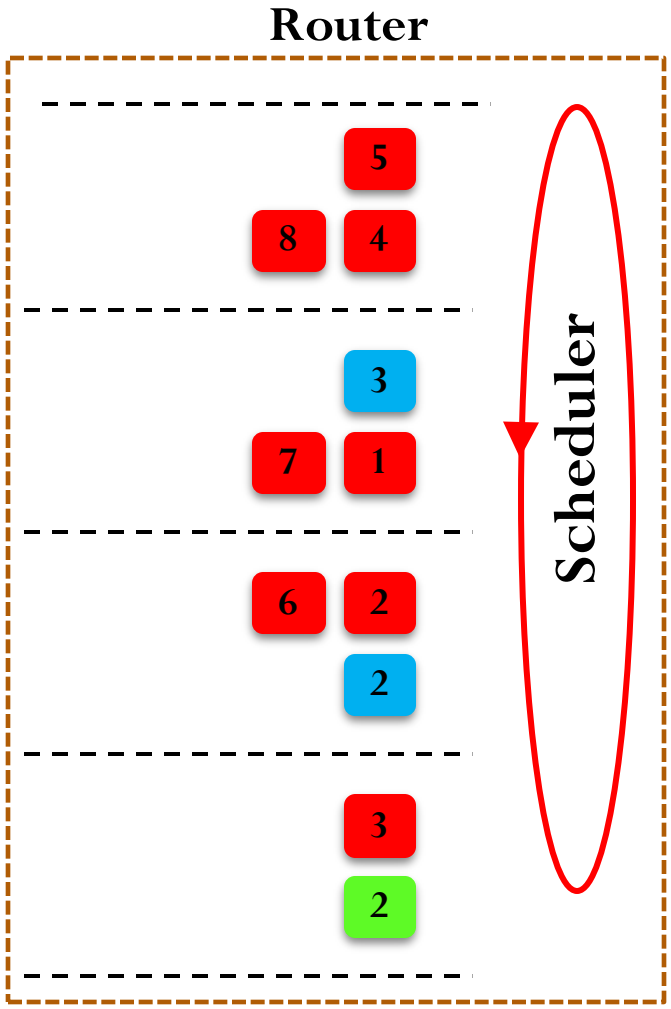


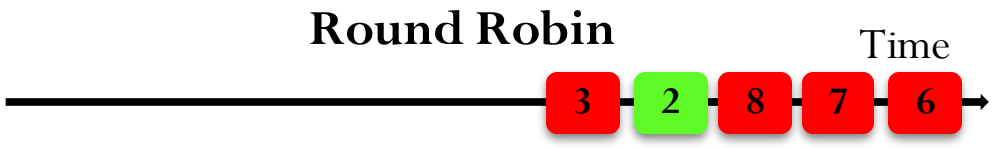
Packet Injection Order at Processor

STC Scheduling Example



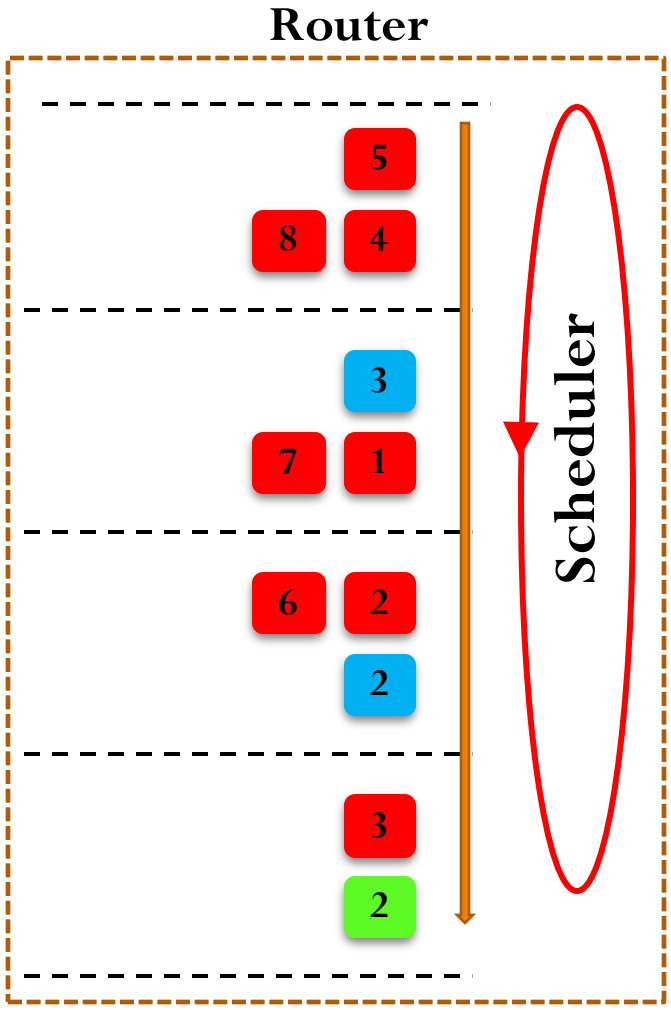
STC Scheduling Example

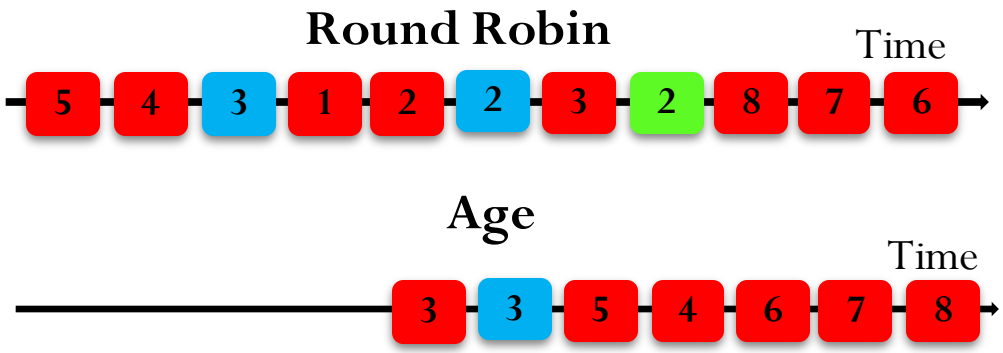




STALL CYCLES				Avg
RR	8	6	11	8.3
Age				
STC				

STC Scheduling Example

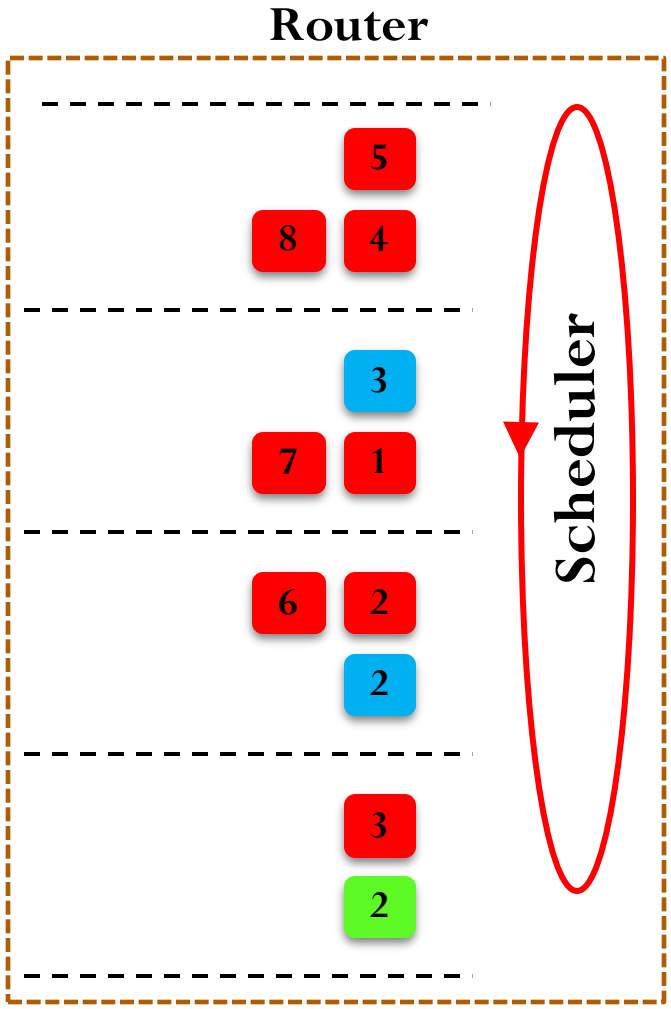


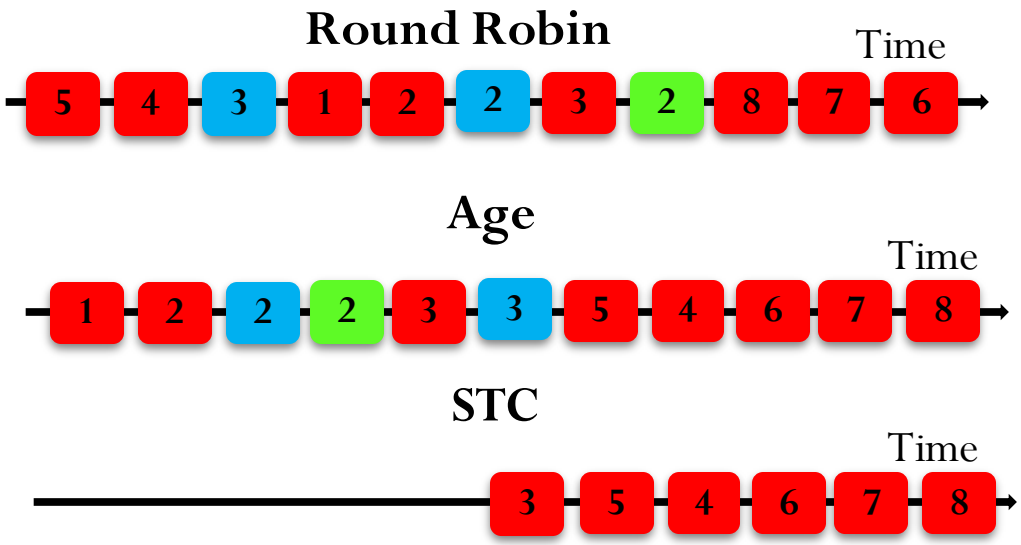


STALL CYCLES				Avg
RR	8	6	11	8.3
Age	4	6	11	7.0
STC				



STC Scheduling Example





STALL CYCLES				Avg
RR	8	6	11	8.3
Age	4	6	11	7.0
STC	1	3	11	5.0

Application-Aware Prioritization in NoCs

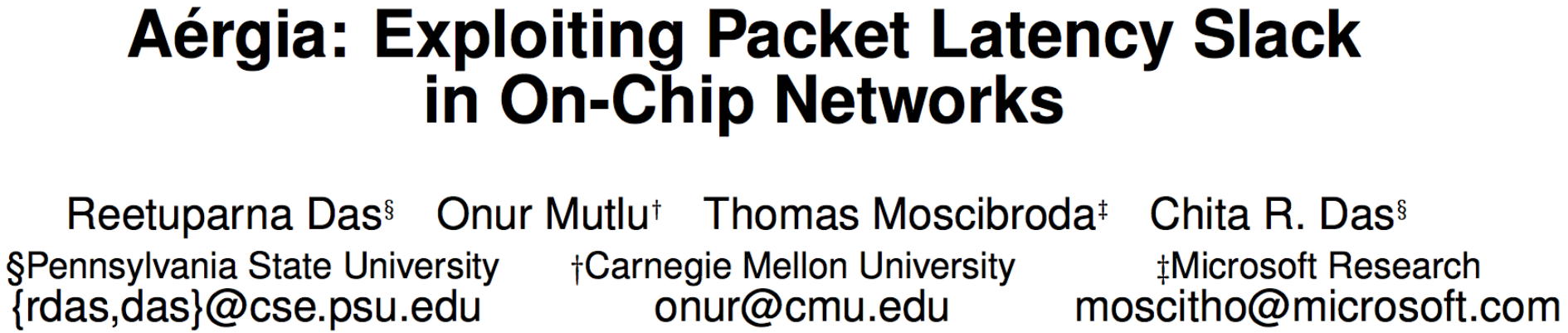
- Reetuparna Das, Onur Mutlu, Thomas Moscibroda, and Chita R. Das, **"Application-Aware Prioritization Mechanisms for On-Chip Networks"**
  *Proceedings of the 42nd International Symposium on Microarchitecture (**MICRO**)*, pages 280-291, New York, NY, December 2009. Slides (pptx)

Application-Aware Prioritization Mechanisms for On-Chip Networks

Reetuparna Das[§] Onur Mutlu[†] Thomas Moscibroda[‡] Chita R. Das[§]
§Pennsylvania State University †Carnegie Mellon University ‡Microsoft Research
{rdas,das}@cse.psu.edu onur@cmu.edu moscitho@microsoft.com

Slack-Based Packet Scheduling

- Reetuparna Das, Onur Mutlu, Thomas Moscibroda, and Chita R. Das, **"Aergia: Exploiting Packet Latency Slack in On-Chip Networks"** *Proceedings of the 37th International Symposium on Computer Architecture (**ISCA**)*, pages 106-116, Saint-Malo, France, June 2010. Slides (pptx)
  One of the 11 computer architecture papers of 2010 selected as Top Picks by IEEE Micro.



Aérgia: Exploiting Packet Latency Slack in On-Chip Networks

Reetuparna Das§ Onur Mutlu† Thomas Moscibroda‡ Chita R. Das§
§Pennsylvania State University {rdas,das}@cse.psu.edu
†Carnegie Mellon University onur@cmu.edu
‡Microsoft Research moscitho@microsoft.com

Slack-Based Packet Scheduling

- The notion of "packet slack"
  - Slack of a packet is the number of cycles it can be delayed in a router without (significantly) reducing application's performance
  - Local network slack

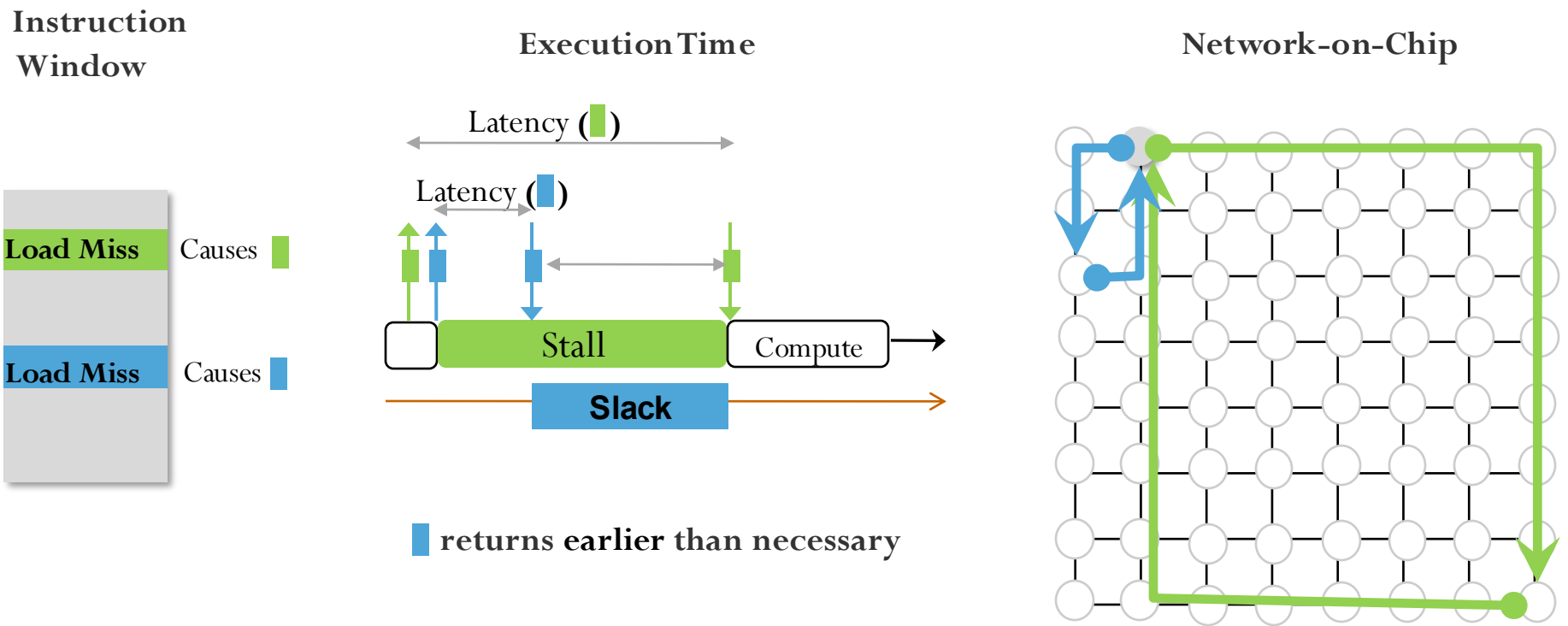
- Source of slack: Memory-Level Parallelism (MLP) or other latency tolerance mechanisms
  - Latency of an application's packet hidden from application due to overlap with latency of pending cache miss requests or other long-latency operations

- Key idea of slack-based packet scheduling:
  - Estimate the slack of each packet
  - Prioritize packets with lower slack

Slack of Packets

- What is slack of a packet?
  - Slack of a packet is number of cycles it can be delayed in a router without (significantly) reducing application's performance
  - Local network slack
- Source of slack: Memory-Level Parallelism (MLP)
  - Latency of an application's packet hidden from application due to overlap with latency of pending cache miss requests
- Prioritize packets with lower slack

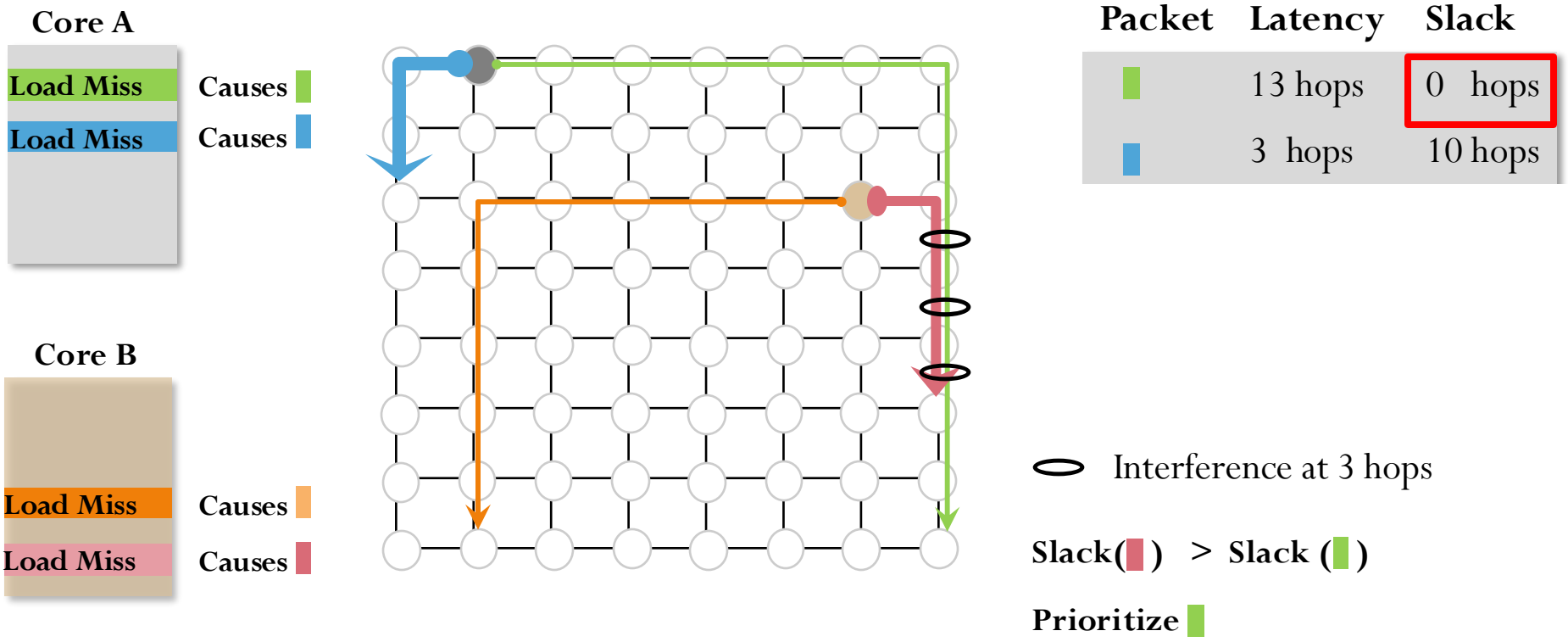
Concept of Slack



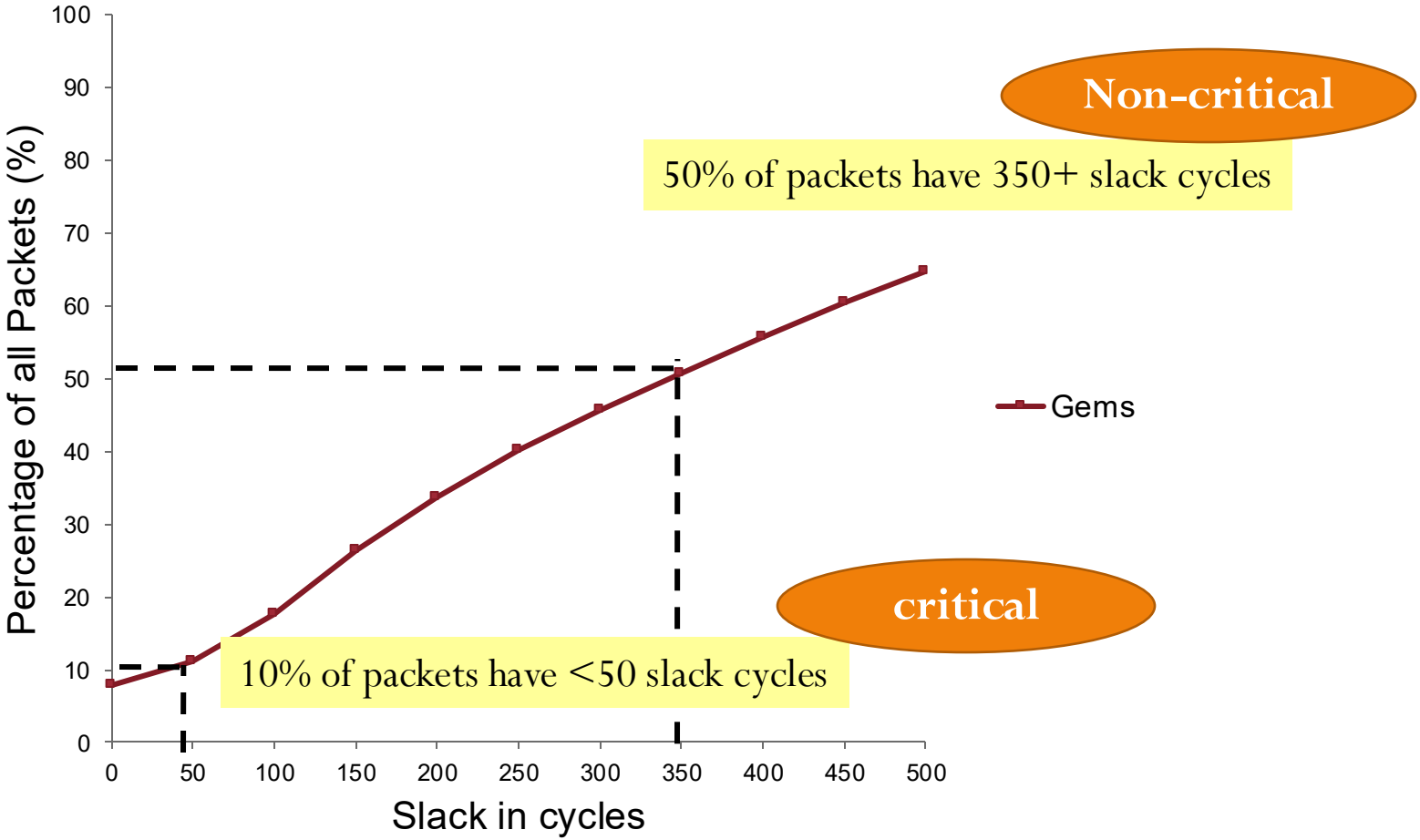
Slack ( ) = Latency ( ) – Latency ( ) = 26 – 6 = 20 hops

Packet( ) can be delayed for available slack cycles without reducing performance!

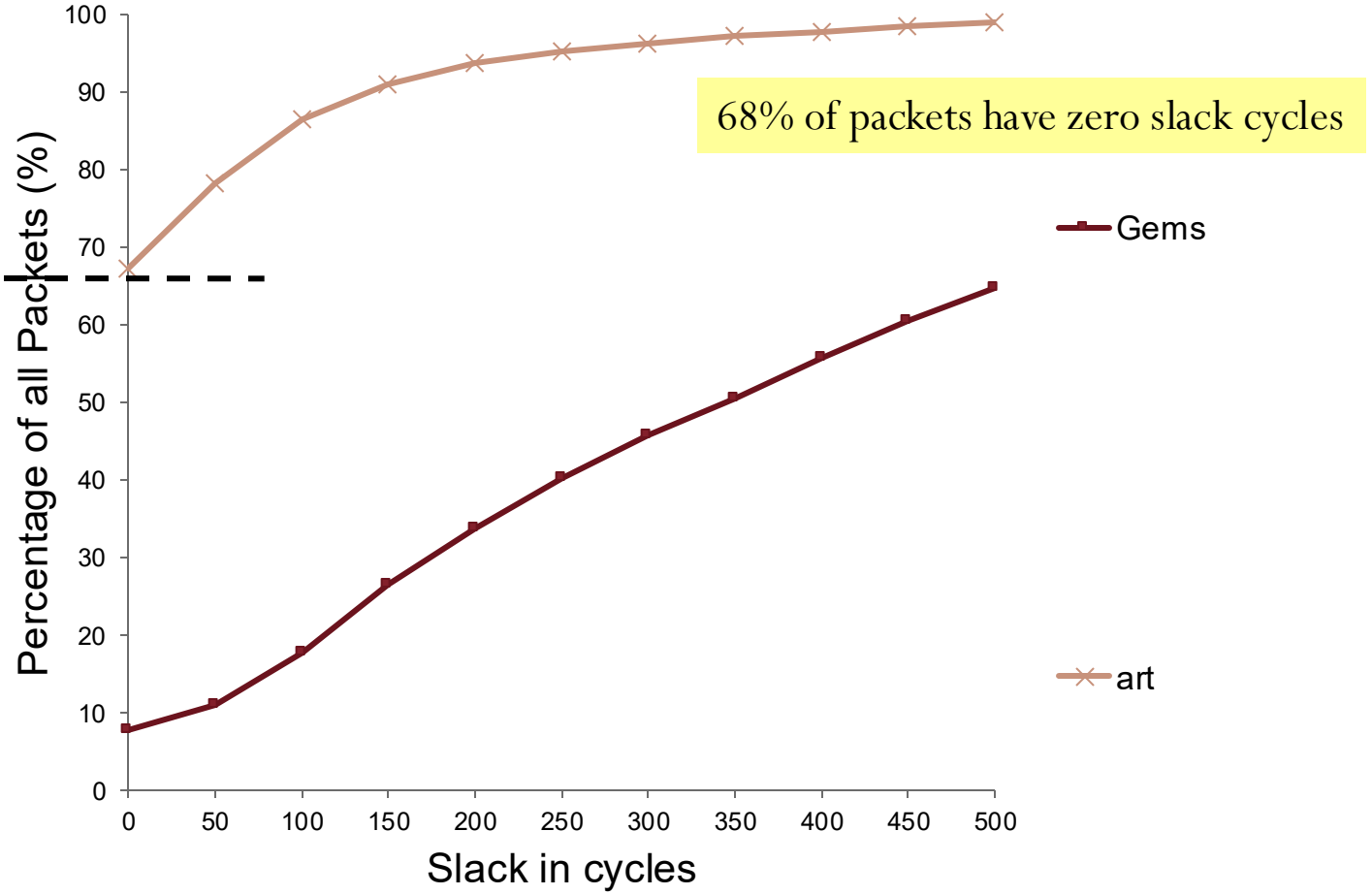
Prioritizing using Slack



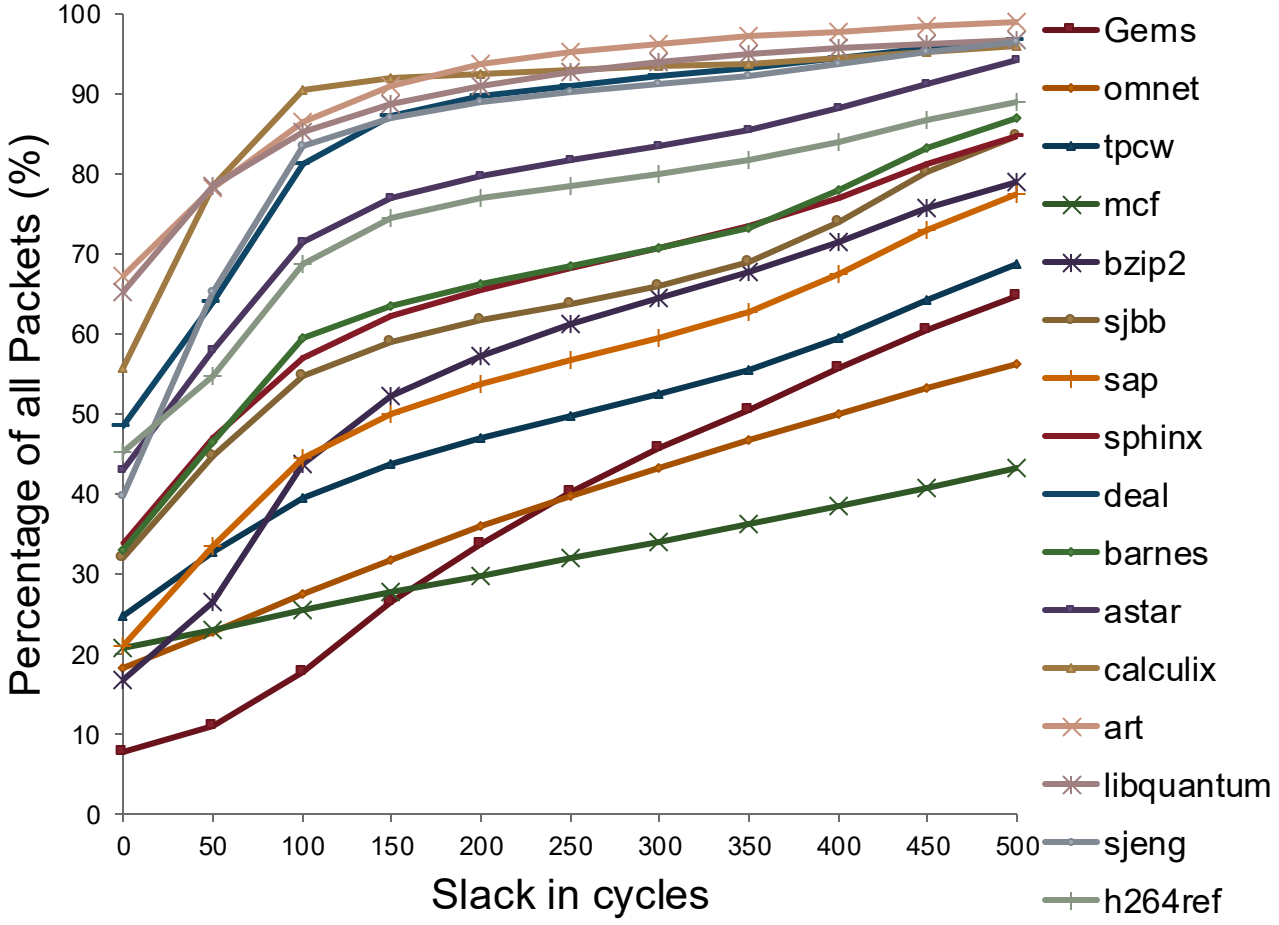
Slack in Applications



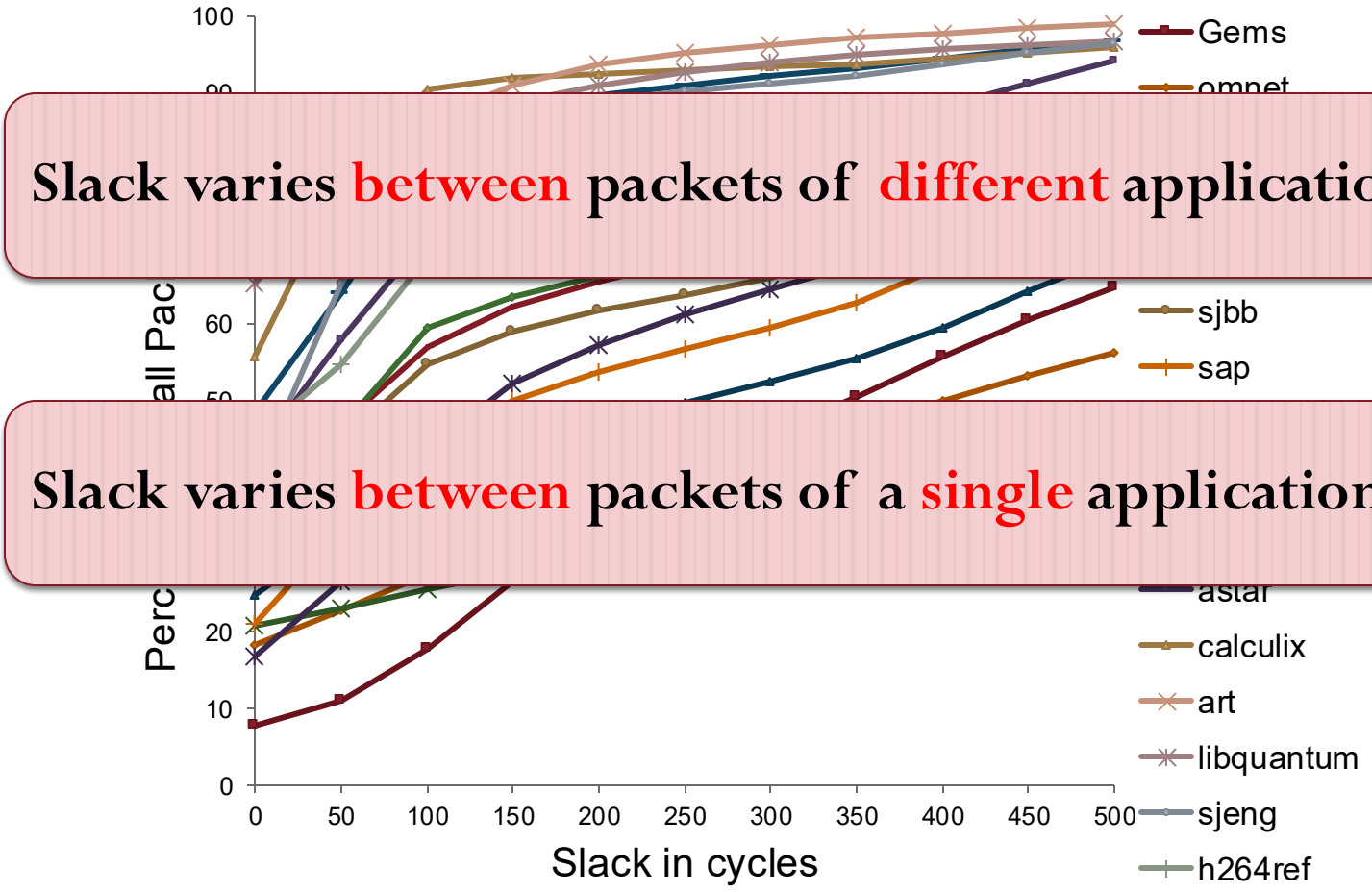
Slack in Applications



Diversity in Slack



Diversity in Slack



Estimating Slack Priority

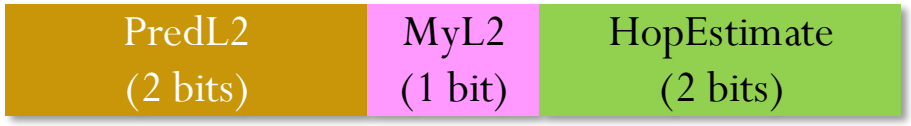
Slack (P) = Max (Latencies of P's Predecessors) – Latency of P

Predecessors(P) are the packets of outstanding cache miss requests when P is issued

- Packet latencies not known when issued

- Predicting latency of any packet Q
  - Higher latency if Q corresponds to an L2 miss
  - Higher latency if Q has to travel farther number of hops

Estimating Slack Priority

- Slack of P = Maximum Predecessor Latency – Latency of P
- Slack(P) = 

PredL2: Set if any predecessor packet is servicing L2 miss

MyL2: Set if P is NOT servicing an L2 miss

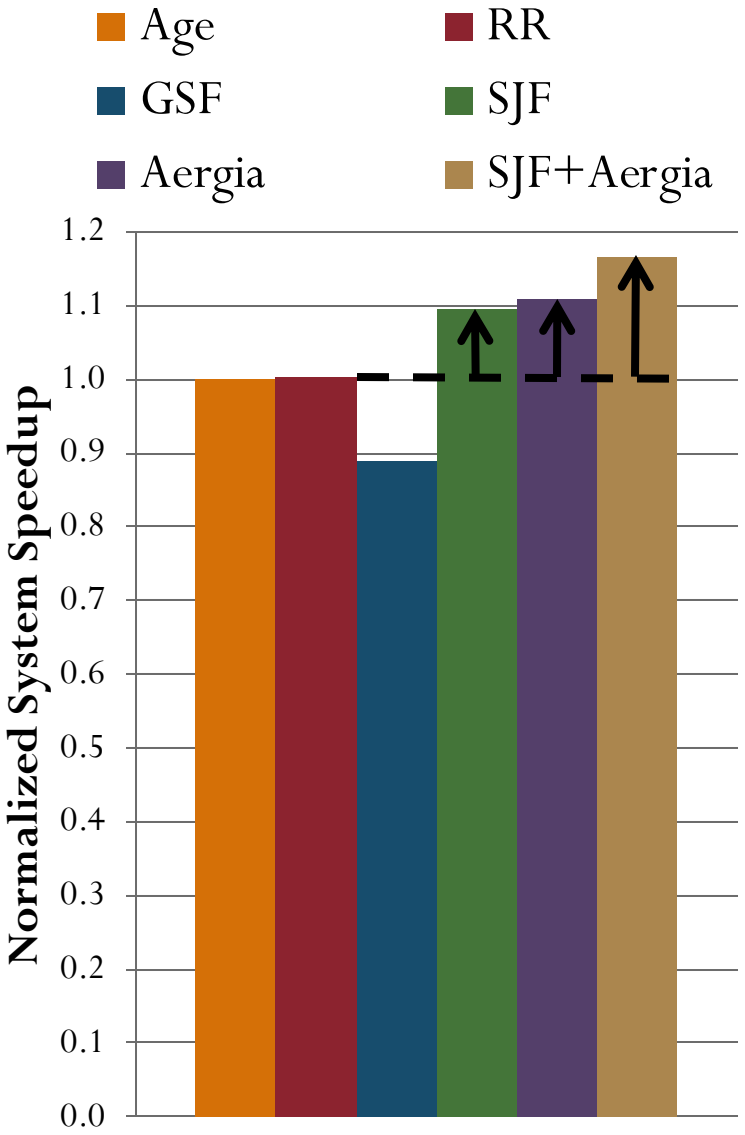
HopEstimate: **Max (# of hops of Predecessors) – hops of P**

Estimating Slack Priority

- How to predict L2 hit or miss at core?
  - *Global Branch Predictor* based L2 Miss Predictor
    - Use Pattern History Table and 2-bit saturating counters
  - *Threshold* based L2 Miss Predictor
    - If #L2 misses in "M" misses >= "T" threshold then next load is a L2 miss.
- Number of miss predecessors?
  - List of outstanding L2 Misses
- Hops estimate?
  - Hops => $\Delta X + \Delta Y$ distance
  - Use predecessor list to calculate slack hop estimate

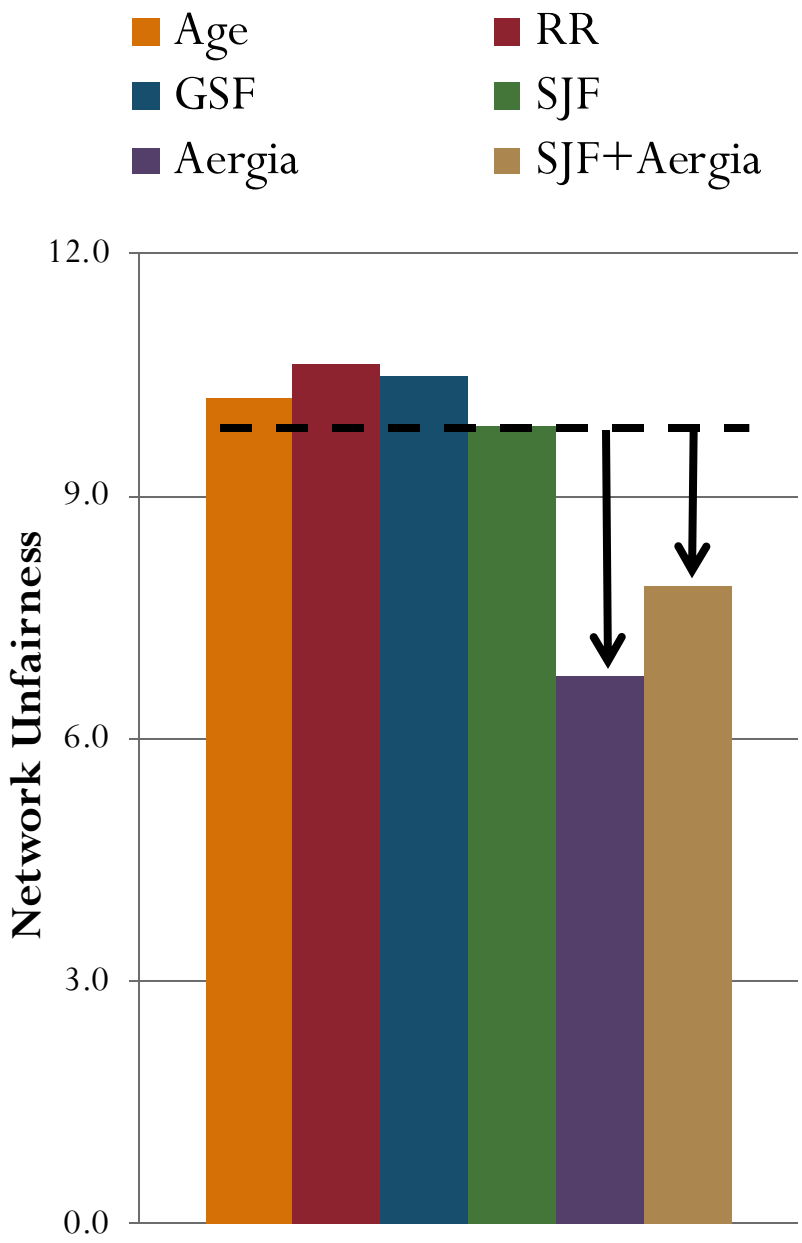
System Performance

- SJF provides 8.9% improvement in weighted speedup
- Aérgia improves system throughput by 10.3%
- Aérgia+SJF improves system throughput by 16.1%



Network Unfairness

- SJF does not imbalance network fairness
- Aergia improves network unfairness by 1.5X
- SJF+Aergia improves network unfairness by 1.3X

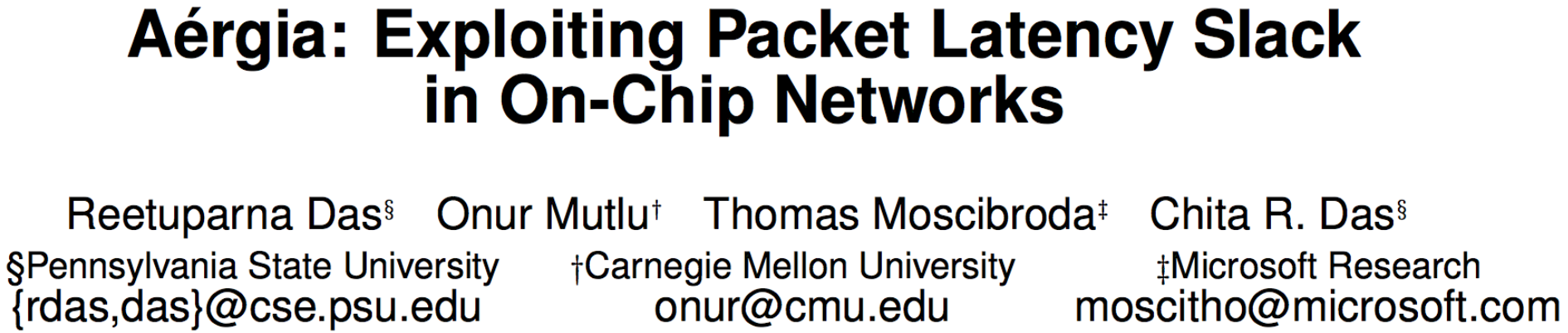


Conclusions & Future Directions

- Packets have different criticality, yet existing packet scheduling policies **treat all packets equally**
- We propose a new approach to packet scheduling in NoCs
  - We define **Slack** as a key measure that characterizes the relative importance of a packet.
  - We propose **Aérgia** a novel architecture to accelerate low slack critical packets
- Result
  - Improves system performance: 16.1%
  - Improves network fairness: 30.8%

Slack-Based Packet Scheduling

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‡Microsoft Research moscitho@microsoft.com

Slowdown Estimation in NoCs

- Xiyue Xiang, Saugata Ghose, Onur Mutlu, and Nian-Feng Tzeng, **"A Model for Application Slowdown Estimation in On-Chip Networks and Its Use for Improving System Fairness and Performance"**
  *Proceedings of the 34th IEEE International Conference on Computer Design (**ICCD**)*, Phoenix, AZ, USA, October 2016.
  [Slides (pptx) (pdf)]

A Model for Application Slowdown Estimation in On-Chip Networks and Its Use for Improving System Fairness and Performance

Xiyue Xiang[†] Saugata Ghose[‡] Onur Mutlu[§‡] Nian-Feng Tzeng[†]
[†]*University of Louisiana at Lafayette* [‡]*Carnegie Mellon University* [§]*ETH Zürich*

Handling Multicast and Hotspot Issues

- Xiyue Xiang, Wentao Shi, Saugata Ghose, Lu Peng, Onur Mutlu, and Nian-Feng Tzeng,
  "Carpool: A Bufferless On-Chip Network Supporting Adaptive Multicast and Hotspot Alleviation"
  *Proceedings of the International Conference on Supercomputing (**ICS**)*, Chicago, IL, USA, June 2017.
  [Slides (pptx) (pdf)]



Heterogeneous Networks

- Asit K. Mishra, Onur Mutlu, and Chita R. Das,
  "A Heterogeneous Multiple Network-on-Chip Design: An Application-Aware Approach"
  *Proceedings of the 50th Design Automation Conference (**DAC**)*,
  Austin, TX, June 2013. Slides (pptx) Slides (pdf)

A Heterogeneous Multiple Network-On-Chip Design: An Application-Aware Approach

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Low-Cost QoS in On-Chip Networks (I)

- Boris Grot, Stephen W. Keckler, and Onur Mutlu, **"Preemptive Virtual Clock: A Flexible, Efficient, and Cost-effective QOS Scheme for Networks-on-Chip"** *Proceedings of the 42nd International Symposium on Microarchitecture (**MICRO**)*, pages 268-279, New York, NY, December 2009. Slides (pdf)

Preemptive Virtual Clock: A Flexible, Efficient, and Cost-effective QOS Scheme for Networks-on-Chip

Boris Grot Stephen W. Keckler Onur Mutlu†

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†Computer Architecture Laboratory (CALCM)
Carnegie Mellon University
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Low-Cost QoS in On-Chip Networks (II)

- Boris Grot, Joel Hestness, Stephen W. Keckler, and Onur Mutlu, **"Kilo-NOC: A Heterogeneous Network-on-Chip Architecture for Scalability and Service Guarantees"**
  *Proceedings of the 38th International Symposium on Computer Architecture (**ISCA**)*, San Jose, CA, June 2011. Slides (pptx)
  One of the 12 computer architecture papers of 2011 selected as Top Picks by IEEE Micro.

Kilo-NOC: A Heterogeneous Network-on-Chip Architecture for Scalability and Service Guarantees

Boris Grot[1] bgrot@cs.utexas.edu
Joel Hestness[1] hestness@cs.utexas.edu
Stephen W. Keckler[1,2] skeckler@nvidia.com
Onur Mutlu[3] onur@cmu.edu

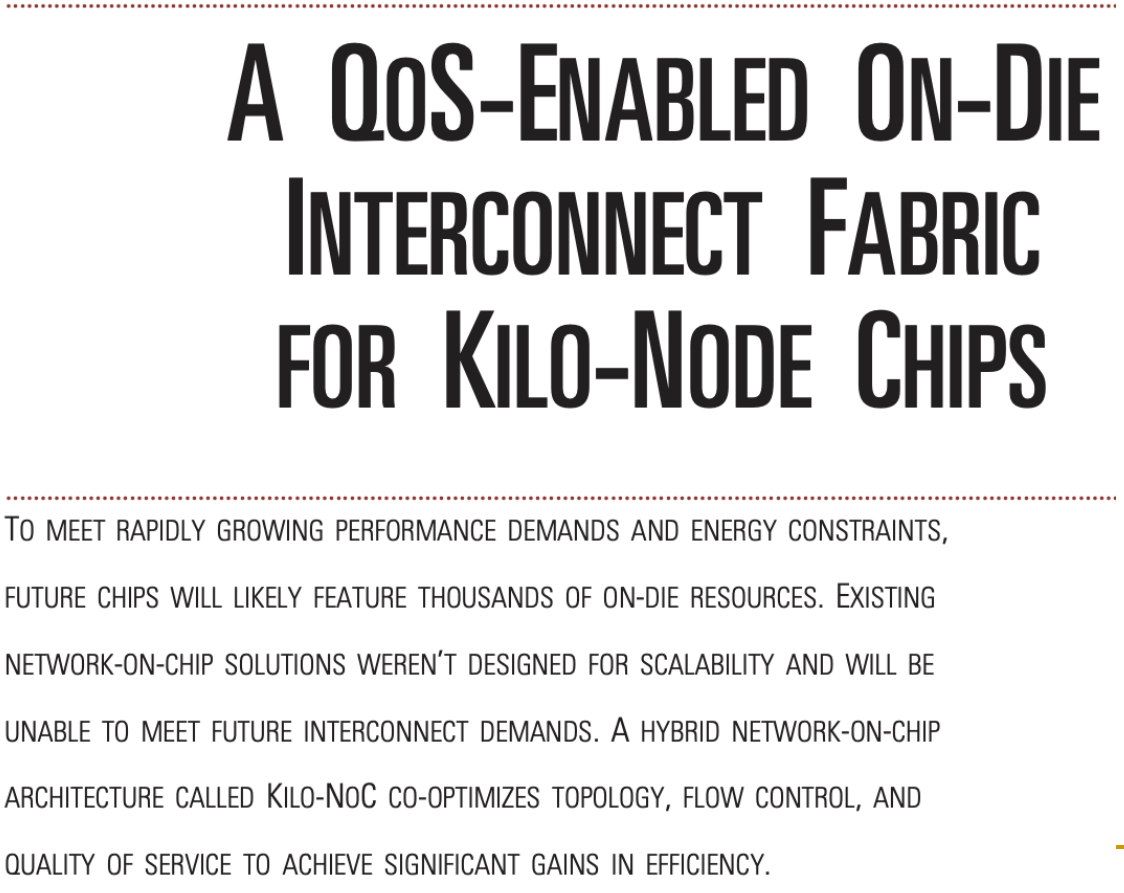
[1]The University of Texas at Austin
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Pittsburgh, PA

Low-Cost QoS in On-Chip Networks (III)

- Boris Grot, Joel Hestness, Stephen W. Keckler, and Onur Mutlu, **"A QoS-Enabled On-Die Interconnect Fabric for Kilo-Node Chips"** *IEEE Micro, Special Issue: Micro's Top Picks from 2011 Computer Architecture Conferences (**MICRO TOP PICKS**)*, Vol. 32, No. 3, May/June 2012.



A QoS-Enabled On-Die Interconnect Fabric for Kilo-Node Chips

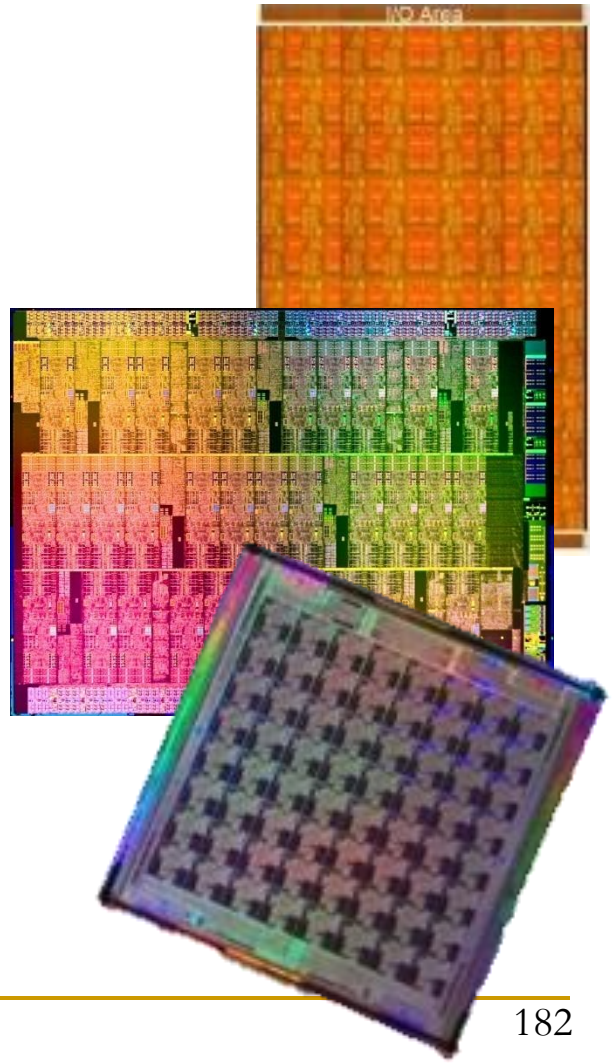
To meet rapidly growing performance demands and energy constraints, future chips will likely feature thousands of on-die resources. Existing network-on-chip solutions weren't designed for scalability and will be unable to meet future interconnect demands. A hybrid network-on-chip architecture called Kilo-NoC co-optimizes topology, flow control, and quality of service to achieve significant gains in efficiency.

Kilo-NoC: Topology-Aware QoS

Boris Grot, Joel Hestness, Stephen W. Keckler, and Onur Mutlu,
"Kilo-NOC: A Heterogeneous Network-on-Chip Architecture for Scalability and Service Guarantees"
*Proceedings of the 38th International Symposium on Computer Architecture (**ISCA**)*, San Jose, CA, June 2011. Slides (pptx)

Motivation

- Extreme-scale chip-level integration
  - Cores
  - Cache banks
  - Accelerators
  - I/O logic
  - Network-on-chip (NOC)
- 10-100 cores today
- 1000+ agents in the near future



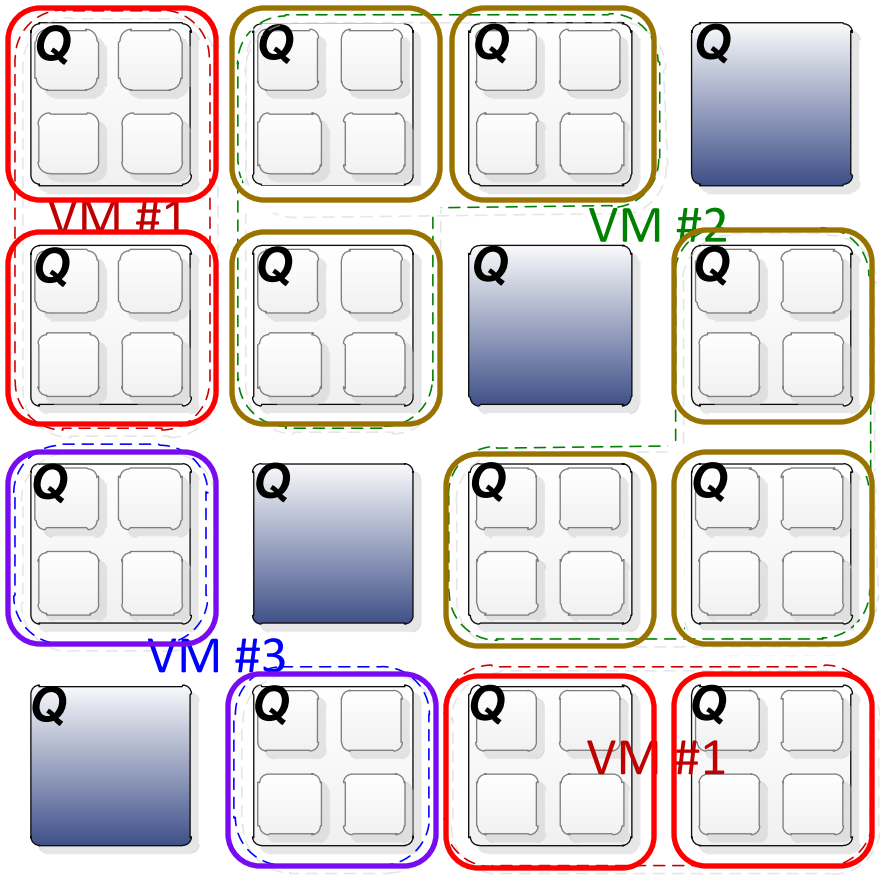
Kilo-NOC requirements

- High efficiency
  - Area
  - Energy
- Good performance
- Strong service guarantees (QoS)

Topology-Aware QoS

- Problem: QoS support in each router is expensive (in terms of buffering, arbitration, bookkeeping)
  - E.g., Grot et al., "Preemptive Virtual Clock: A Flexible, Efficient, and Cost-effective QOS Scheme for Networks-on-Chip," MICRO 2009.
- Goal: Provide QoS guarantees at low area and power cost
- Idea:
  - Isolate shared resources in a region of the network, support QoS within that area
  - Design the topology so that applications can access the region without interference

Baseline QOS-enabled CMP



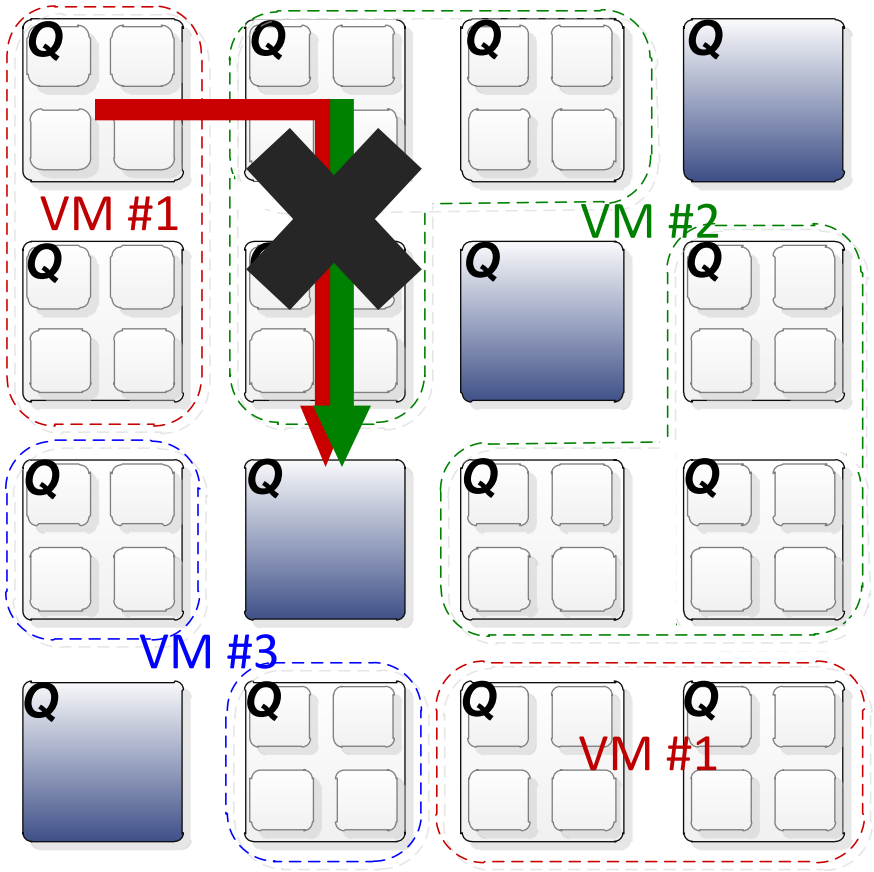
Multiple VMs sharing a die

Shared resources (e.g., memory controllers)

VM-private resources (cores, caches)

Q QOS-enabled router

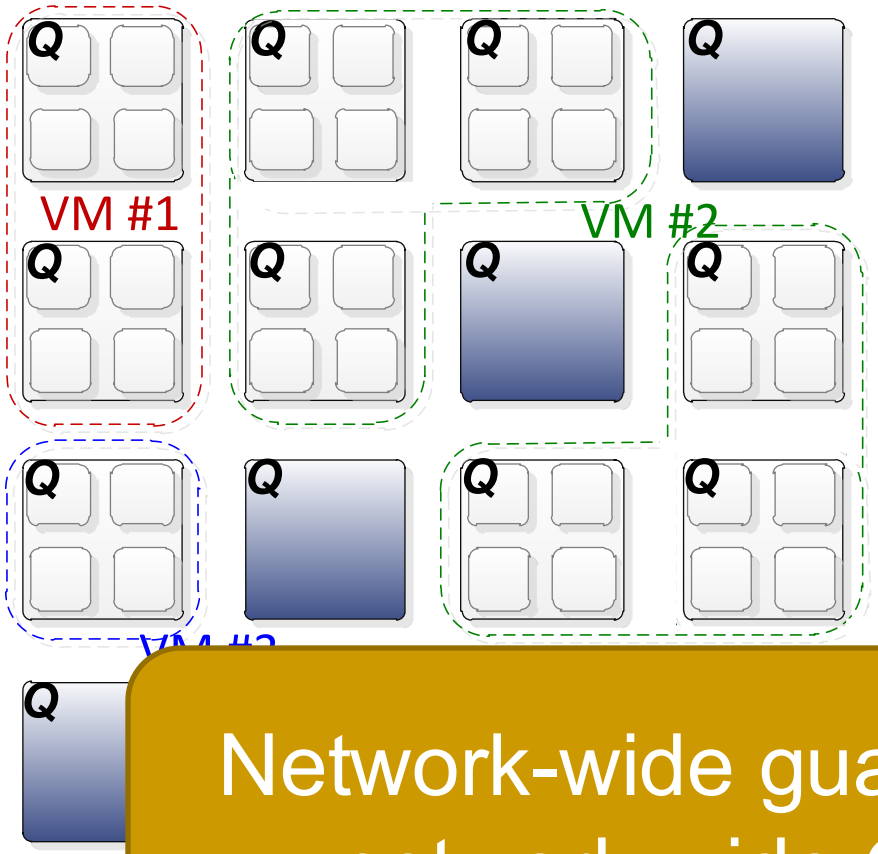
Conventional NOC QOS



Contention scenarios:

- Shared resources
  - memory access
- Intra-VM traffic
  - shared cache access
- Inter-VM traffic
  - VM page sharing

Conventional NOC QOS



Contention scenarios:

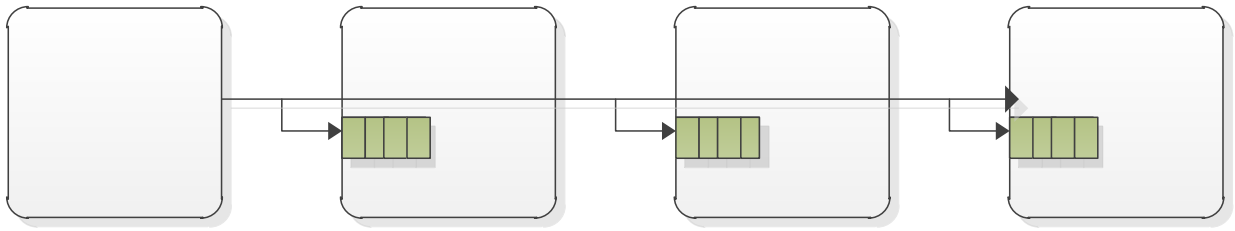
- Shared resources
  - memory access
- Intra-VM traffic
  - shared cache access
- Inter-VM traffic
  - VM page sharing

Network-wide guarantees *without* network-wide QOS support

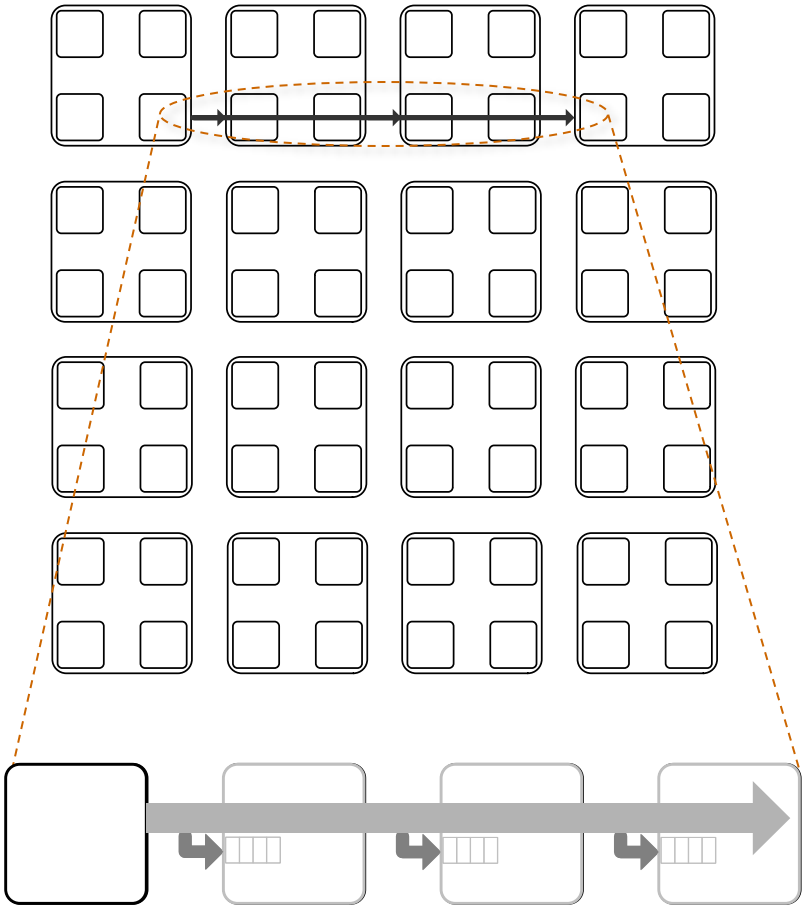
Kilo-NOC QOS

- Insight: leverage rich network connectivity
  - Naturally reduce interference among flows
  - Limit the extent of hardware QOS support
- Requires a low-diameter topology
  - This work: Multidrop Express Channels (MECS)

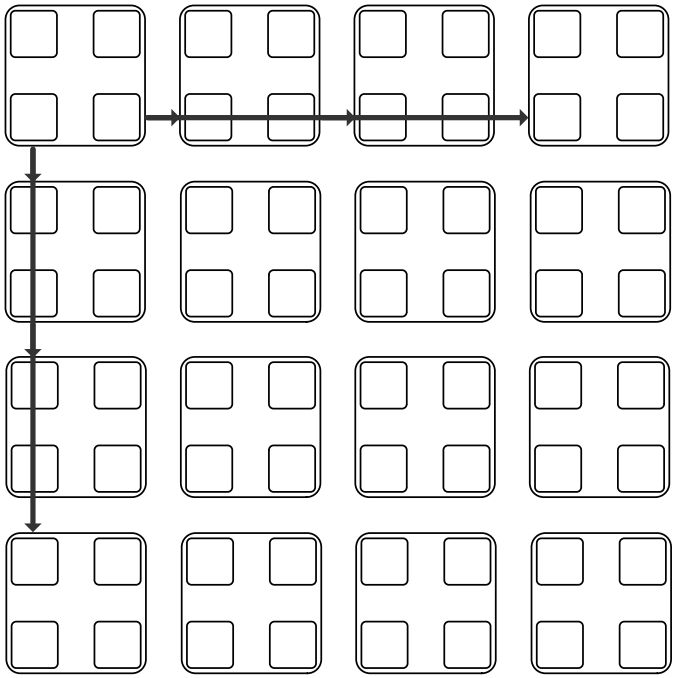
Grot et al., HPCA 2009

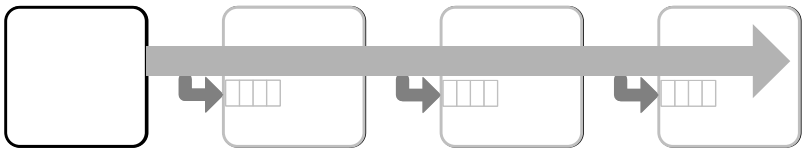


Multidrop Express Channels (MECS)

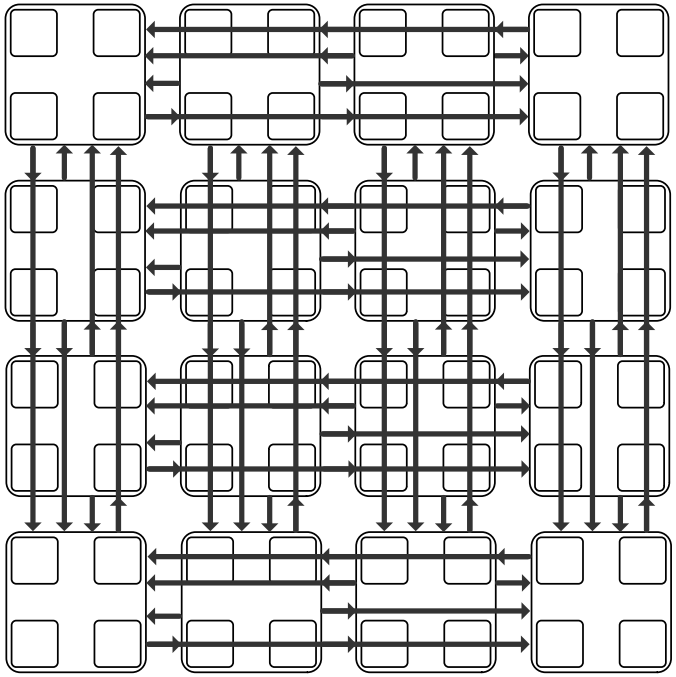


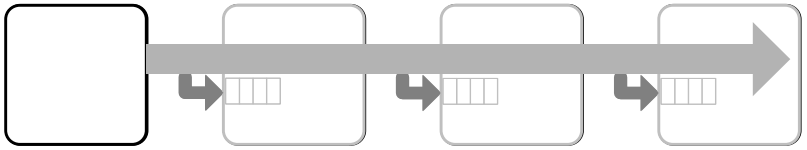
Multidrop Express Channels (MECS)



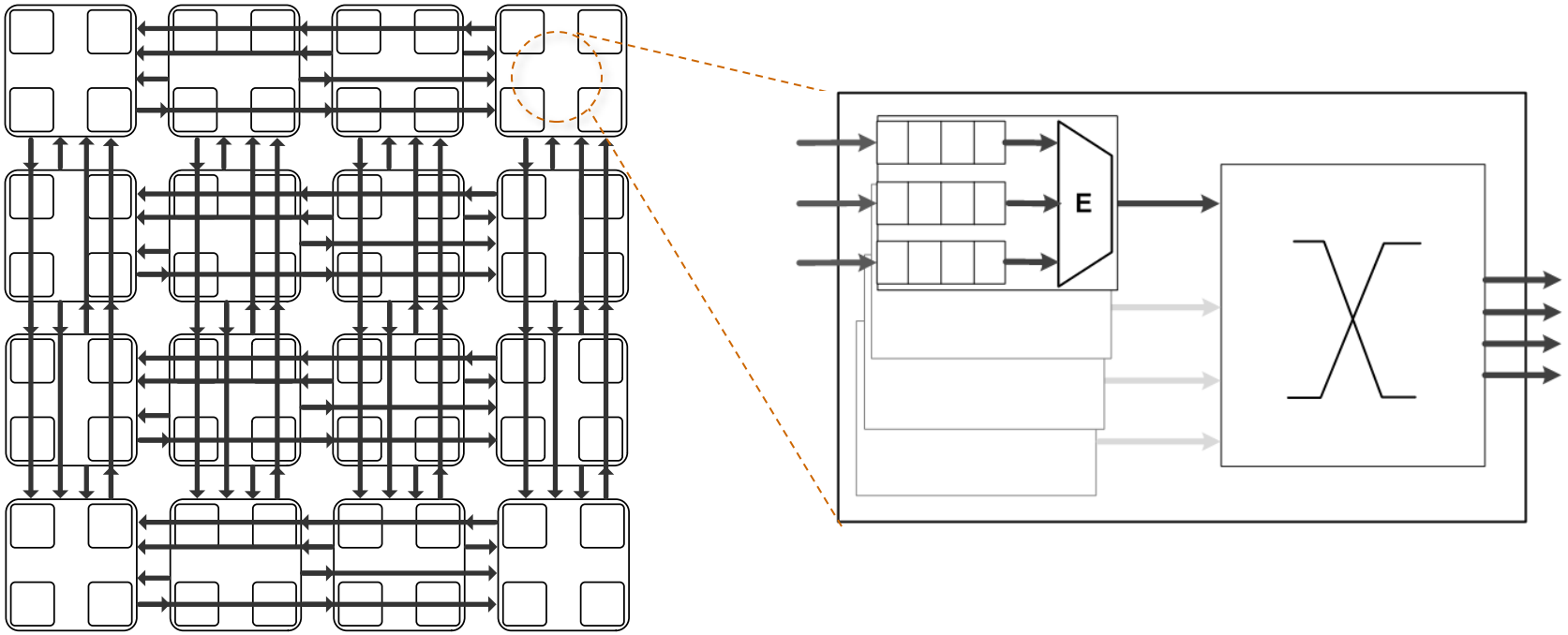


Multidrop Express Channels (MECS)

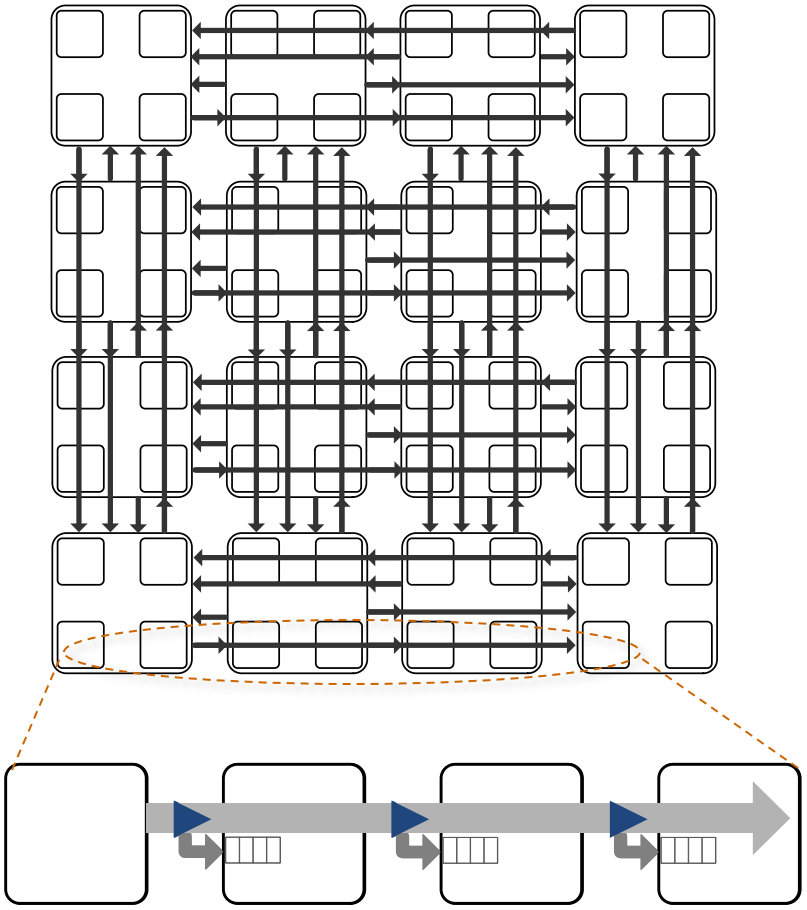




Multidrop Express Channels (MECS)



Multidrop Express Channels (MECS)



- Pros
  - One-to-many topology
  - Low diameter: 2 hops
  - *k* channels row/column
  - Asymmetric
- Cons
  - Asymmetric
  - Increased control (arbitration) complexity

Multi-Drop Express Channels (MECS)

- Boris Grot, Joel Hestness, Stephen W. Keckler, and Onur Mutlu, **"Express Cube Topologies for On-Chip Interconnects"** *Proceedings of the 15th International Symposium on High-Performance Computer Architecture (**HPCA**)*, pages 163-174, Raleigh, NC, February 2009. Slides (ppt)

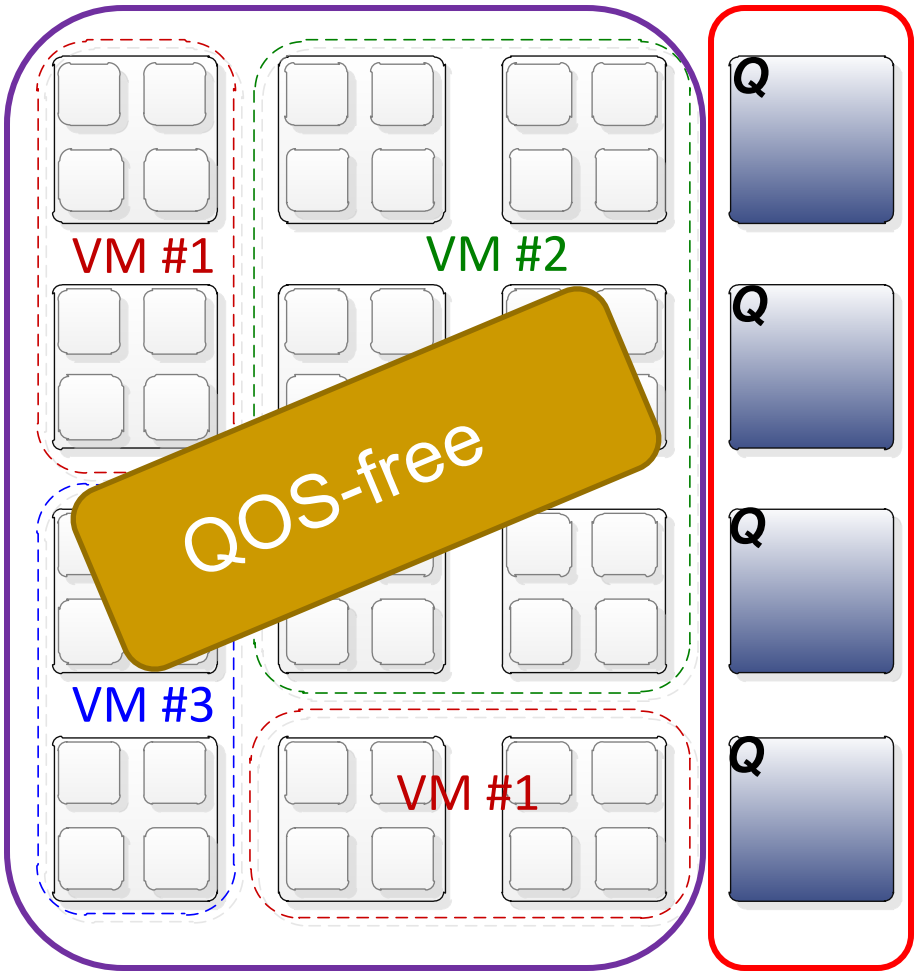
Express Cube Topologies for On-Chip Interconnects

Boris Grot Joel Hestness Stephen W. Keckler Onur Mutlu†

Department of Computer Sciences
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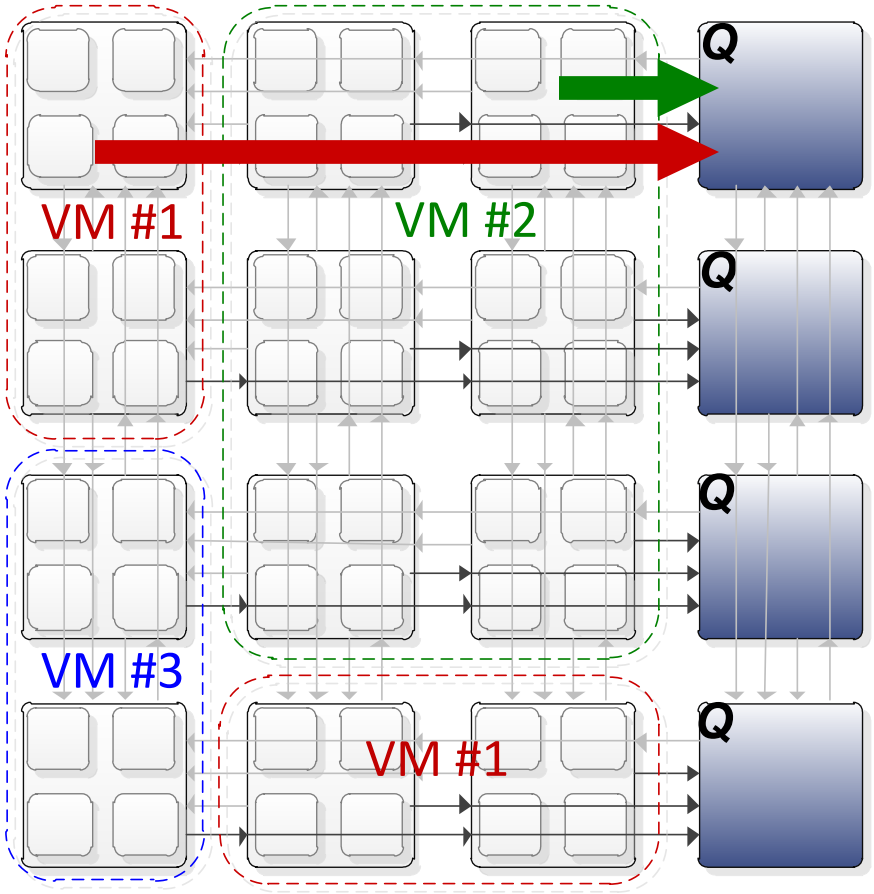
†Computer Architecture Laboratory (CALCM)
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Topology-Aware QOS



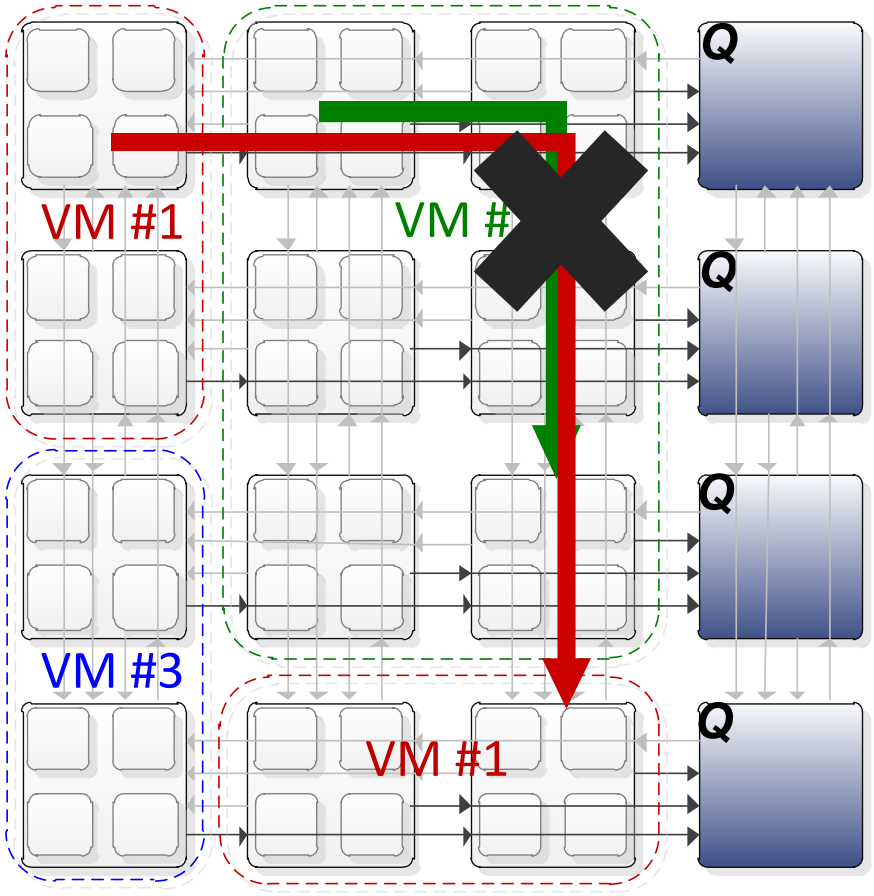
- Dedicated, QOS-enabled regions
  - Rest of die: QOS-free
- Richly-connected topology
  - Traffic isolation
- Special routing rules
  - Manage interference

Topology-Aware QOS



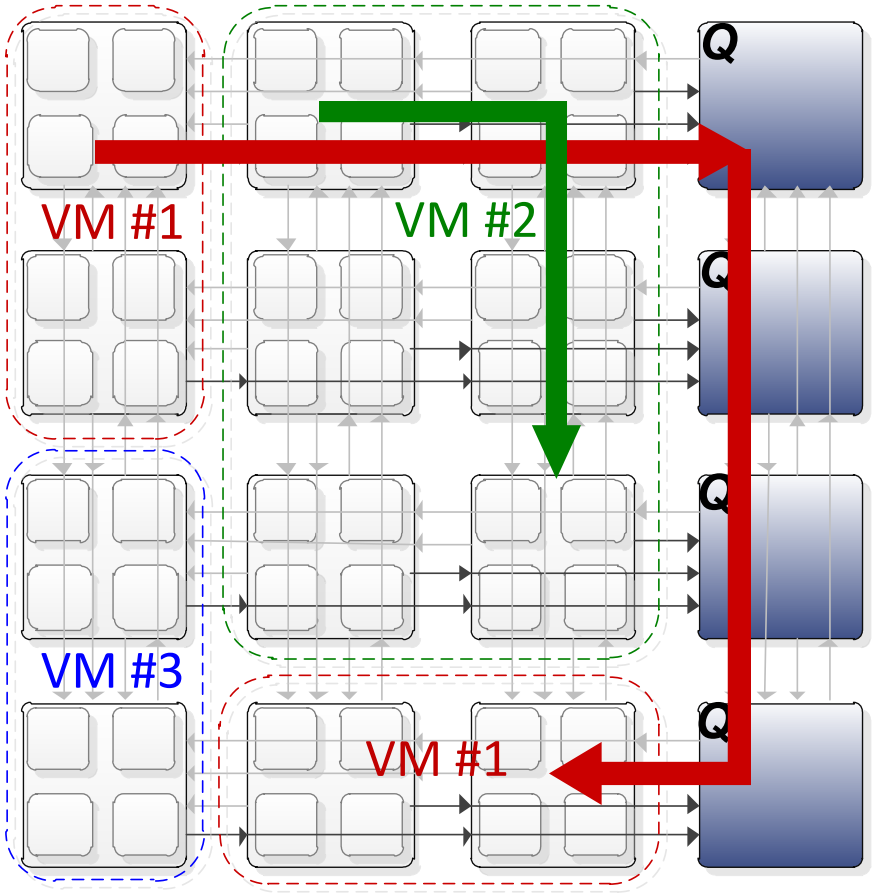
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Topology-Aware QOS



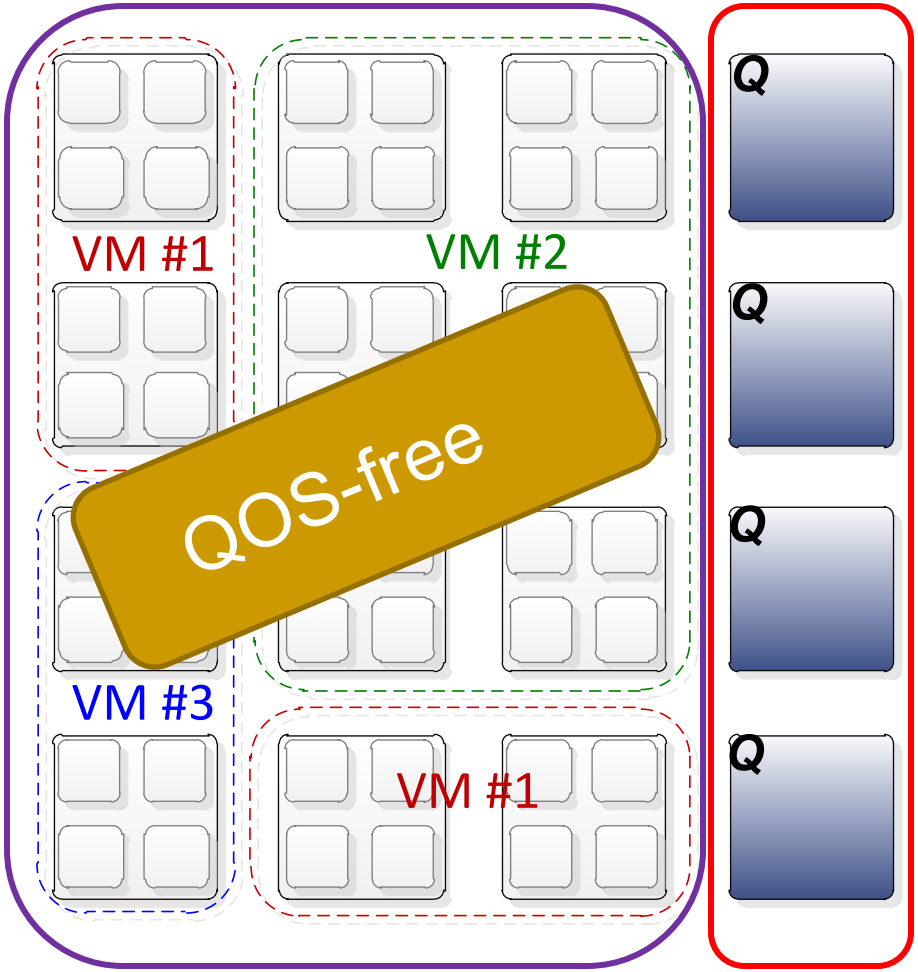
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Topology-Aware QOS



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Kilo-NOC

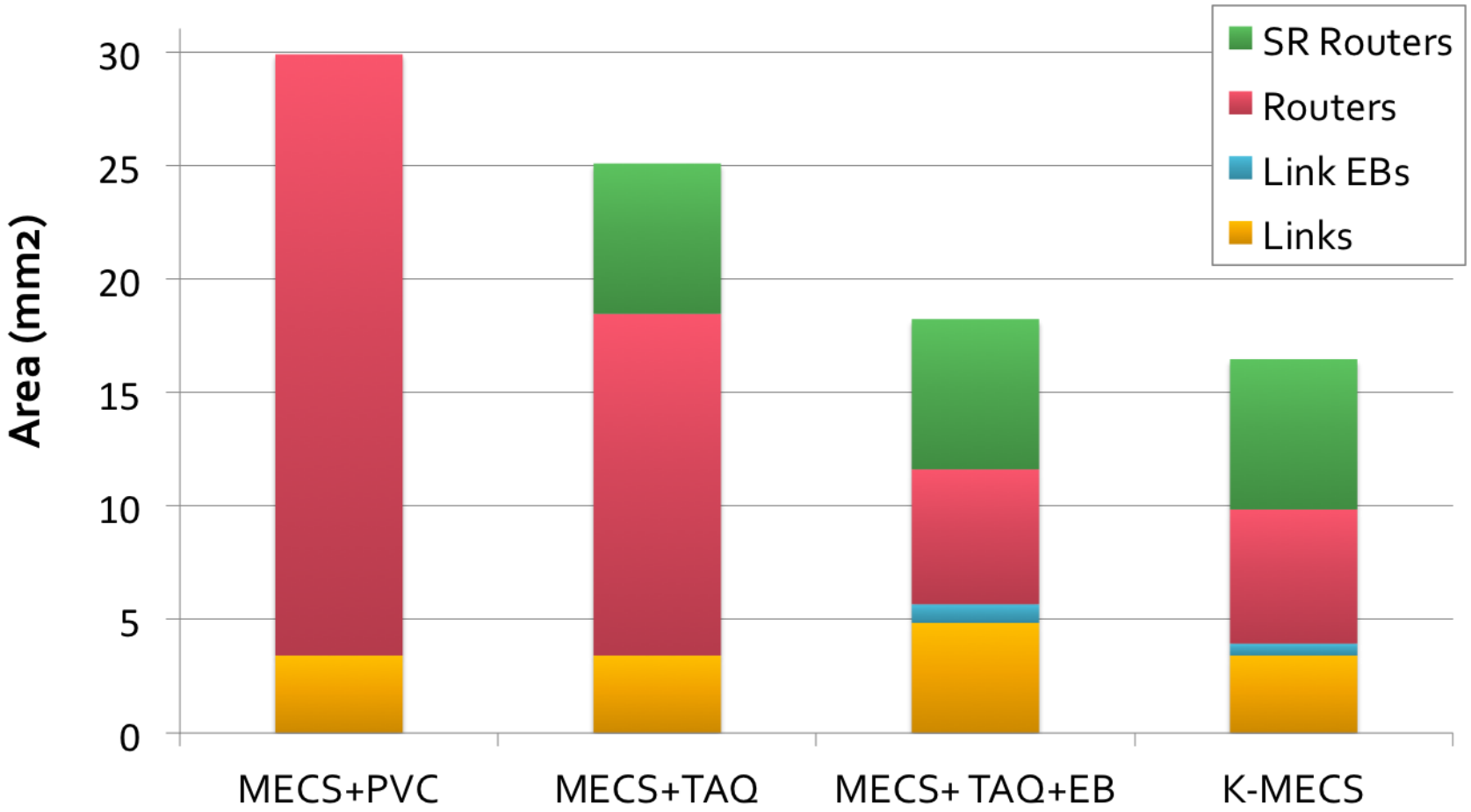


- Topology-aware QOS support
  - Limit QOS complexity to a fraction of the die
- Optimized flow control
  - Reduce buffer requirements in QOS-free regions

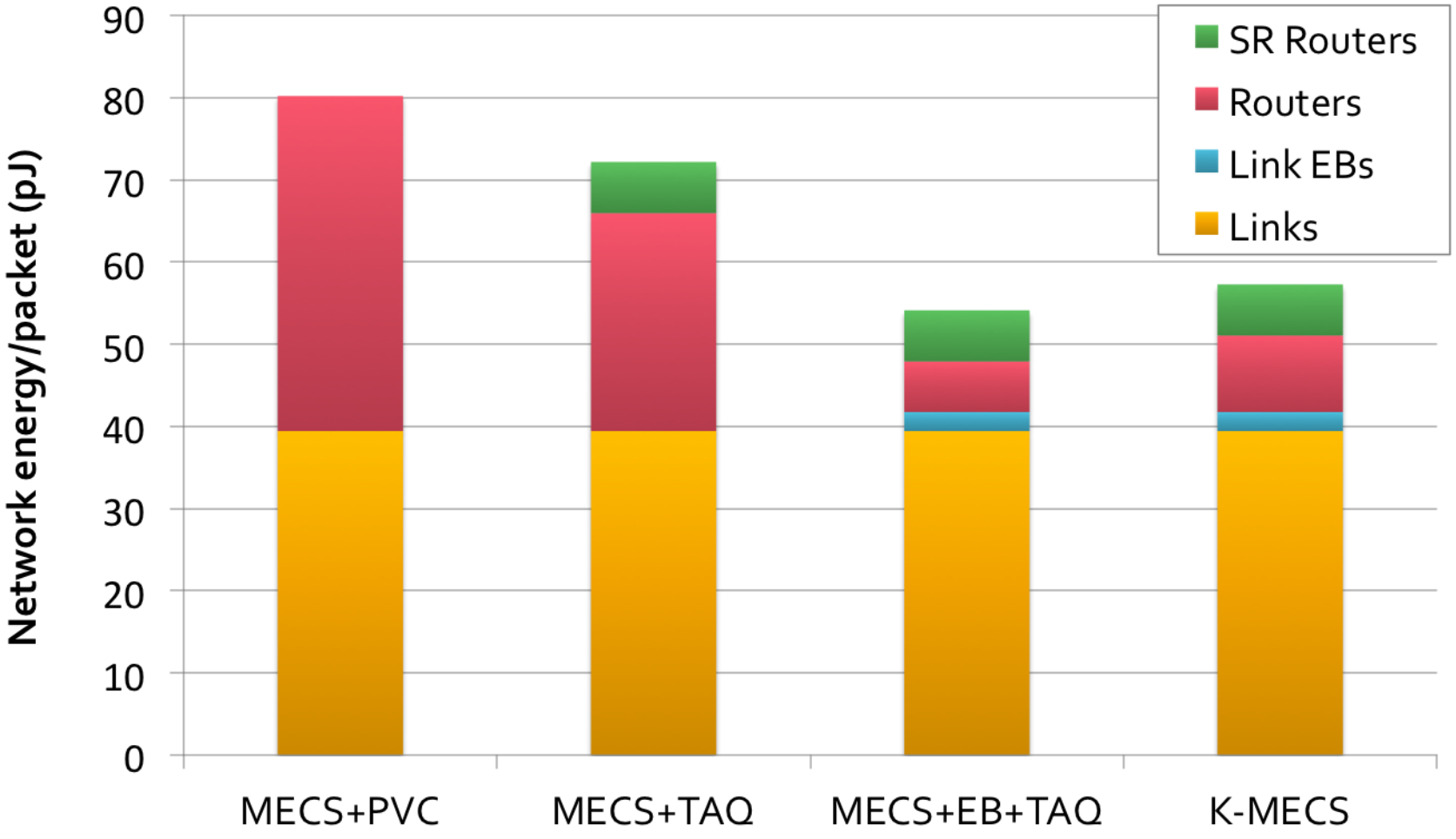
Evaluation Methodology

Parameter	Value
Technology	15 nm
Vdd	0.7 V
System	1024 tiles: 256 concentrated nodes (64 shared resources)
Networks:	
MECS+PVC	VC flow control, QOS support (PVC) at each node
MECS+TAQ	VC flow control, QOS support only in shared regions
MECS+TAQ+EB	EB flow control outside of SRs, Separate *Request* and *Reply* networks
K-MECS	Proposed organization: TAQ + hybrid flow control

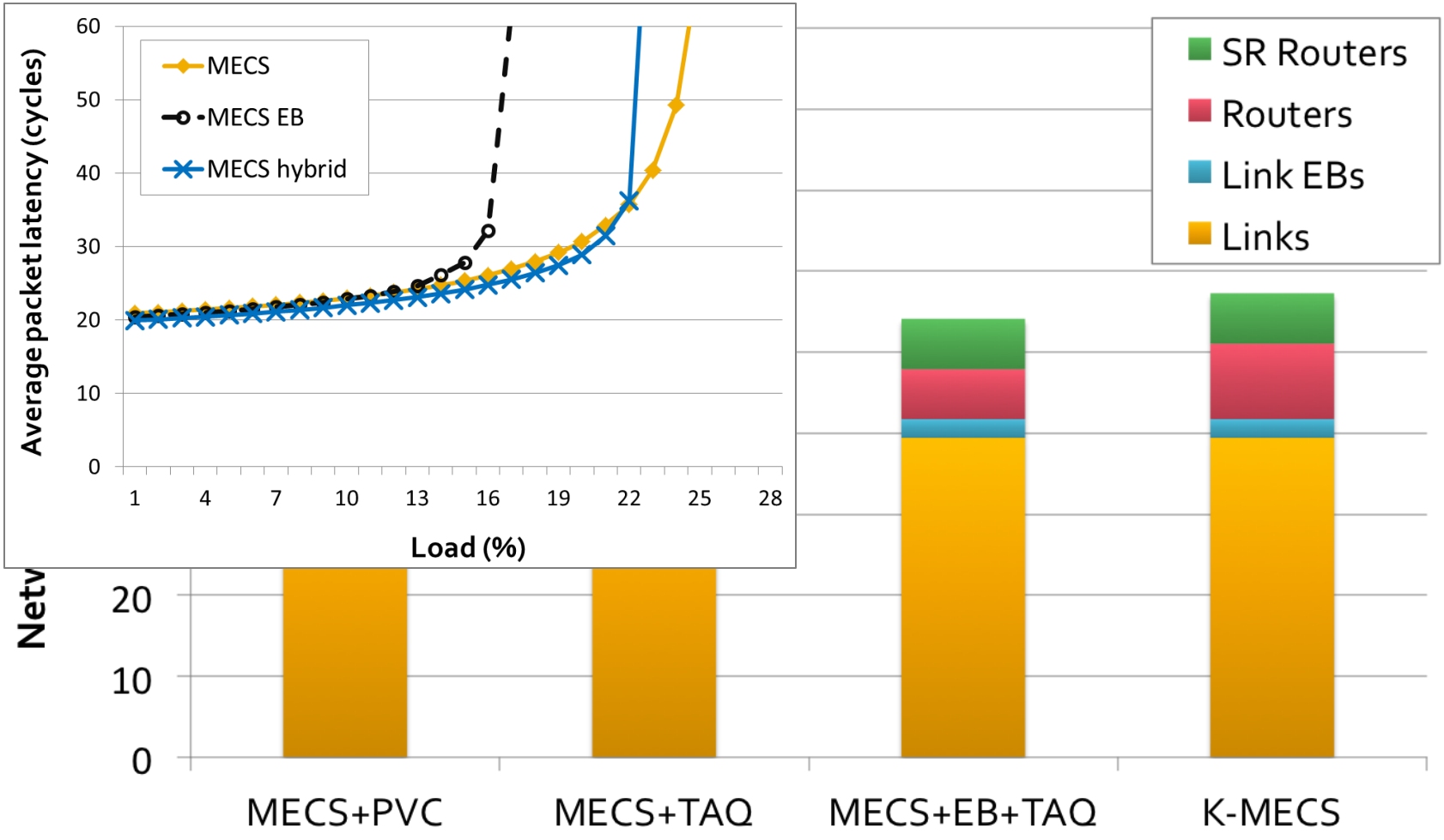
Area comparison



Energy comparison



Energy comparison



Summary

Kilo-NOC: a heterogeneous NOC architecture for kilo-node substrates

- Topology-aware QOS
  - Limits QOS support to a fraction of the die
  - Leverages low-diameter topologies
  - Improves NOC area- and energy-efficiency
  - Provides strong guarantees

More on Kilo-NoC (I)

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More on Kilo-NoC (II)

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A QoS-Enabled On-Die Interconnect Fabric for Kilo-Node Chips

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Multi-Drop Express Channels

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Express Cube Topologies for On-Chip Interconnects

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Computer Architecture

Lecture 24: On-Chip Networks & Router Design

Prof. Onur Mutlu
ETH Zürich

Fall 2025

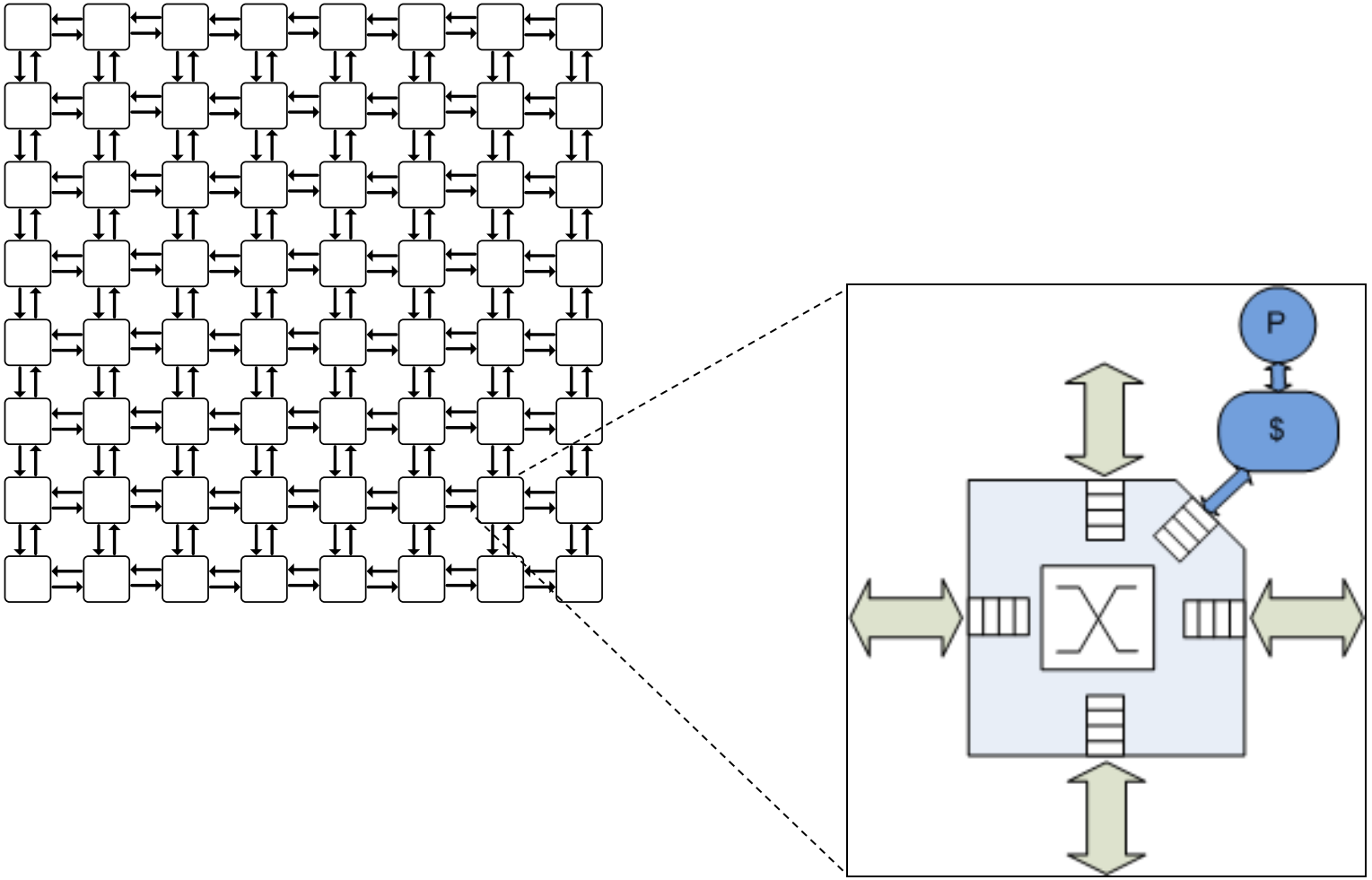
12 December 2025

Backup Slides for Additional Information

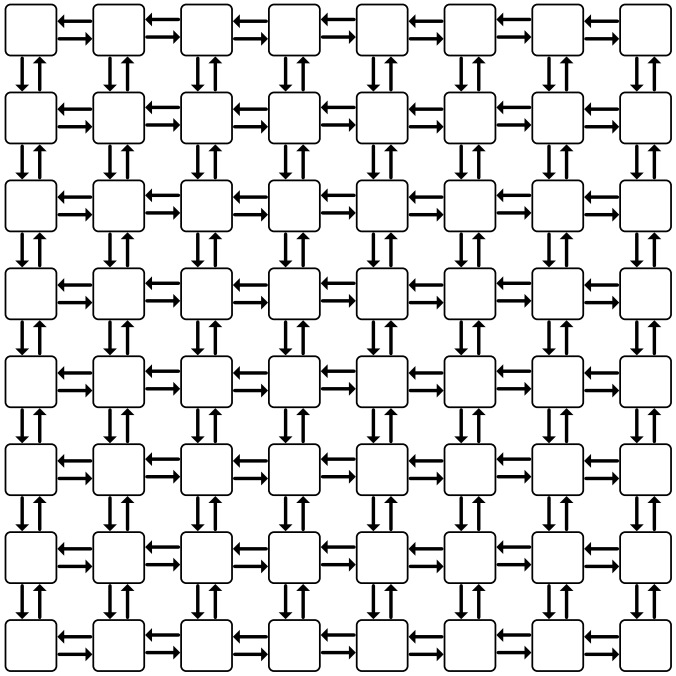
Express-Cube Topologies

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2-D Mesh

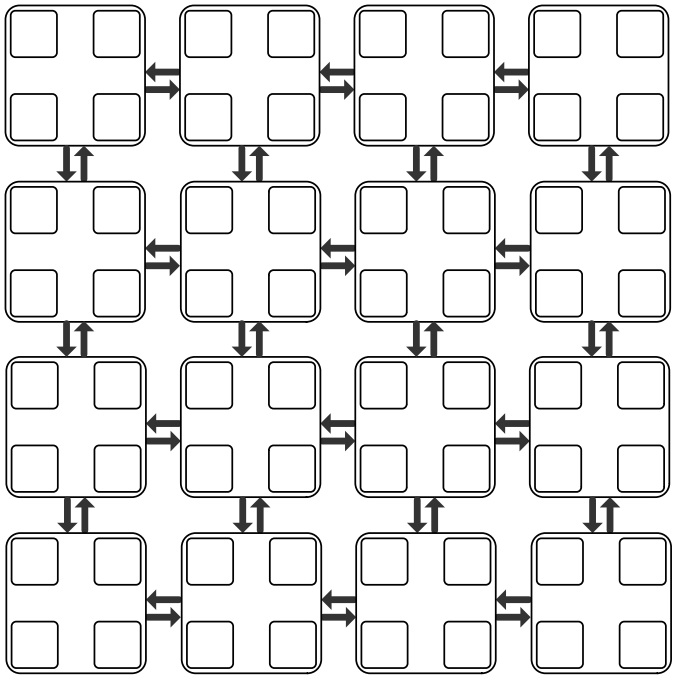


2-D Mesh



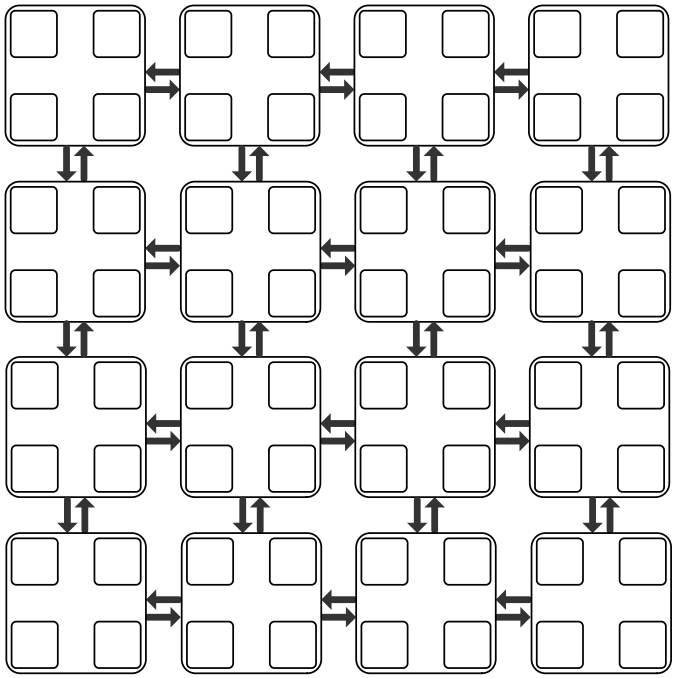
- Pros
  - Low design & layout complexity
  - Simple, fast routers
- Cons
  - Large diameter
  - Energy & latency impact

Concentration *(Balfour & Dally, ICS '06)*



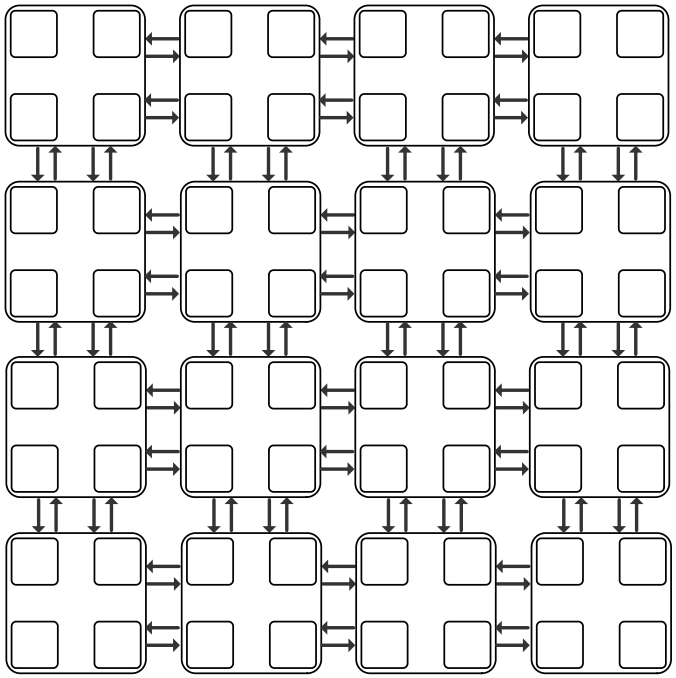
- Pros
  - Multiple *terminals* attached to a router node
  - Fast nearest-neighbor communication via the crossbar
  - Hop count reduction proportional to *concentration* degree
- Cons
  - Benefits limited by crossbar complexity

Concentration



- Side-effects
  - Fewer channels
  - Greater channel width

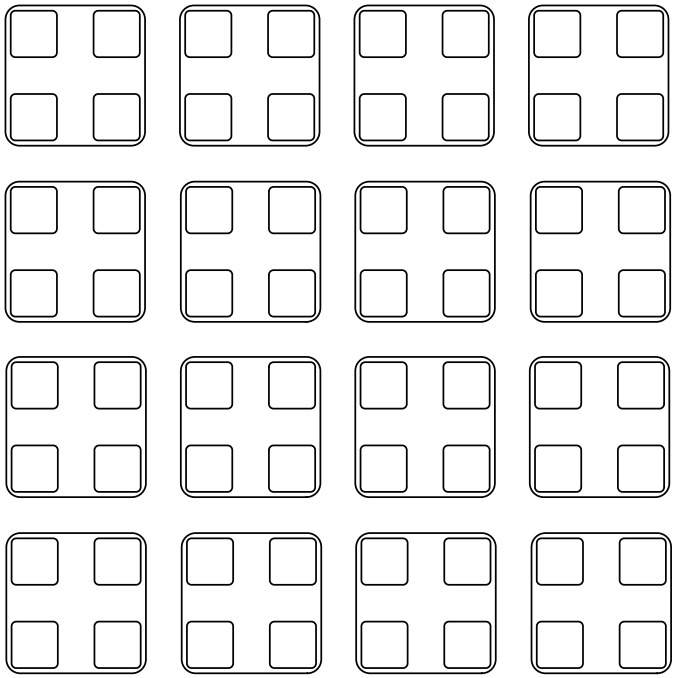
Replication



CMesh-X2

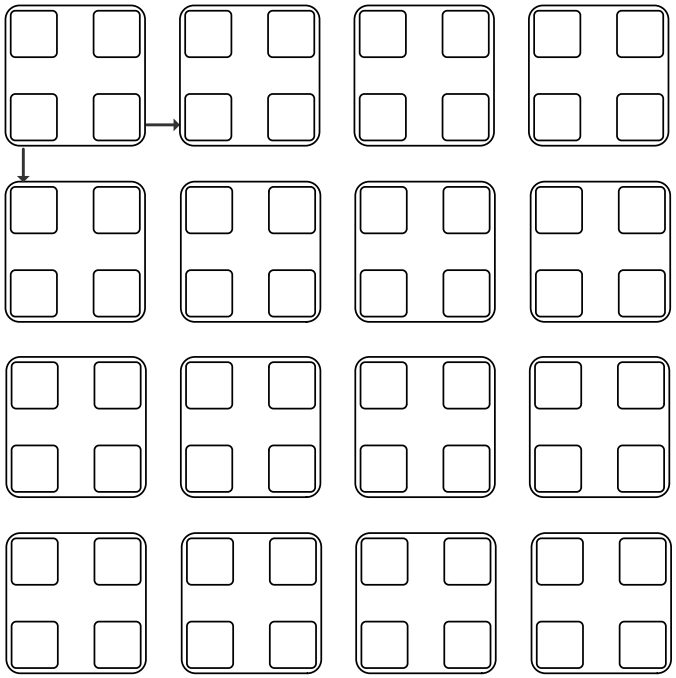
- Benefits
  - Restores bisection channel count
  - Restores channel width
  - Reduced crossbar complexity

Flattened Butterfly *(Kim et al., Micro '07)*

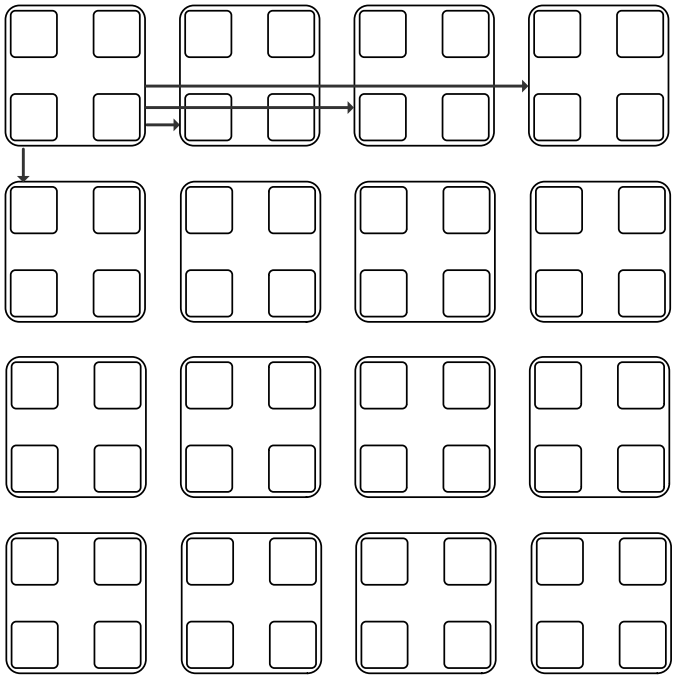


- Objectives:
  - Improve connectivity
  - Exploit the wire budget

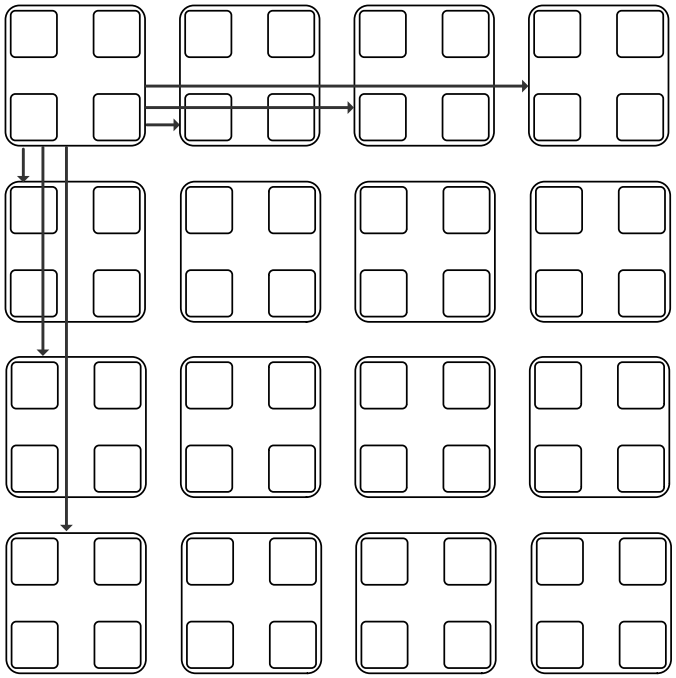
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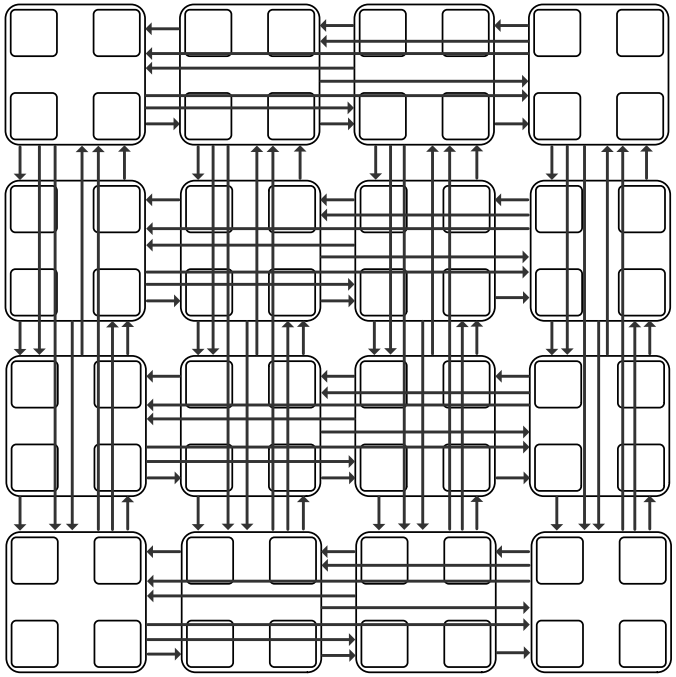
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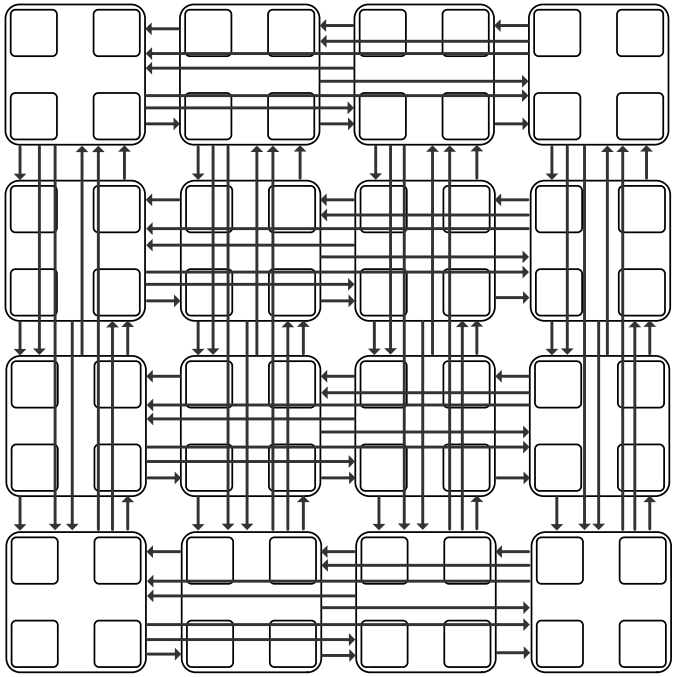
Flattened Butterfly *(Kim et al., Micro '07)*



Flattened Butterfly *(Kim et al., Micro '07)*

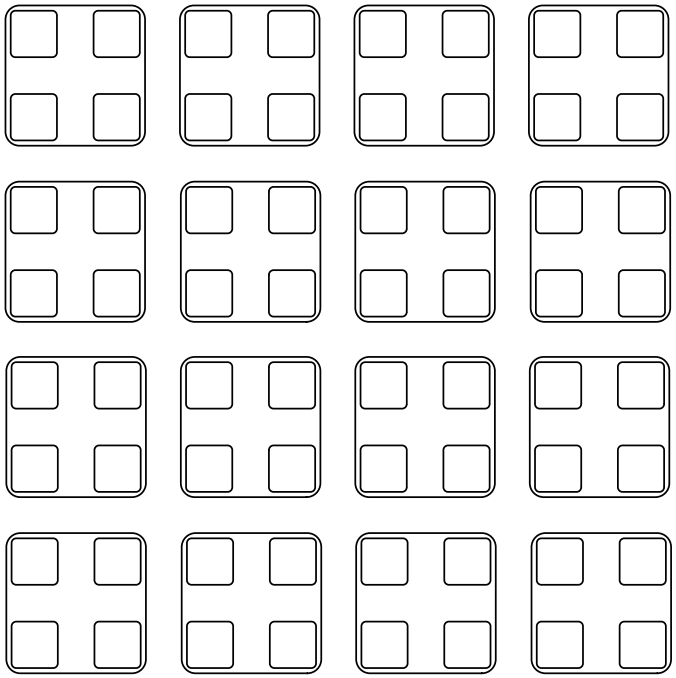


Flattened Butterfly *(Kim et al., Micro '07)*



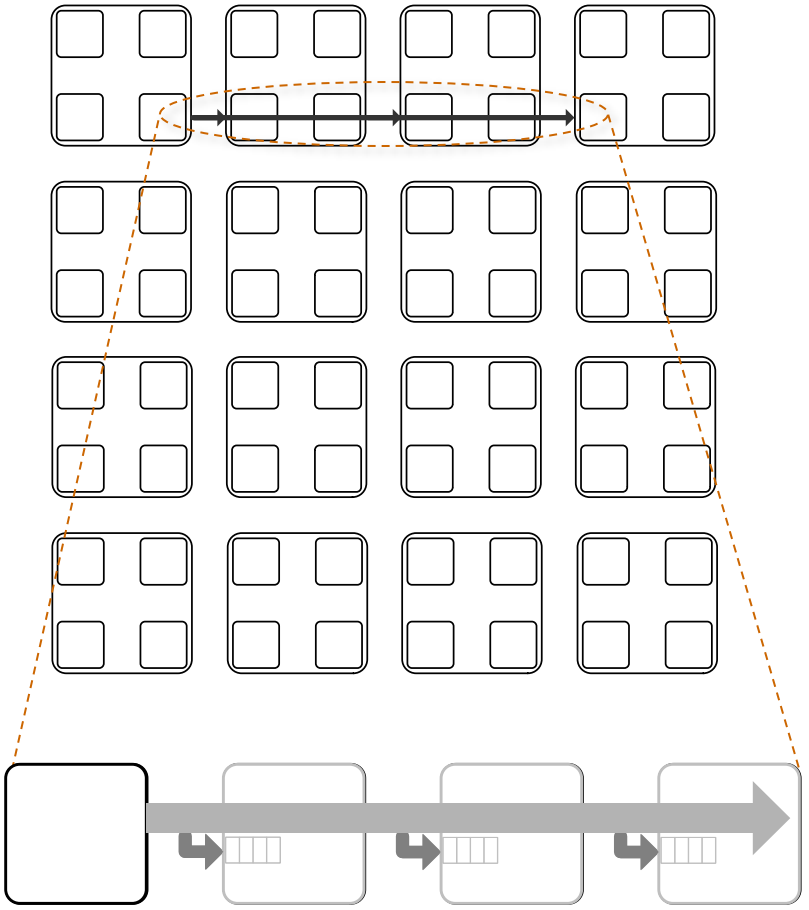
- Pros
  - Excellent connectivity
  - Low diameter: 2 hops
- Cons
  - High channel count: $k^2/2$ per row/column
  - Low channel utilization
  - Increased control (arbitration) complexity

Multidrop Express Channels (MECS)

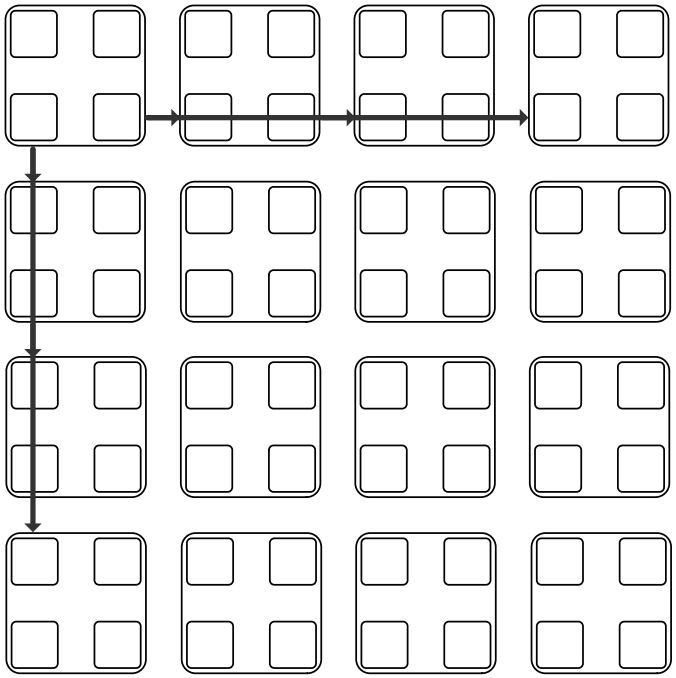


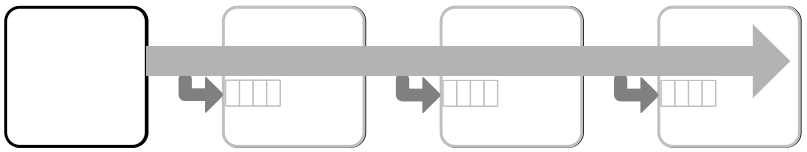
- Objectives:
  - Connectivity
  - More scalable channel count
  - Better channel utilization

Multidrop Express Channels (MECS)

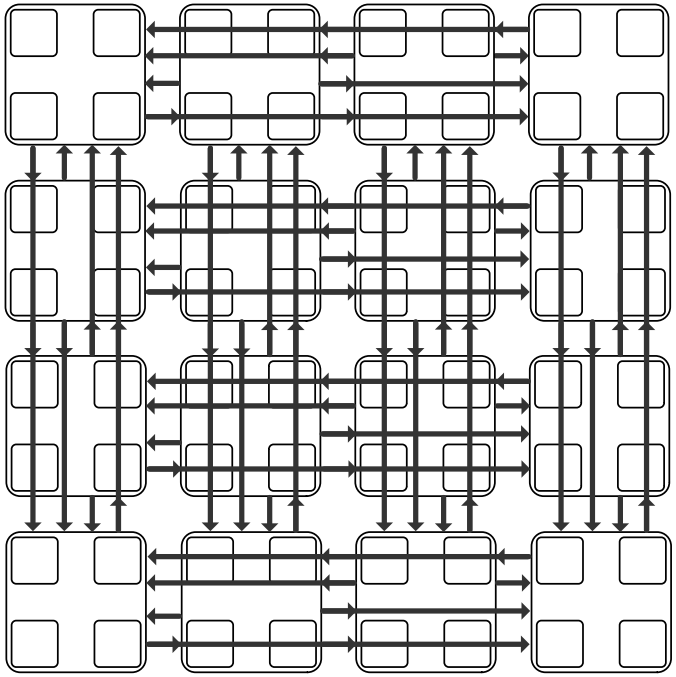


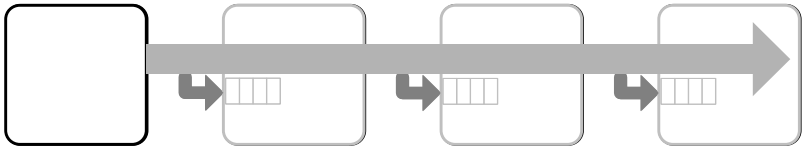
Multidrop Express Channels (MECS)



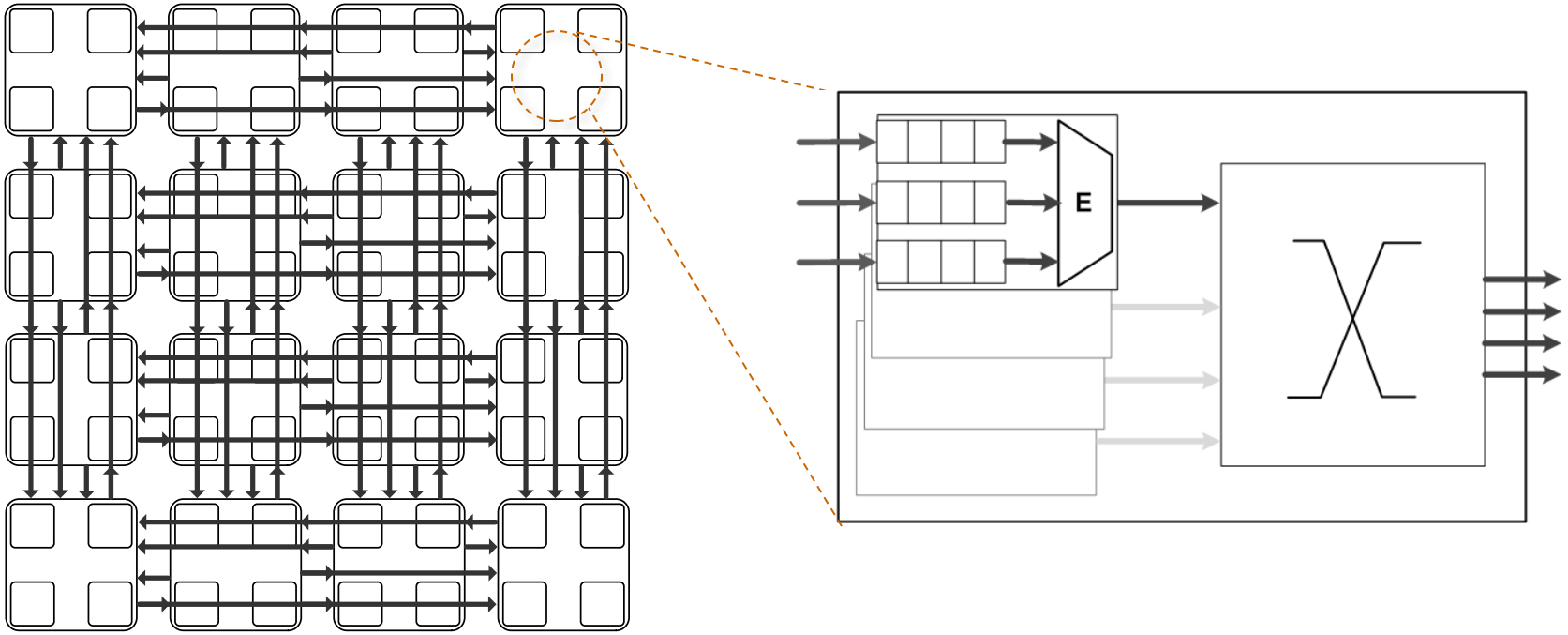


Multidrop Express Channels (MECS)

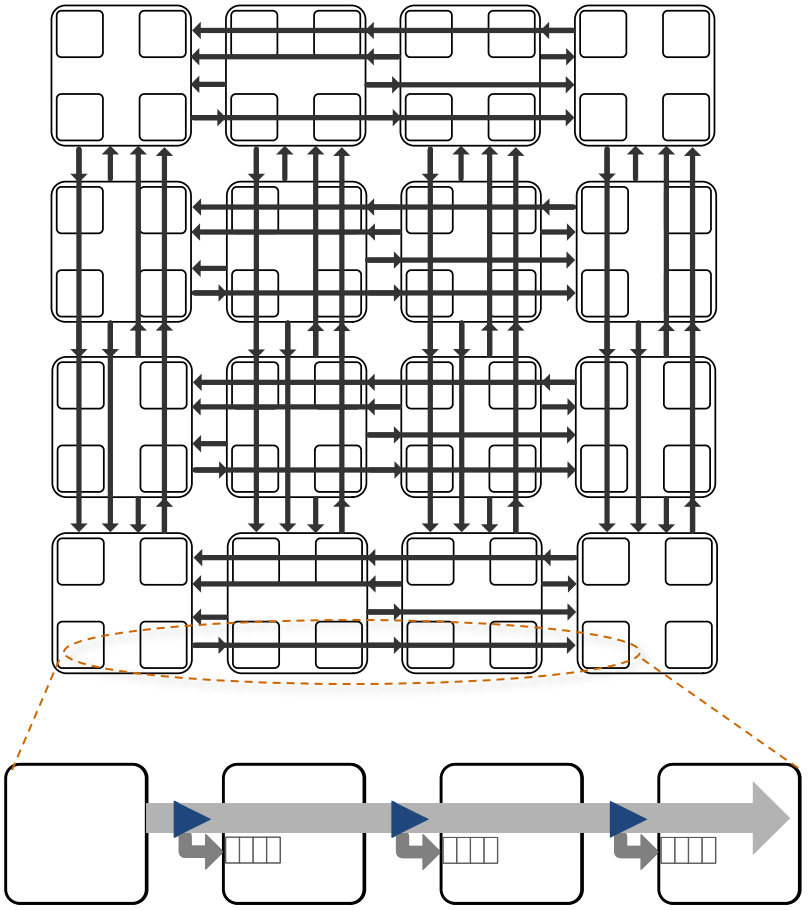




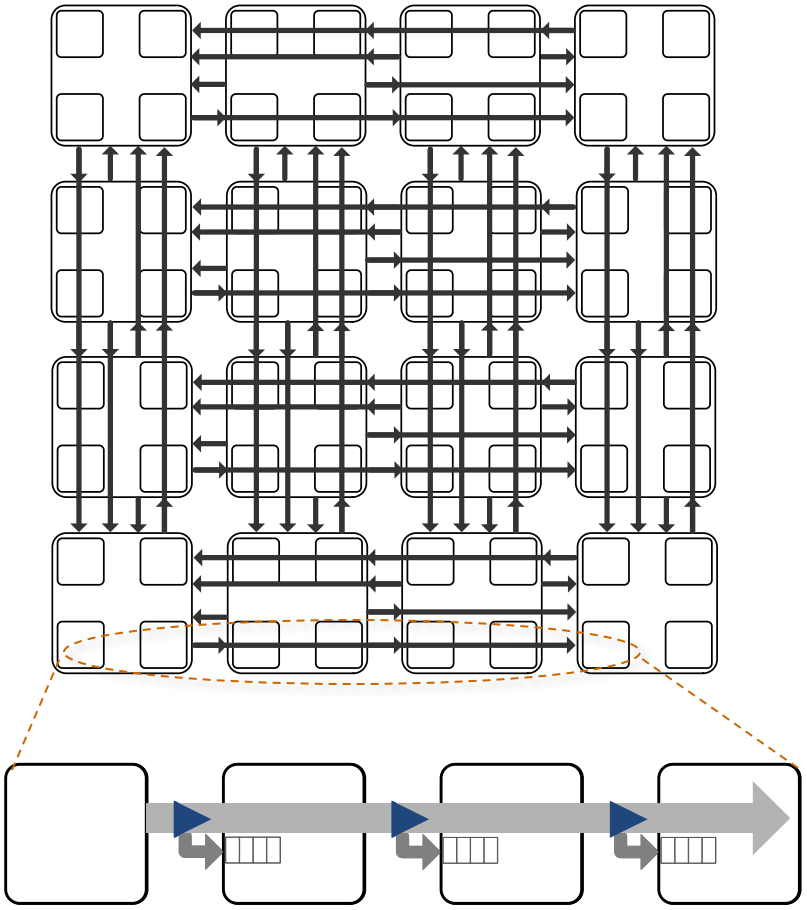
Multidrop Express Channels (MECS)



Multidrop Express Channels (MECS)

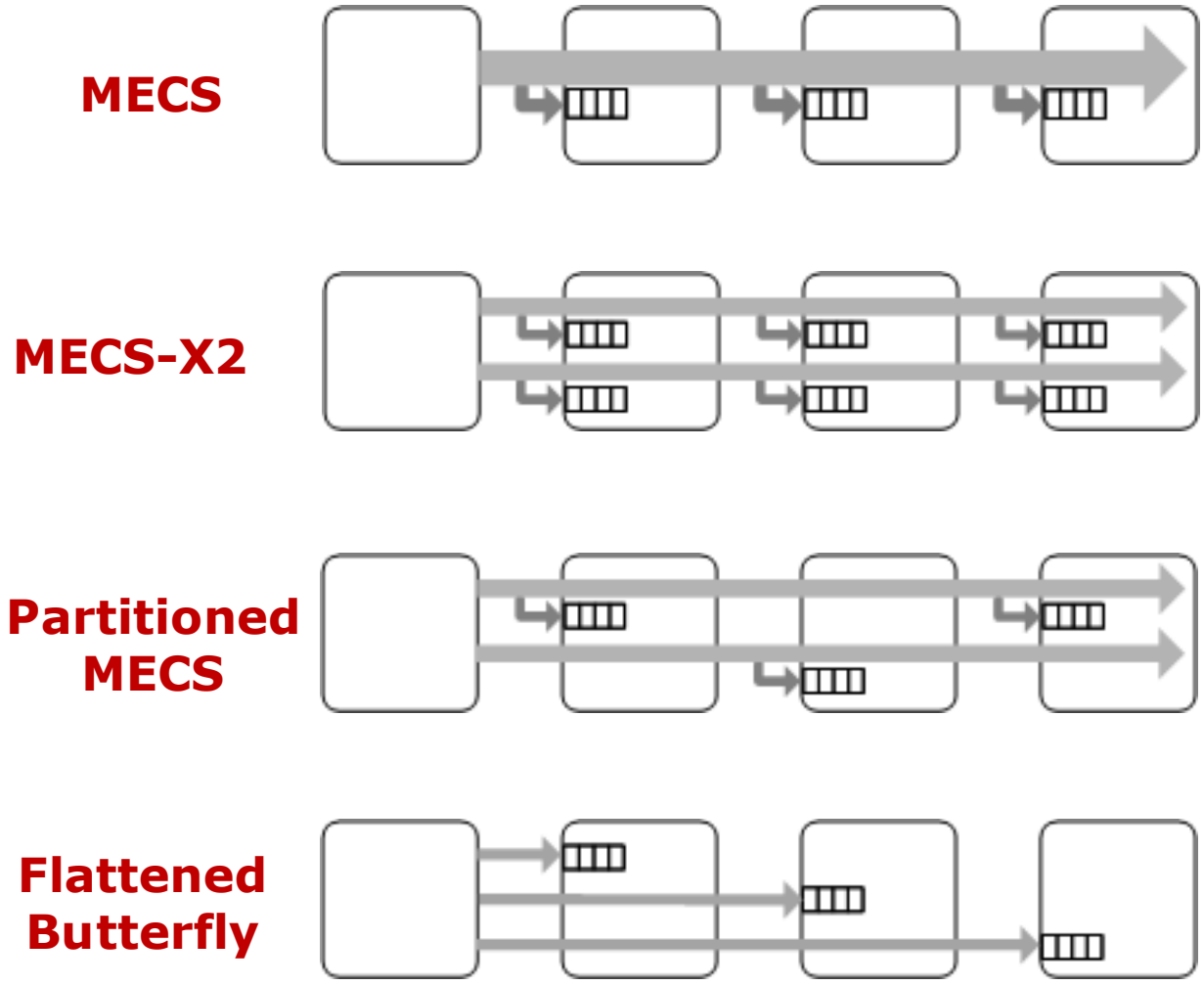


Multidrop Express Channels (MECS)



- Pros
  - One-to-many topology
  - Low diameter: 2 hops
  - *k* channels row/column
  - Asymmetric
- Cons
  - Asymmetric
  - Increased control (arbitration) complexity

Partitioning: a GEC Example



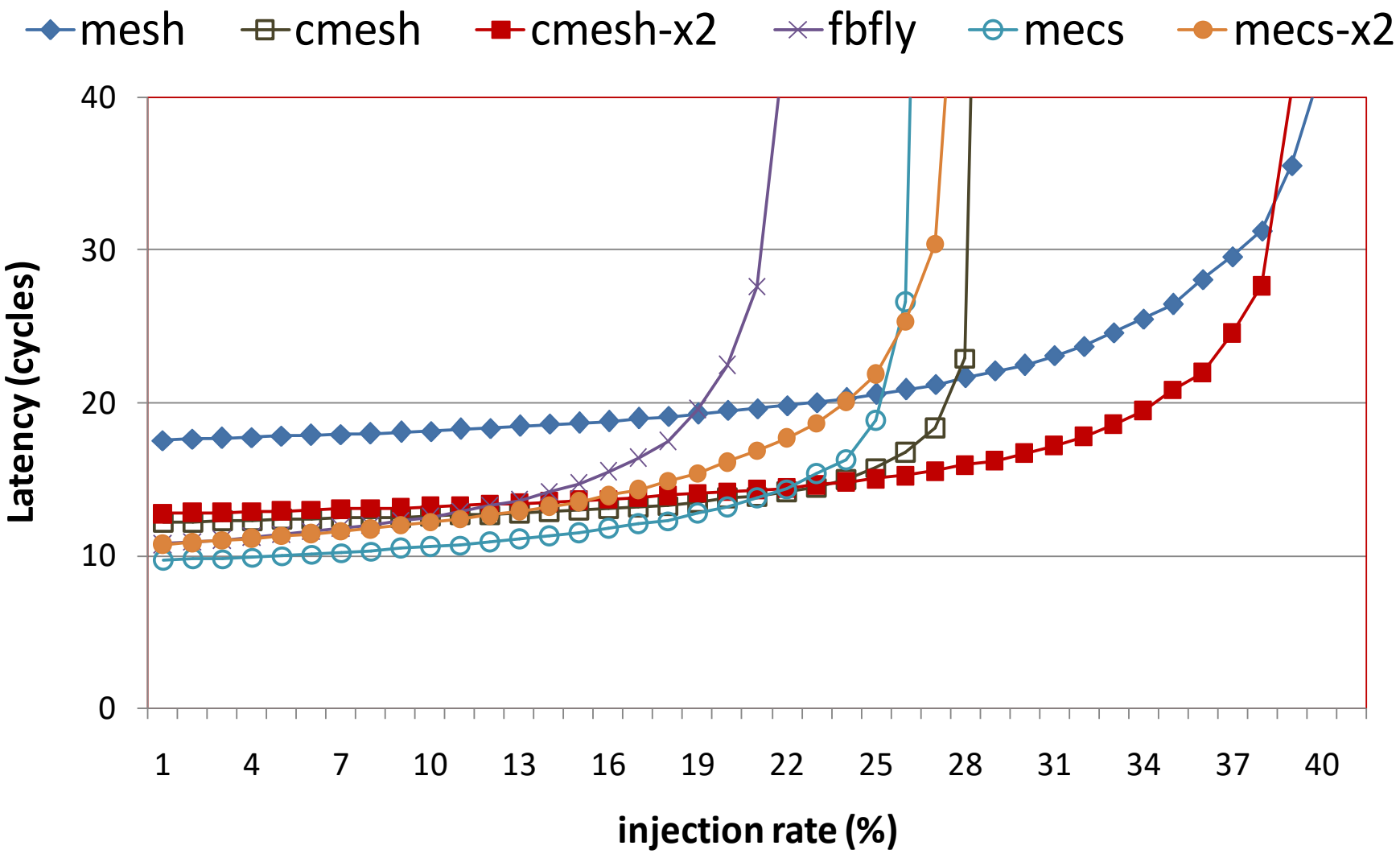
Analytical Comparison

	CMesh		FBfly		MECS	
Network Size	**64**	**256**	**64**	**256**	**64**	**256**
Radix (conctr'd)	4	8	4	8	4	8
Diameter	6	14	2	2	2	2
Channel count	2	2	8	32	4	8
Channel width	576	1152	144	72	288	288
Router inputs	4	4	6	14	6	14
Router outputs	4	4	6	14	4	4

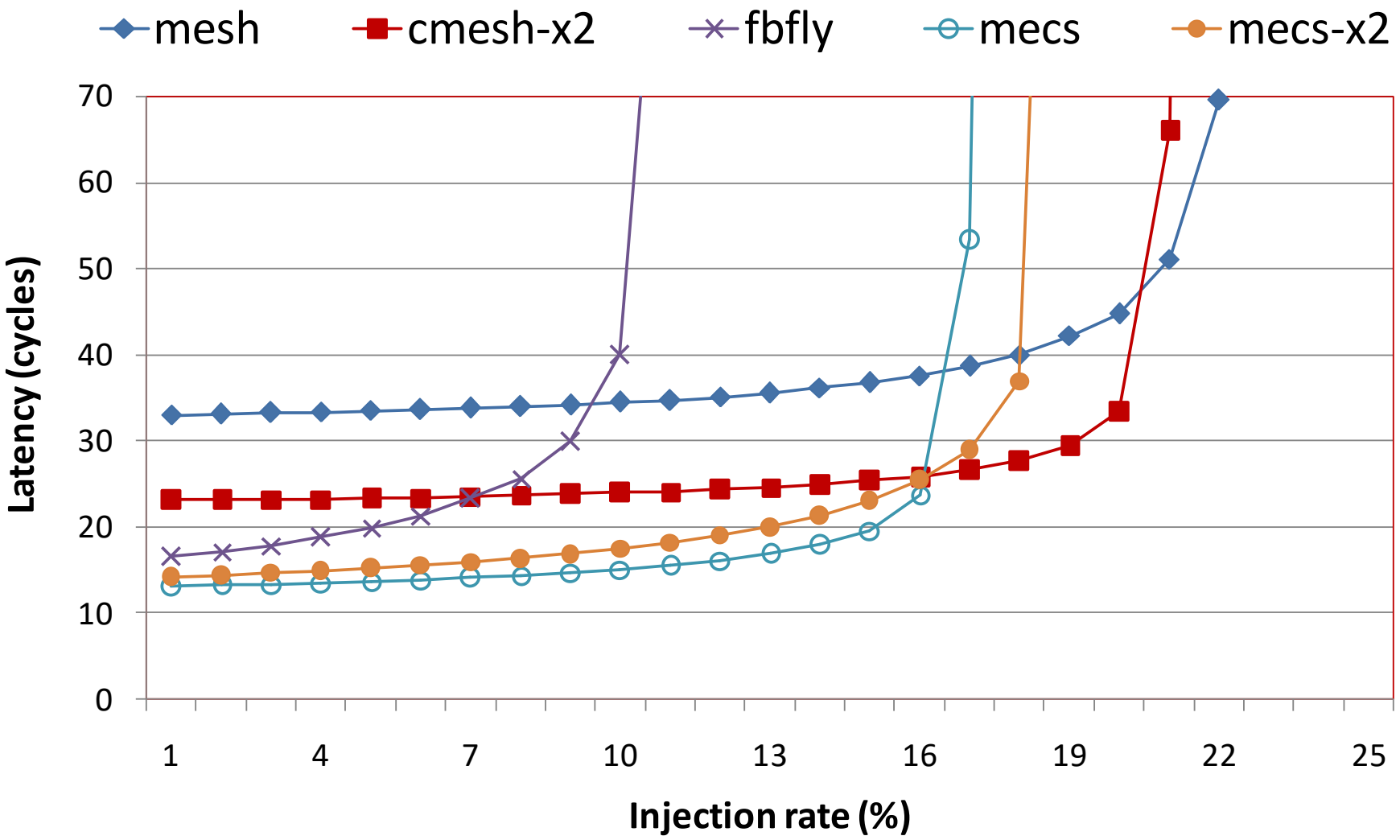
Experimental Methodology

Topologies	Mesh, CMesh, CMesh-X2, FBFly, MECS, MECS-X2
Network sizes	64 & 256 terminals
Routing	DOR, adaptive
Messages	64 & 576 bits
Synthetic traffic	Uniform random, bit complement, transpose, self-similar
PARSEC benchmarks	Blackscholes, Bodytrack, Canneal, Ferret, Fluidanimate, Freqmine, Vip, x264
Full-system config	M5 simulator, Alpha ISA, 64 OOO cores
Energy evaluation	Orion + CACTI 6

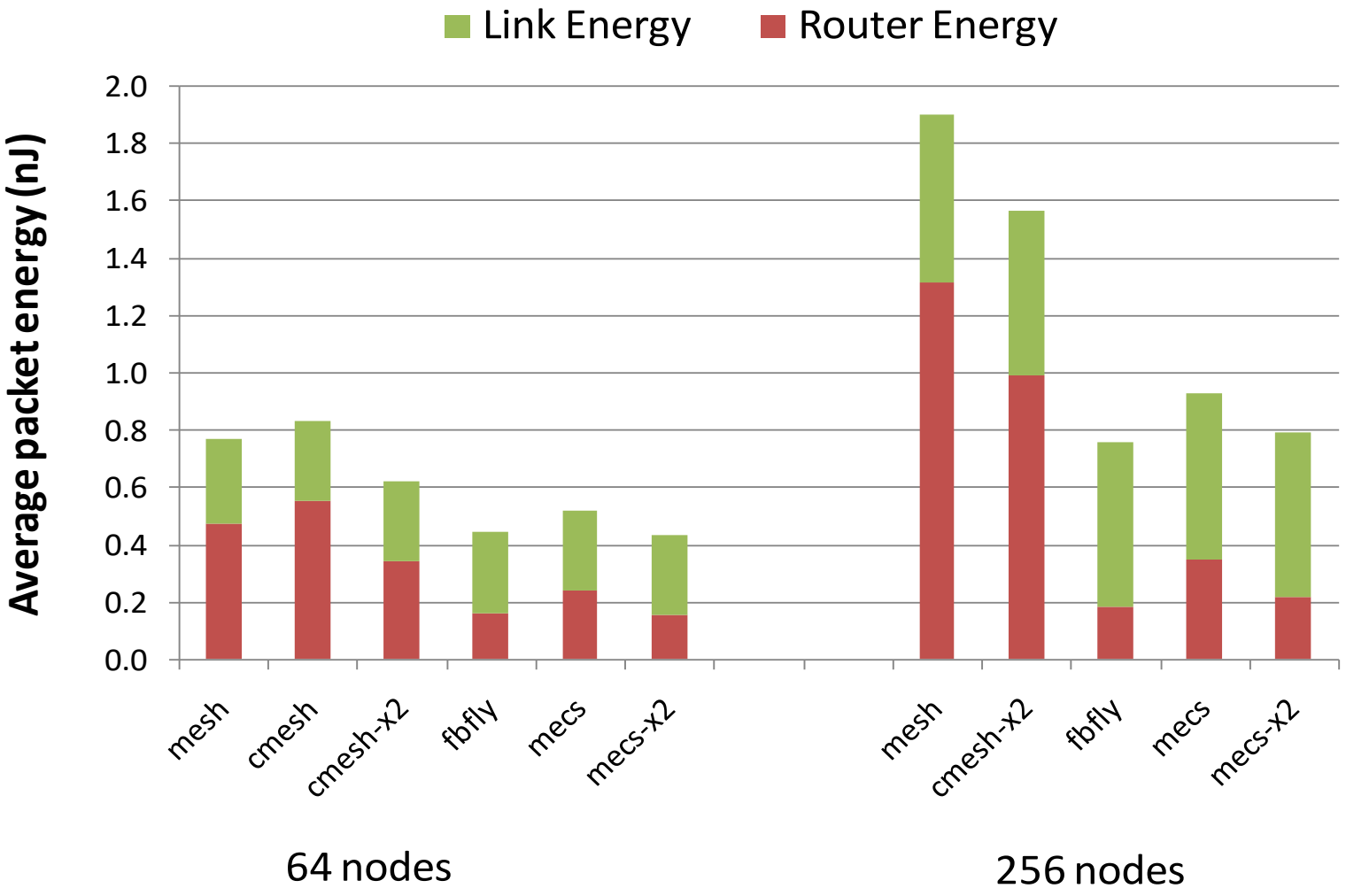
64 nodes: Uniform Random



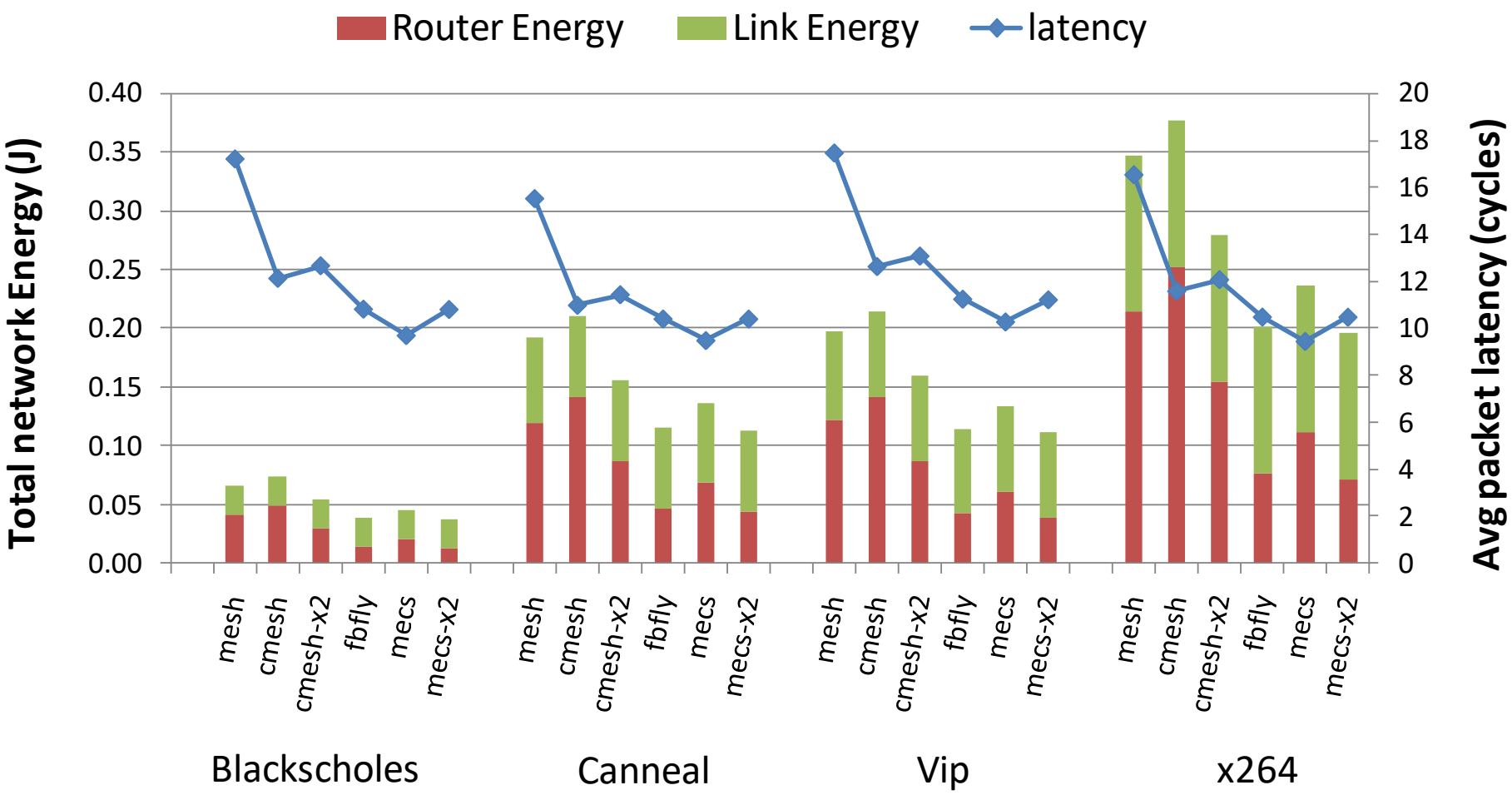
256 nodes: Uniform Random



Energy (100K pkts, Uniform Random)



64 Nodes: PARSEC



Summary

- MECS
  - A new one-to-many topology
  - Good fit for planar substrates
  - Excellent connectivity
  - Effective wire utilization
- Generalized Express Cubes
  - Framework & taxonomy for NOC topologies
  - Extension of the k-ary n-cube model
  - Useful for understanding and exploring on-chip interconnect options
  - Future: expand & formalize

Scalability: Express Cube Topologies

- Boris Grot, Joel Hestness, Stephen W. Keckler, and Onur Mutlu, **"Express Cube Topologies for On-Chip Interconnects"** *Proceedings of the 15th International Symposium on High-Performance Computer Architecture (**HPCA**)*, pages 163-174, Raleigh, NC, February 2009. Slides (ppt)

Express Cube Topologies for On-Chip Interconnects

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Interconnect Readings

Application-Aware Prioritization in NoCs

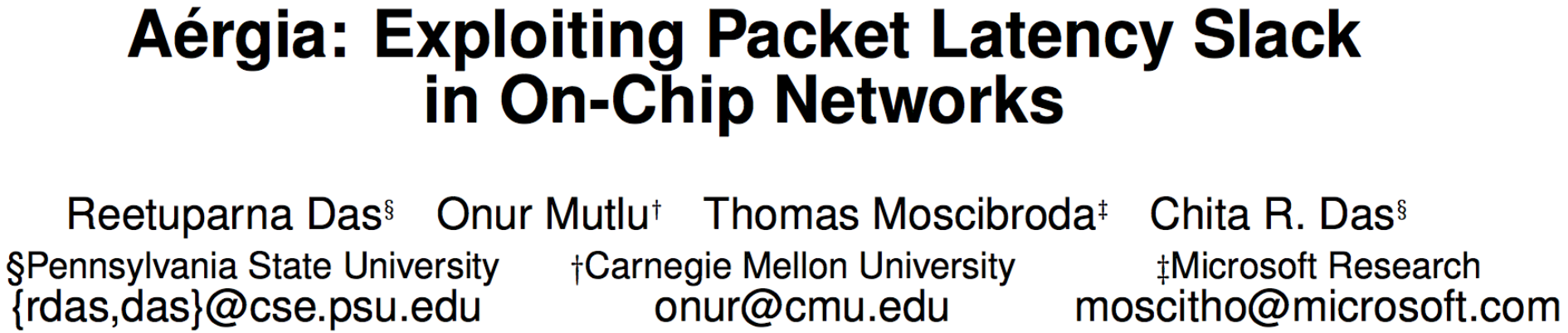
- Reetuparna Das, Onur Mutlu, Thomas Moscibroda, and Chita R. Das, **"Application-Aware Prioritization Mechanisms for On-Chip Networks"**
  *Proceedings of the 42nd International Symposium on Microarchitecture (**MICRO**)*, pages 280-291, New York, NY, December 2009. Slides (pptx)

Application-Aware Prioritization Mechanisms for On-Chip Networks

Reetuparna Das[§] Onur Mutlu[†] Thomas Moscibroda[‡] Chita R. Das[§]
§Pennsylvania State University †Carnegie Mellon University ‡Microsoft Research
{rdas,das}@cse.psu.edu onur@cmu.edu moscitho@microsoft.com

Slack-Based Packet Scheduling

- Reetuparna Das, Onur Mutlu, Thomas Moscibroda, and Chita R. Das, **"Aergia: Exploiting Packet Latency Slack in On-Chip Networks"** *Proceedings of the 37th International Symposium on Computer Architecture (**ISCA**)*, pages 106-116, Saint-Malo, France, June 2010. Slides (pptx)
  One of the 11 computer architecture papers of 2010 selected as Top Picks by IEEE Micro.



Aérgia: Exploiting Packet Latency Slack in On-Chip Networks

Reetuparna Das§ Onur Mutlu† Thomas Moscibroda‡ Chita R. Das§
§Pennsylvania State University {rdas,das}@cse.psu.edu
†Carnegie Mellon University onur@cmu.edu
‡Microsoft Research moscitho@microsoft.com

Low-Cost QoS in On-Chip Networks (I)

- Boris Grot, Stephen W. Keckler, and Onur Mutlu, **"Preemptive Virtual Clock: A Flexible, Efficient, and Cost-effective QOS Scheme for Networks-on-Chip"** *Proceedings of the 42nd International Symposium on Microarchitecture (**MICRO**)*, pages 268-279, New York, NY, December 2009. Slides (pdf)

Preemptive Virtual Clock: A Flexible, Efficient, and Cost-effective QOS Scheme for Networks-on-Chip

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†Computer Architecture Laboratory (CALCM)
Carnegie Mellon University
onur@cmu.edu

Low-Cost QoS in On-Chip Networks (II)

- Boris Grot, Joel Hestness, Stephen W. Keckler, and Onur Mutlu, **"Kilo-NOC: A Heterogeneous Network-on-Chip Architecture for Scalability and Service Guarantees"**
  *Proceedings of the 38th International Symposium on Computer Architecture (**ISCA**)*, San Jose, CA, June 2011. Slides (pptx)
  One of the 12 computer architecture papers of 2011 selected as Top Picks by IEEE Micro.

Kilo-NOC: A Heterogeneous Network-on-Chip Architecture for Scalability and Service Guarantees

Boris Grot[1] bgrot@cs.utexas.edu
Joel Hestness[1] hestness@cs.utexas.edu
Stephen W. Keckler[1,2] skeckler@nvidia.com
Onur Mutlu[3] onur@cmu.edu

[1]The University of Texas at Austin, Austin, TX
[2]NVIDIA, Santa Clara, CA
[3]Carnegie Mellon University, Pittsburgh, PA

Low-Cost QoS in On-Chip Networks (III)

- Boris Grot, Joel Hestness, Stephen W. Keckler, and Onur Mutlu, **"A QoS-Enabled On-Die Interconnect Fabric for Kilo-Node Chips"** *IEEE Micro, Special Issue: Micro's Top Picks from 2011 Computer Architecture Conferences (**MICRO TOP PICKS**)*, Vol. 32, No. 3, May/June 2012.



A QoS-Enabled On-Die Interconnect Fabric for Kilo-Node Chips

To meet rapidly growing performance demands and energy constraints, future chips will likely feature thousands of on-die resources. Existing network-on-chip solutions weren't designed for scalability and will be unable to meet future interconnect demands. A hybrid network-on-chip architecture called Kilo-NoC co-optimizes topology, flow control, and quality of service to achieve significant gains in efficiency.

Throttling Based Fairness in NoCs

- Kevin Chang, Rachata Ausavarungnirun, Chris Fallin, and Onur Mutlu, **"HAT: Heterogeneous Adaptive Throttling for On-Chip Networks"** *Proceedings of the 24th International Symposium on Computer Architecture and High Performance Computing* (***SBAC-PAD***), New York, NY, October 2012. Slides (pptx) (pdf)

HAT: Heterogeneous Adaptive Throttling for On-Chip Networks

Kevin Kai-Wei Chang, Rachata Ausavarungnirun, Chris Fallin, Onur Mutlu
Carnegie Mellon University
{kevincha,rachata,cfallin,onur}@cmu.edu

Scalability: Express Cube Topologies

- Boris Grot, Joel Hestness, Stephen W. Keckler, and Onur Mutlu, **"Express Cube Topologies for On-Chip Interconnects"** *Proceedings of the 15th International Symposium on High-Performance Computer Architecture* (***HPCA***), pages 163-174, Raleigh, NC, February 2009. Slides (ppt)

Express Cube Topologies for On-Chip Interconnects

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onur@cmu.edu

Scalability: Slim NoC

- Maciej Besta, Syed Minhaj Hassan, Sudhakar Yalamanchili, Rachata Ausavarungnirun, Onur Mutlu, Torsten Hoefler, **"Slim NoC: A Low-Diameter On-Chip Network Topology for High Energy Efficiency and Scalability"**
  *Proceedings of the 23rd International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**)*, Williamsburg, VA, USA, March 2018.
  [Slides (pptx) (pdf)] [Lightning Session Slides (pptx) (pdf)]
  [Poster (pdf)]

Slim NoC: A Low-Diameter On-Chip Network Topology for High Energy Efficiency and Scalability

Maciej Besta[1] Syed Minhaj Hassan[2] Sudhakar Yalamanchili[2]
Rachata Ausavarungnirun[3] Onur Mutlu[1,3] Torsten Hoefler[1]

[1]ETH Zürich [2]Georgia Institute of Technology [3]Carnegie Mellon University

Bufferless Deflection Routing in NoCs

- Thomas Moscibroda and Onur Mutlu,
  "A Case for Bufferless Routing in On-Chip Networks"
  *Proceedings of the 36th International Symposium on Computer Architecture (**ISCA**)*, pages 196-207, Austin, TX, June 2009. Slides (pptx)

A Case for Bufferless Routing in On-Chip Networks

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onur@cmu.edu

Minimally-Buffered Deflection Routing

- Chris Fallin, Greg Nazario, Xiangyao Yu, Kevin Chang, Rachata Ausavarungnirun, and Onur Mutlu,
  "MinBD: Minimally-Buffered Deflection Routing for Energy-Efficient Interconnect"
  *Proceedings of the 6th ACM/IEEE International Symposium on Networks on Chip (**NOCS**)*, Lyngby, Denmark, May 2012. Slides (pptx) (pdf)
  One of the five papers nominated for the Best Paper Award by the Program Committee.

MinBD: Minimally-Buffered Deflection Routing for Energy-Efficient Interconnect

Chris Fallin, Greg Nazario, Xiangyao Yu†, Kevin Chang, Rachata Ausavarungnirun, Onur Mutlu

Carnegie Mellon University
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yxythu@gmail.com

"Bufferless" Hierarchical Rings

- Rachata Ausavarungnirun, Chris Fallin, Xiangyao Yu, Kevin Chang, Greg Nazario, Reetuparna Das, Gabriel Loh, and Onur Mutlu,
  "Design and Evaluation of Hierarchical Rings with Deflection Routing"
  *Proceedings of the 26th International Symposium on Computer Architecture and High Performance Computing (**SBAC-PAD**)*, Paris, France, October 2014. [Slides (pptx) (pdf)] [Source Code]

- Describes the design and implementation of a mostly-bufferless hierarchical ring

Design and Evaluation of Hierarchical Rings with Deflection Routing

Rachata Ausavarungnirun Chris Fallin Xiangyao Yu† Kevin Kai-Wei Chang
Greg Nazario Reetuparna Das§ Gabriel H. Loh‡ Onur Mutlu

Carnegie Mellon University §University of Michigan †MIT ‡Advanced Micro Devices, Inc.

"Bufferless" Hierarchical Rings (II)

- Rachata Ausavarungnirun, Chris Fallin, Xiangyao Yu, Kevin Chang, Greg Nazario, Reetuparna Das, Gabriel Loh, and Onur Mutlu, **"A Case for Hierarchical Rings with Deflection Routing: An Energy-Efficient On-Chip Communication Substrate"** ***Parallel Computing*** *(**PARCO**)*, 2016.
  - arXiv.org version, February 2016.

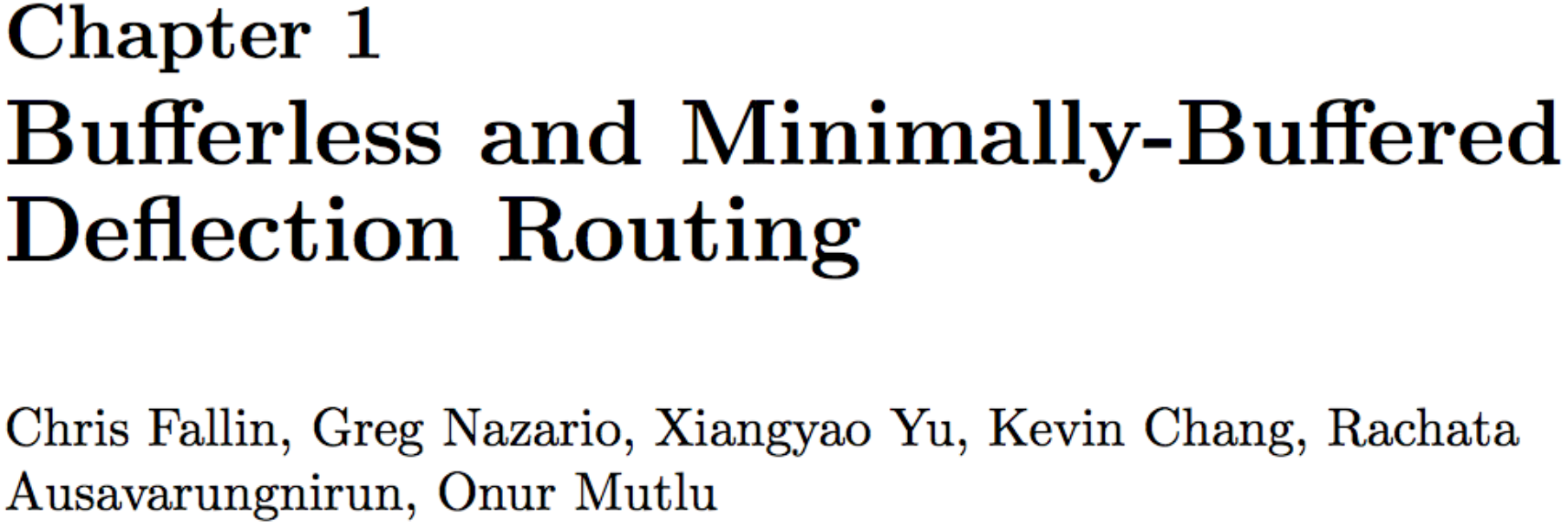
Achieving both High Energy Efficiency
and High Performance in On-Chip Communication
using Hierarchical Rings with Deflection Routing

Rachata Ausavarungnirun Chris Fallin Xiangyao Yu† Kevin Kai-Wei Chang
Greg Nazario Reetuparna Das§ Gabriel H. Loh‡ Onur Mutlu

Carnegie Mellon University §University of Michigan †MIT ‡AMD

A Review of Bufferless Interconnects

- Chris Fallin, Greg Nazario, Xiangyao Yu, Kevin Chang, Rachata Ausavarungnirun, and Onur Mutlu,
  "Bufferless and Minimally-Buffered Deflection Routing"
  Invited Book Chapter in Routing Algorithms in Networks-on-Chip, pp. 241-275, Springer, 2014.


Chapter 1

Bufferless and Minimally-Buffered Deflection Routing

Chris Fallin, Greg Nazario, Xiangyao Yu, Kevin Chang, Rachata Ausavarungnirun, Onur Mutlu

Summary of Eight Years of Research

Energy-Efficient Deflection-based On-chip Networks: Topology, Routing, Flow Control

Rachata Ausavarungnirun[b], Onur Mutlu[a]

SAFARI Research Group

[a]*ETH Zürich*
[b]*King Mongkut's University of Technology North Bangkok*

Abstract

As the number of cores scales to tens and hundreds, the energy consumption of routers across various types of on-chip networks in chip muiltiprocessors (CMPs) increases significantly. A major source of this energy consumption comes from the input buffers inside Network-on-Chip (NoC) routers, which are traditionally designed to maximize performance. To mitigate this high energy cost, many works propose bufferless router designs that utilize deflection routing to resolve port contention. While this approach is able to maintain high performance relative to its buffered counterparts at low network traffic, the bufferless router design suffers performance degradation under high network load.

In order to maintain high performance and energy efficiency under *both* low and high network loads, this chapter discusses critical drawbacks of traditional bufferless designs and describes recent research works focusing on two major modifications to improve the overall performance of the traditional bufferless network-on-chip design. The first modification is a minimally-buffered design that introduces limited buffering inside critical parts of the on-chip network in order to reduce the number of deflections. The second modification is a hierarchical bufferless interconnect design that aims to further improve performance by limiting the number of hops each packet needs to travel while in the network. In both approaches, we discuss design tradeoffs and provide evaluation results based on common CMP configurations with various network topologies to show the effectiveness of each proposal.

Keywords: network-on-chip, deflection routing, topology, bufferless router, energy efficiency, high-performance computing, computer architecture, emerging technologies, latency, low-latency computing

https://arxiv.org/pdf/2112.02516.pdf

On-Chip vs. Off-Chip Congestion Control

- George Nychis, Chris Fallin, Thomas Moscibroda, Onur Mutlu, and Srinivasan Seshan,
  "On-Chip Networks from a Networking Perspective: Congestion and Scalability in Many-core Interconnects"
  *Proceedings of the 2012 ACM SIGCOMM Conference (**SIGCOMM**)*, Helsinki, Finland, August 2012. Slides (pptx)

On-Chip Networks from a Networking Perspective: Congestion and Scalability in Many-Core Interconnects

George Nychis†, Chris Fallin†, Thomas Moscibroda§, Onur Mutlu†, Srinivasan Seshan†

† Carnegie Mellon University
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§ Microsoft Research Asia
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On-Chip vs. Off-Chip Congestion Control

- George Nychis, Chris Fallin, Thomas Moscibroda, and Onur Mutlu, **"Next Generation On-Chip Networks: What Kind of Congestion Control Do We Need?"** *Proceedings of the 9th ACM Workshop on Hot Topics in Networks (**HOTNETS**)*, Monterey, CA, October 2010. Slides (ppt) (key)

Next Generation On-Chip Networks: What Kind of Congestion Control Do We Need?

George Nychis†, Chris Fallin†, Thomas Moscibroda§, Onur Mutlu†

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Summary of Study [SIGCOMM 2012]

- Highlighted a traditional networking problem in a new context
  - Unique design requires novel solution
- Showed congestion limits efficiency and scalability, and that self-throttling nature of cores prevents congestion collapse
- Showed on-chip congestion control requires application-awareness
- Our application-aware congestion controller provided:
  - A more efficient network-layer (reduced latency)
  - Improvements in system throughput (by 27%)
  - Effectively scale the CMP (shown for up to 4096 cores)

Slowdown Estimation in NoCs

- Xiyue Xiang, Saugata Ghose, Onur Mutlu, and Nian-Feng Tzeng, **"A Model for Application Slowdown Estimation in On-Chip Networks and Its Use for Improving System Fairness and Performance"**
  *Proceedings of the 34th IEEE International Conference on Computer Design (**ICCD**)*, Phoenix, AZ, USA, October 2016.
  [Slides (pptx) (pdf)]

A Model for Application Slowdown Estimation in On-Chip Networks and Its Use for Improving System Fairness and Performance

Xiyue Xiang[†] Saugata Ghose[‡] Onur Mutlu[§‡] Nian-Feng Tzeng[†]
[†]*University of Louisiana at Lafayette* [‡]*Carnegie Mellon University* [§]*ETH Zürich*

Handling Multicast and Hotspot Issues

- Xiyue Xiang, Wentao Shi, Saugata Ghose, Lu Peng, Onur Mutlu, and Nian-Feng Tzeng,
  "Carpool: A Bufferless On-Chip Network Supporting Adaptive Multicast and Hotspot Alleviation"
  *Proceedings of the International Conference on Supercomputing (**ICS**)*, Chicago, IL, USA, June 2017.
  [Slides (pptx) (pdf)]

Carpool: A Bufferless On-Chip Network Supporting Adaptive Multicast and Hotspot Alleviation

Xiyue Xiang[†] Wentao Shi[⋆] Saugata Ghose[‡] Lu Peng[⋆] Onur Mutlu[§‡] Nian-Feng Tzeng[†]
[†]University of Louisiana at Lafayette [⋆]Louisiana State University [‡]Carnegie Mellon University [§]ETH Zürich

Heterogeneous Networks

- Asit K. Mishra, Onur Mutlu, and Chita R. Das,
  "A Heterogeneous Multiple Network-on-Chip Design: An Application-Aware Approach"
  *Proceedings of the 50th Design Automation Conference (**DAC**)*,
  Austin, TX, June 2013. Slides (pptx) Slides (pdf)

A Heterogeneous Multiple Network-On-Chip Design: An Application-Aware Approach

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Pittsburgh, PA 15213, USA
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Chita R. Das
The Pennsylvania State University
University Park, PA 16802, USA
das@cse.psu.edu

More Readings

- Studies of congestion and congestion control in on-chip vs. internet-like networks

- George Nychis, Chris Fallin, Thomas Moscibroda, Onur Mutlu, and Srinivasan Seshan,
"On-Chip Networks from a Networking Perspective: Congestion and Scalability in Many-core Interconnects"
*Proceedings of the 2012 ACM SIGCOMM Conference (**SIGCOMM**)*, Helsinki, Finland, August 2012. Slides (pptx)

- George Nychis, Chris Fallin, Thomas Moscibroda, and Onur Mutlu,
"Next Generation On-Chip Networks: What Kind of Congestion Control Do We Need?"
*Proceedings of the 9th ACM Workshop on Hot Topics in Networks (**HOTNETS**)*, Monterey, CA, October 2010. Slides (ppt) (key)

Source Throttling in Bufferless NoCs

- Kevin Chang, Rachata Ausavarungnirun, Chris Fallin, and Onur Mutlu, **"HAT: Heterogeneous Adaptive Throttling for On-Chip Networks"** *Proceedings of the 24th International Symposium on Computer Architecture and High Performance Computing (**SBAC-PAD**)*, New York, NY, October 2012. Slides (pptx) (pdf)

HAT: Heterogeneous Adaptive Throttling for On-Chip Networks

Kevin Kai-Wei Chang, Rachata Ausavarungnirun, Chris Fallin, Onur Mutlu
Carnegie Mellon University
{kevincha,rachata,cfallin,onur}@cmu.edu

HAT: Heterogeneous Adaptive Throttling for On-Chip Networks

Kevin Chang, Rachata Ausavarungnirun, Chris Fallin, and Onur Mutlu,

"HAT: Heterogeneous Adaptive Throttling for On-Chip Networks"

*Proceedings of the 24th International Symposium on Computer Architecture and High Performance Computing (**SBAC-PAD**)*, New York, NY, October 2012. Slides (pptx) (pdf)

Carnegie Mellon University SAFARI

Executive Summary

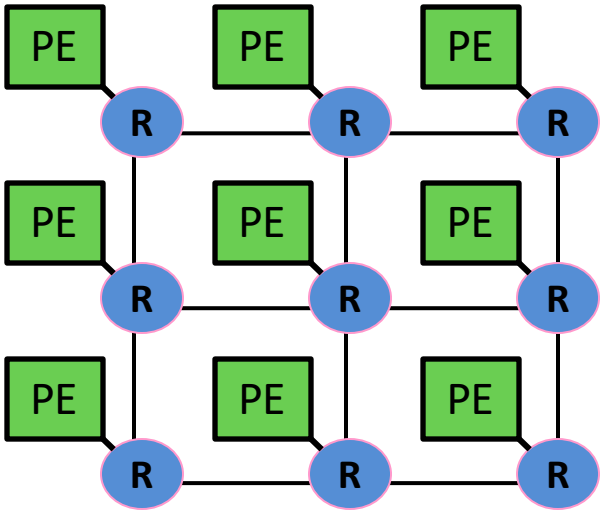
- **Problem:** Packets contend in on-chip networks (NoCs), causing congestion, thus reducing performance
- **Observations:**

  1) Some applications are more sensitive to network latency than others
  2) Applications must be throttled differently to achieve peak performance
- **Key Idea: Heterogeneous Adaptive Throttling (HAT)**
  1) Application-aware source throttling
  2) Network-load-aware throttling rate adjustment
- **Result:** Improves performance and energy efficiency over state-of-the-art source throttling policies

Outline

- **Background and Motivation**
- Mechanism
- Prior Works
- Results

On-Chip Networks

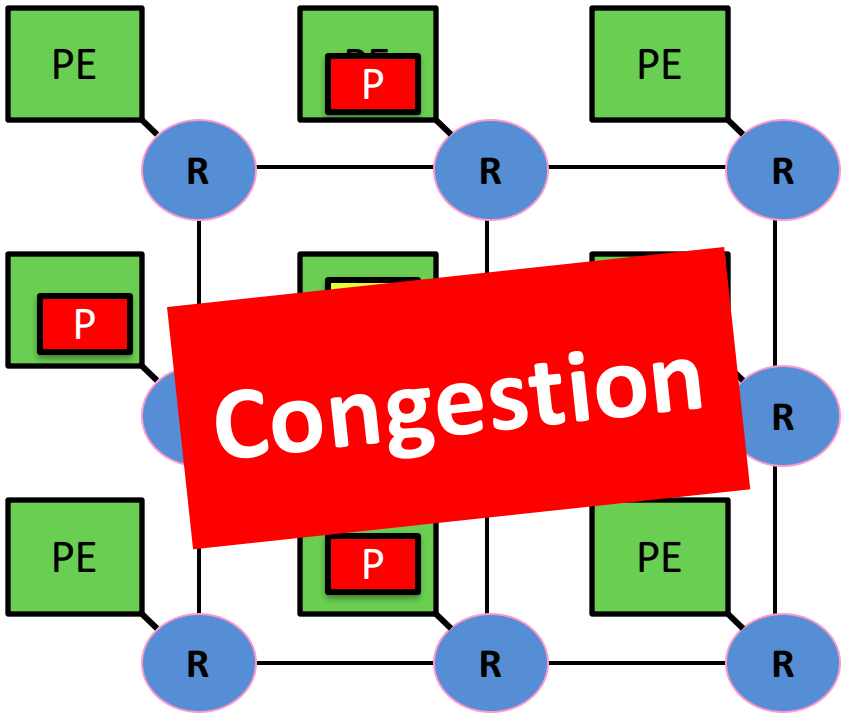


- Connect **cores, caches, memory controllers, etc**
- **Packet switched**
- **2D mesh:** Most commonly used topology
- Primarily serve **cache misses** and **memory requests**
- **Router designs**
  - Buffered: **Input buffers** to hold contending packets
  - Bufferless: **Misroute (deflect)** contending packets

R — Router

PE — Processing Element
(Cores, L2 Banks, Memory Controllers, etc)

Network Congestion Reduces Performance



R Router

P Packet

PE Processing Element
(Cores, L2 Banks, Memory Controllers, etc)

Limited shared resources (buffers and links)

- **Design constraints: power, chip area**, and **timing**

Network congestion:
↓Network throughput
↓Application performance

Goal

- **Improve performance in a highly congested NoC**

- Reducing network load decreases network congestion, hence improves performance

- **Approach: source throttling to reduce network load**
  - Temporarily delay new traffic injection

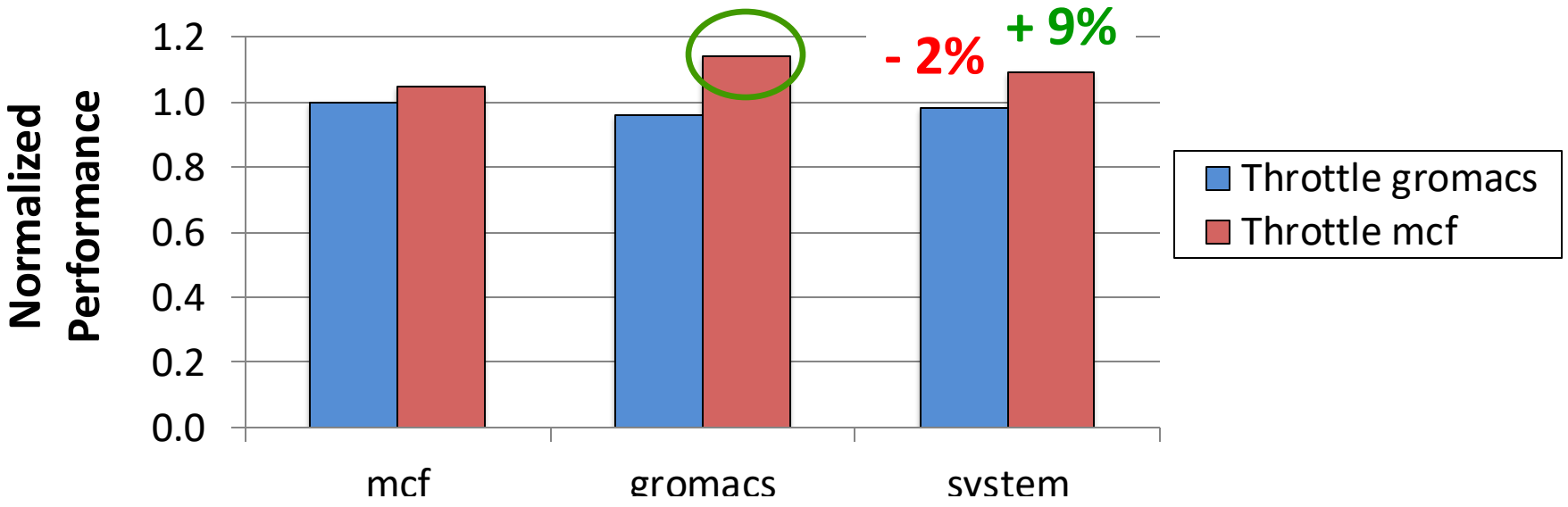
- **Naïve mechanism: throttle every single node**

Key Observation #1

Different applications respond differently to changes in **network latency**

gromacs: network-**non**-intensive

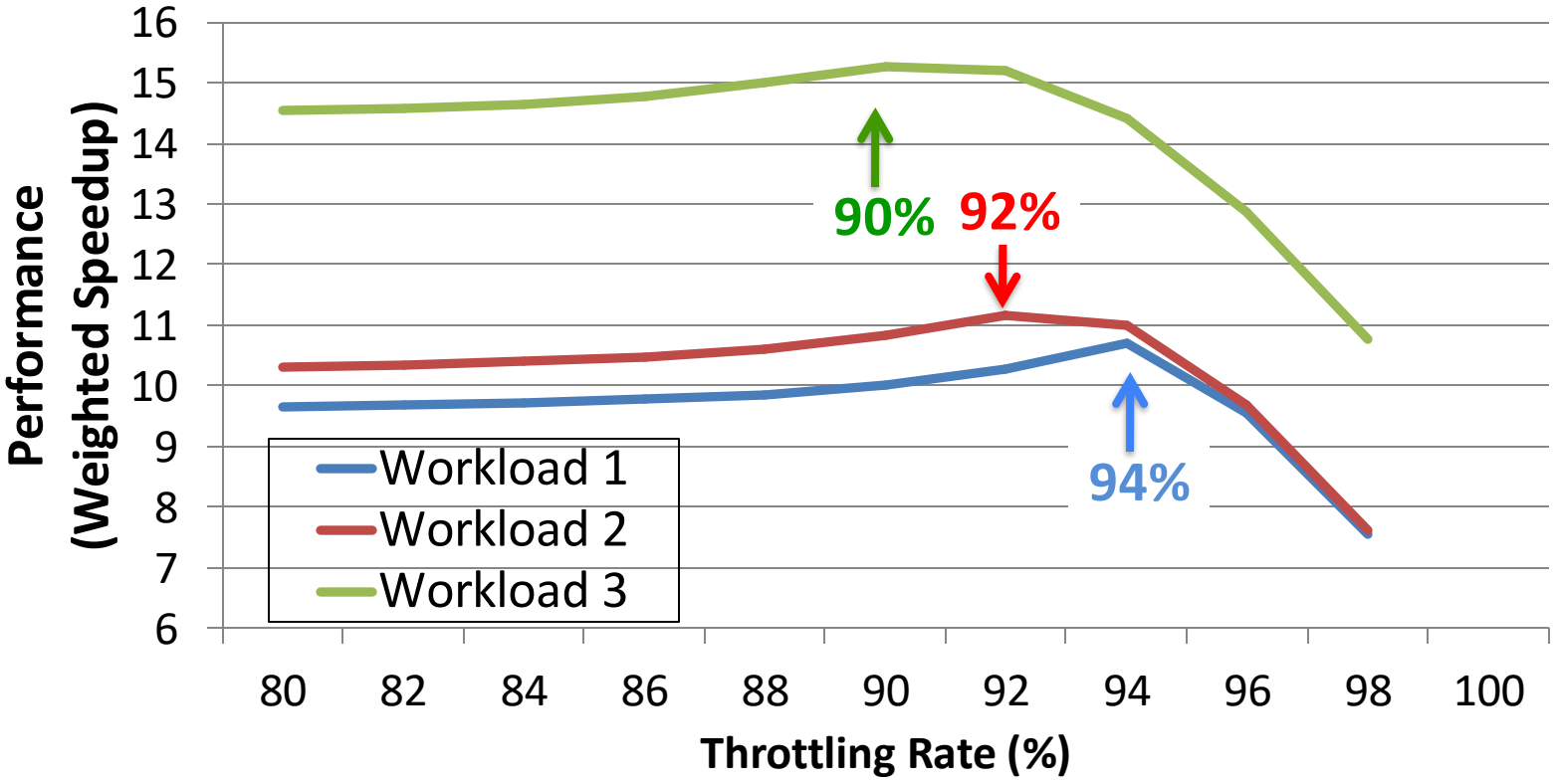
mcf: network-intensive



Throttling **network-intensive** applications benefits system performance more

Key Observation #2

Different workloads achieve peak performance at different throttling rates



Dynamically adjusting throttling rate yields better performance than a single static rate

Outline

- Background and Motivation
- **Mechanism**
- Prior Works
- Results

Heterogeneous Adaptive Throttling (HAT)

1. **Application-aware throttling:**
   Throttle **network-intensive** applications that interfere with **network-non-intensive** applications

2. **Network-load-aware throttling rate adjustment:**
   Dynamically adjusts throttling rate to adapt to different workloads

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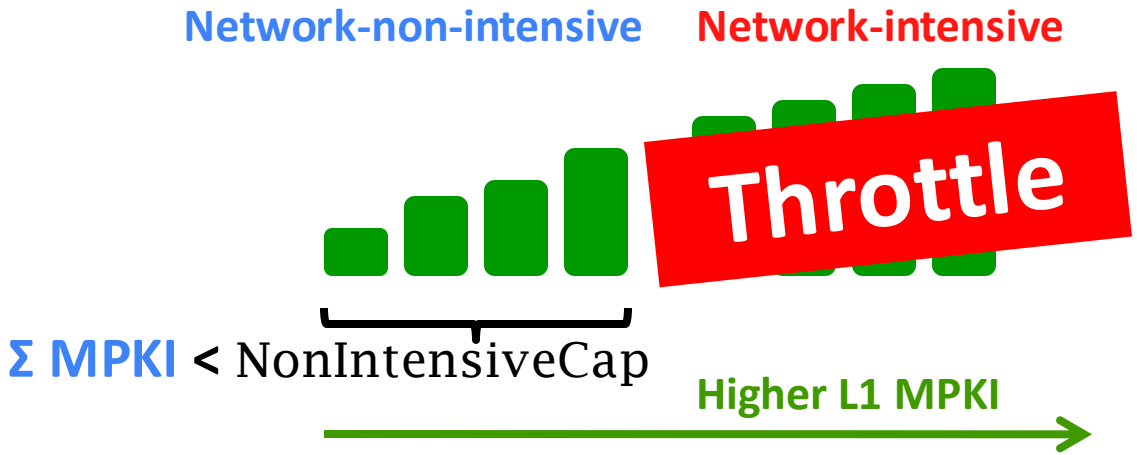
Application-Aware Throttling

1. **Measure Network Intensity**

   Use **L1 MPKI** (misses per thousand instructions) to estimate network intensity

2. **Classify Application**

   Sort applications by L1 MPKI



3. **Throttle network-intensive applications**

Heterogeneous Adaptive Throttling (HAT)

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Dynamic Throttling Rate Adjustment

- For a given **network design**, peak performance tends to occur at a **fixed network load point**

- **Dynamically** adjust throttling rate to achieve that network load point

Dynamic Throttling Rate Adjustment

- **Goal:** maintain network load at a peak performance point

1. **Measure network load**
2. **Compare and adjust throttling rate**

If **network load** > **peak point**:

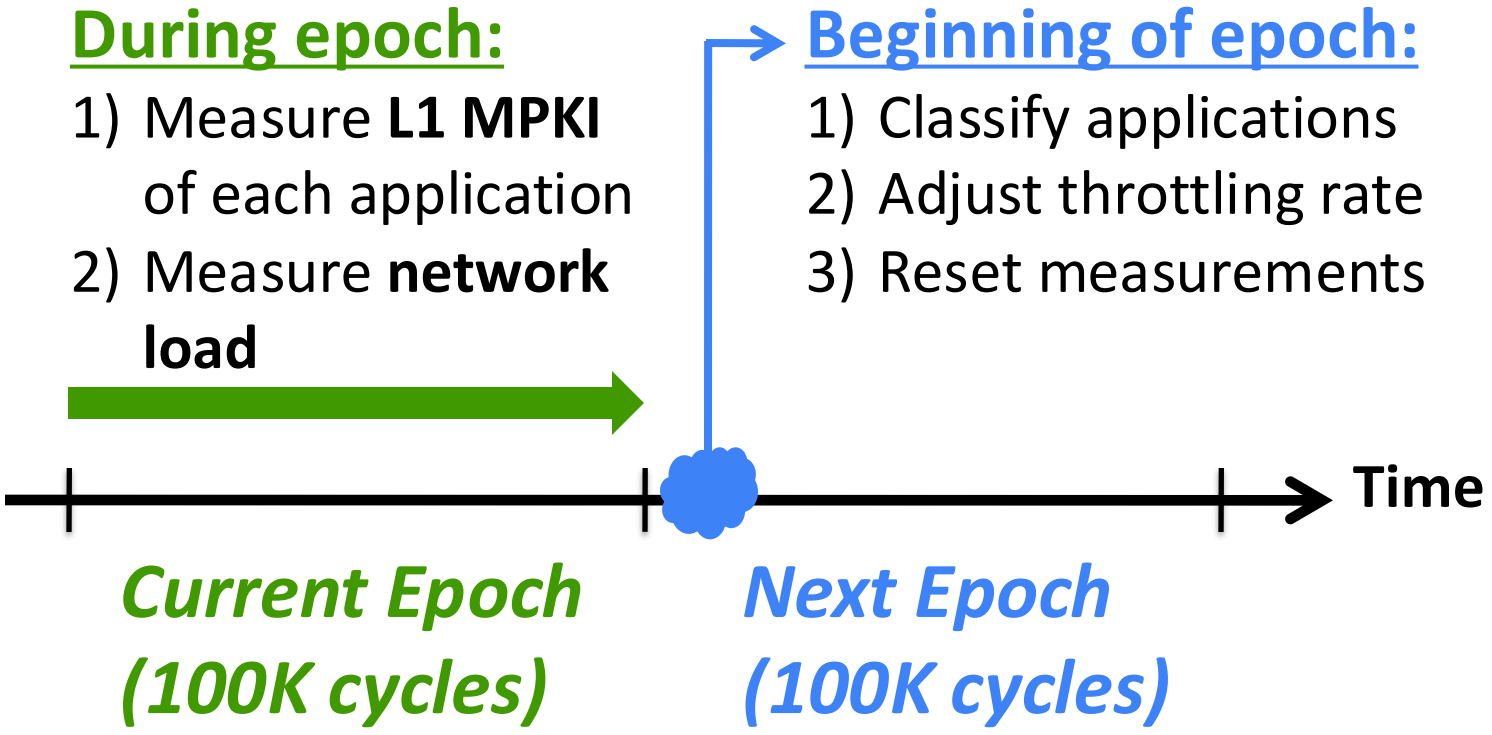
    Increase throttling rate

elif **network load** ≤ **peak point**:

    Decrease throttling rate

Epoch-Based Operation

- Continuous **HAT** operation is expensive
- **Solution:** performs **HAT** at epoch granularity



Outline

- Background and Motivation
- Mechanism
- **Prior Works**
- Results

Prior Source Throttling Works

- **Source throttling for bufferless NoCs**
  [Nychis+ Hotnets'10, SIGCOMM'12]
  - Application-aware throttling based on starvation rate
  - Does not adaptively adjust throttling rate
  - "Heterogeneous Throttling"
- **Source throttling off-chip buffered networks**
  [Thottethodi+ HPCA'01]
  - Dynamically trigger throttling based on fraction of buffer occupancy
  - Not application-aware: fully block packet injections of every node
  - "Self-tuned Throttling"

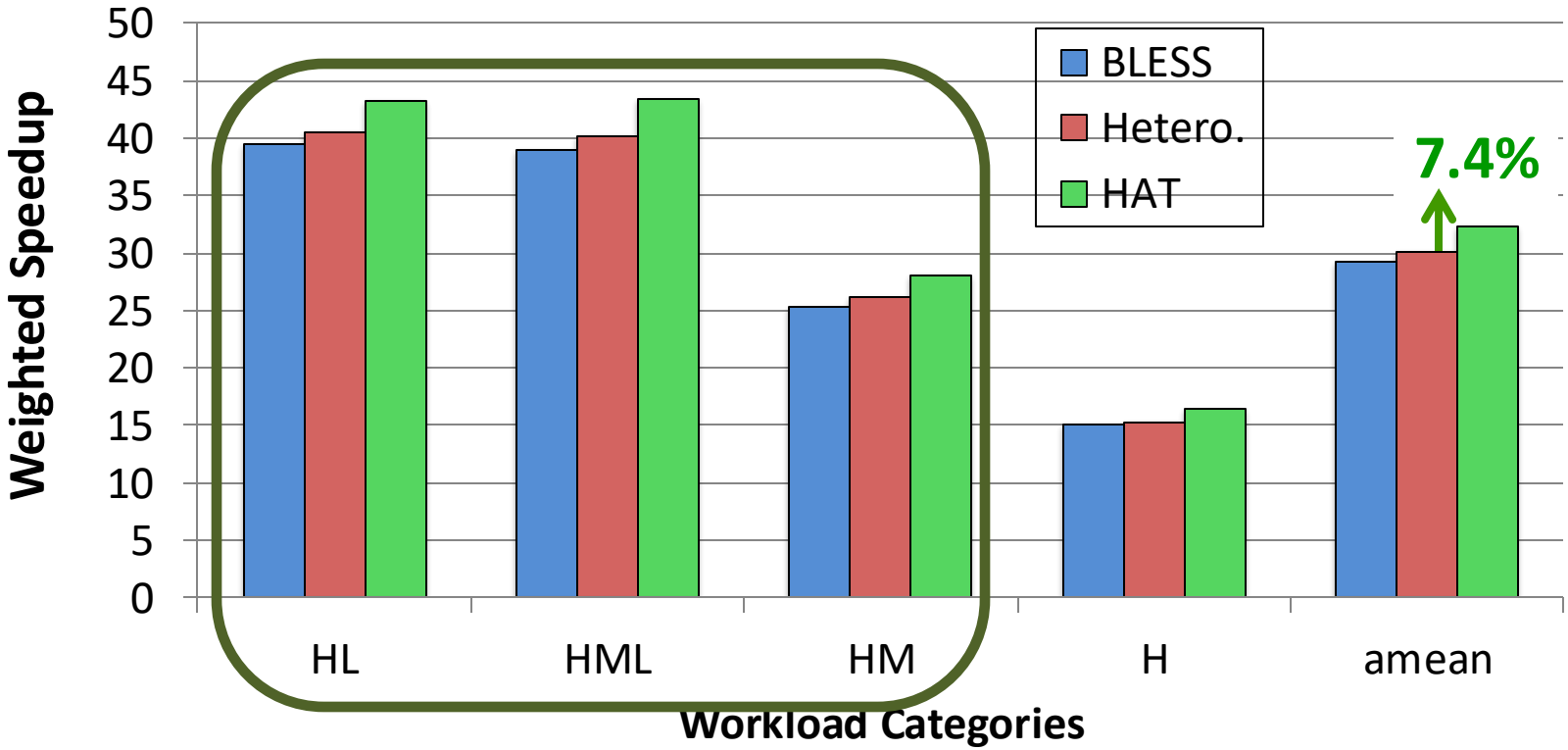
Outline

- Background and Motivation
- Mechanism
- Prior Works
- **Results**

Methodology

- **Chip Multiprocessor Simulator**
  - **64-node** multi-core systems with a **2D-mesh topology**
  - Closed-loop core/cache/NoC cycle-level model
  - 64KB L1, perfect L2 (always hits to stress NoC)
- **Router Designs**
  - **Virtual-channel buffered** router**:** 4 VCs, 4 flits/VC [Dally+ IEEE TPDS'92]
  - **Bufferless deflection** routers**: BLESS** [Moscibroda+ ISCA'09]
- **Workloads**
  - 60 multi-core workloads: SPEC CPU2006 benchmarks
  - Categorized based on their network intensity
    - **L**ow/**M**edium/**H**igh intensity categories
- **Metrics:** Weighted Speedup (perf.), perf./Watt (energy eff.), and maximum slowdown (fairness)

Performance: Bufferless NoC (BLESS)

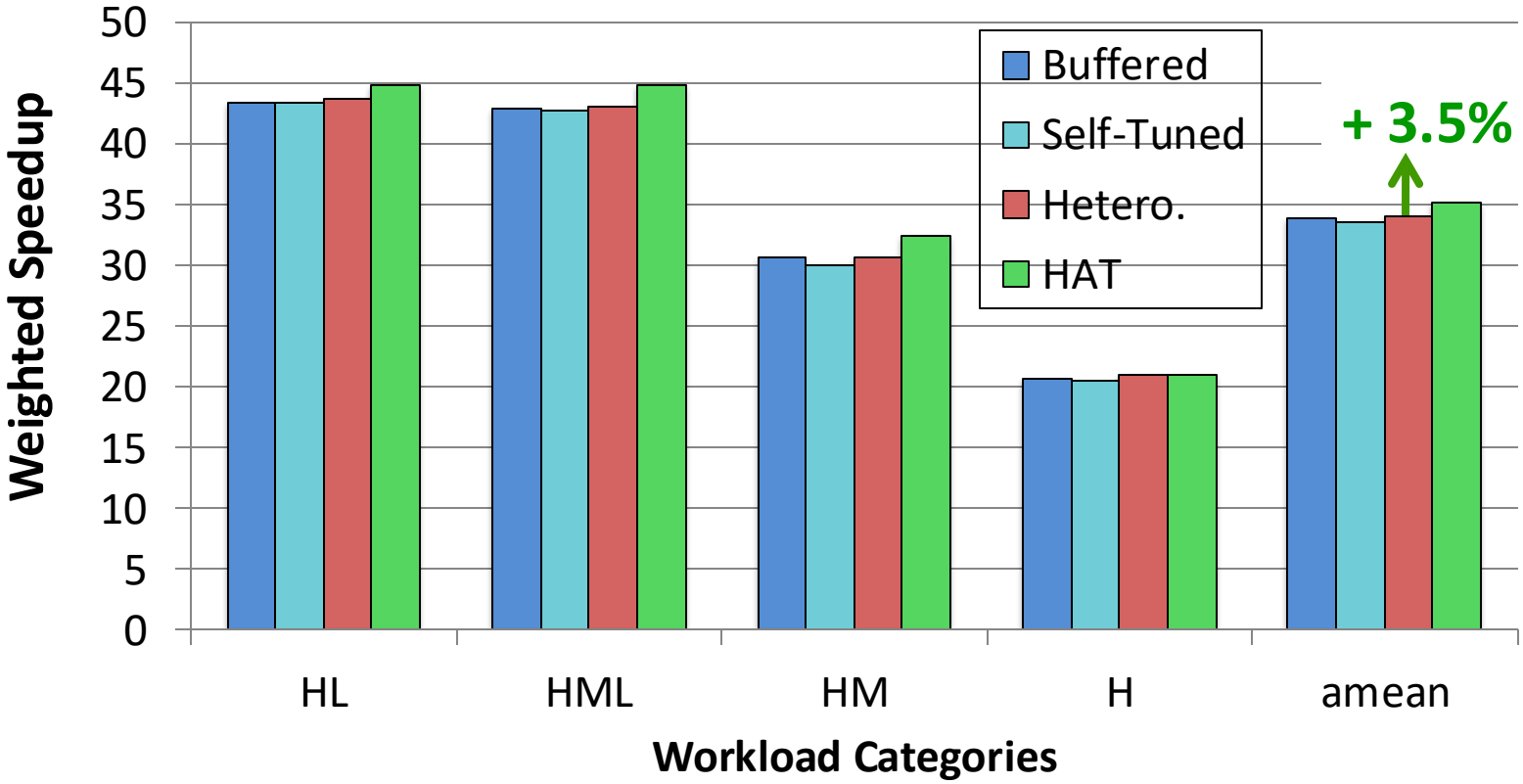


HAT provides better performance improvement than past work

Highest improvement on **heterogeneous** workload mixes

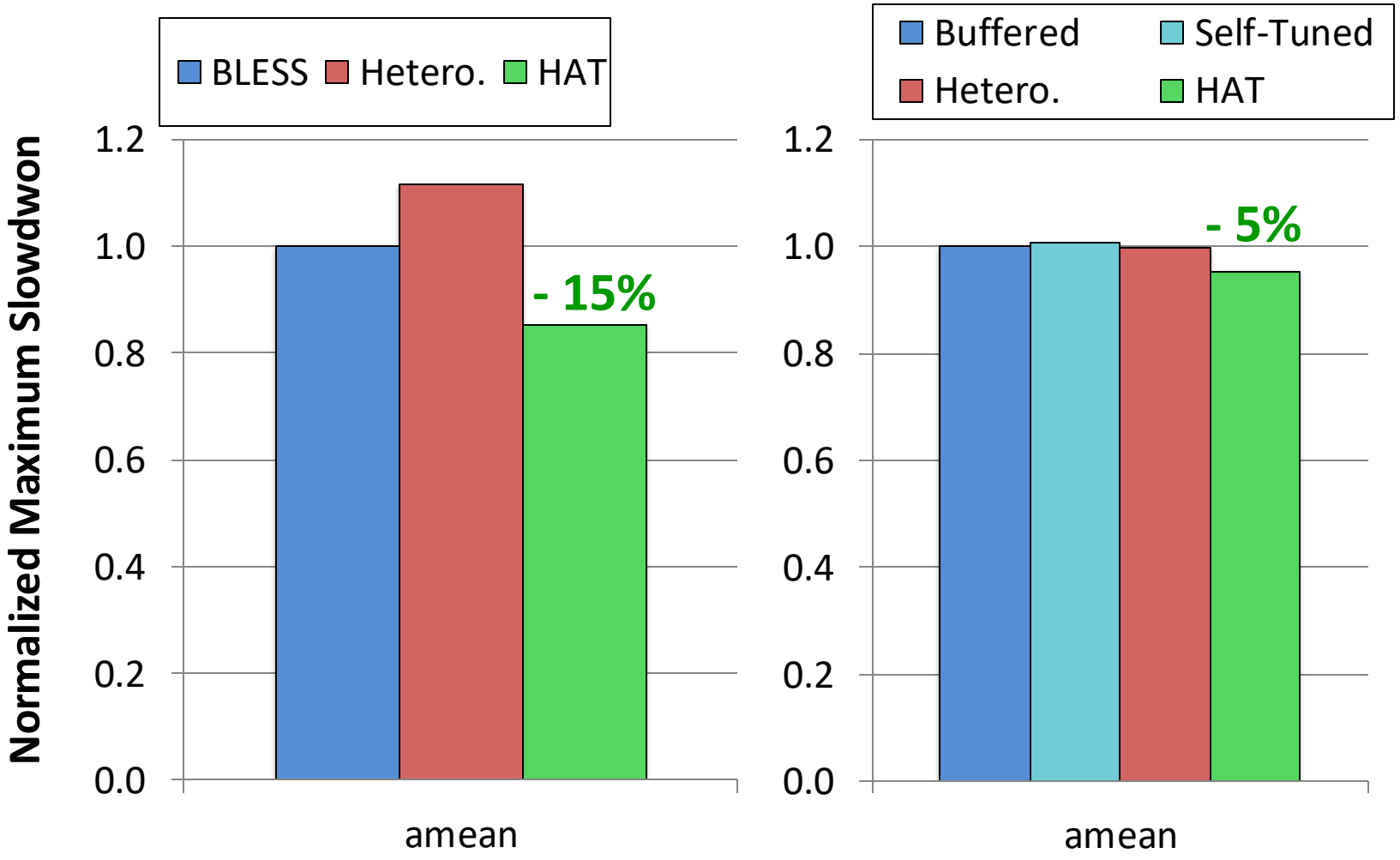
- L and **M** are more **sensitive** to network latency

Performance: Buffered NoC



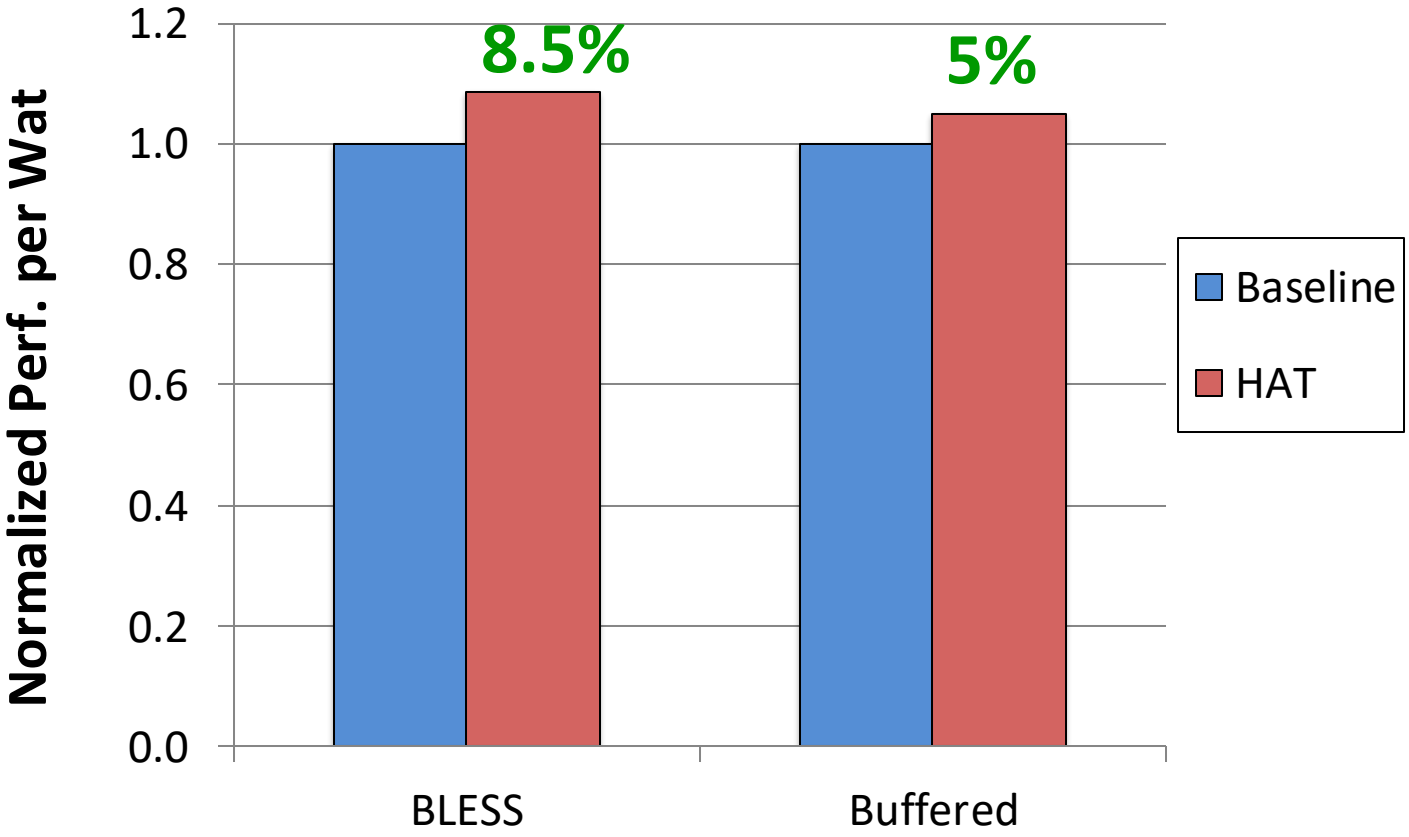
Congestion is much lower in Buffered NoC, but **HAT** still provides performance benefit

Application Fairness



HAT provides better fairness than prior works

Network Energy Efficiency



HAT increases energy efficiency by reducing congestion

Other Results in Paper

- Performance on **CHIPPER**
- Performance on **multithreaded** workloads
- Parameters sensitivity sweep of **HAT**

Conclusion

- **Problem:** Packets contend in on-chip networks (NoCs), causing congestion, thus reducing performance
- **Observations:**

  1) Some applications are more sensitive to network latency than others
  2) Applications must be throttled differently to achieve peak performance
- **Key Idea: Heterogeneous Adaptive Throttling (HAT)**
  1) Application-aware source throttling
  2) Network-load-aware throttling rate adjustment
- **Result:** Improves performance and energy efficiency over state-of-the-art source throttling policies

Source Throttling in Bufferless NoCs

- Kevin Chang, Rachata Ausavarungnirun, Chris Fallin, and Onur Mutlu, **"HAT: Heterogeneous Adaptive Throttling for On-Chip Networks"** *Proceedings of the 24th International Symposium on Computer Architecture and High Performance Computing (**SBAC-PAD**)*, New York, NY, October 2012. Slides (pptx) (pdf)

HAT: Heterogeneous Adaptive Throttling for On-Chip Networks

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